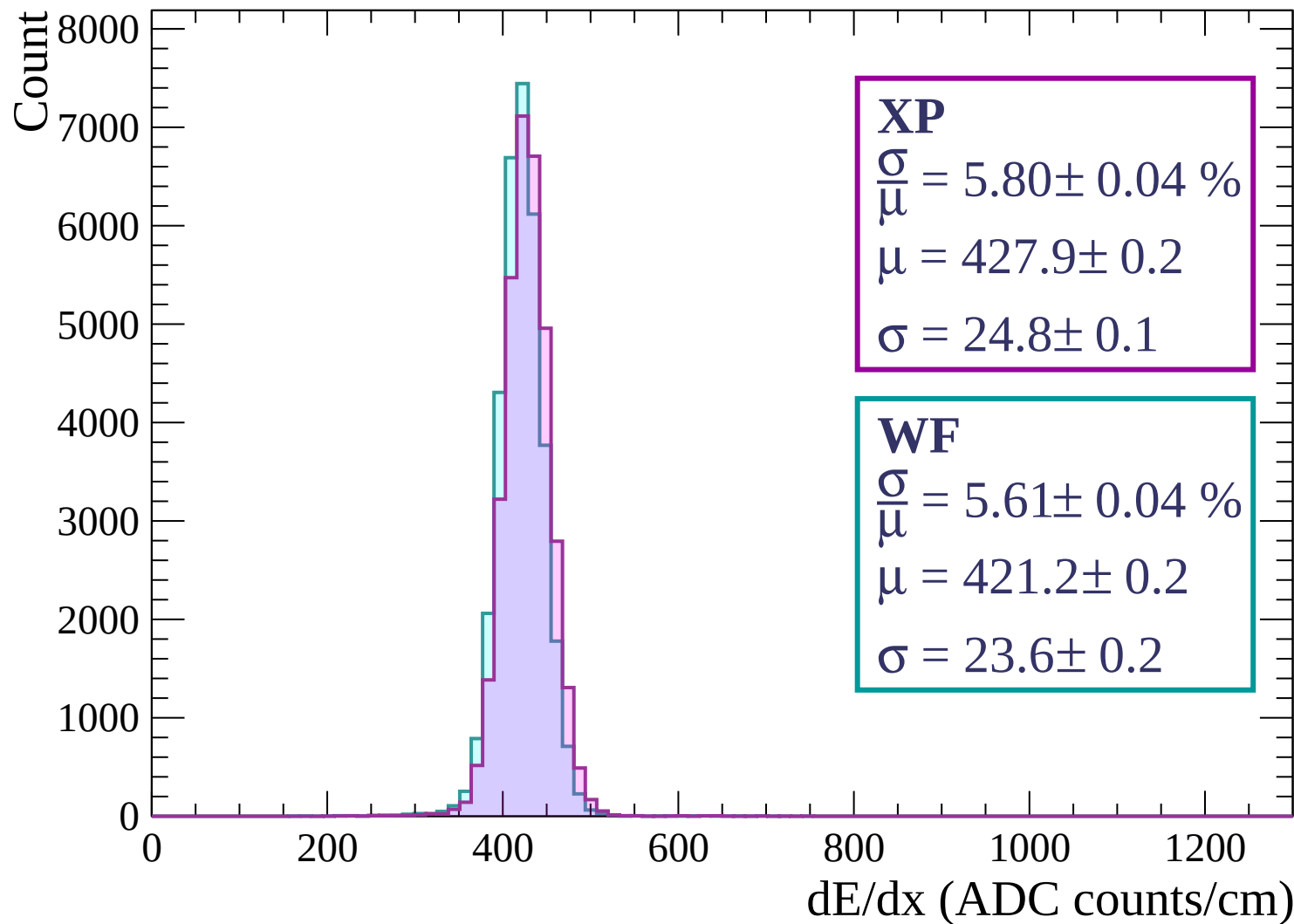
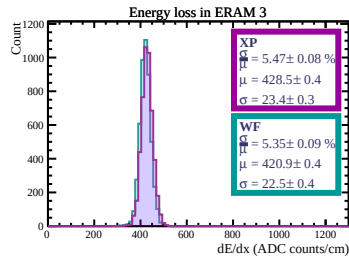
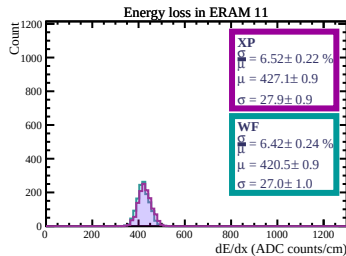
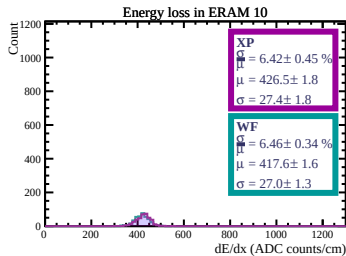
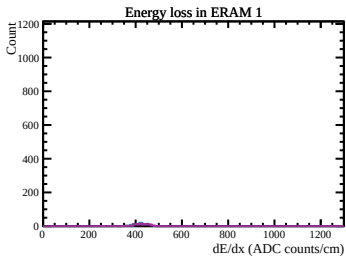
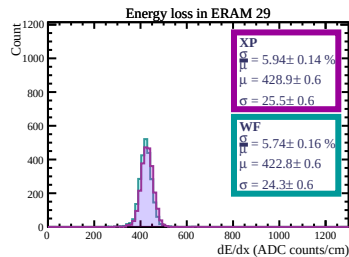
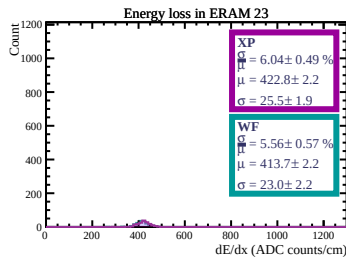
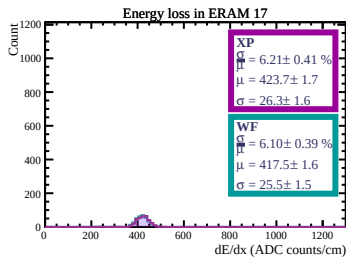
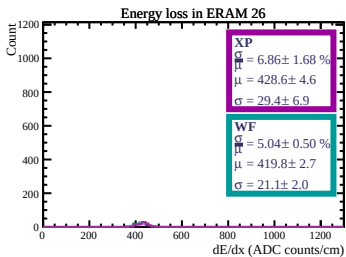
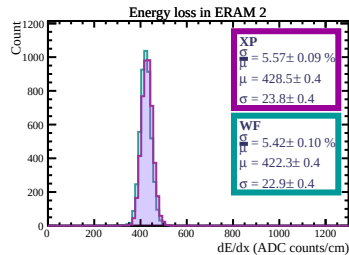
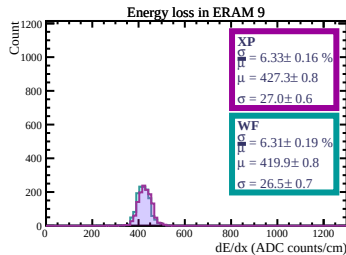
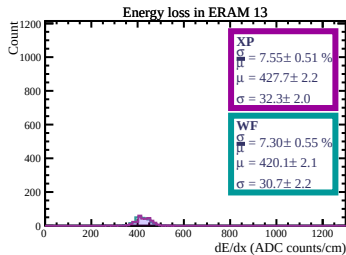
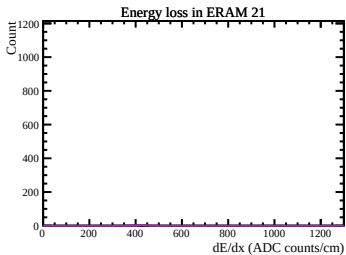
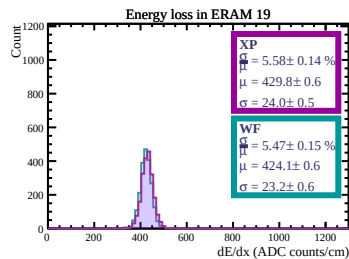
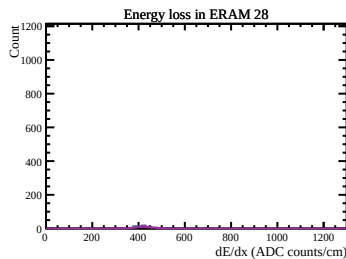
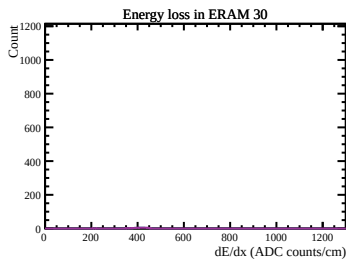
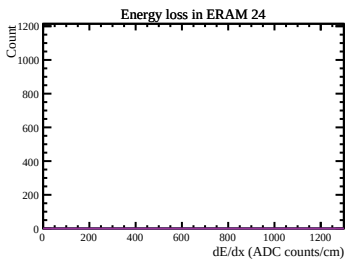
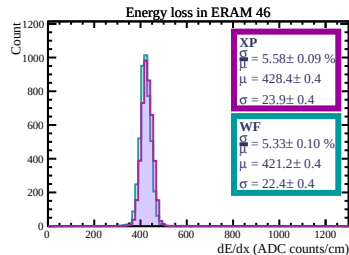
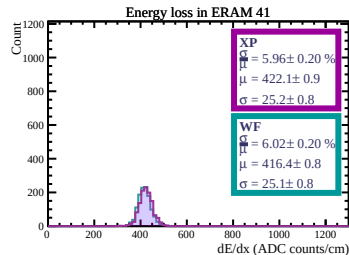
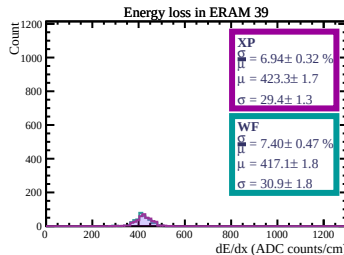
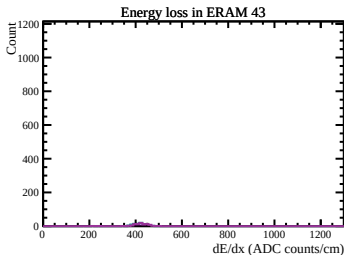
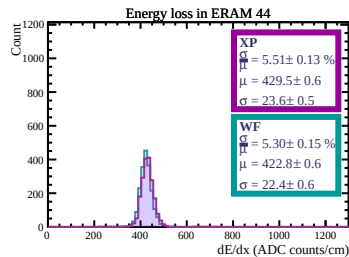
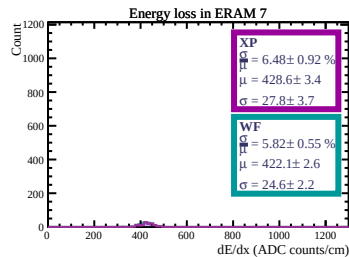
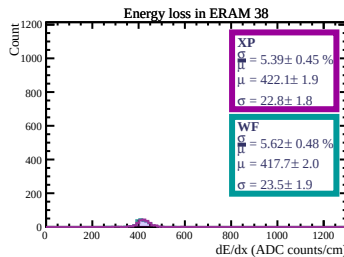
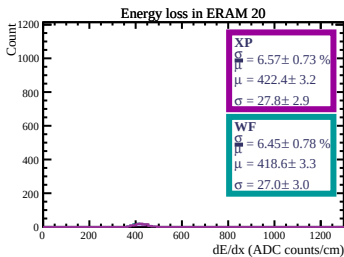
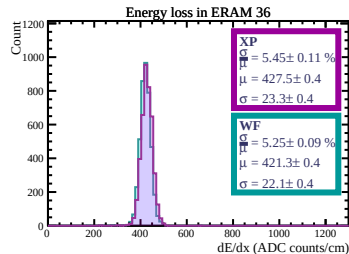
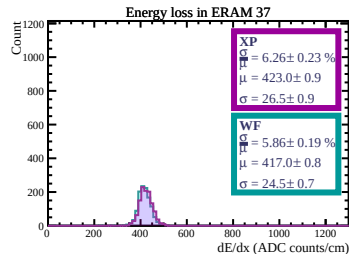
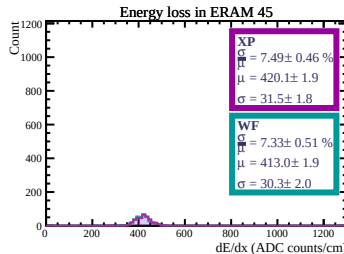
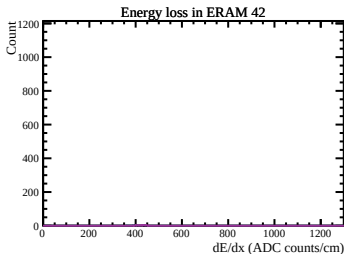
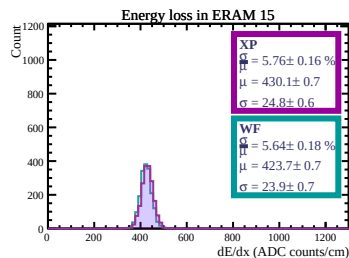
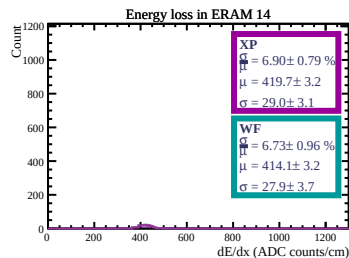
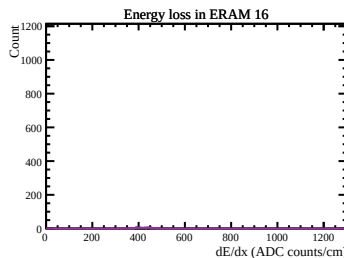
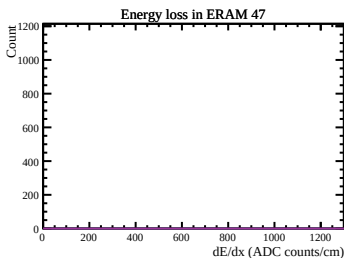
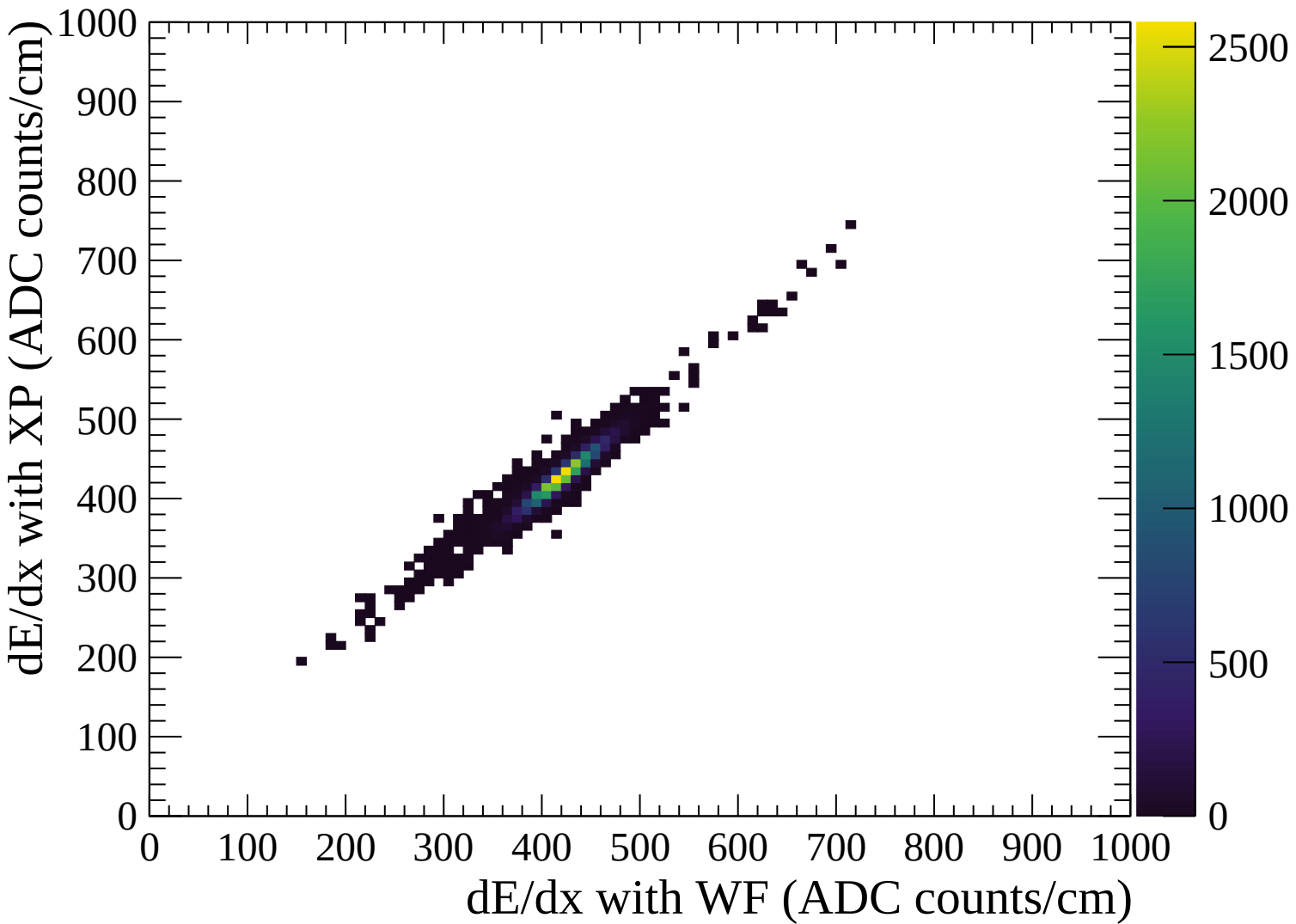
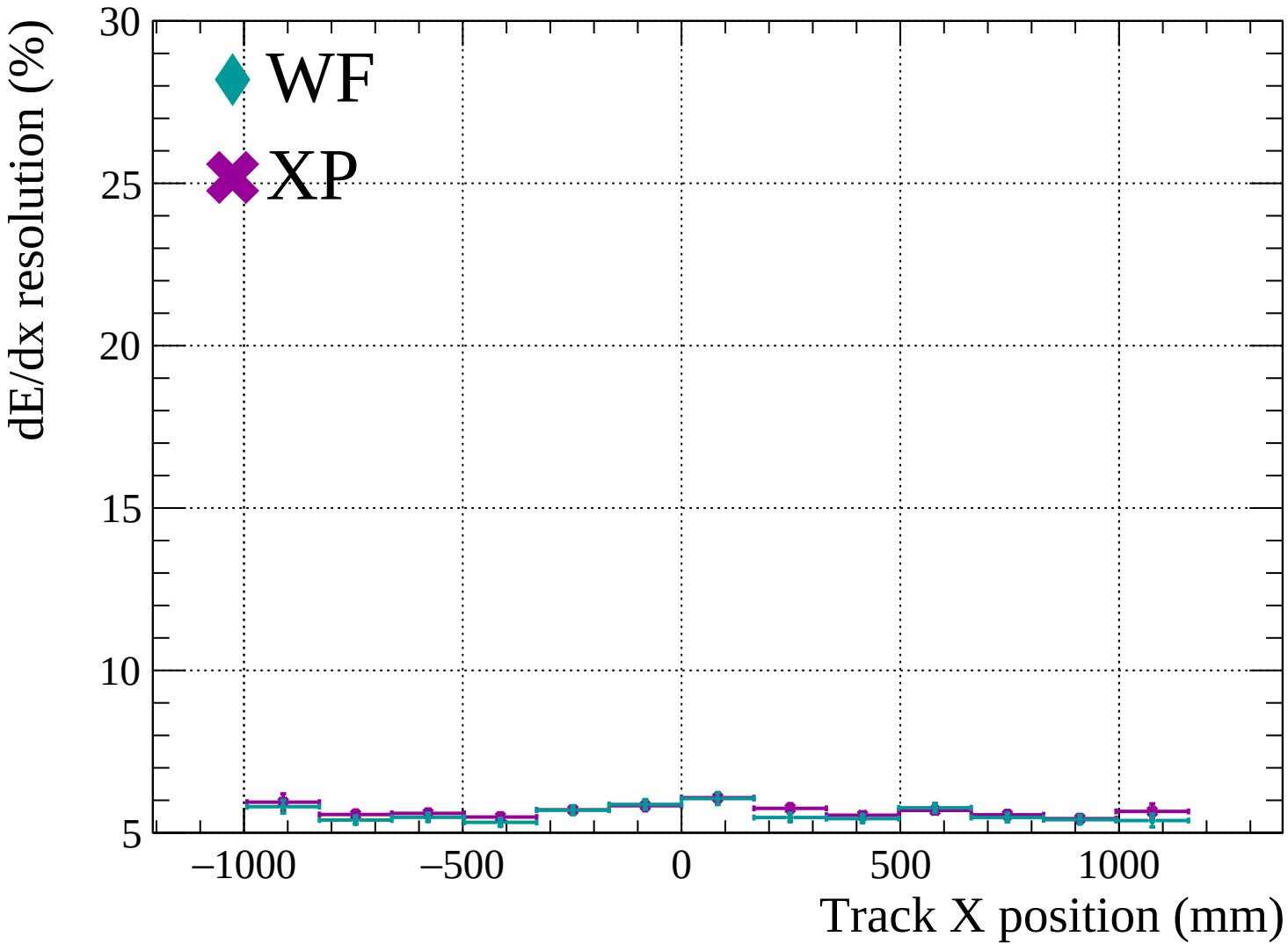


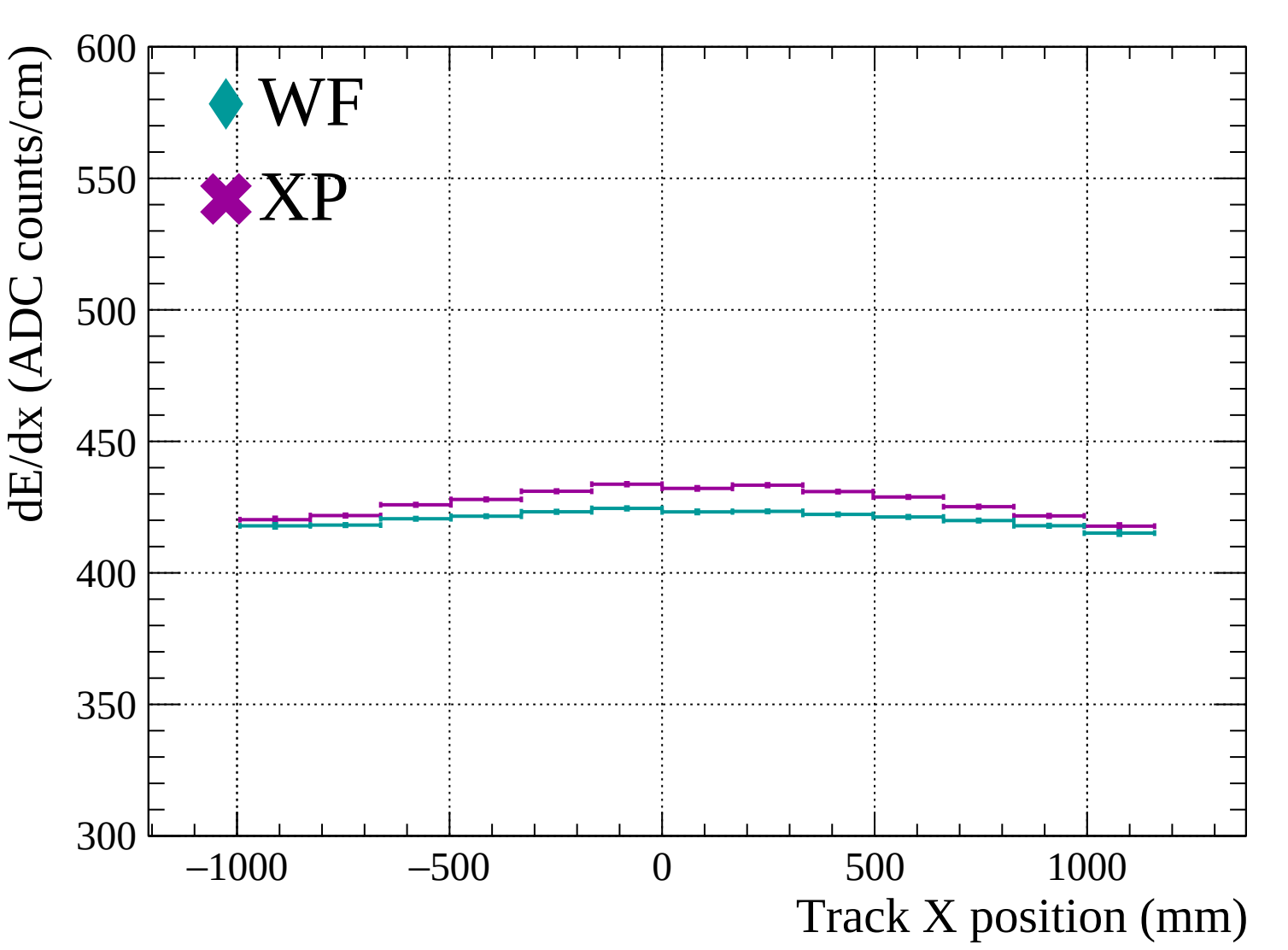


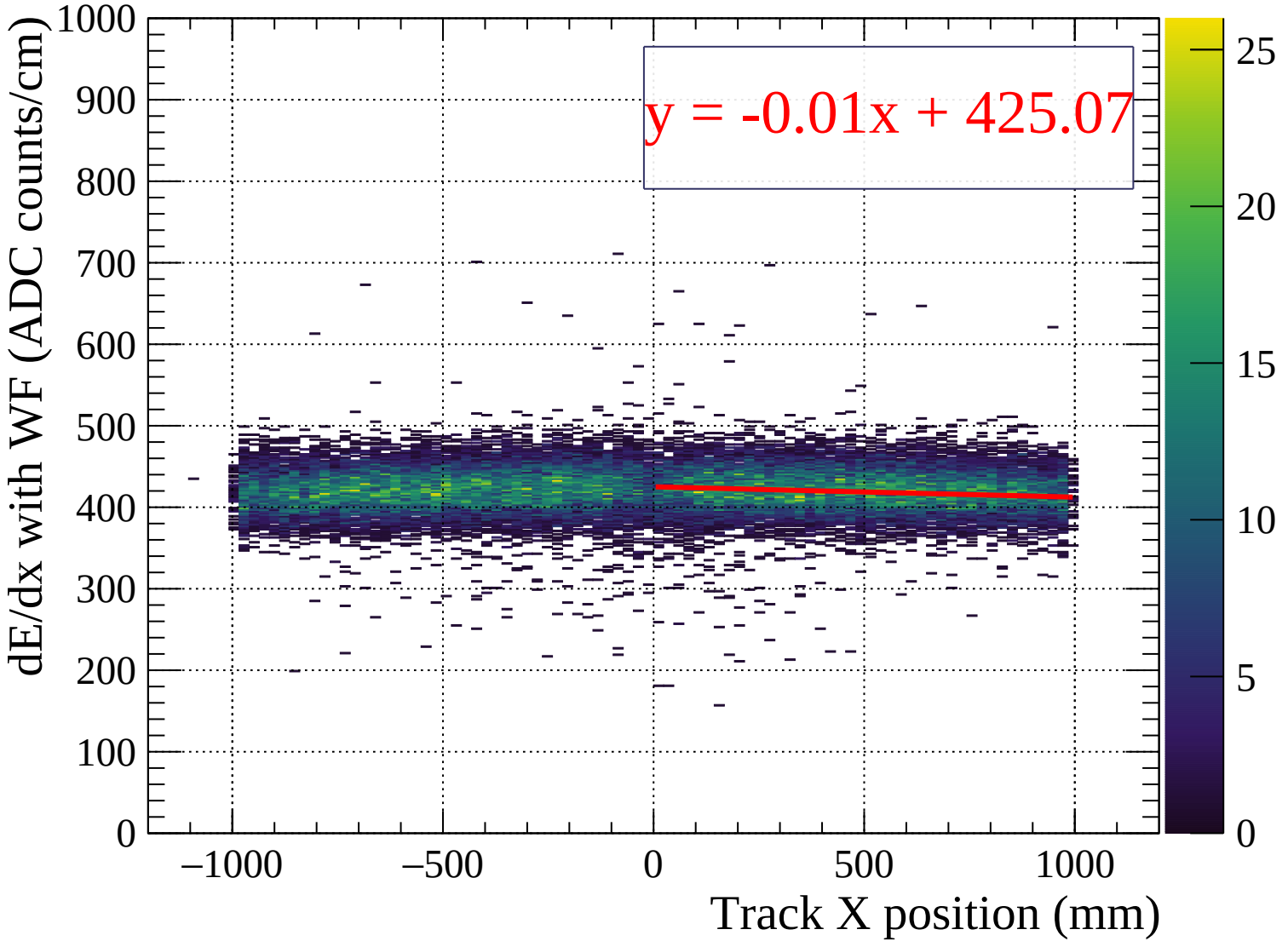


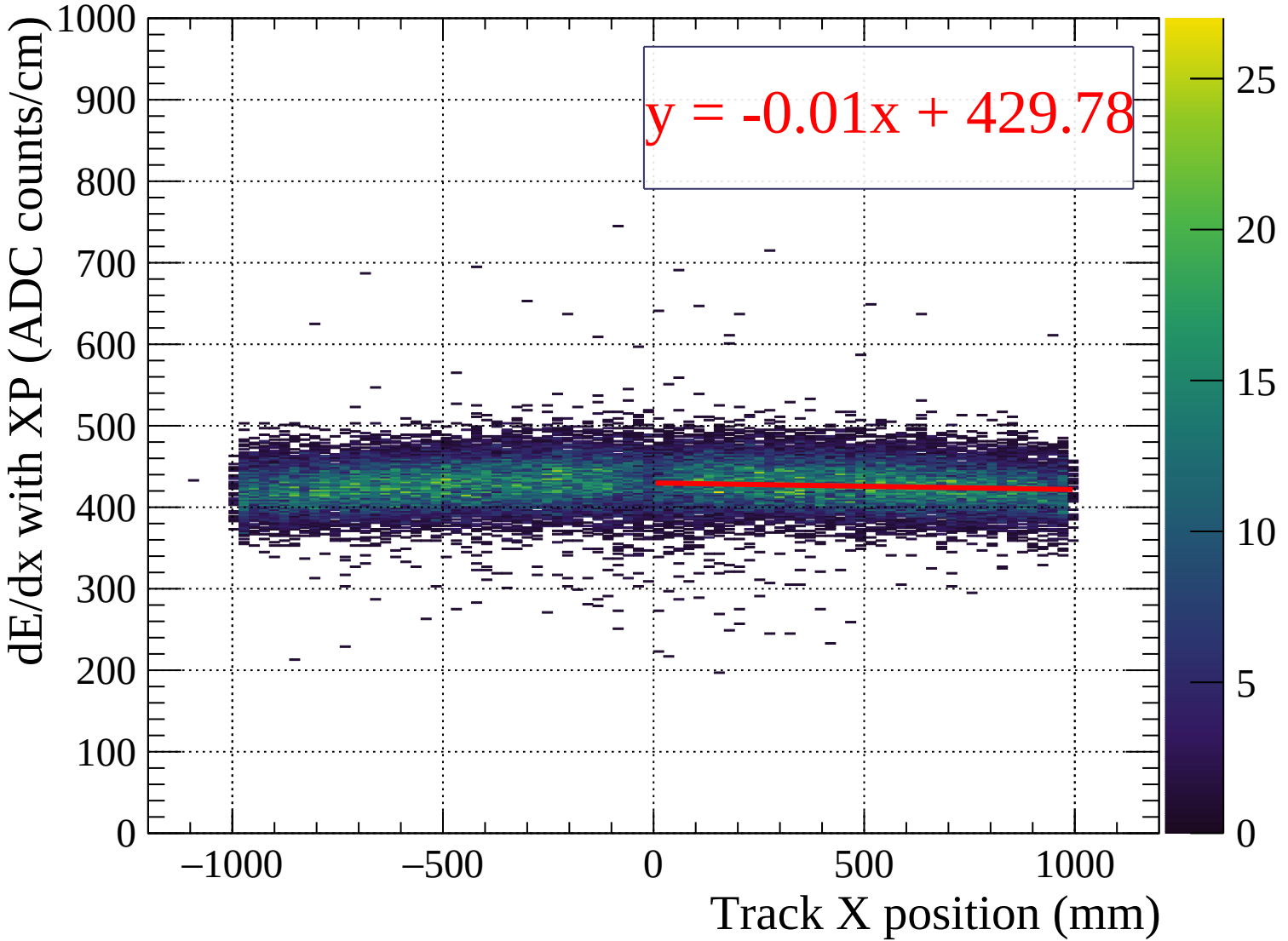


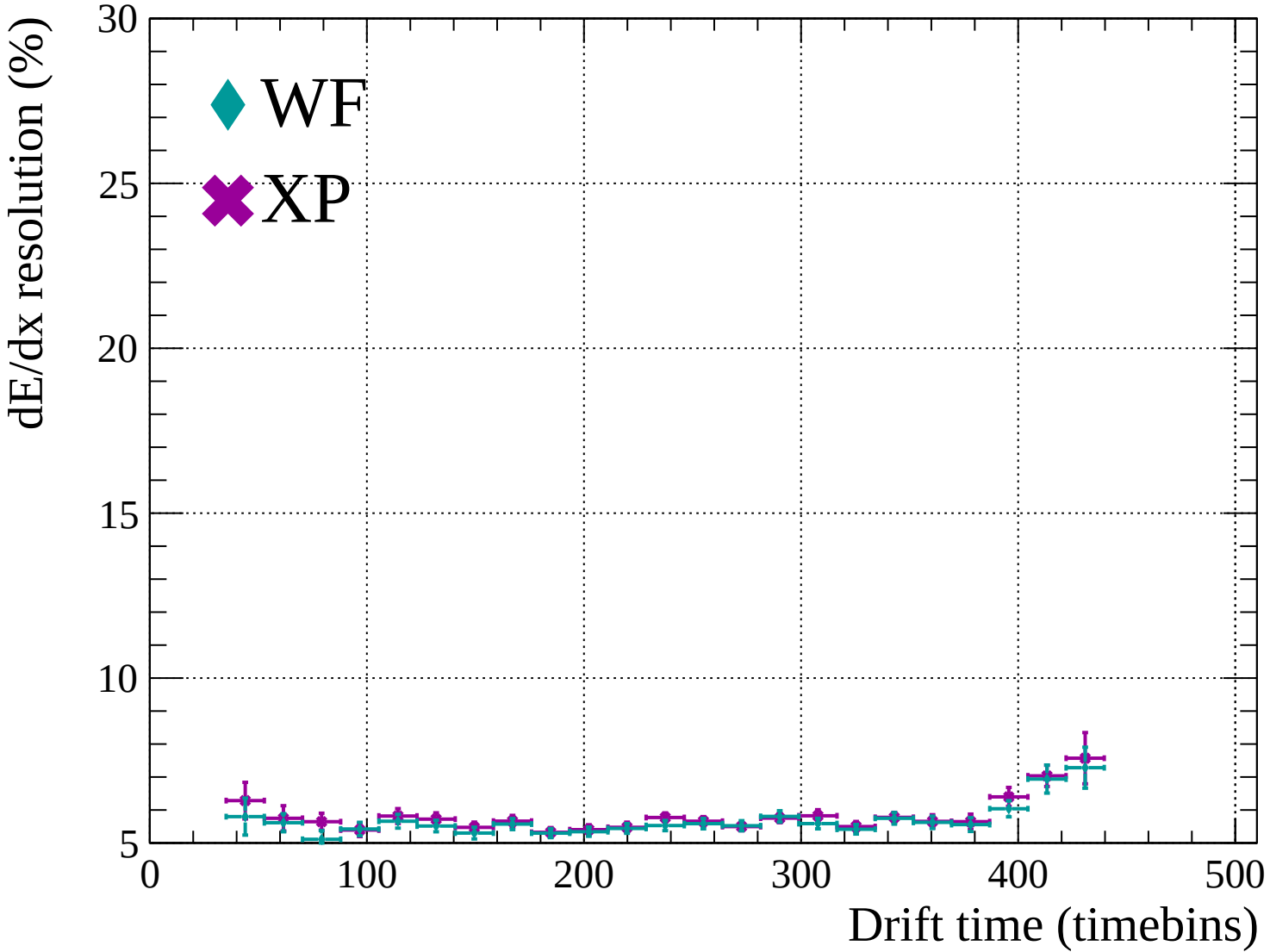


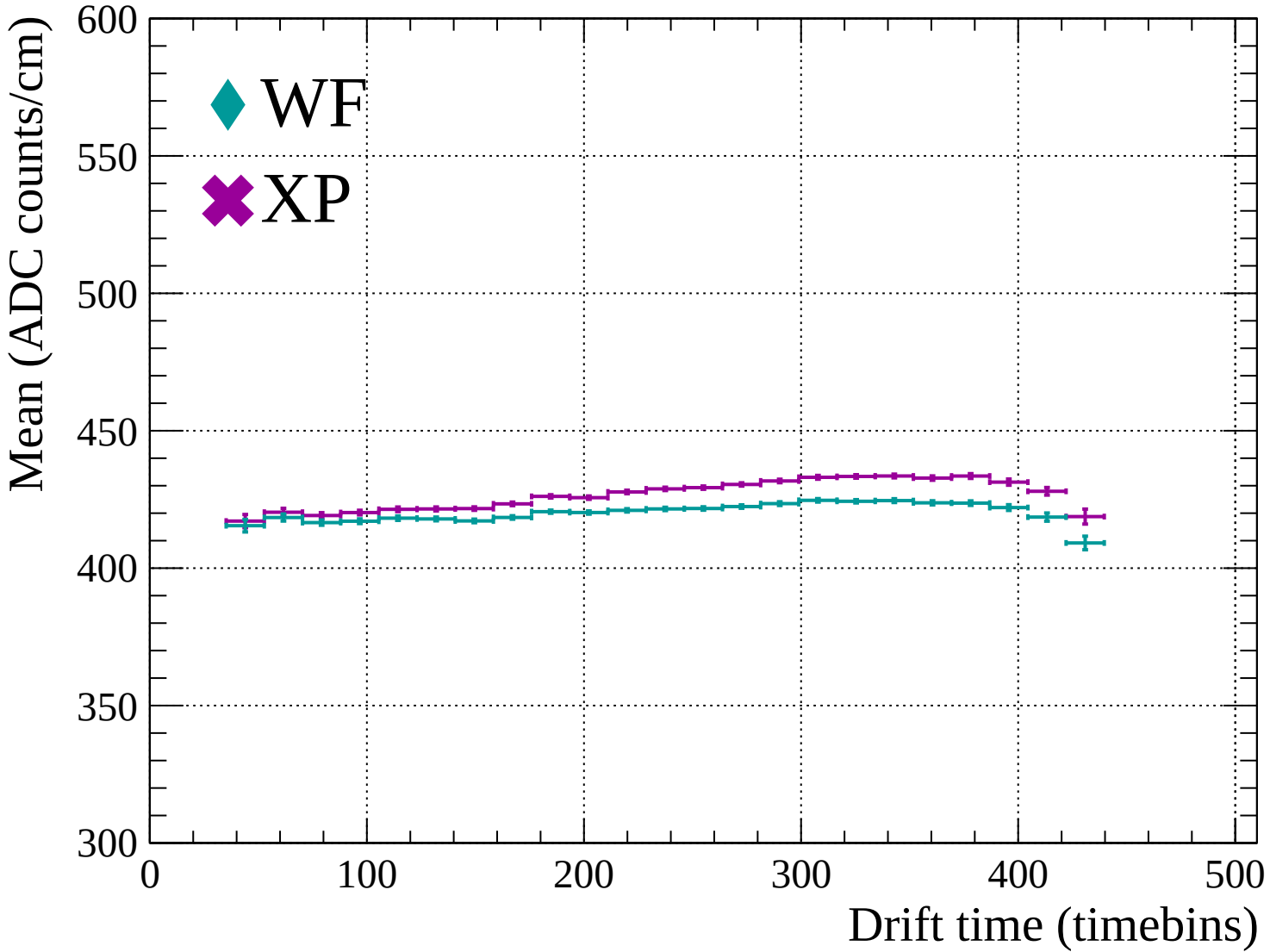


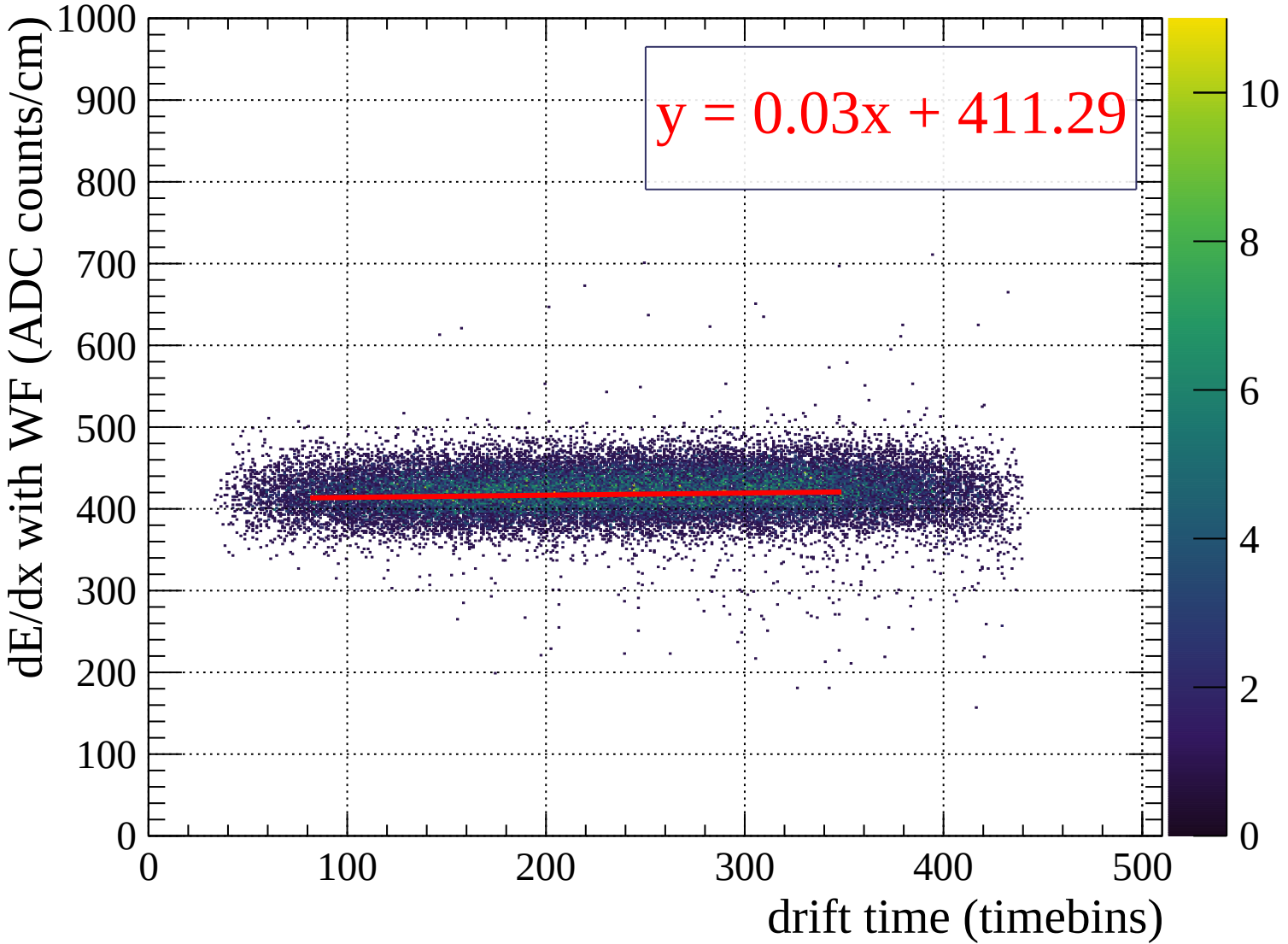


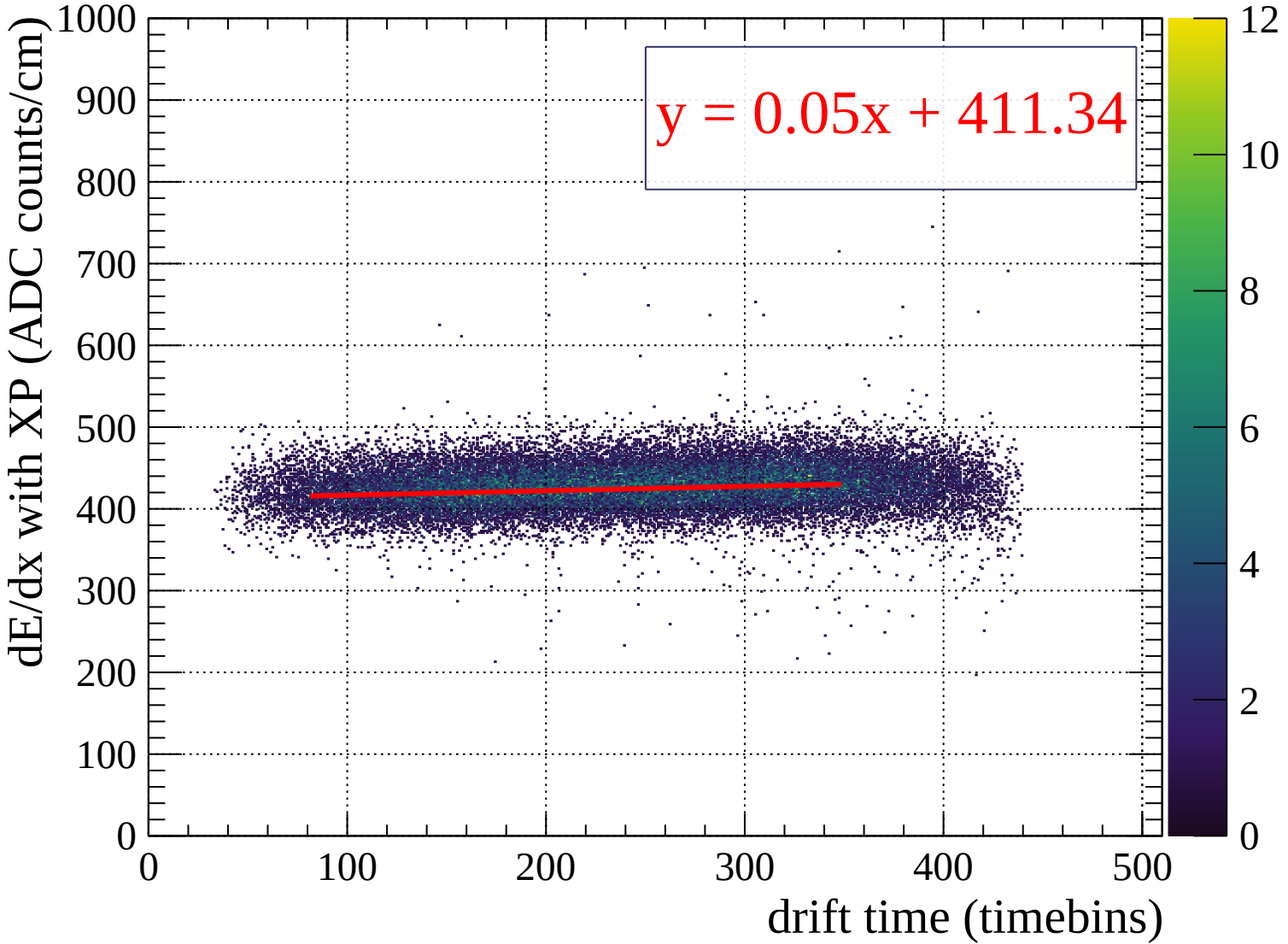


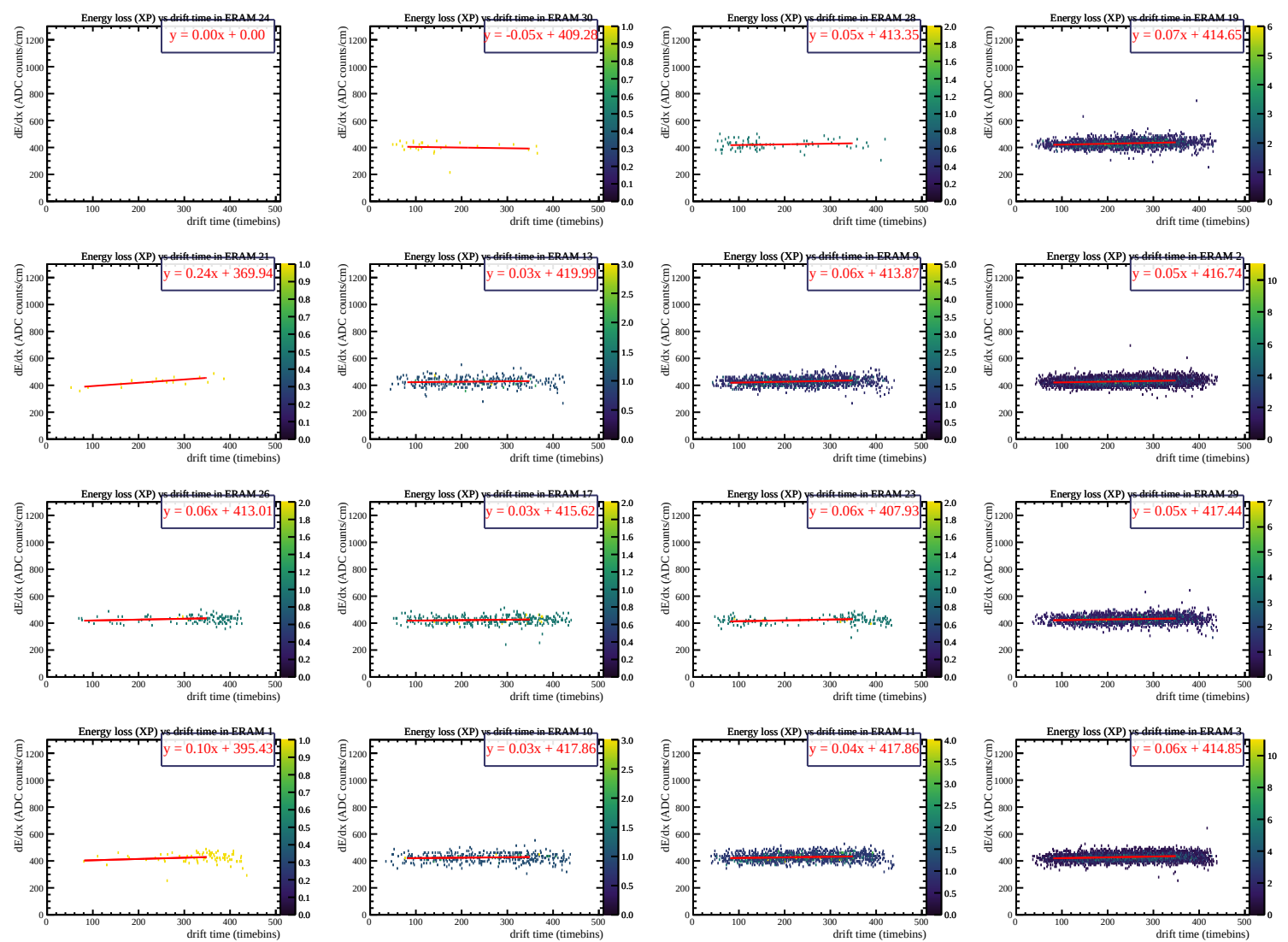


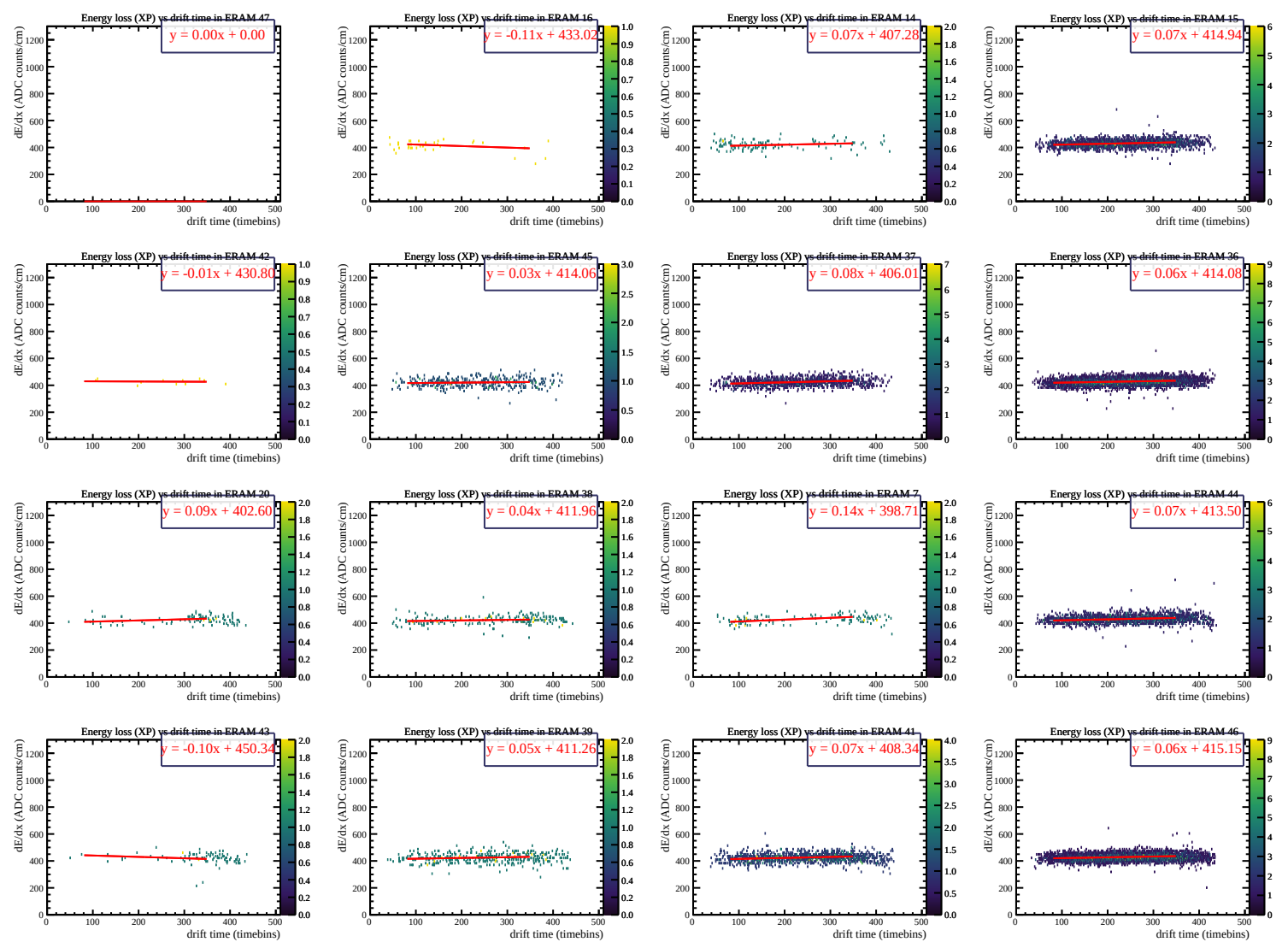


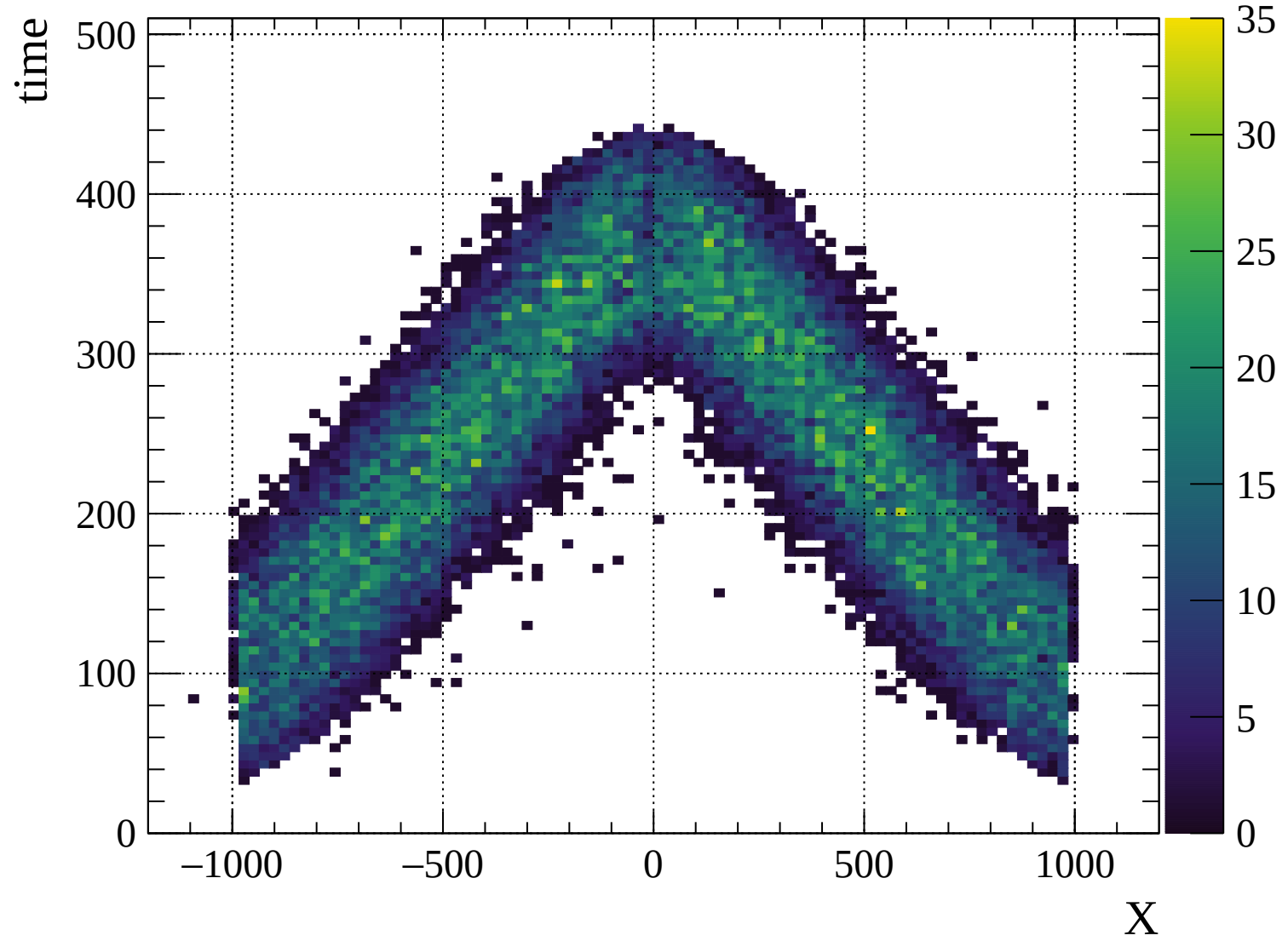


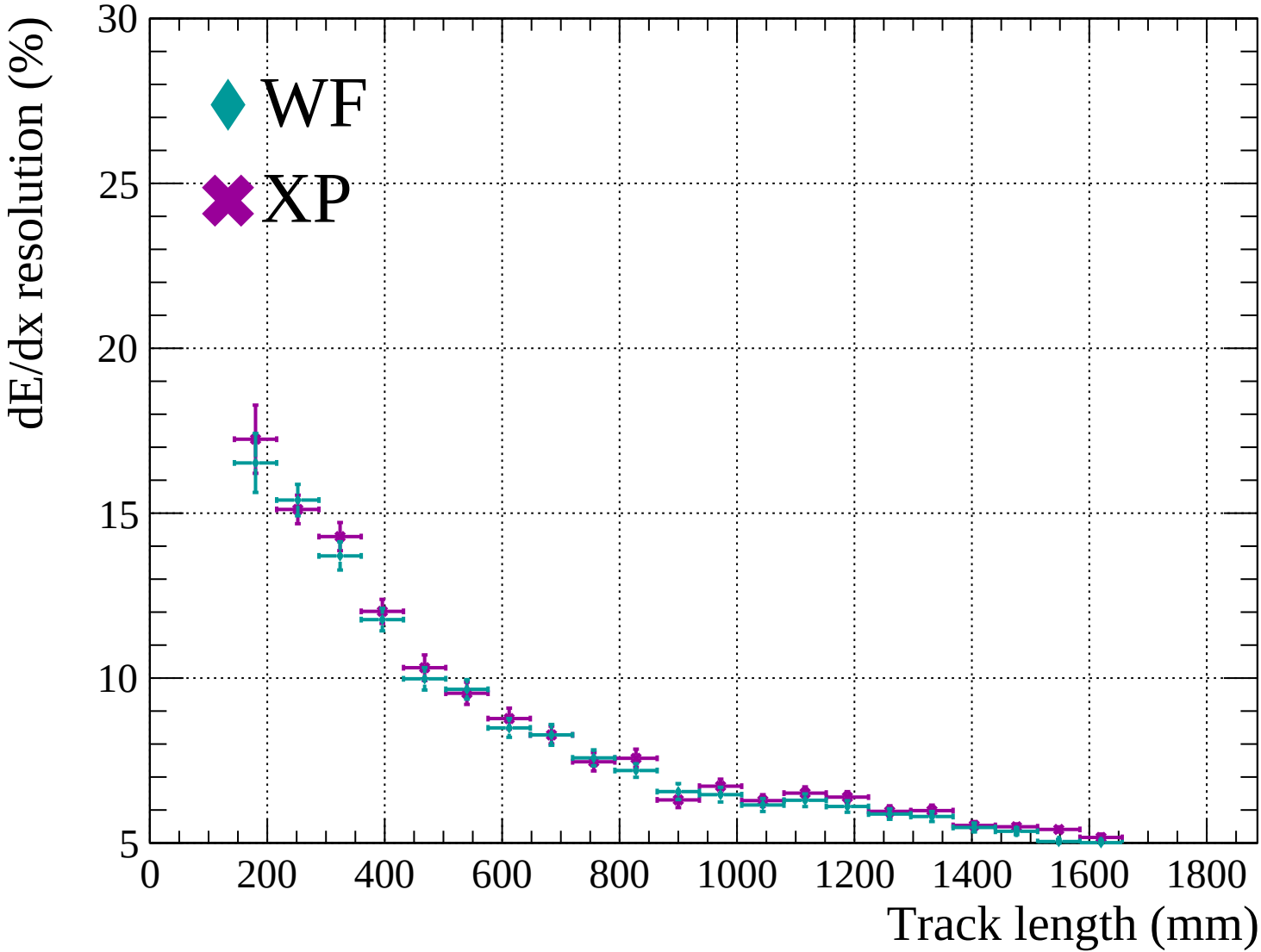


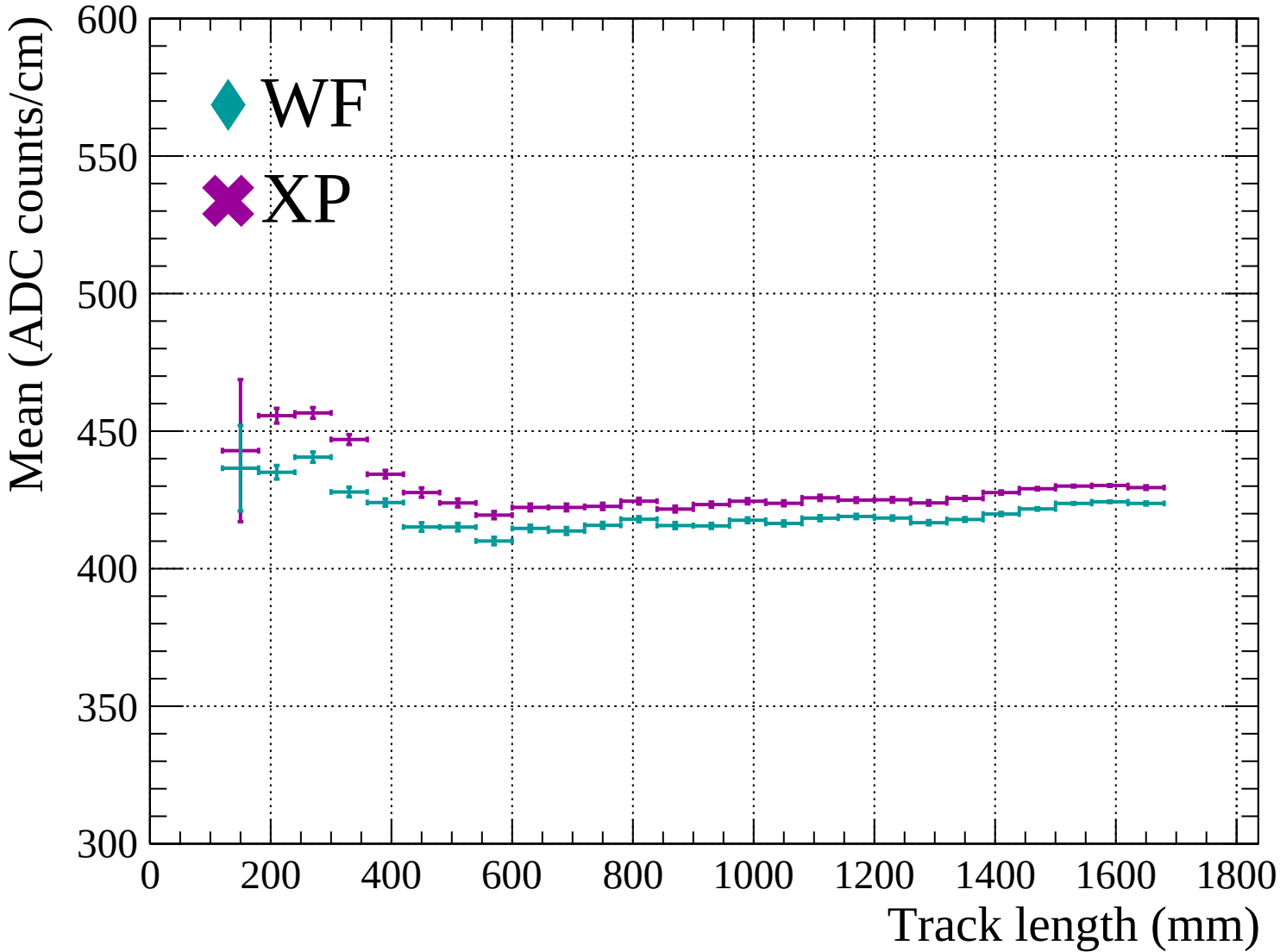


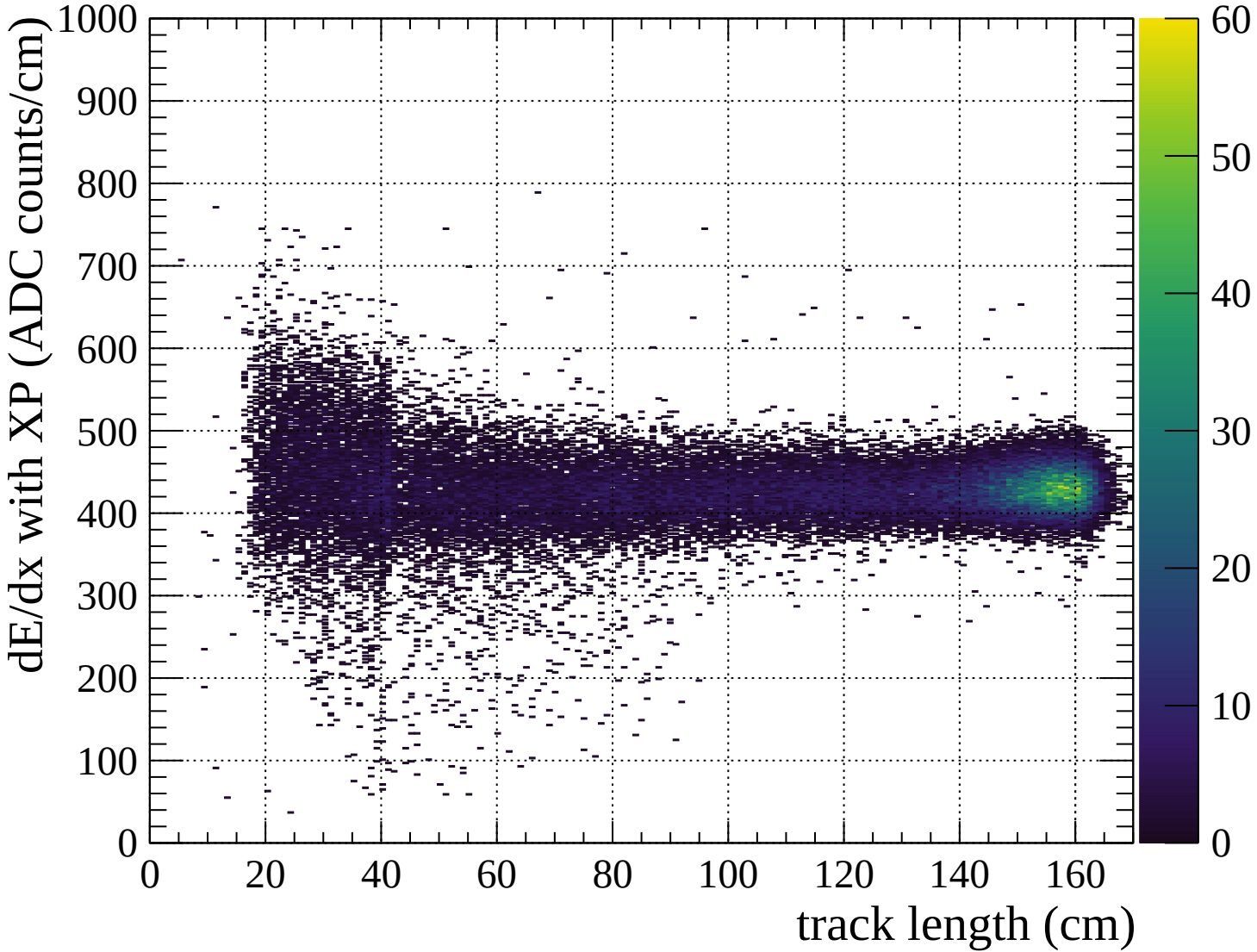


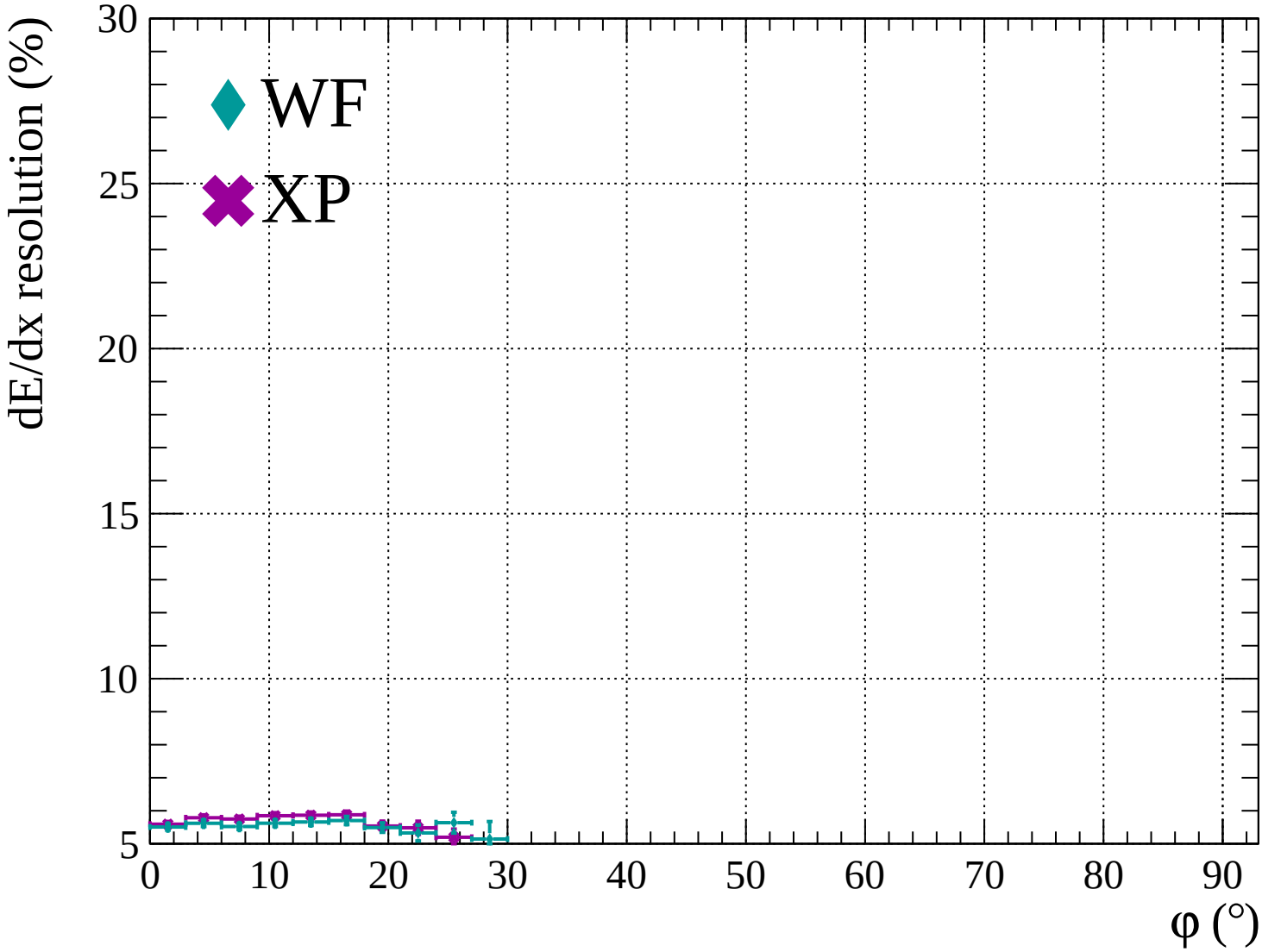


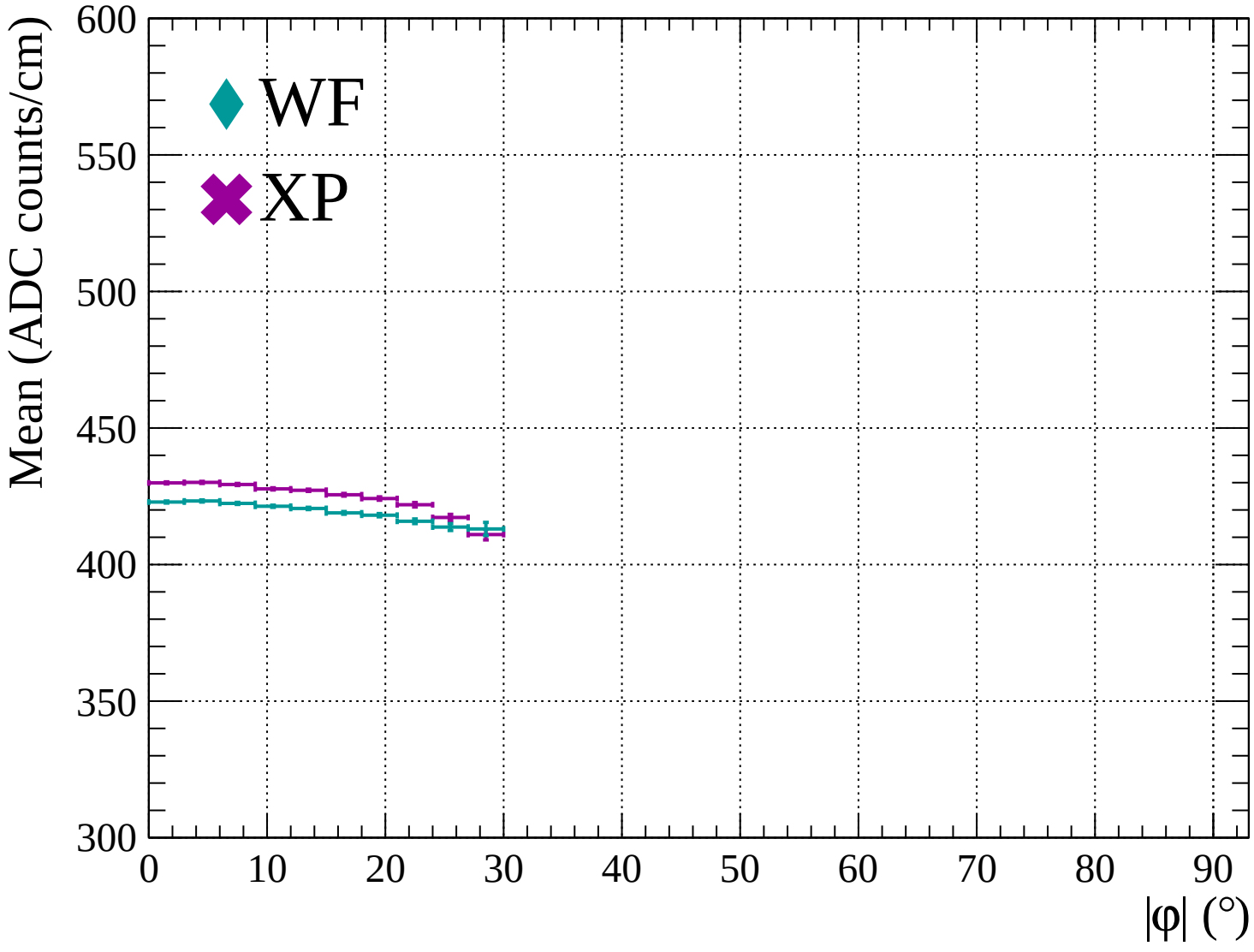


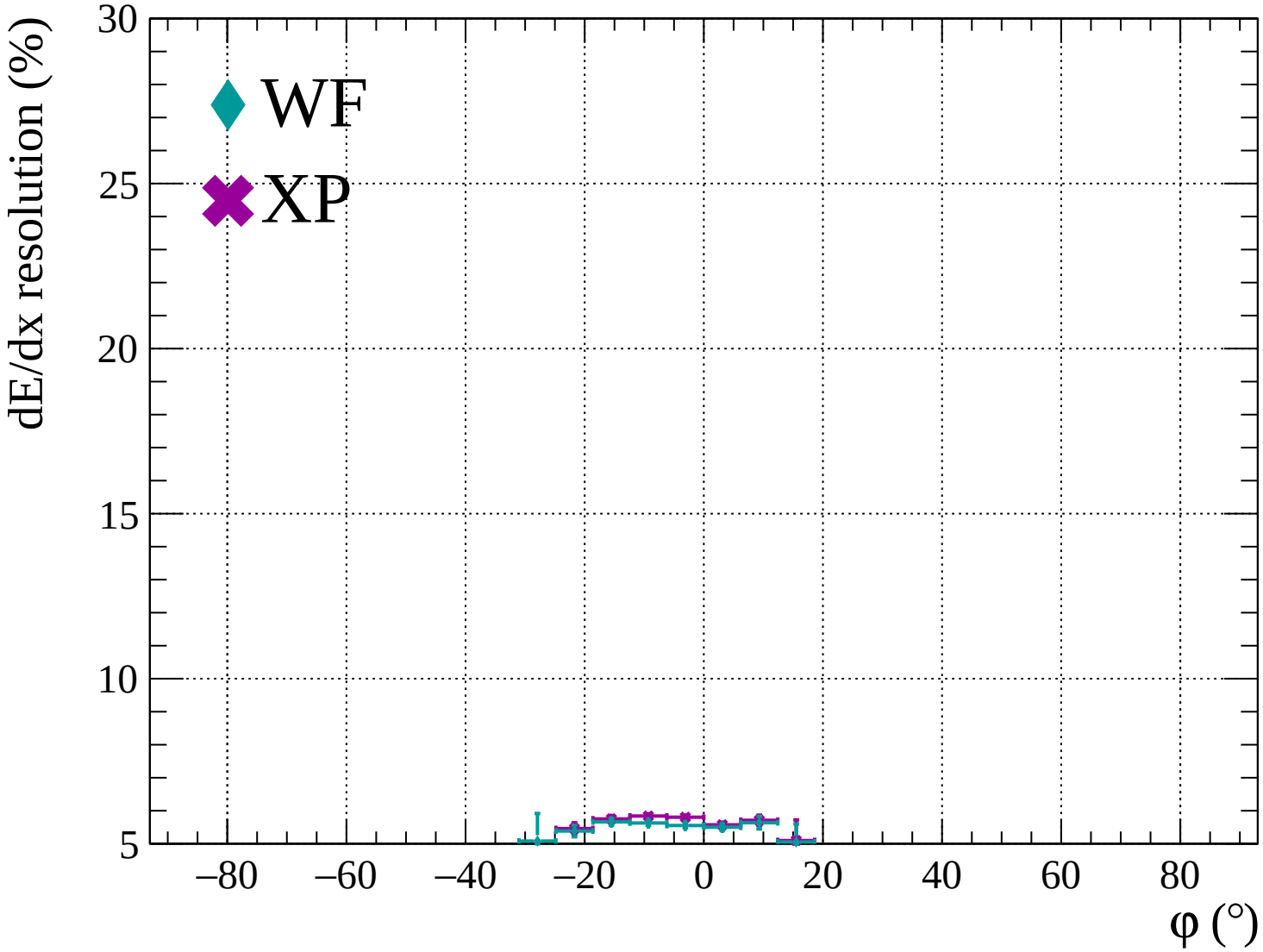


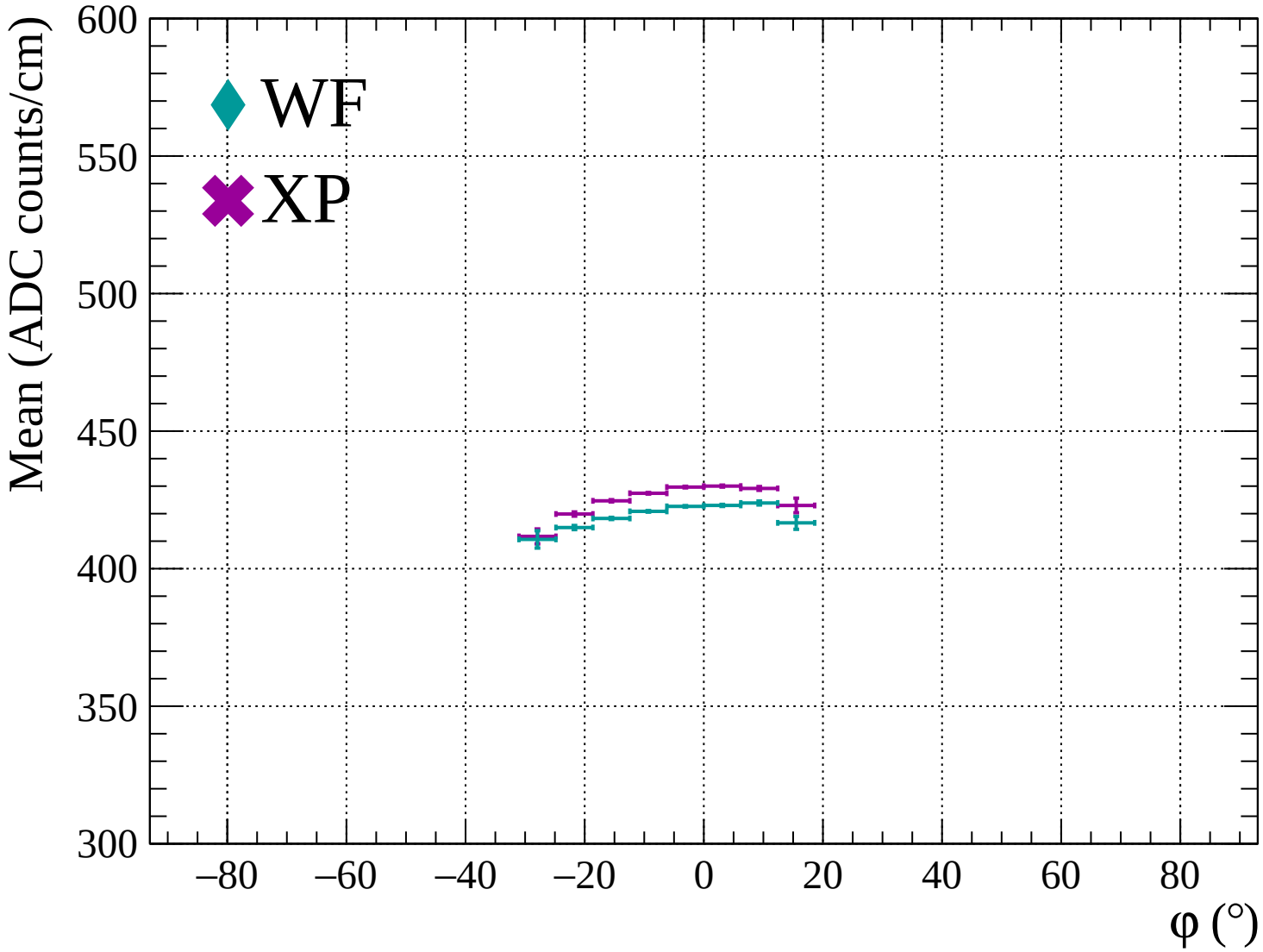


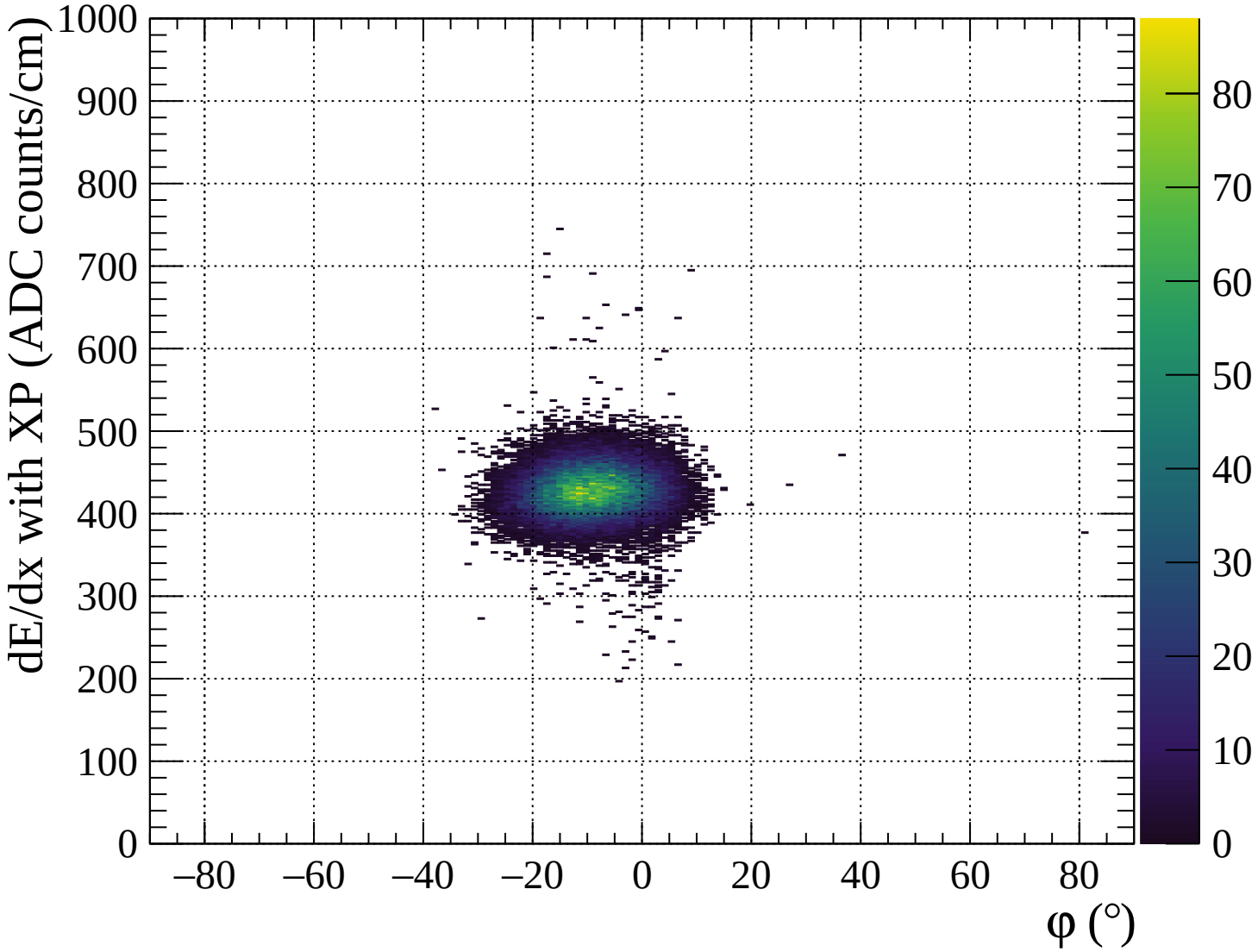


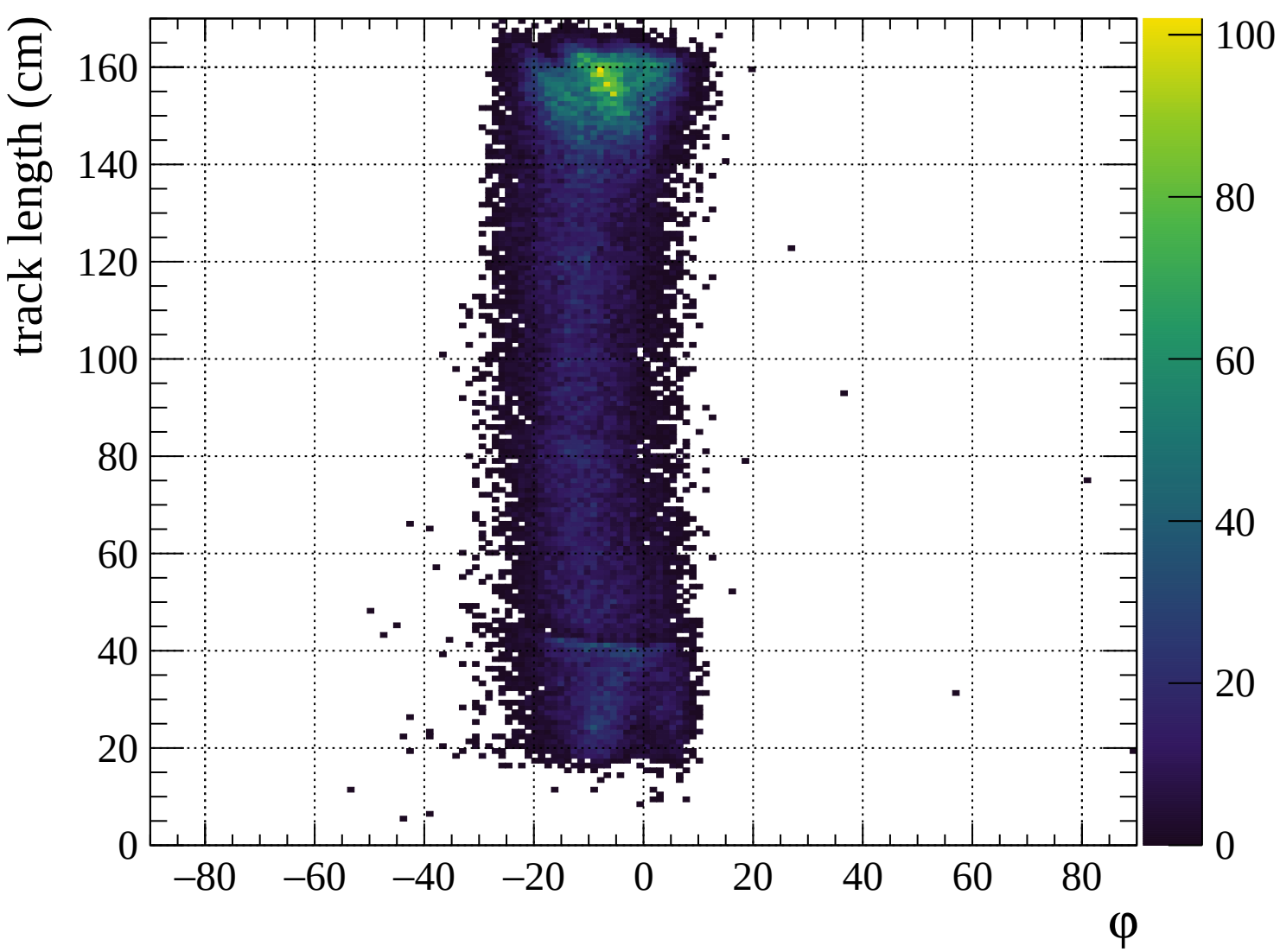


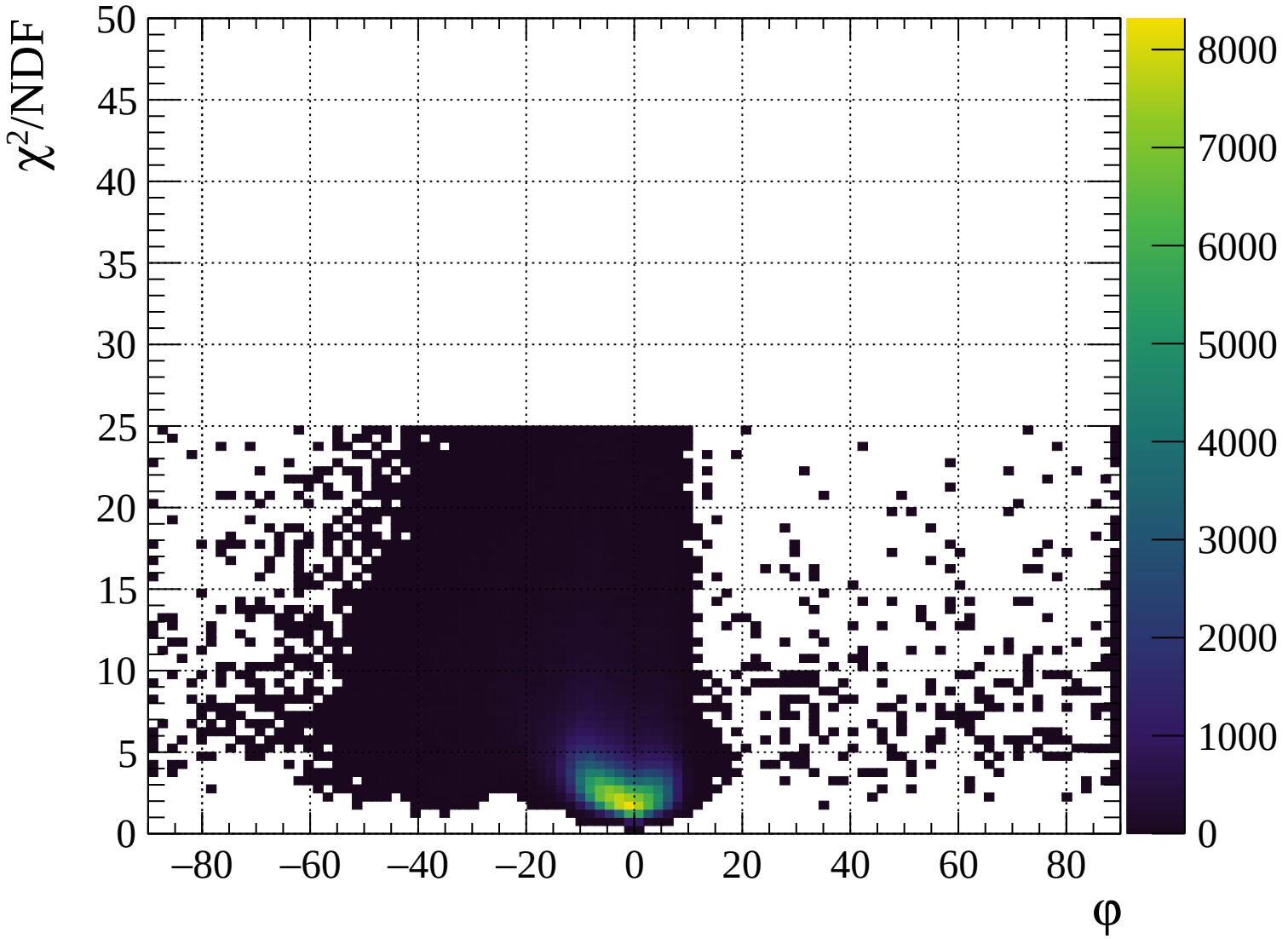


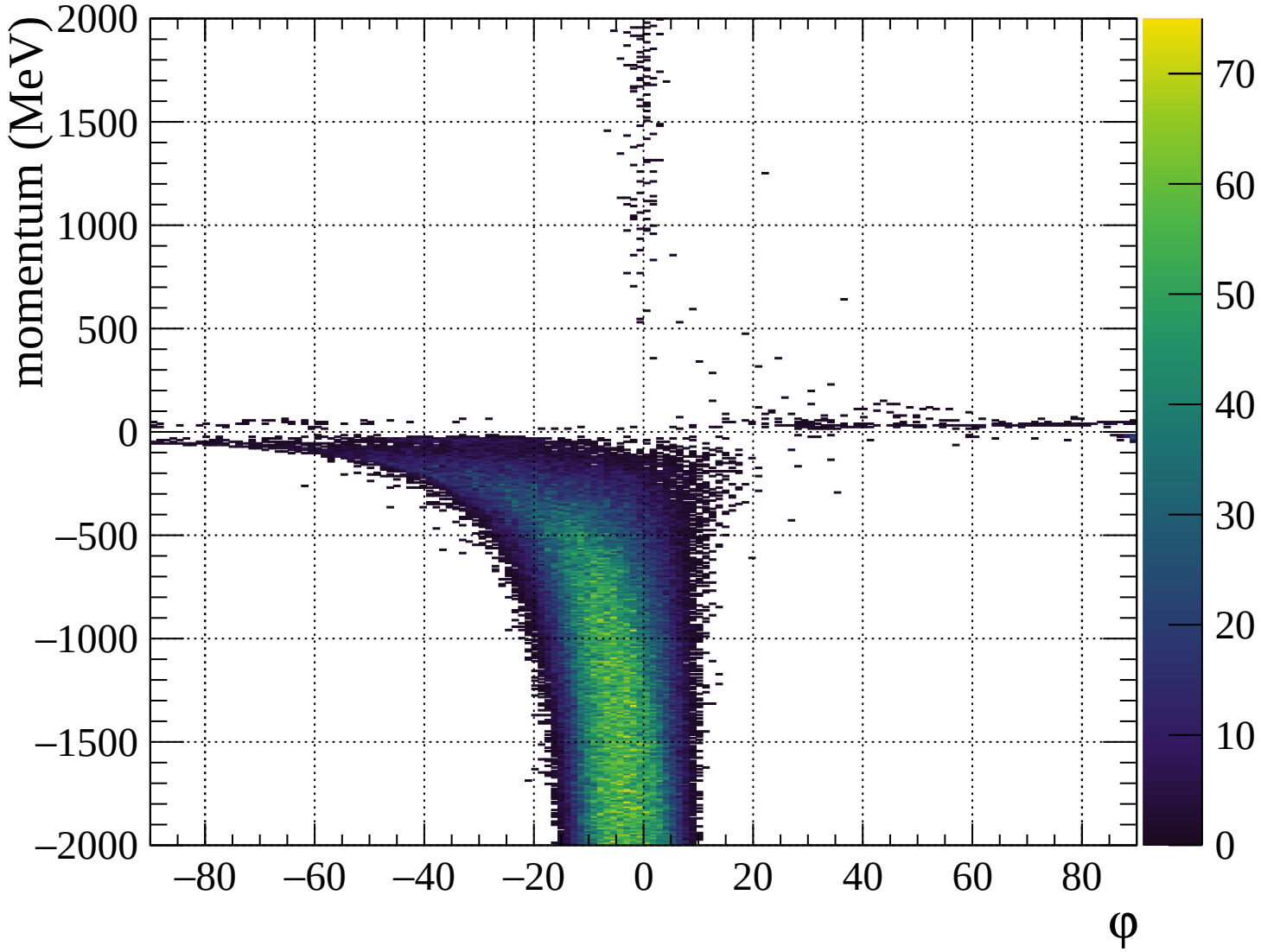


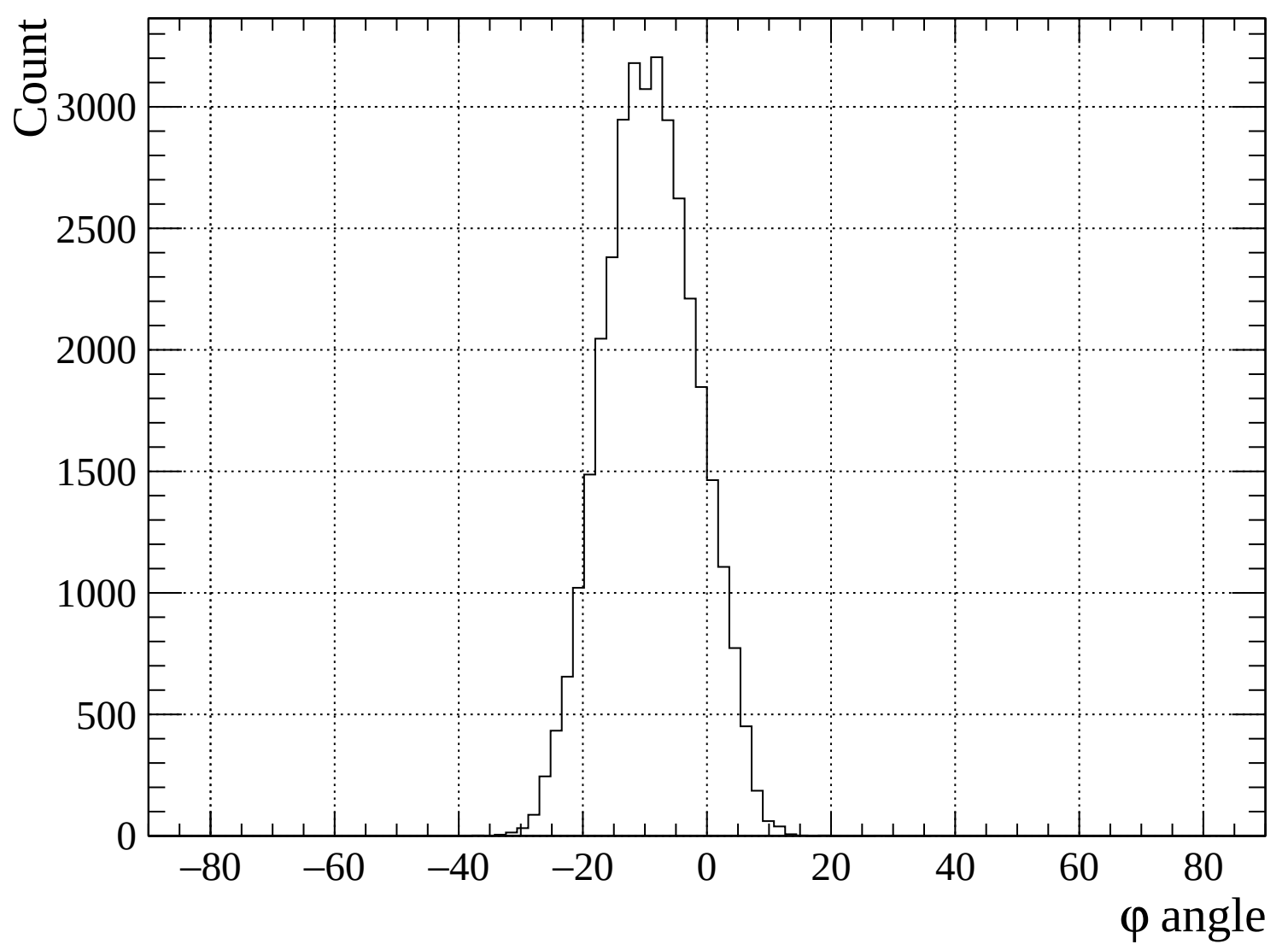


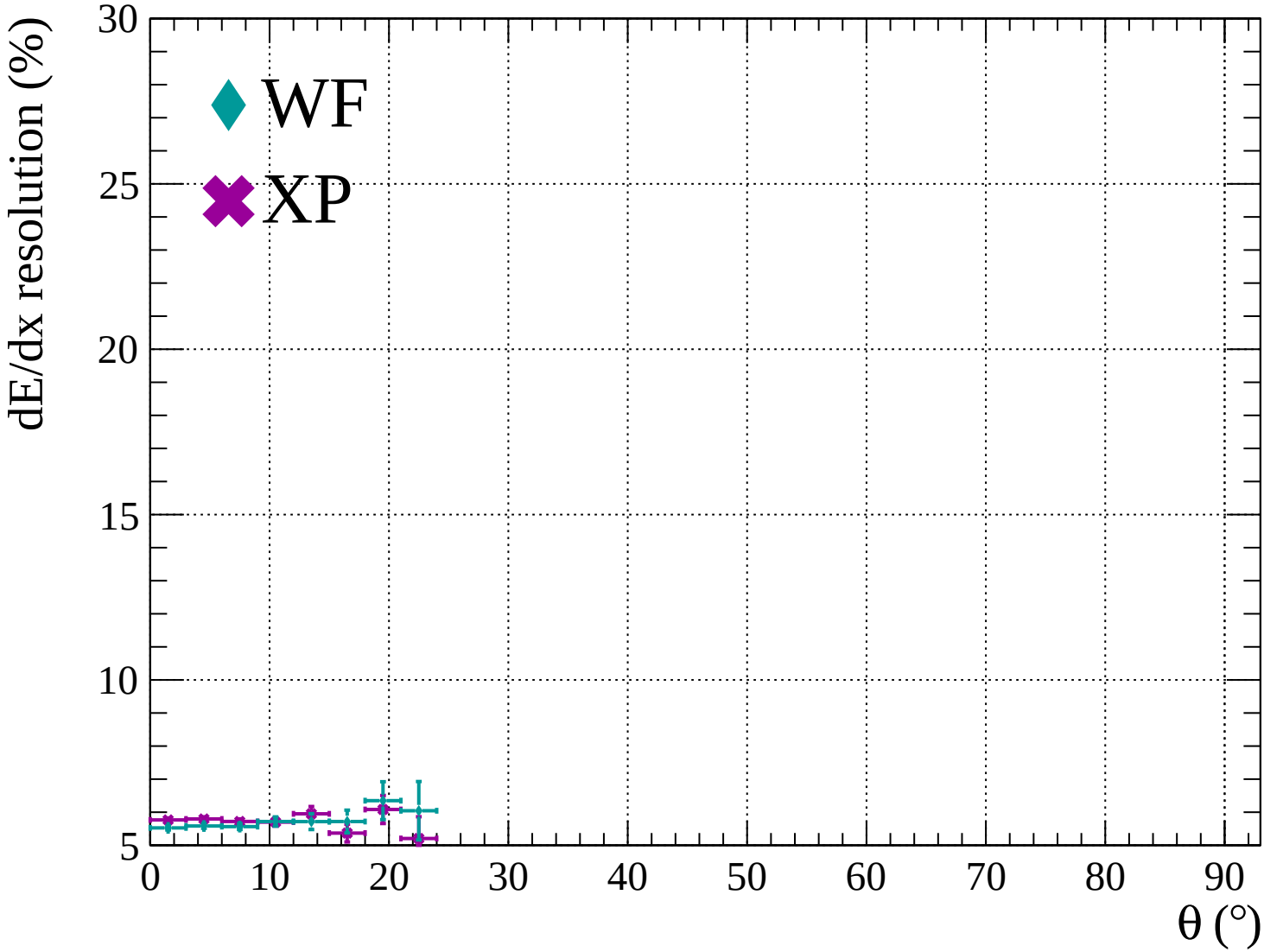


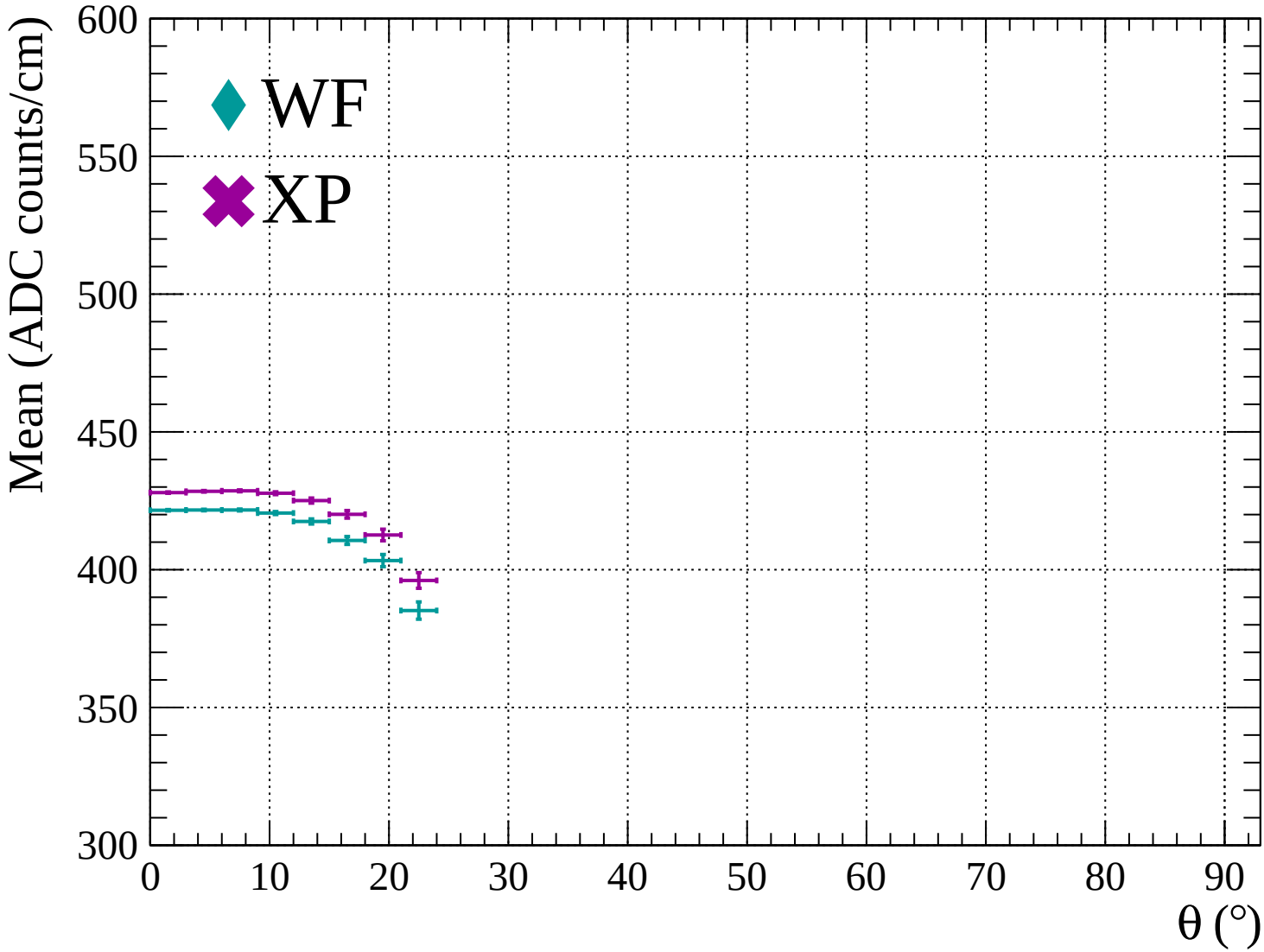


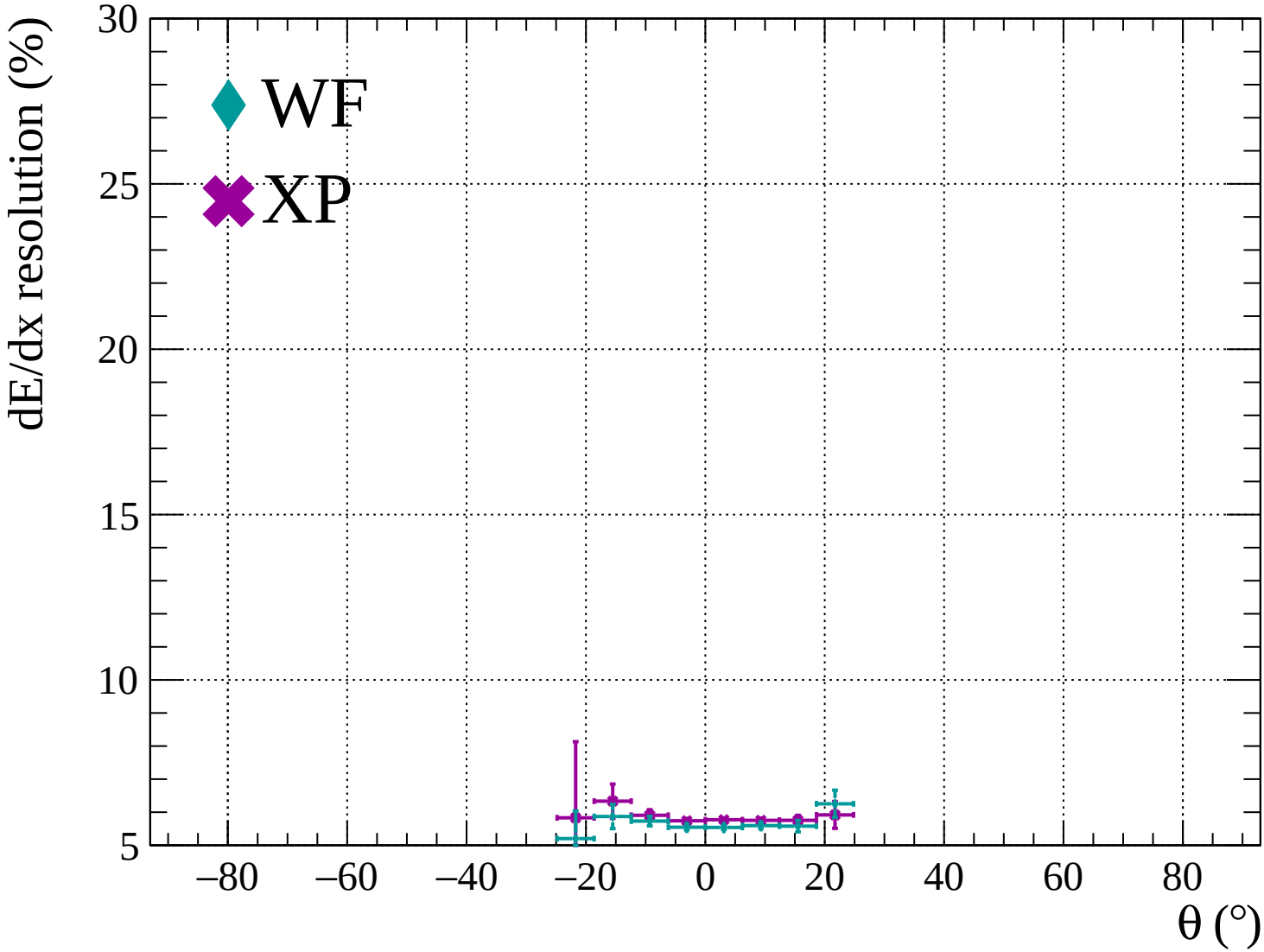


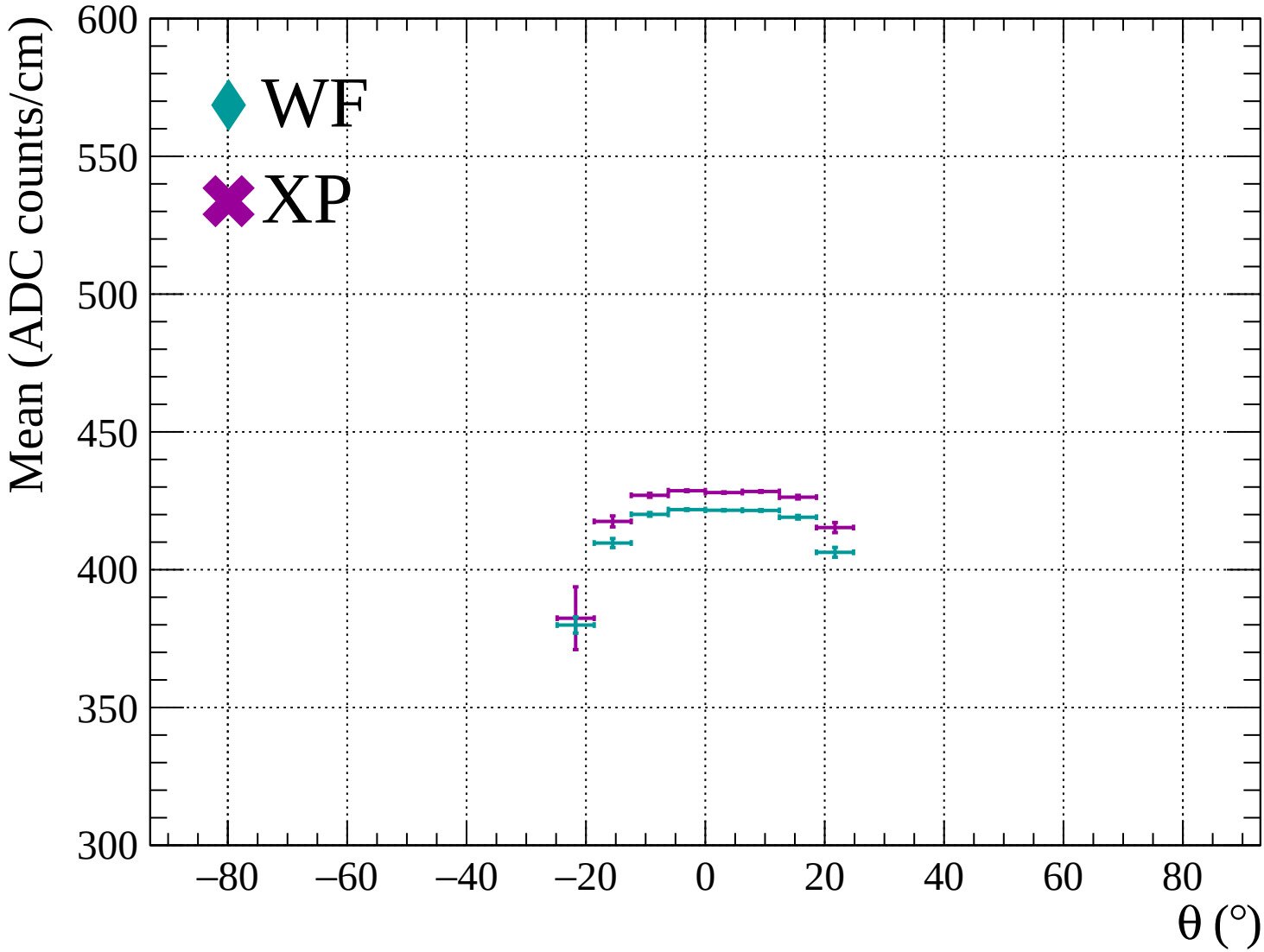


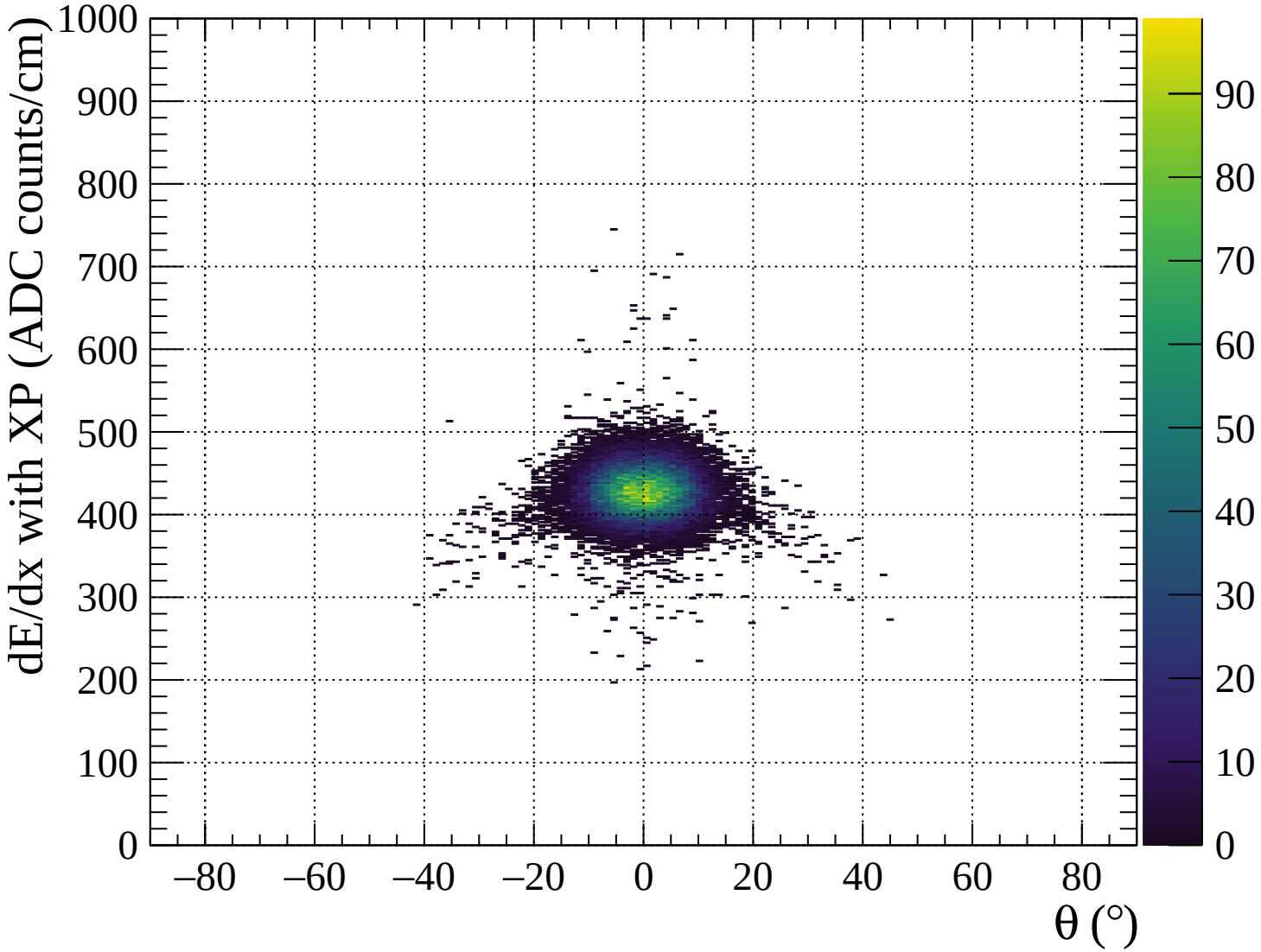


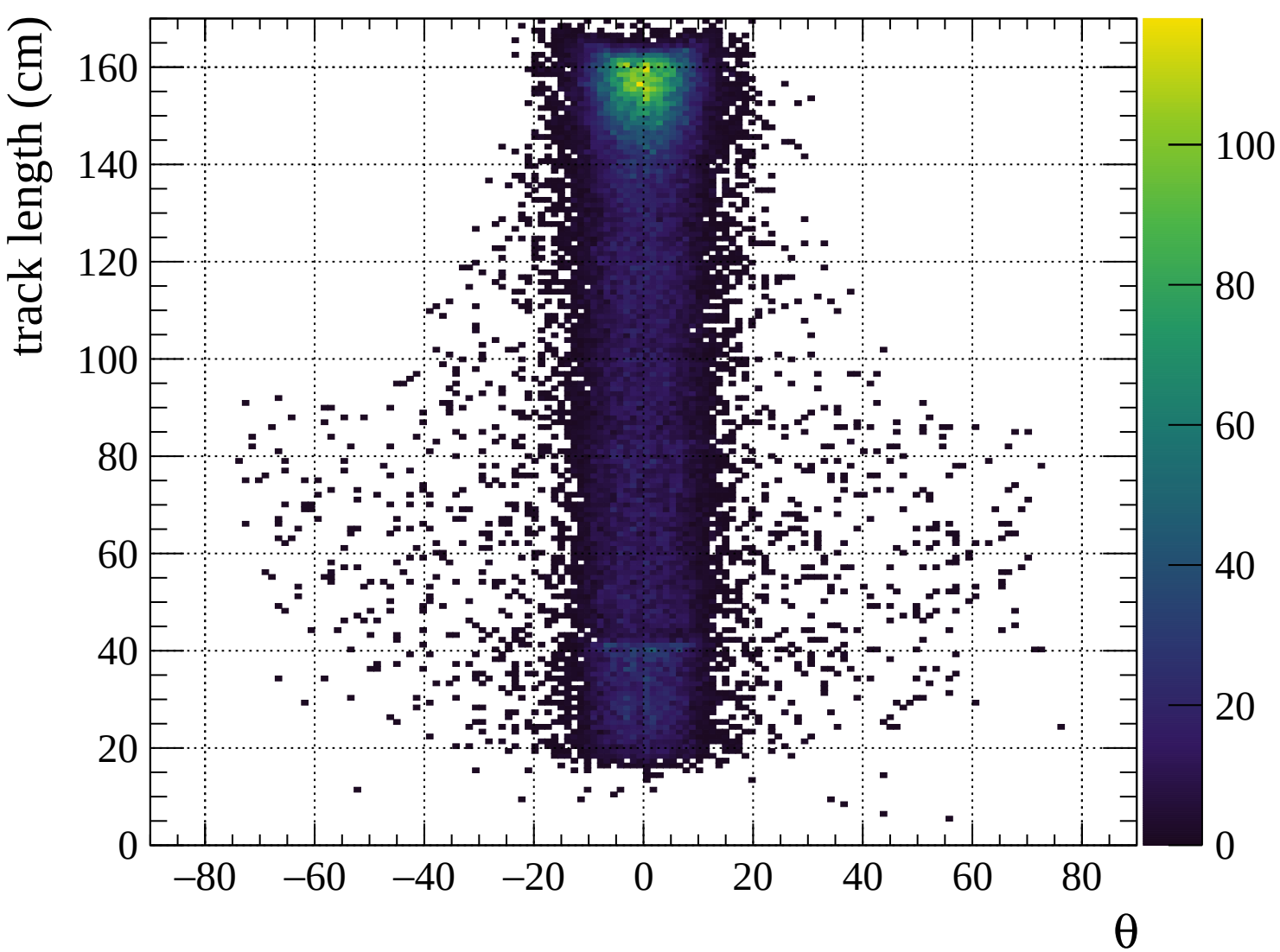


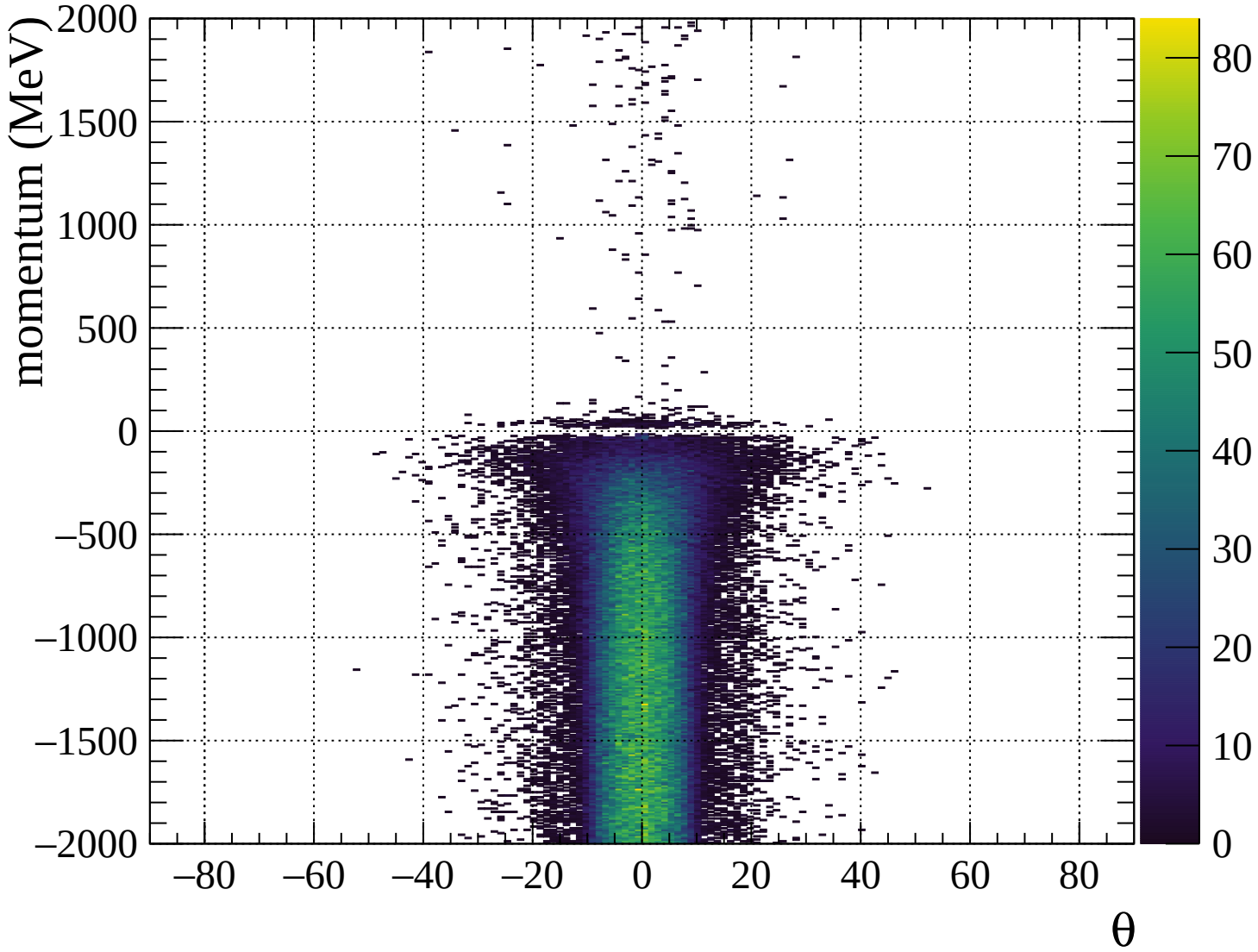


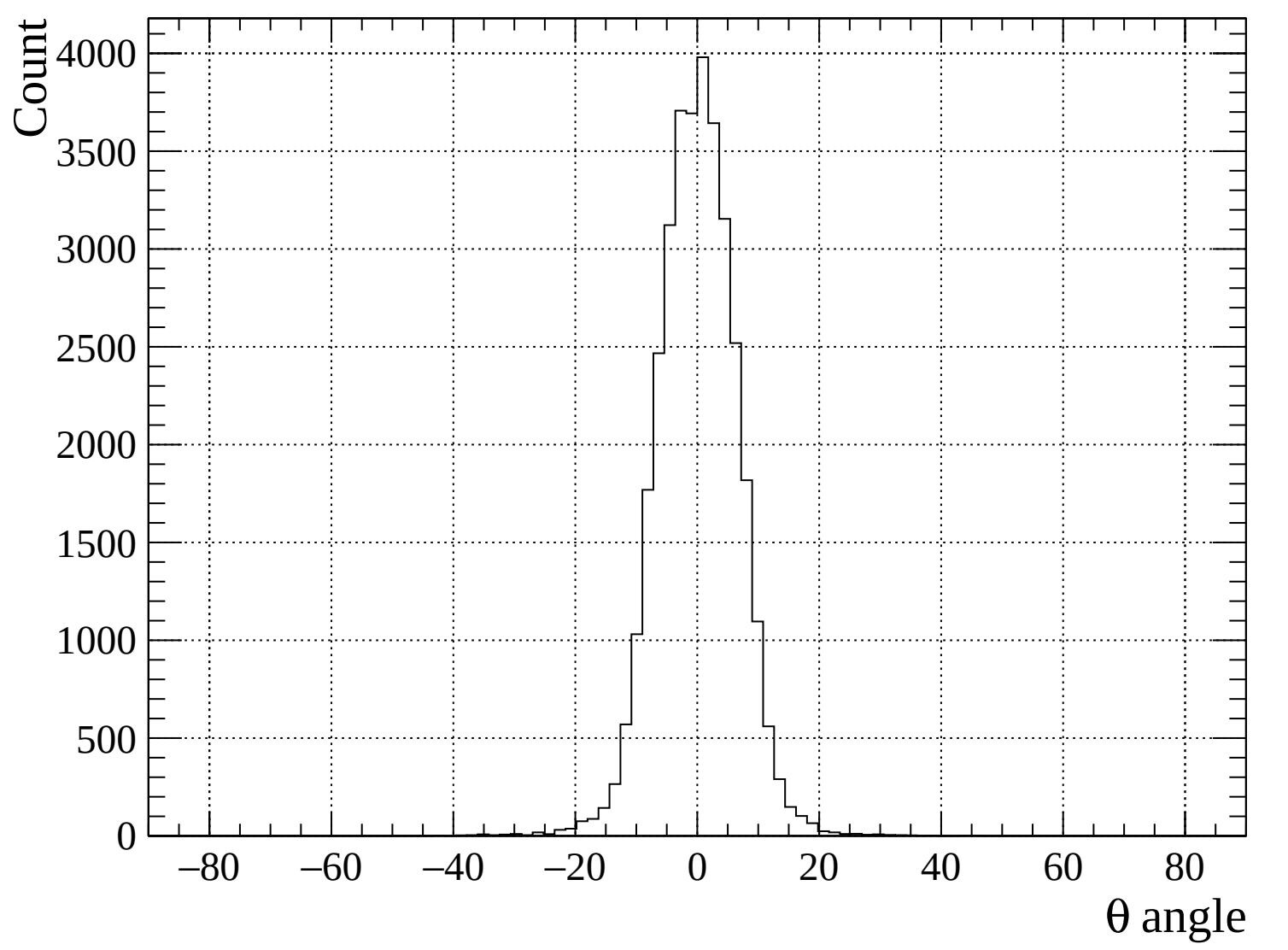


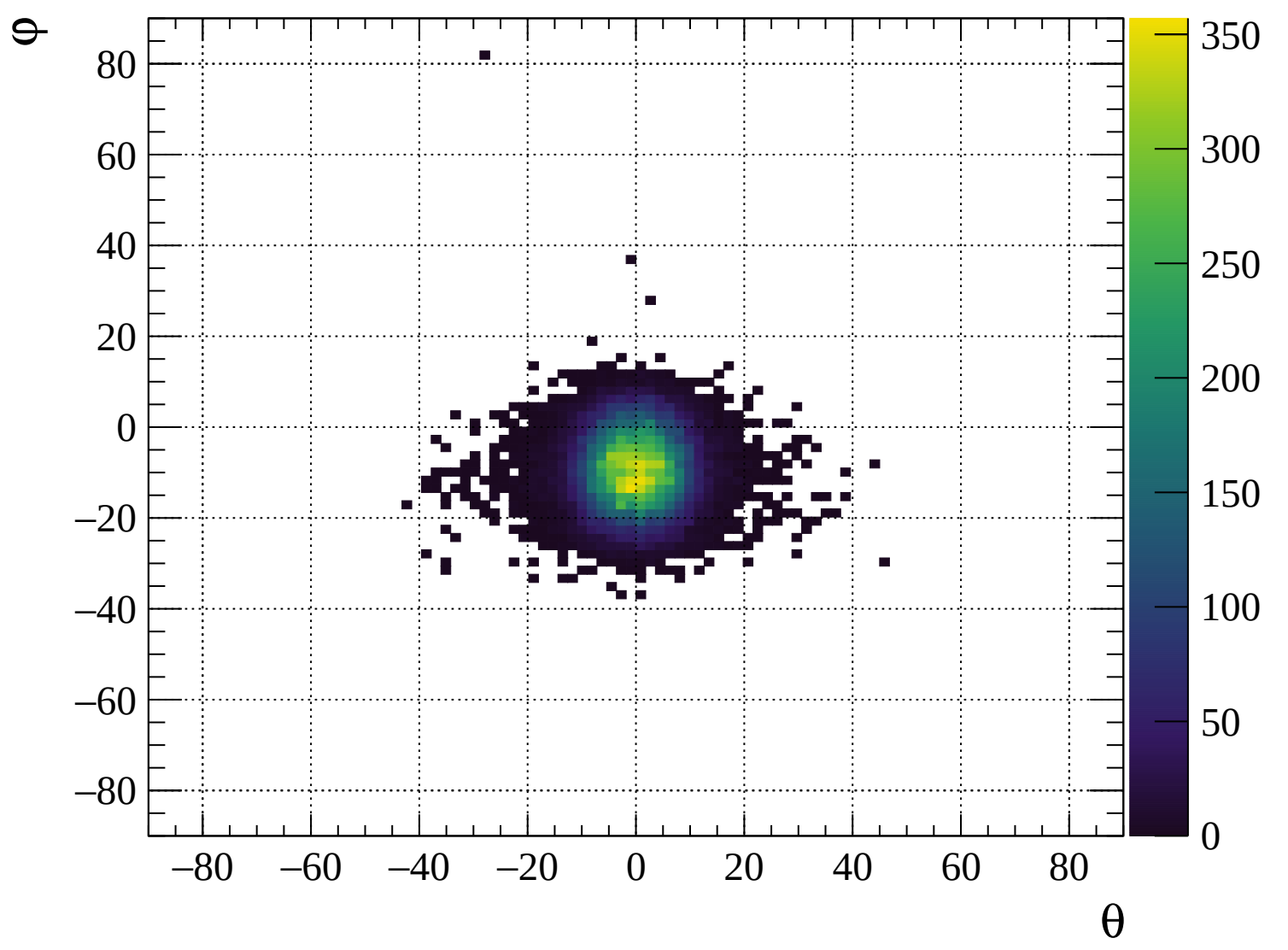


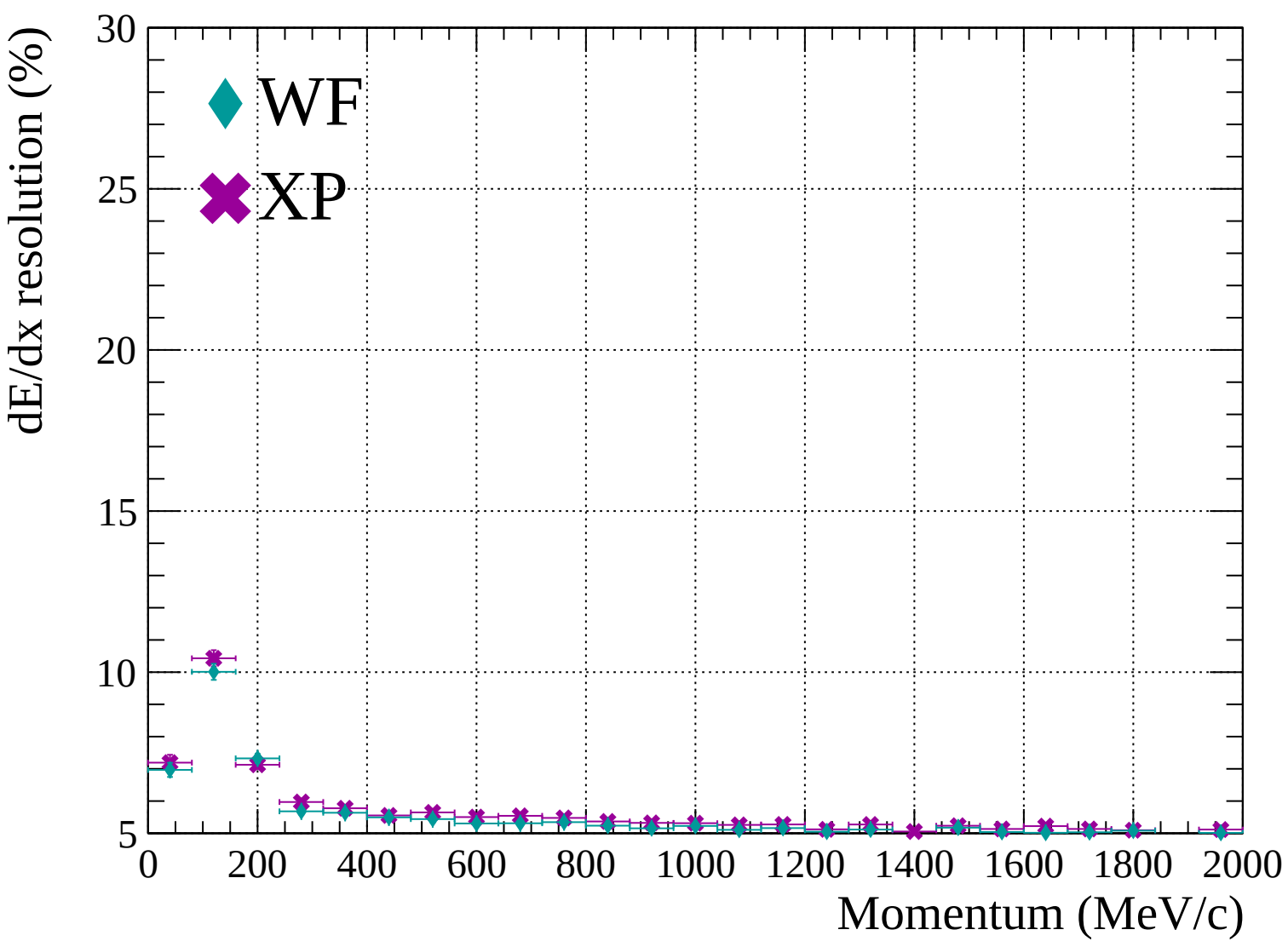


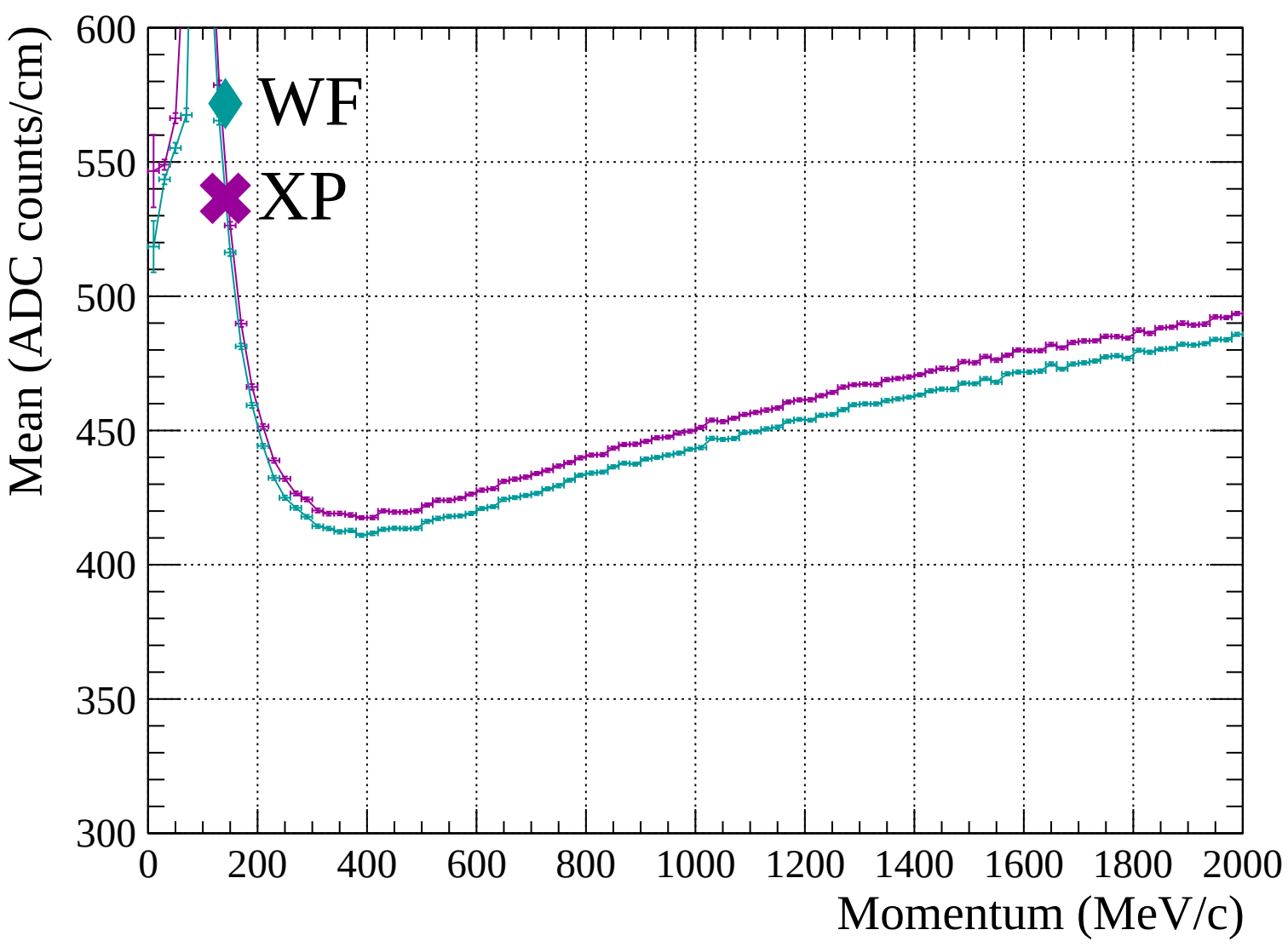


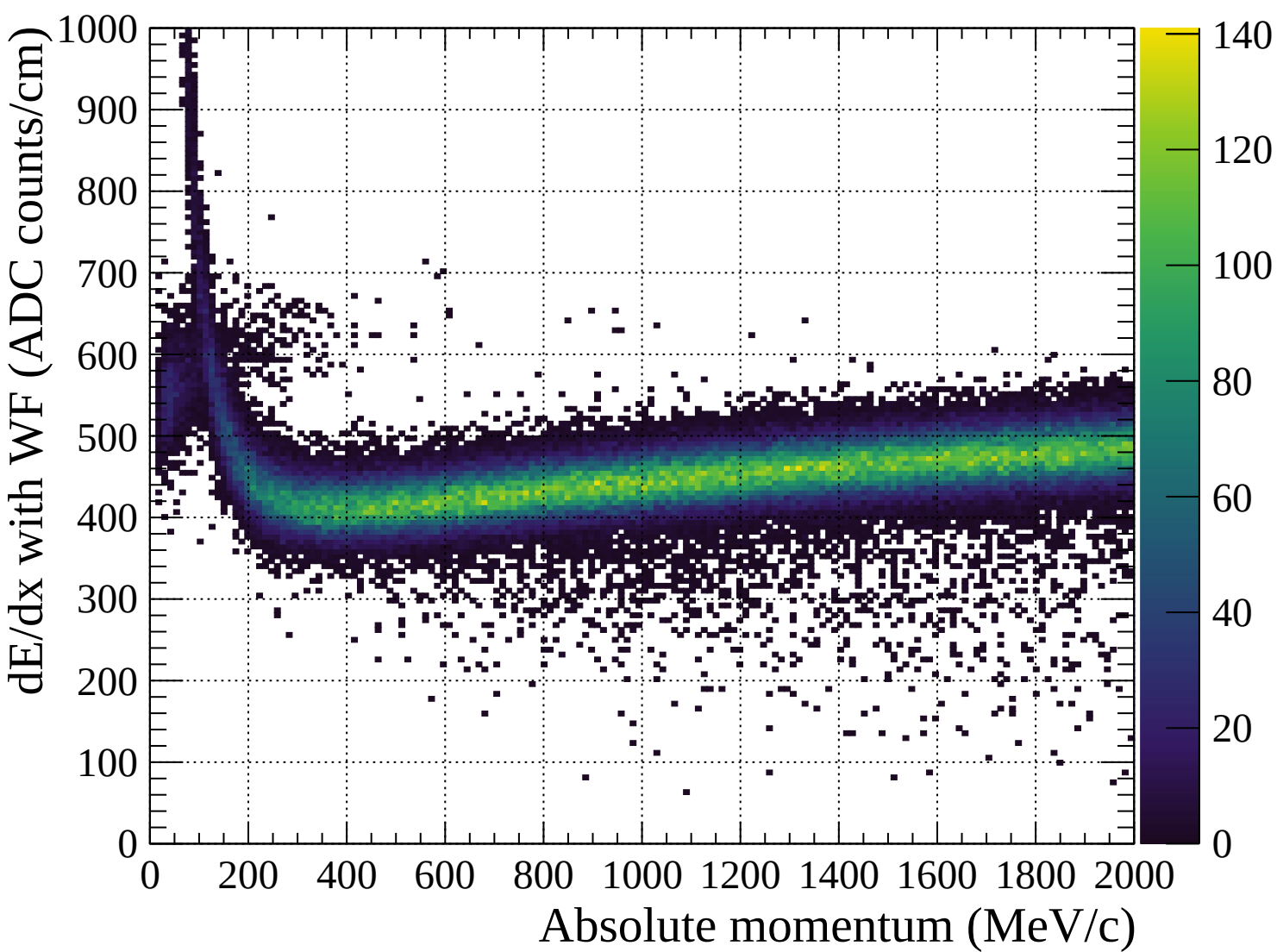


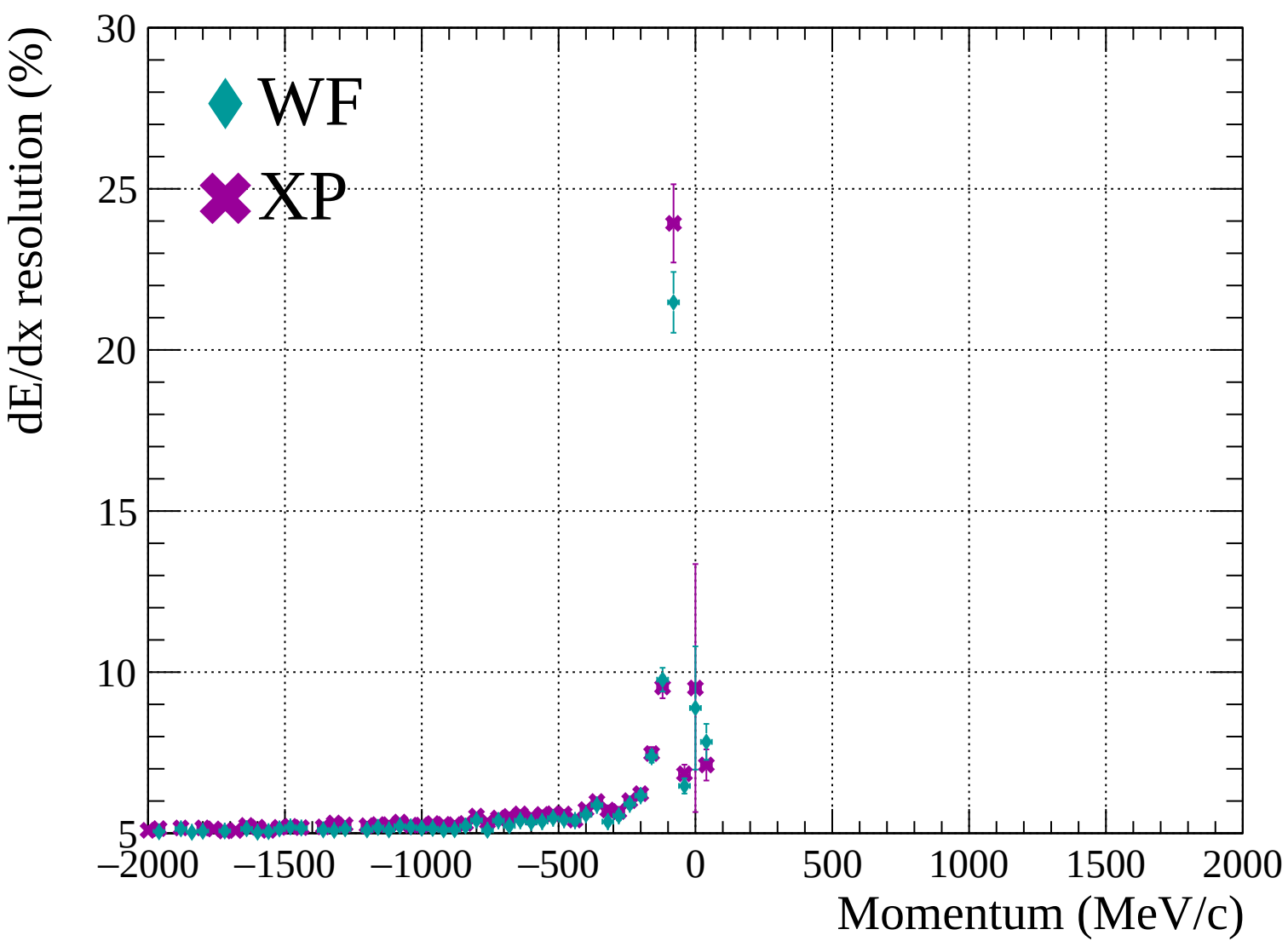


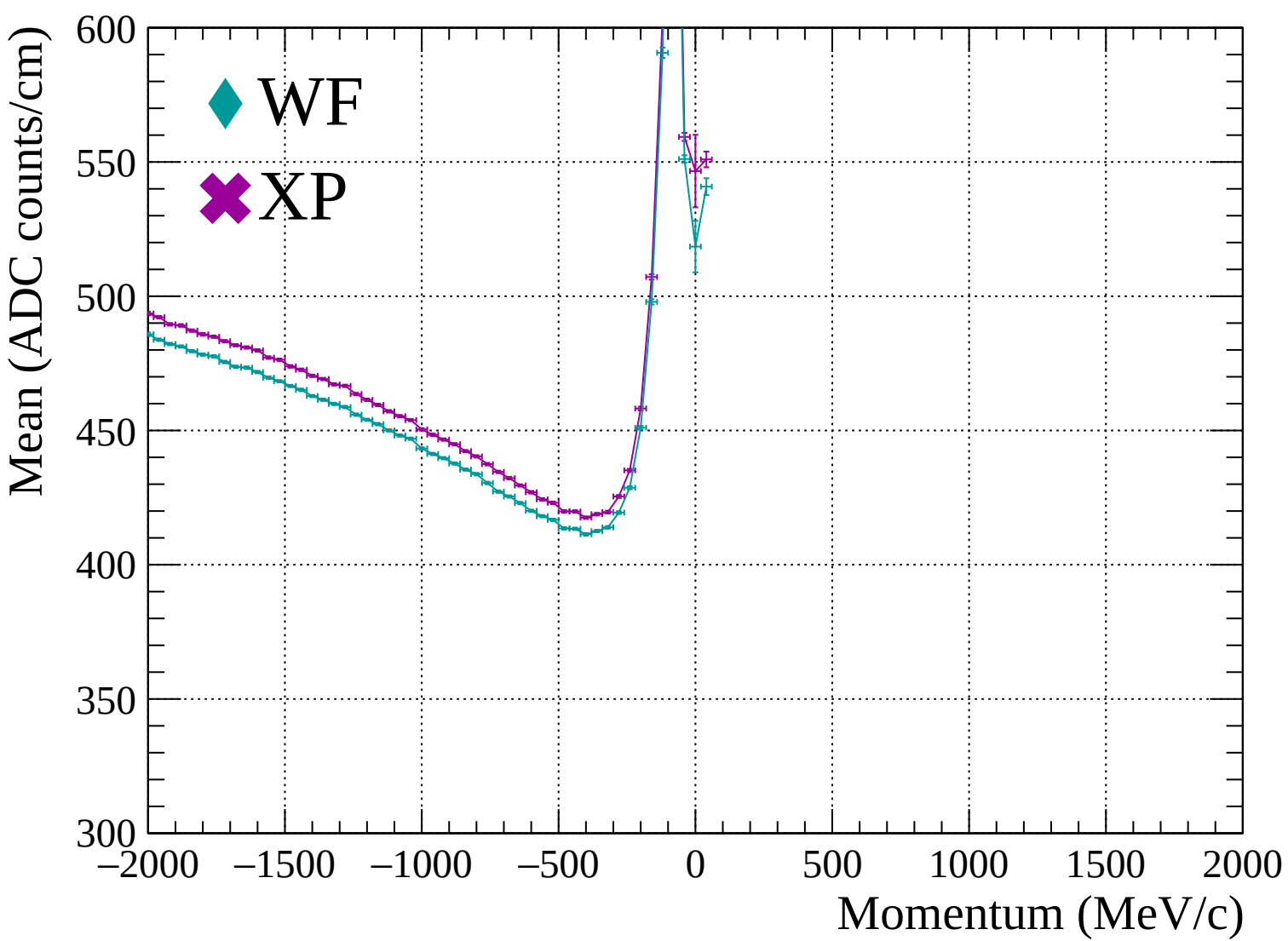


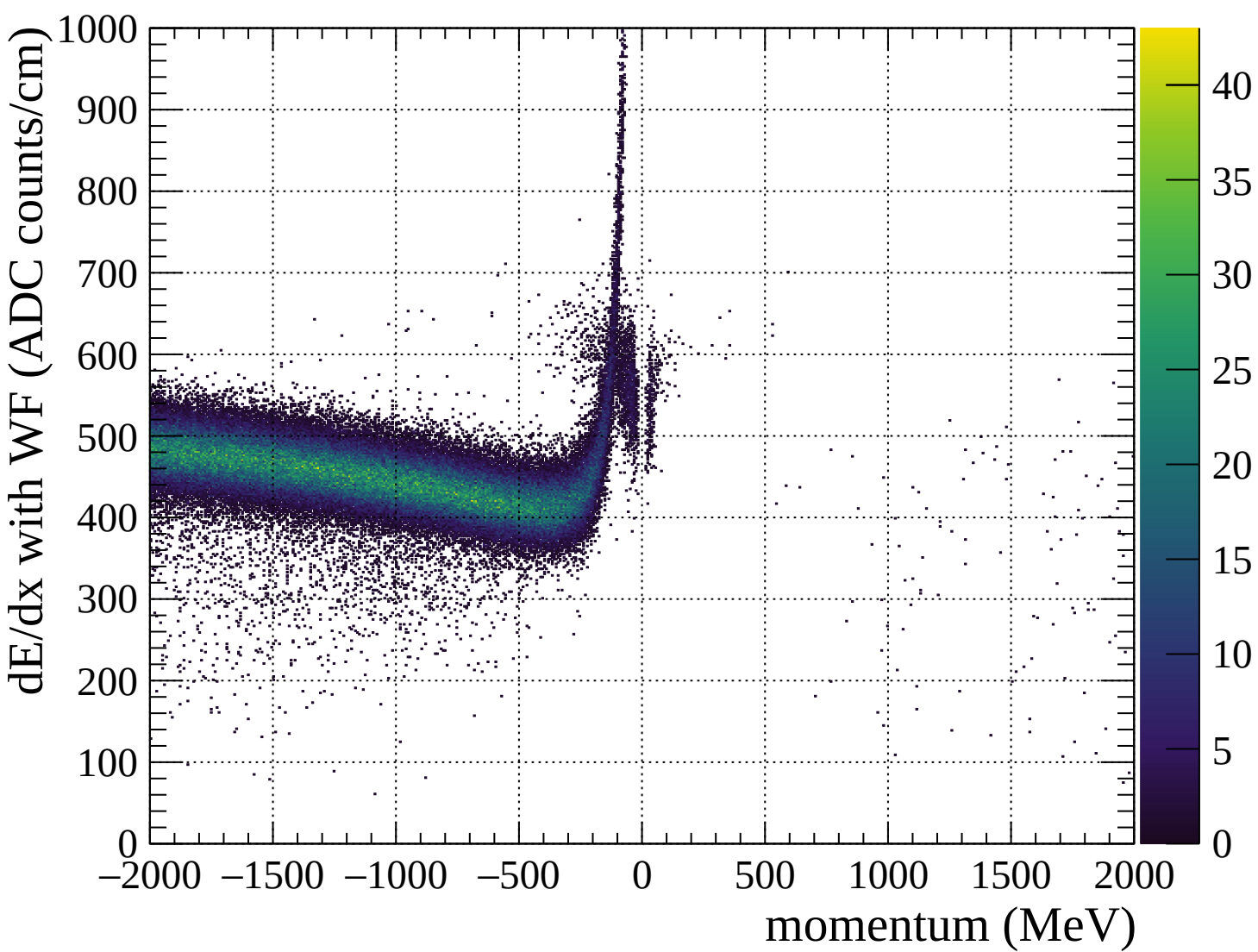


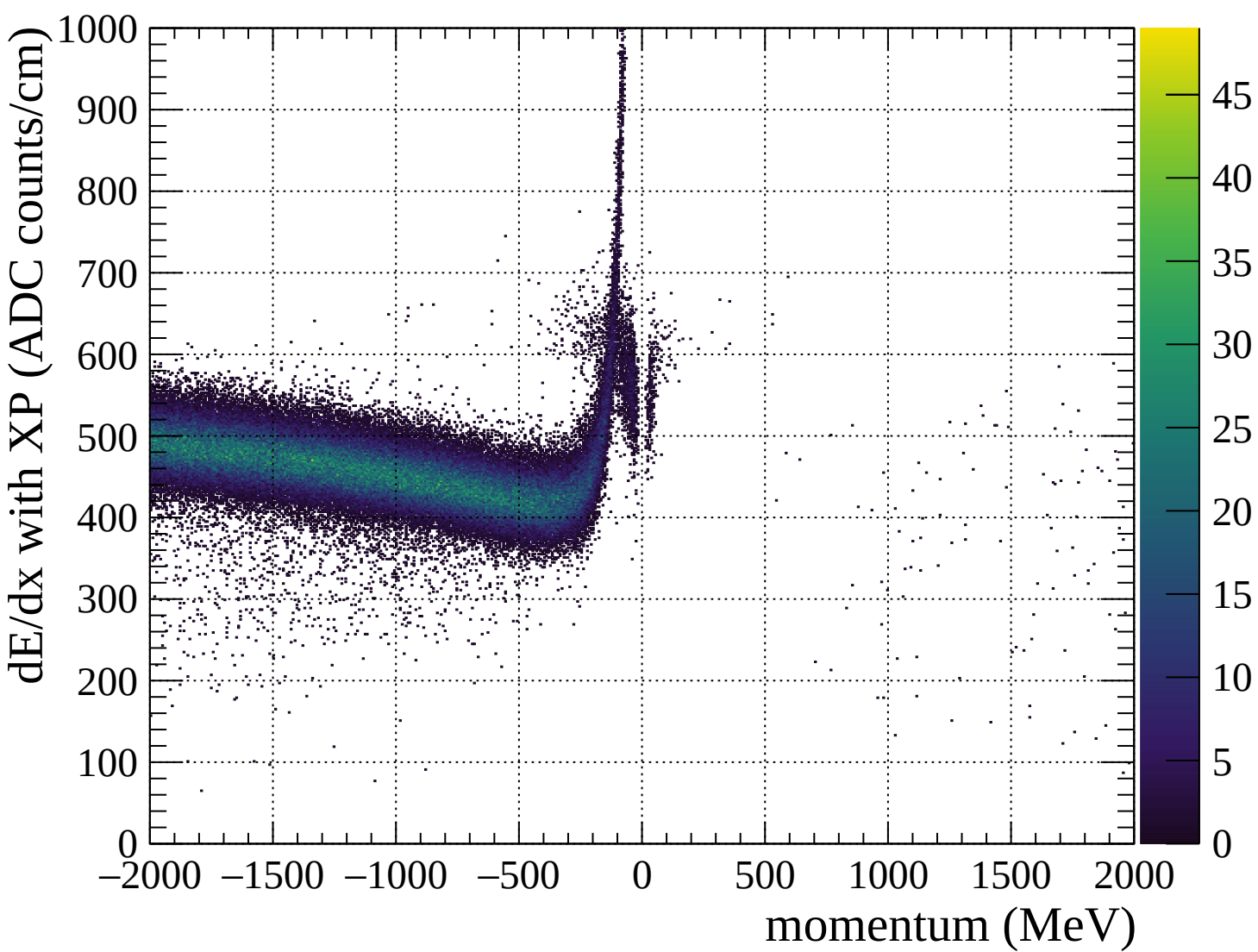


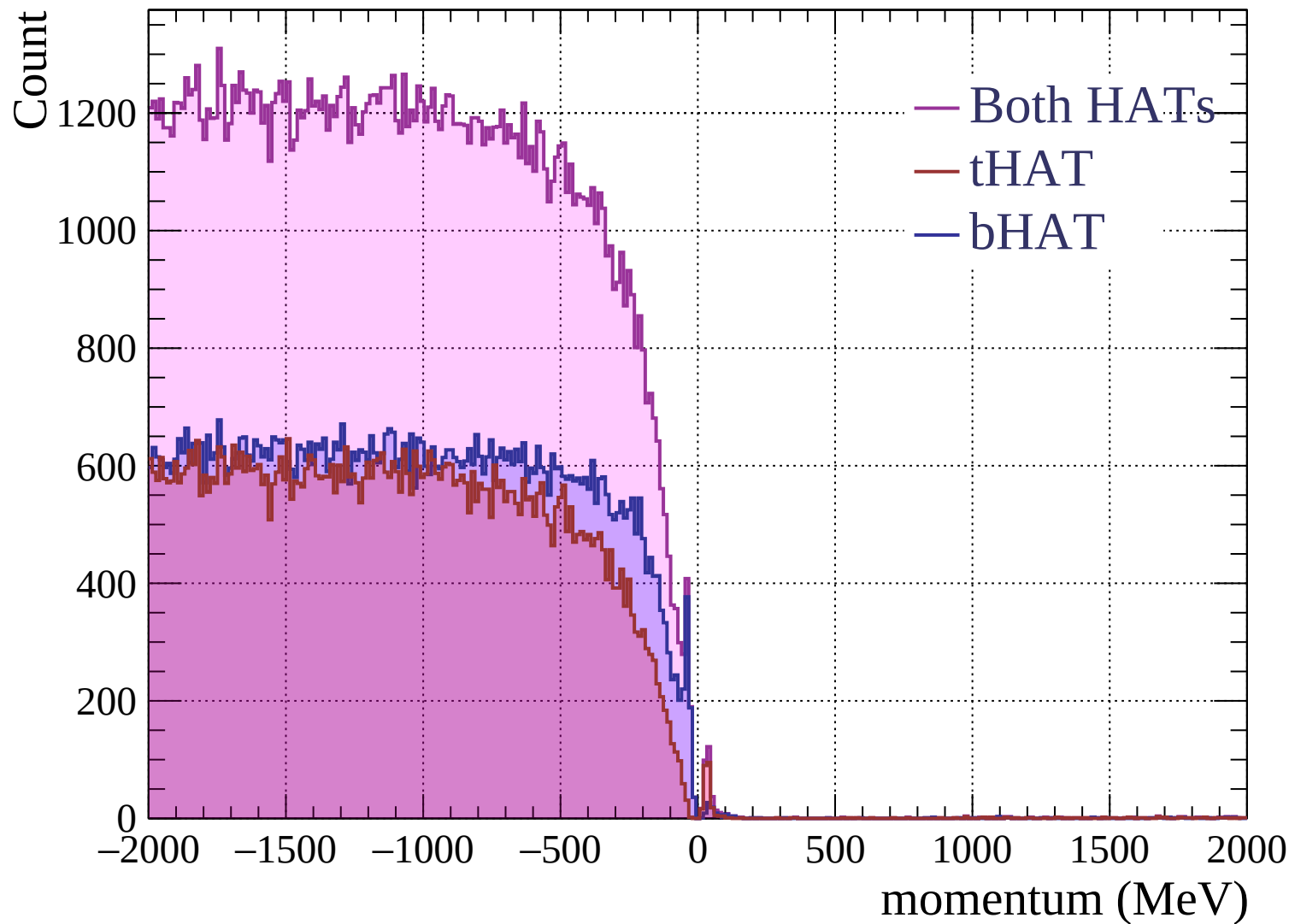


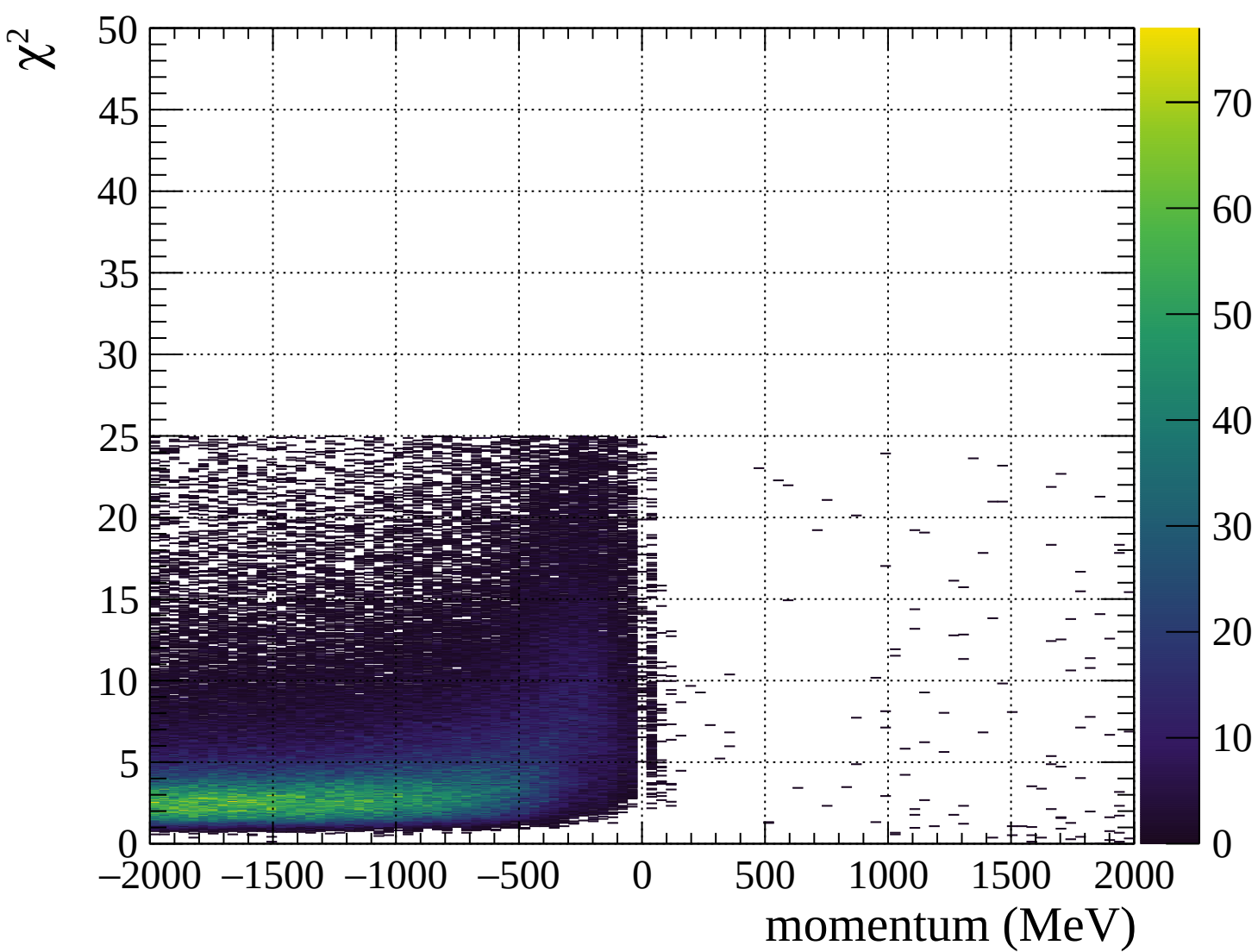


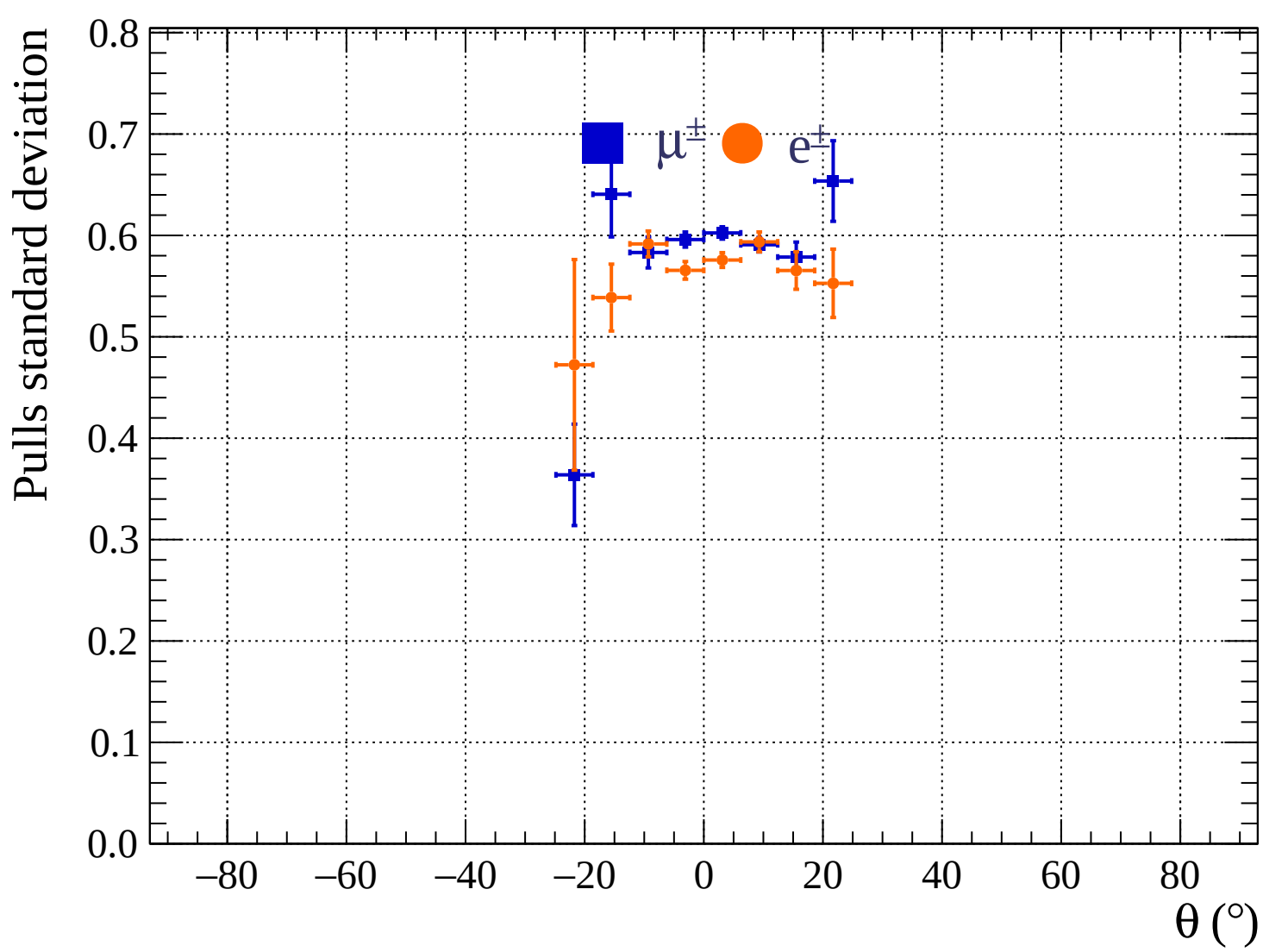


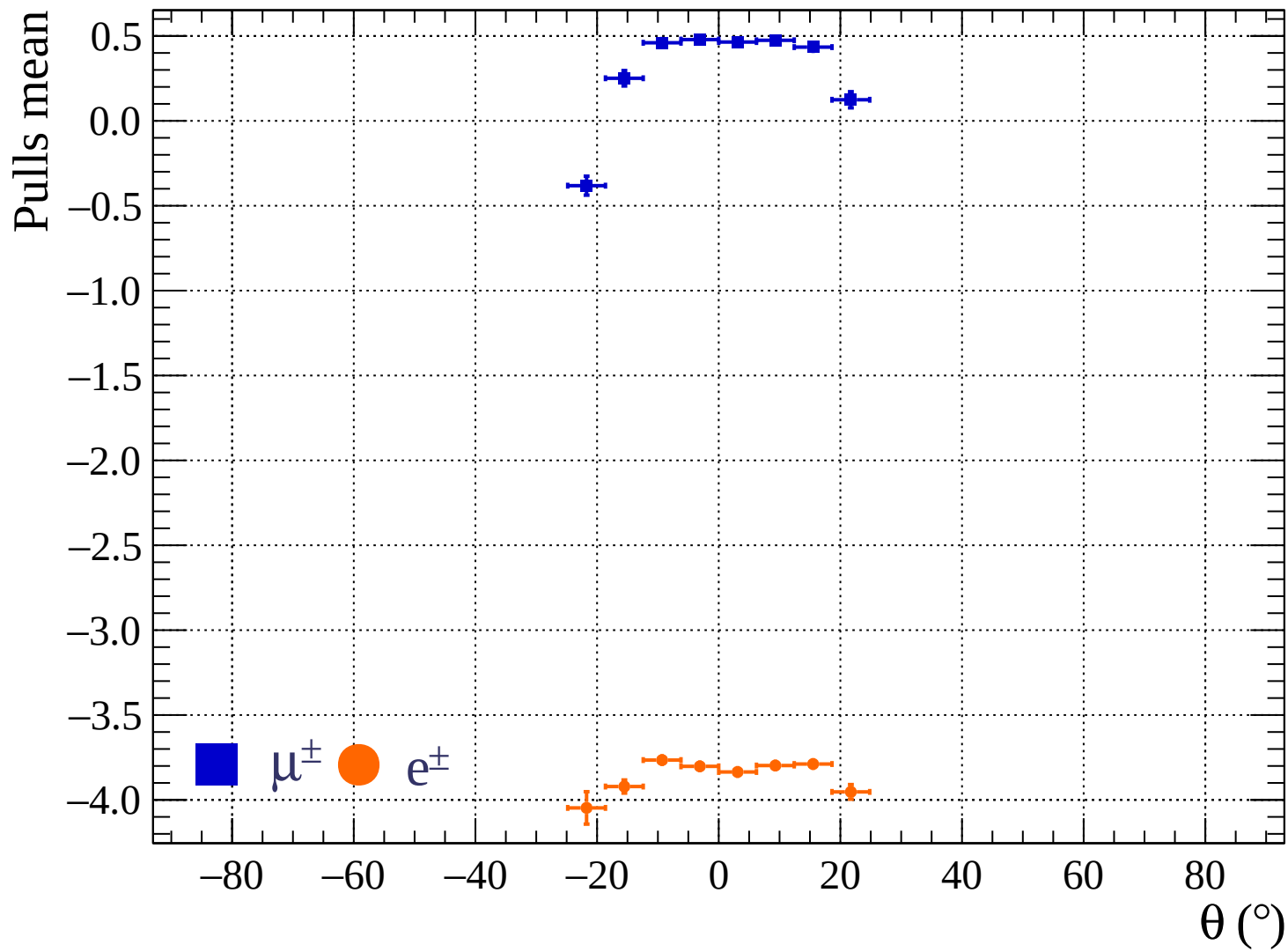




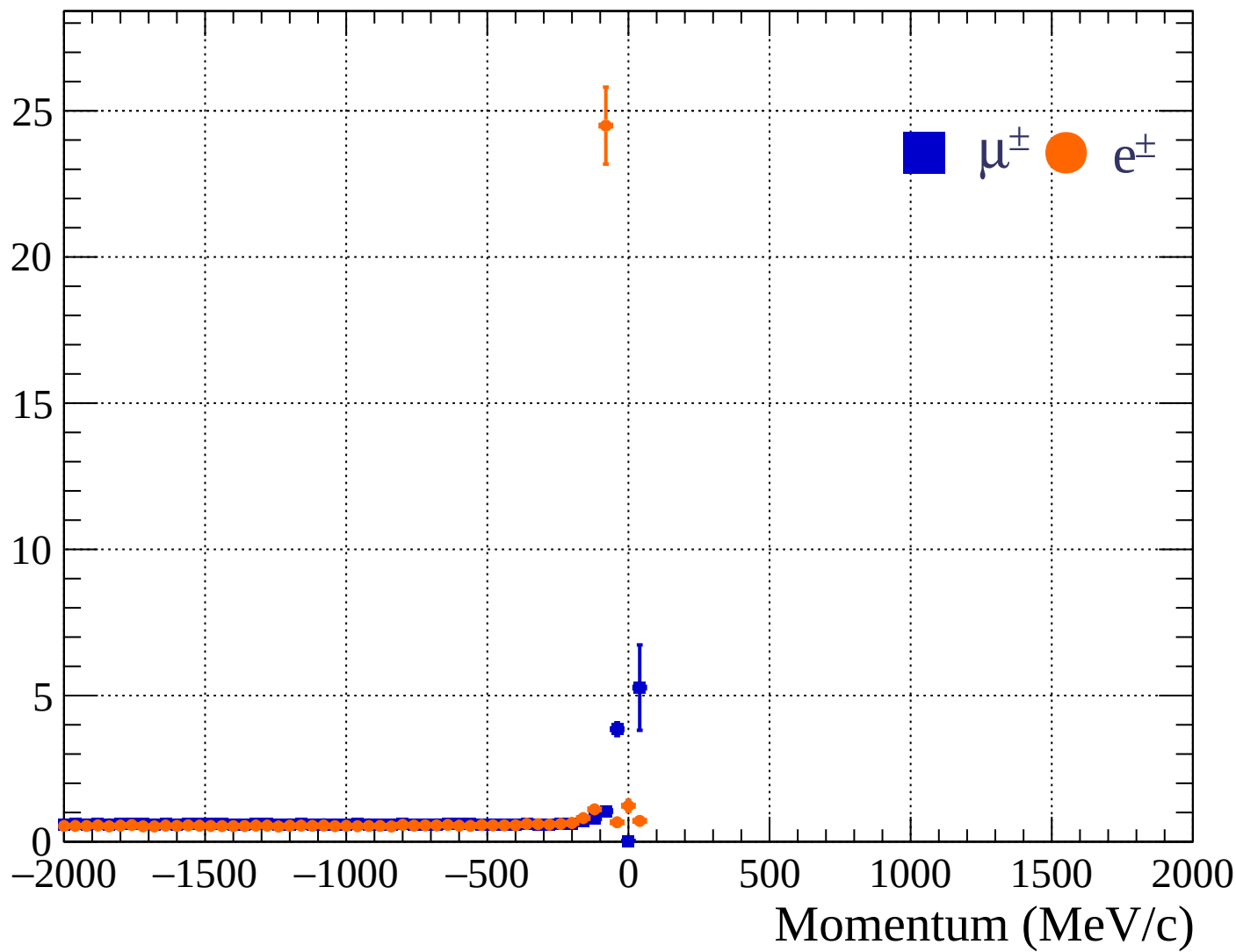




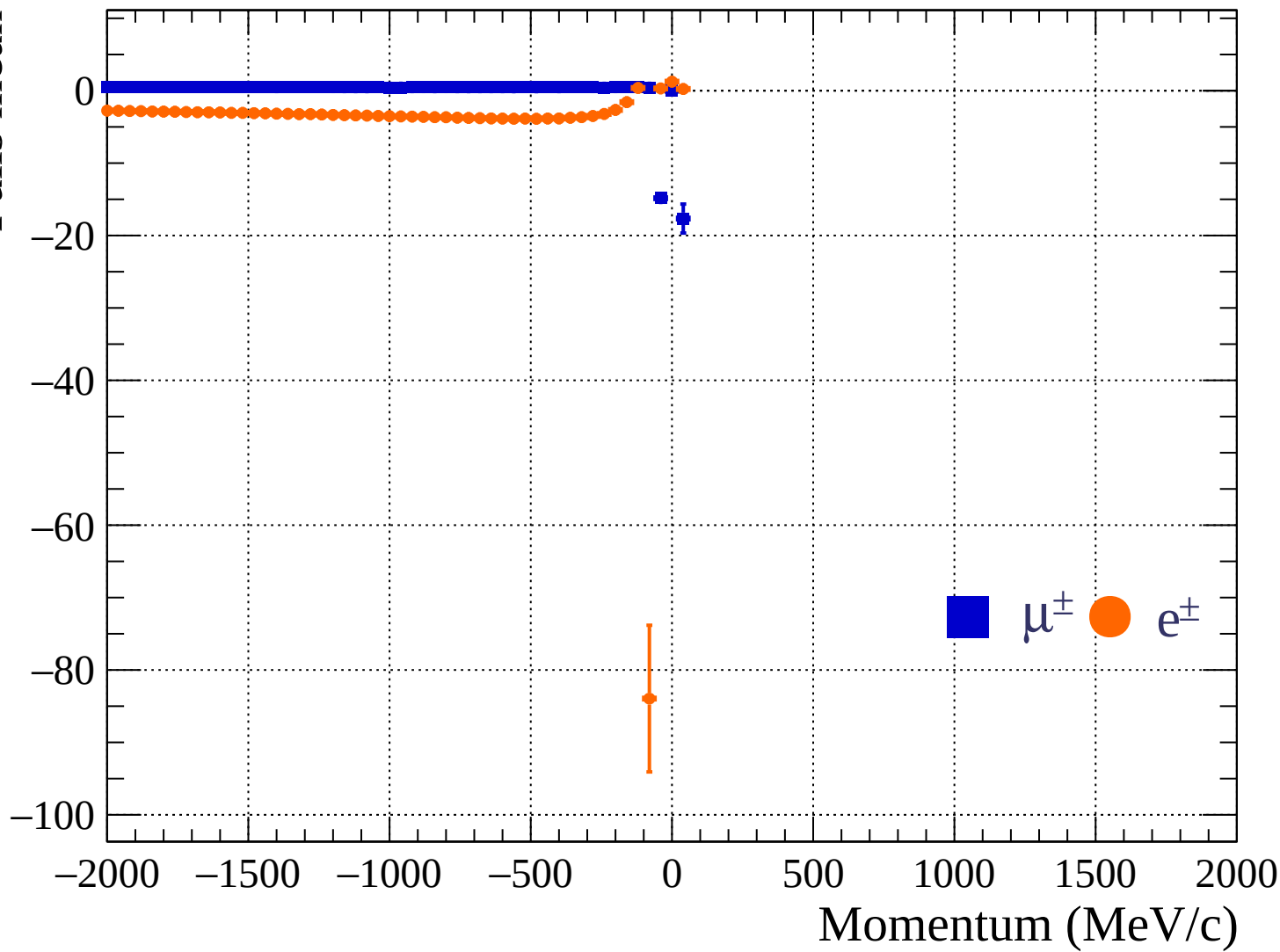


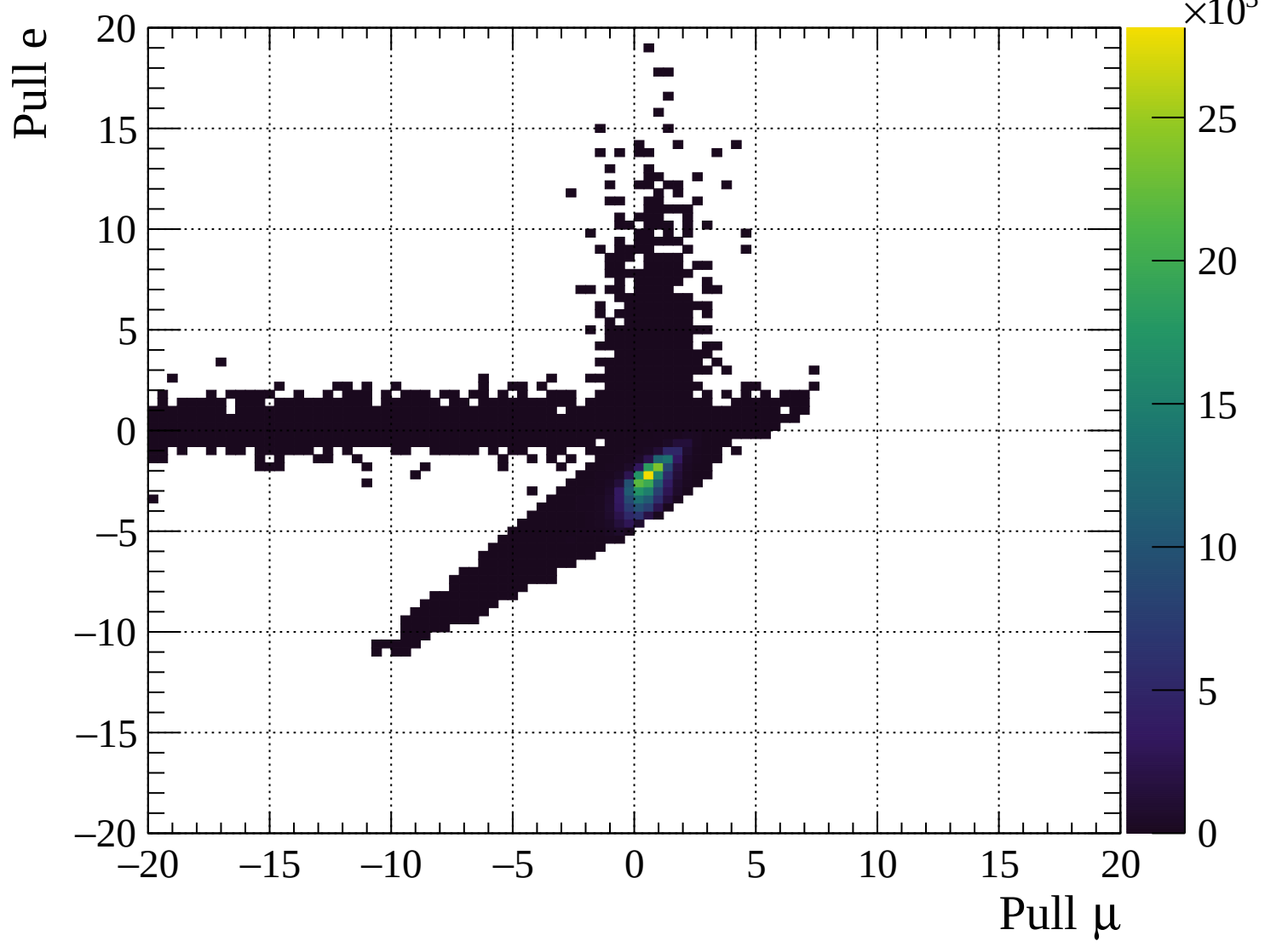


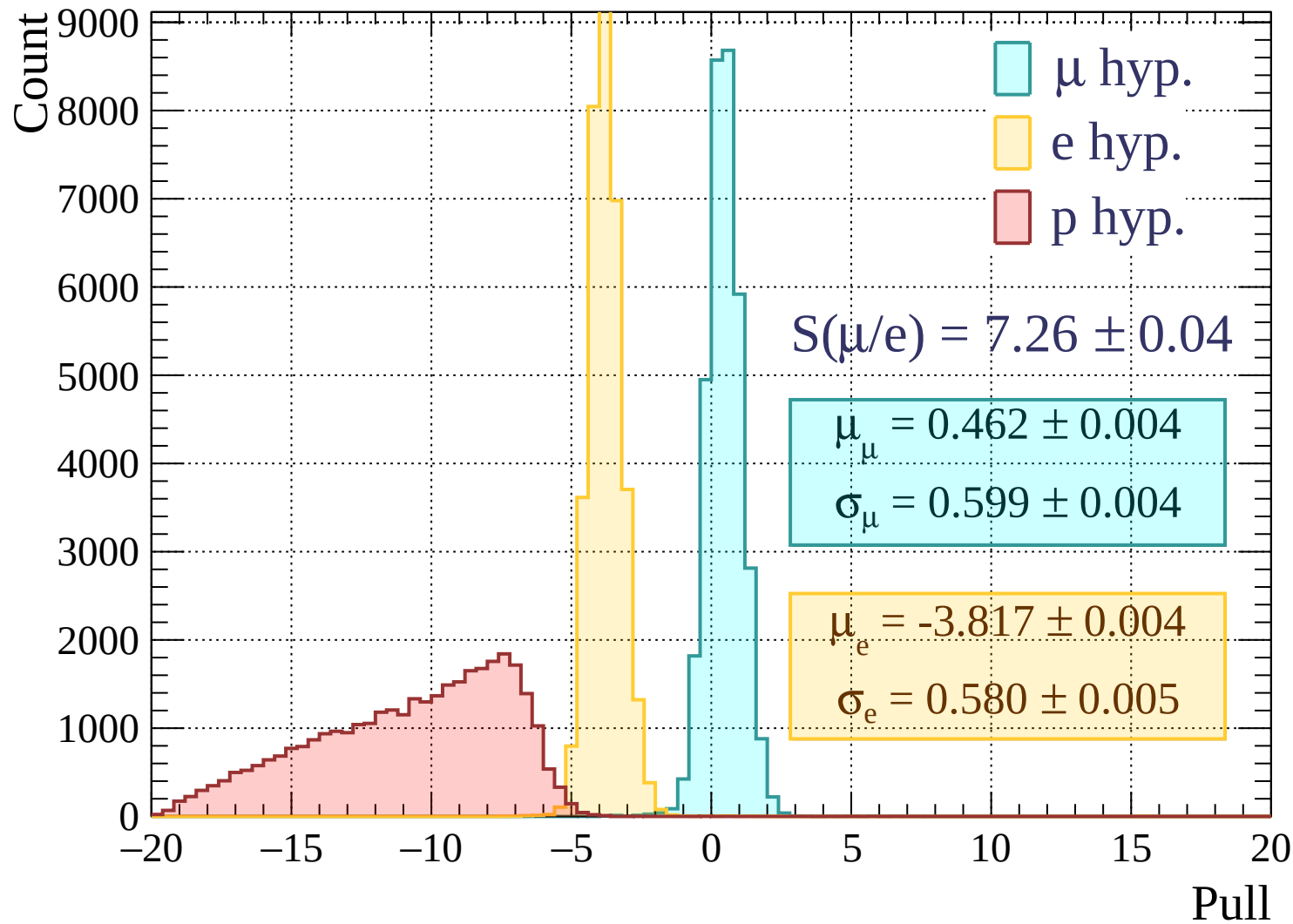
Pulls standard deviation

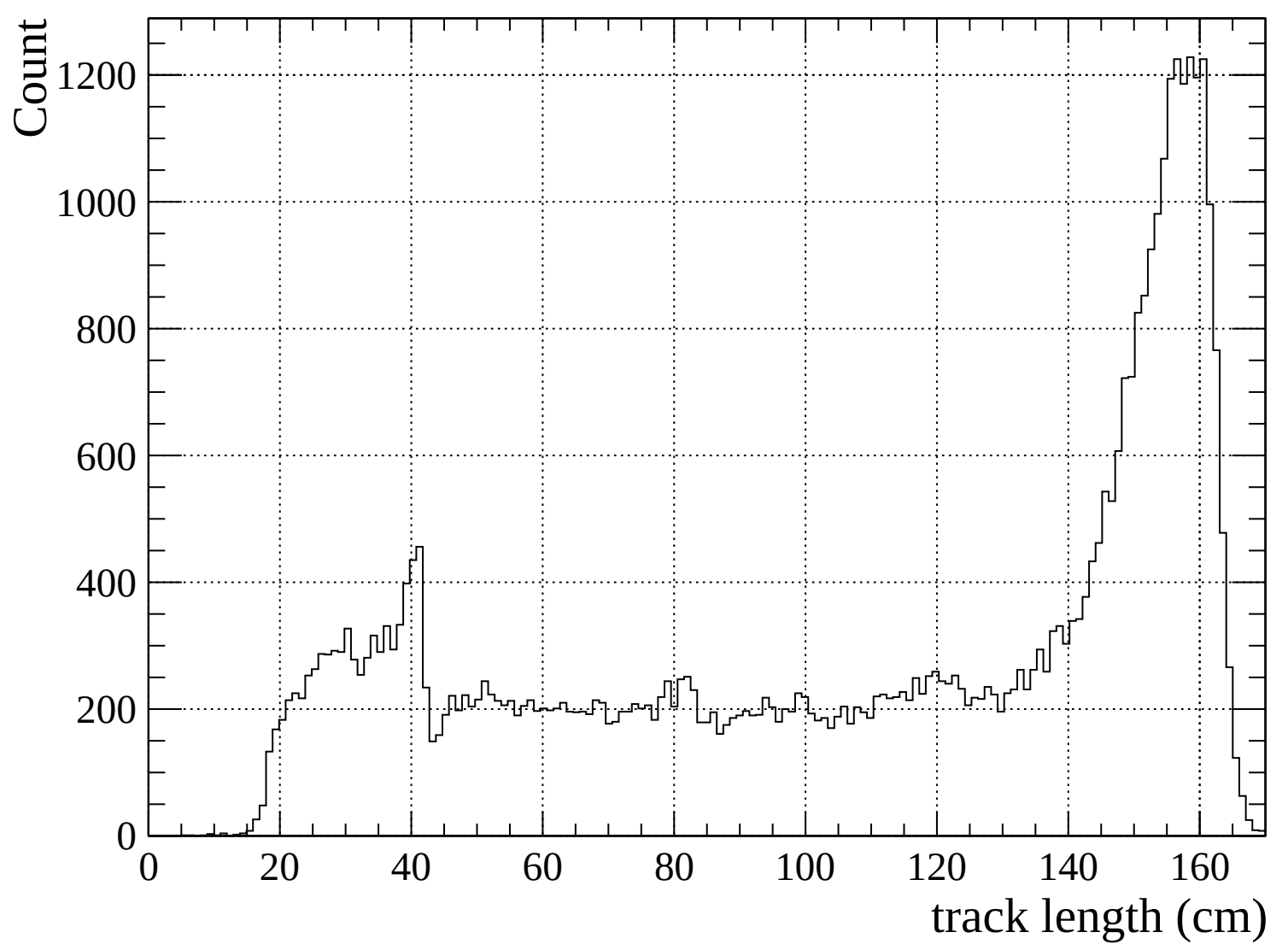


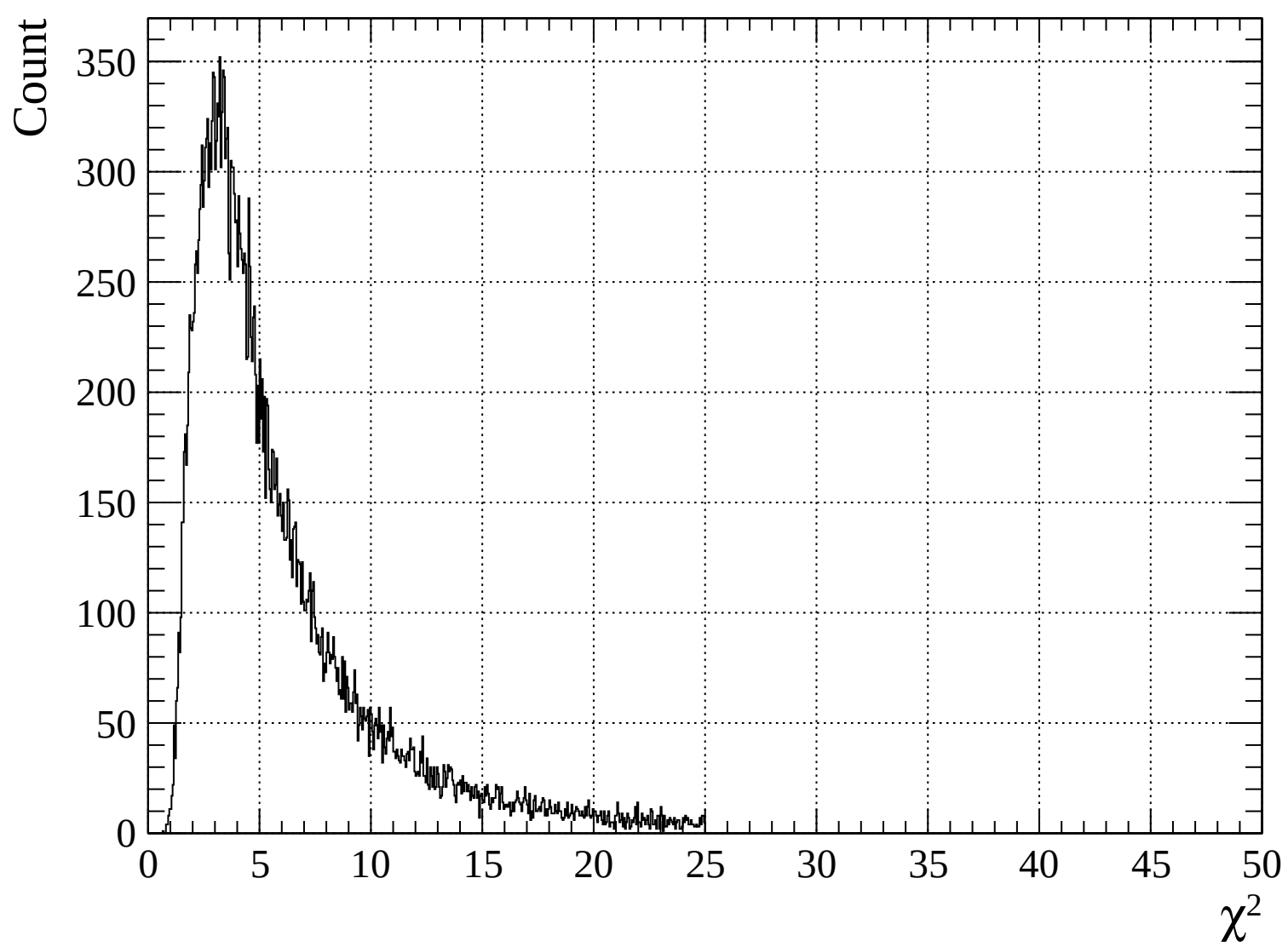
Pulls mean

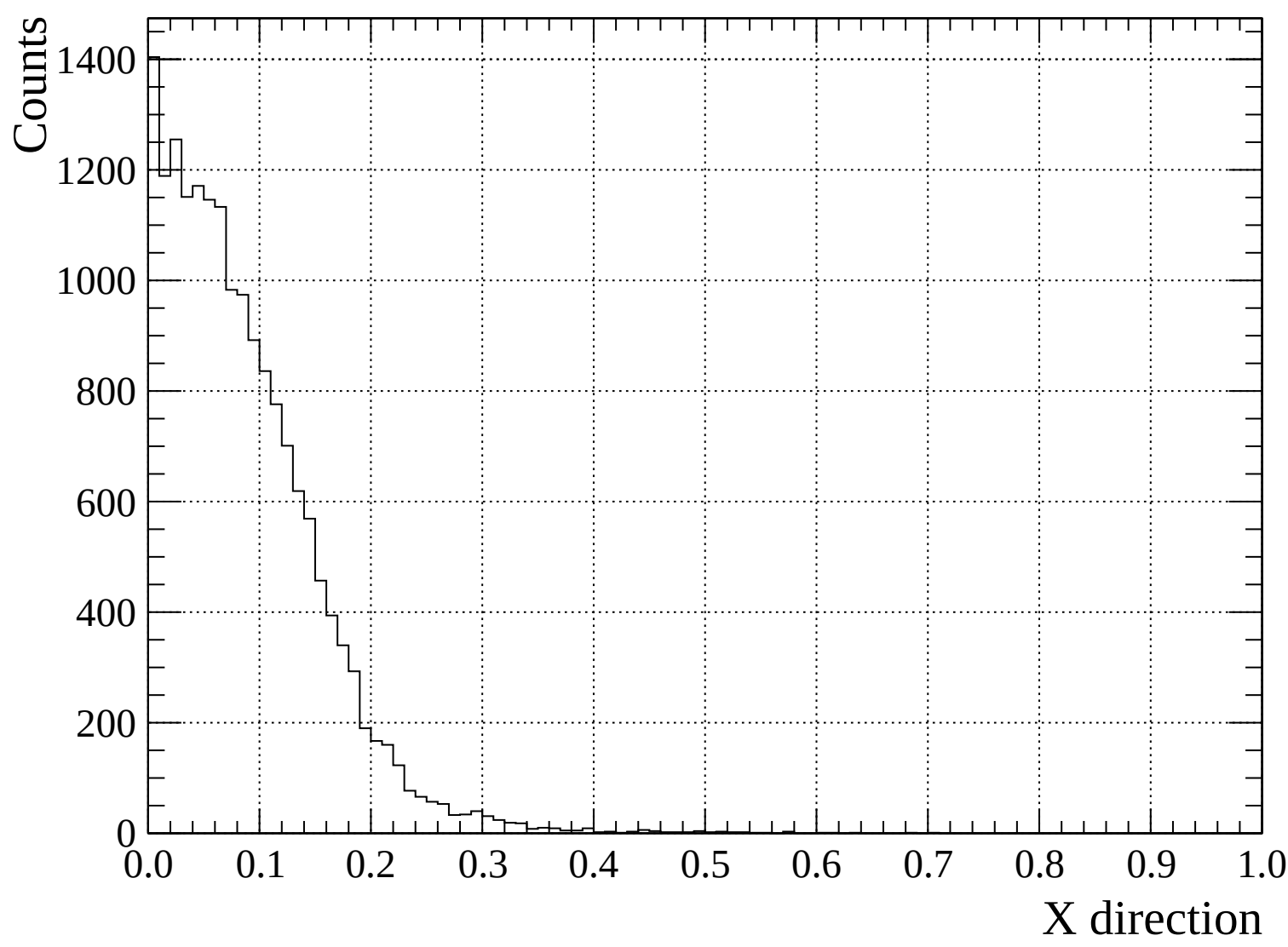


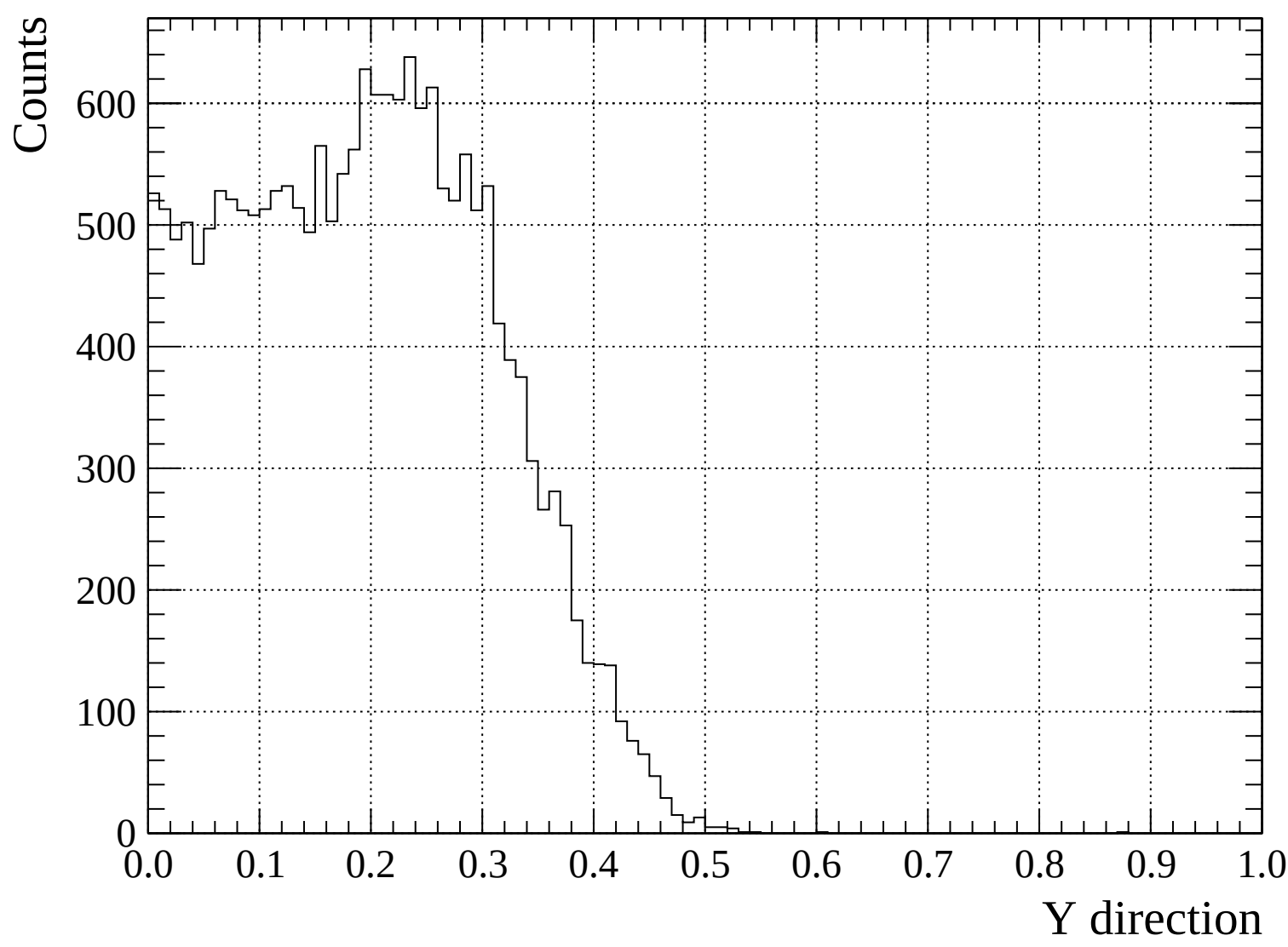


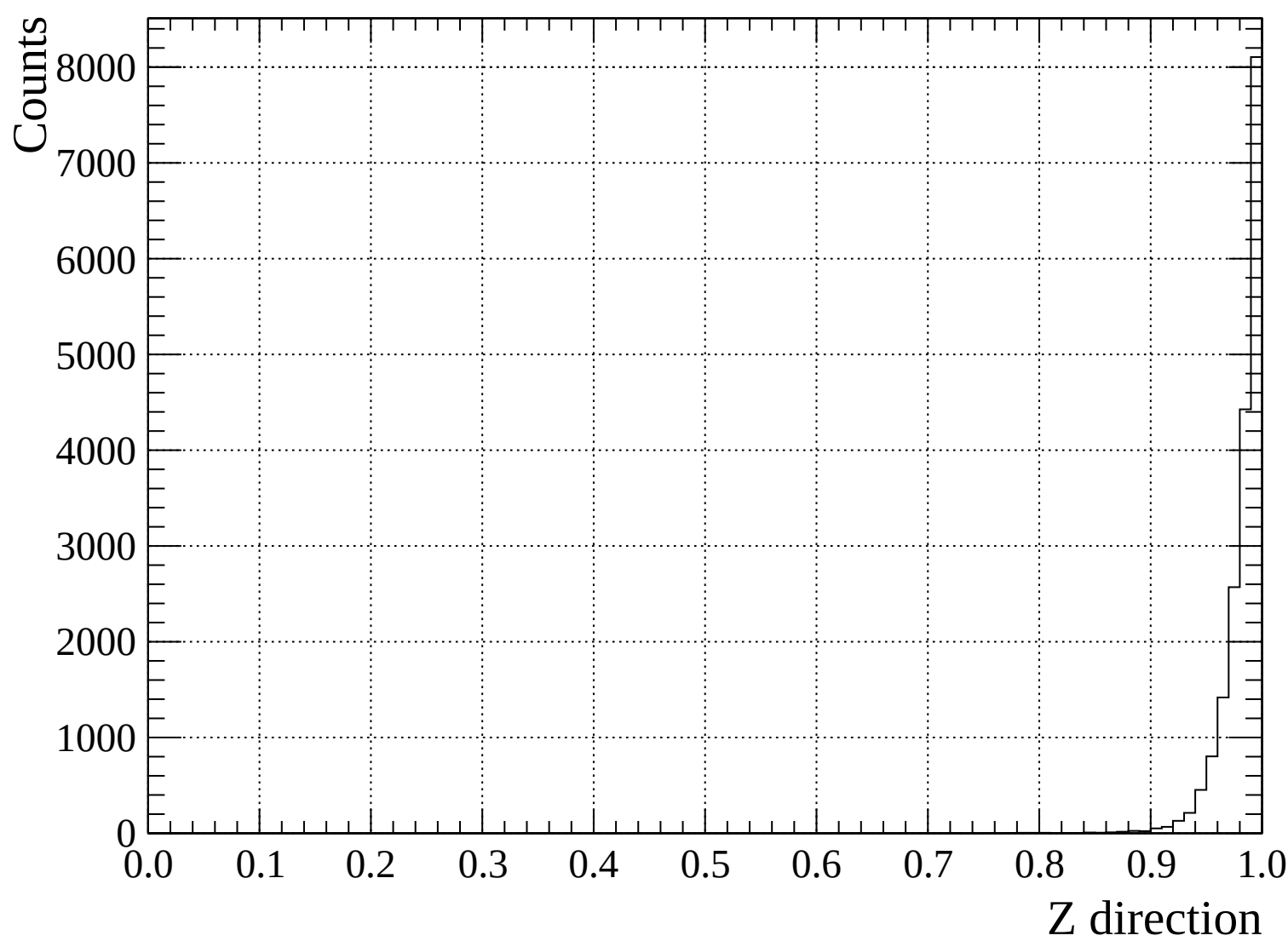


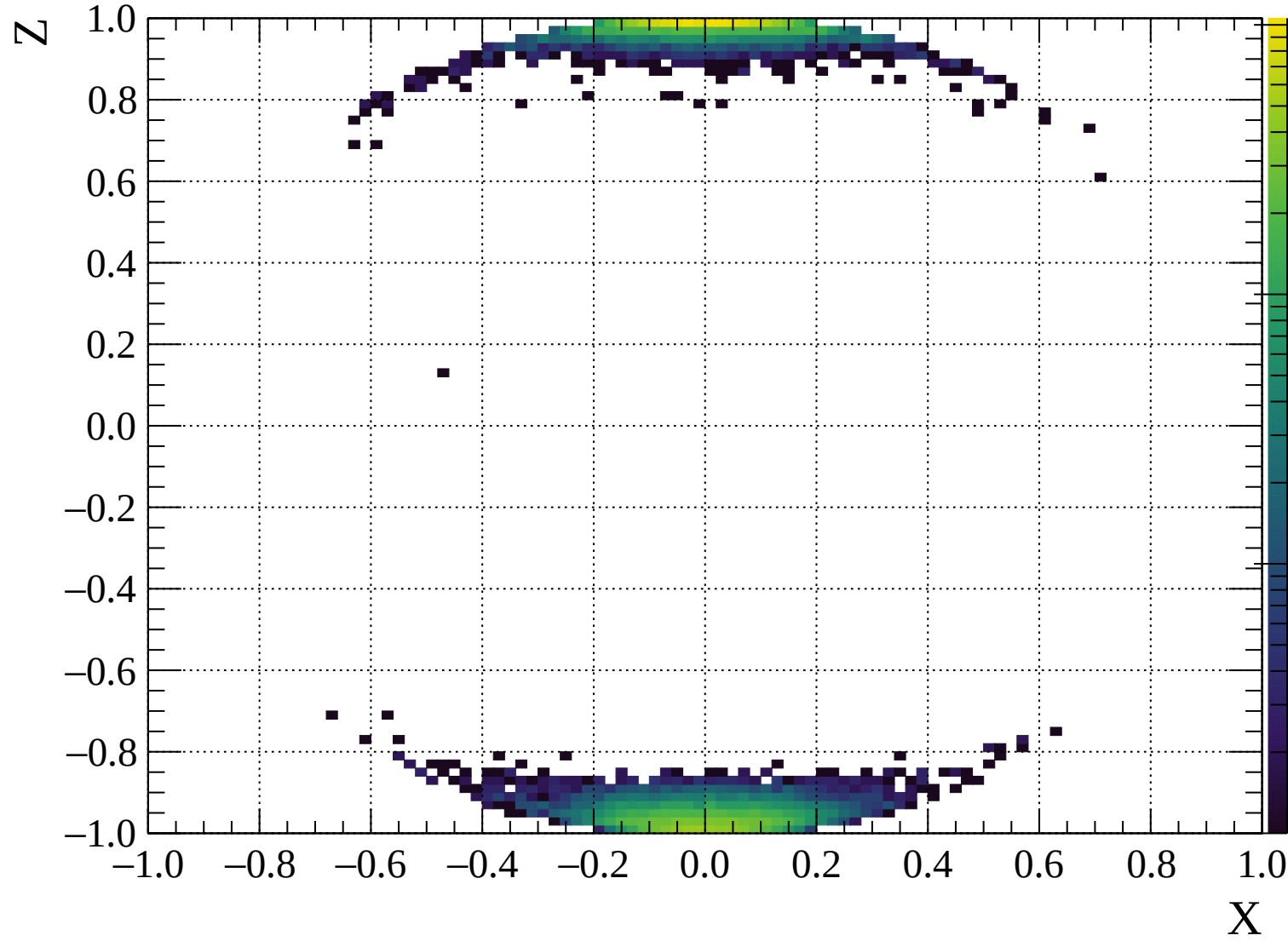


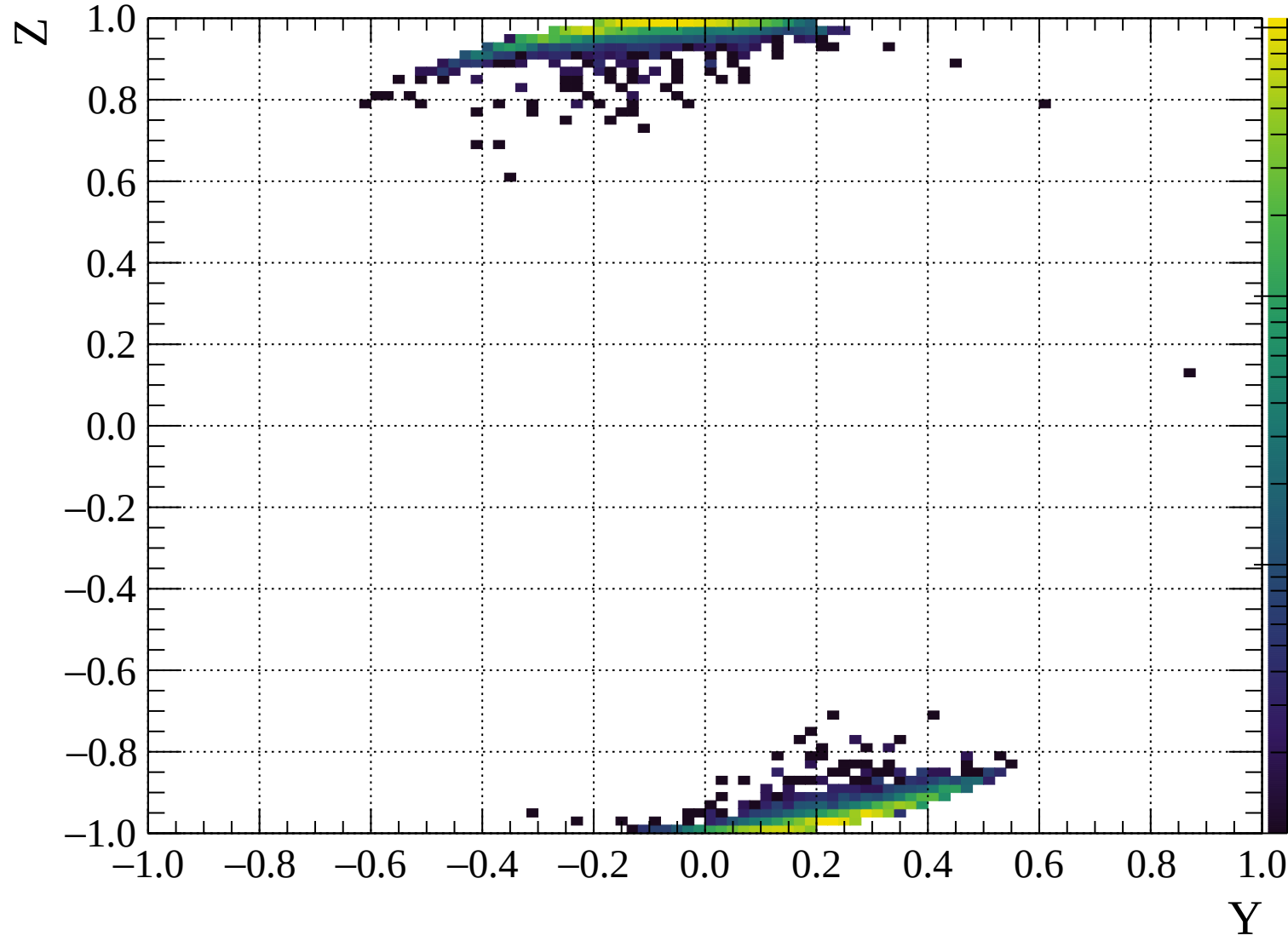


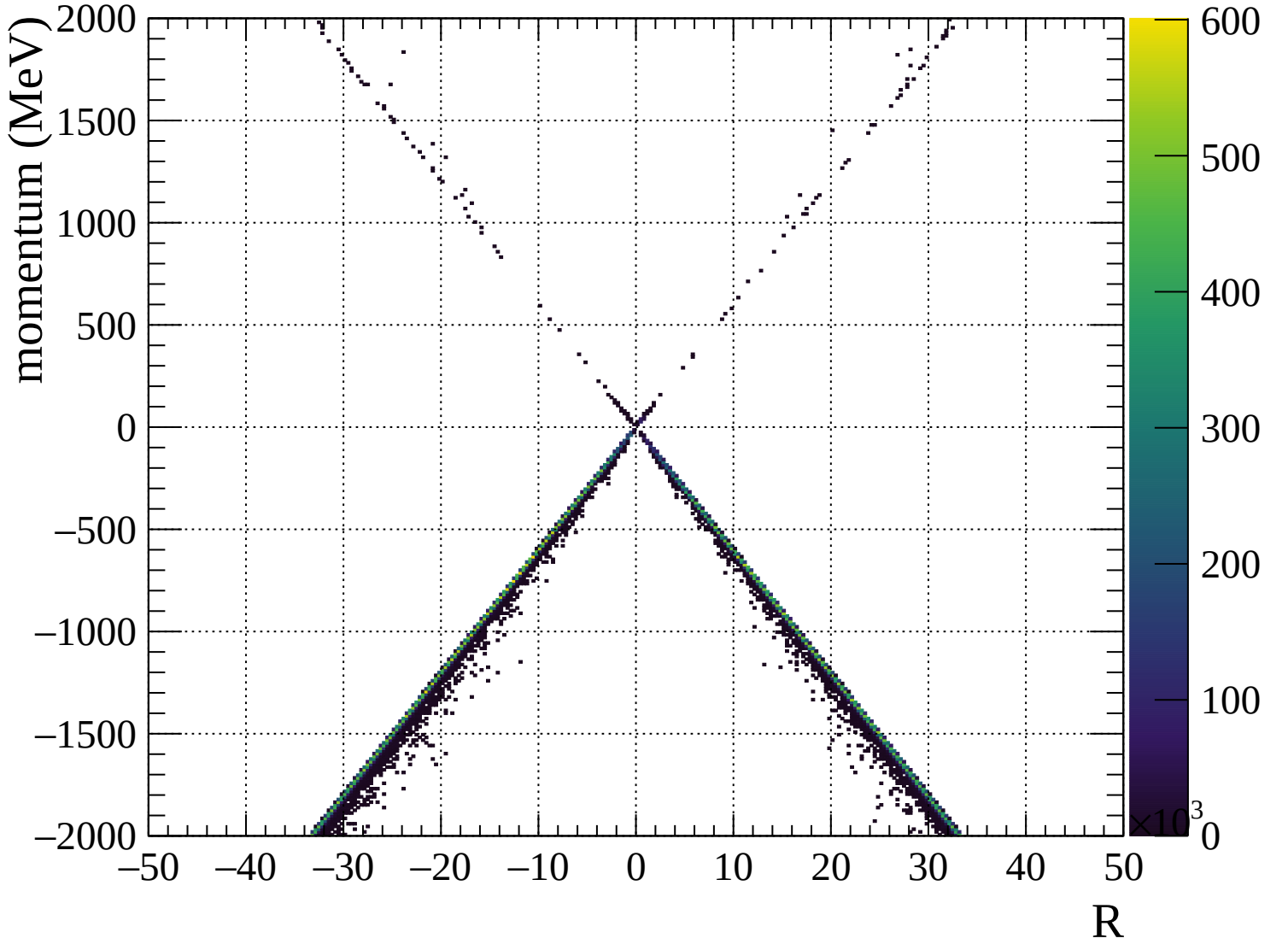


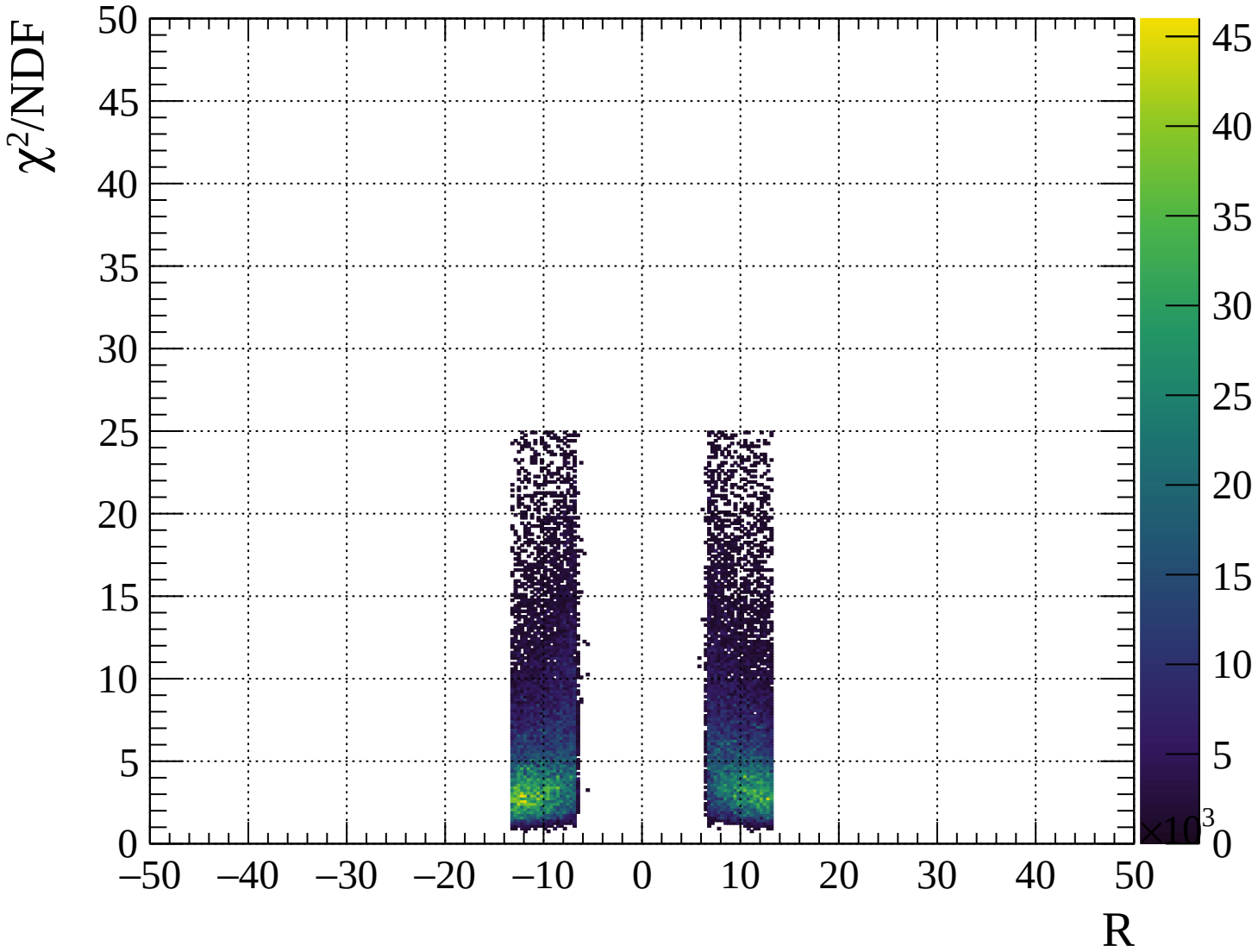






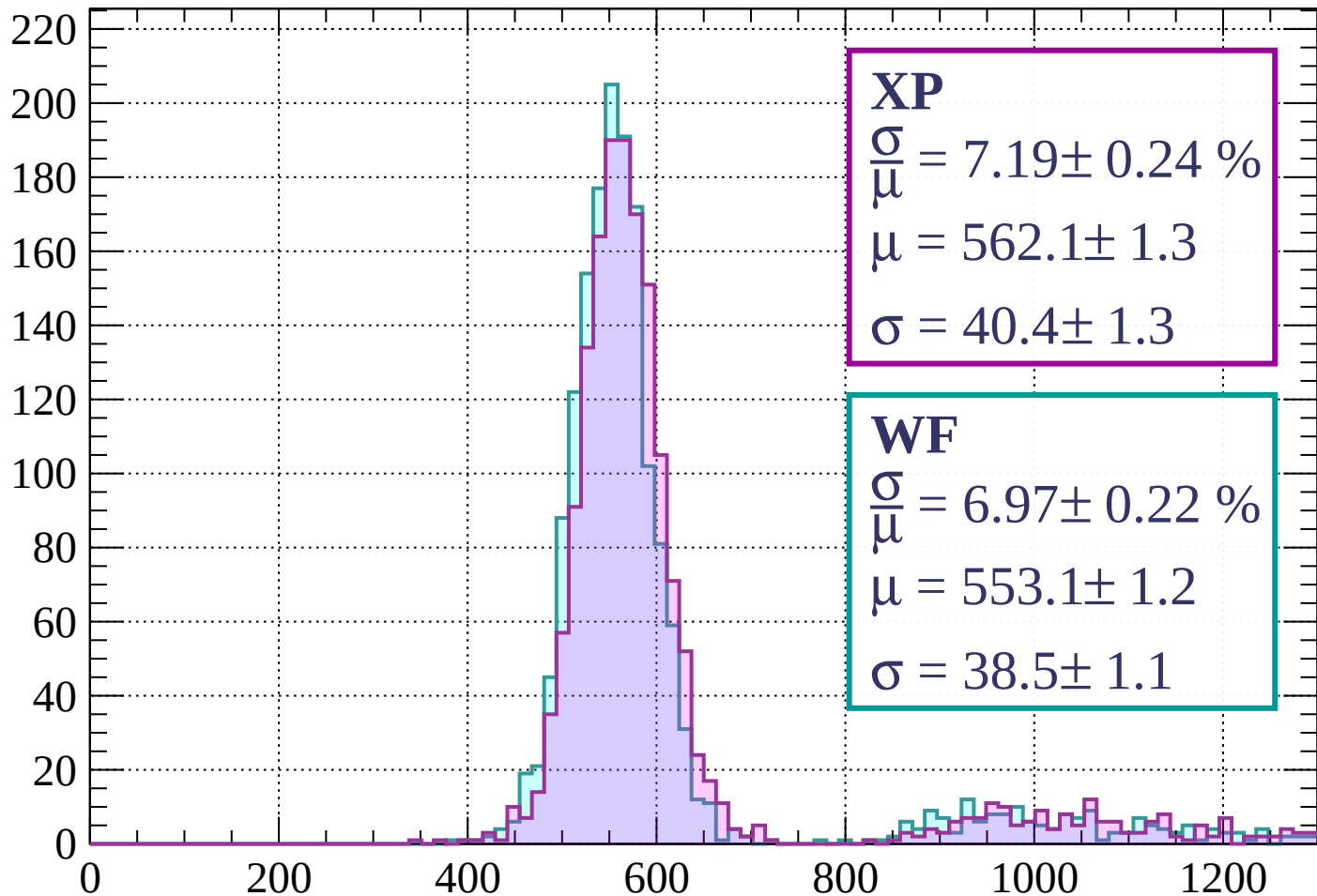






Energy loss | $0 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 7.19 \pm 0.24 \%$$

$$\mu = 562.1 \pm 1.3$$

$$\sigma = 40.4 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.97 \pm 0.22 \%$$

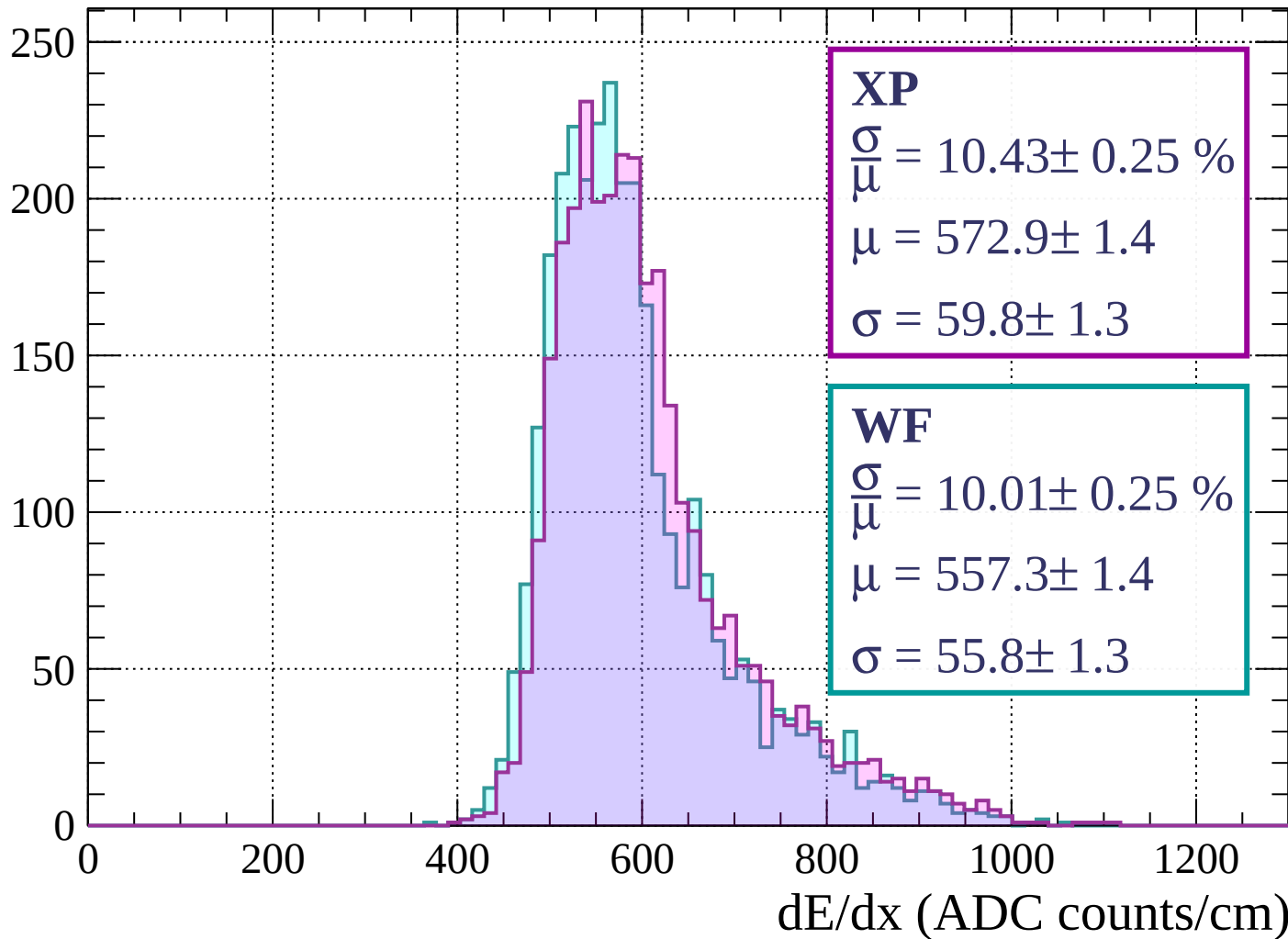
$$\mu = 553.1 \pm 1.2$$

$$\sigma = 38.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $80 < p < 160$

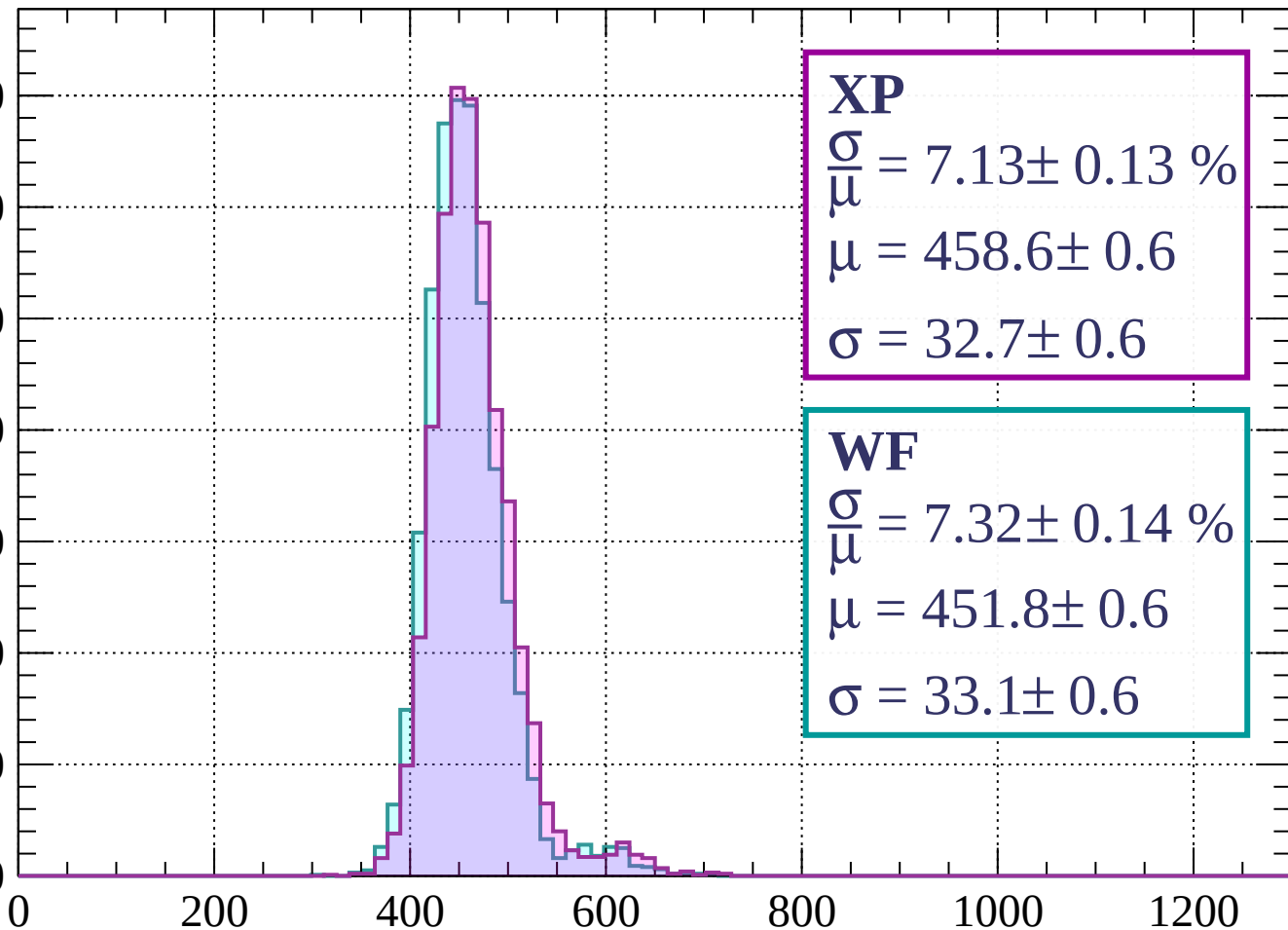
Count



Energy loss | $160 < p < 240$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 7.13 \pm 0.13 \%$$

$$\mu = 458.6 \pm 0.6$$

$$\sigma = 32.7 \pm 0.6$$

WF

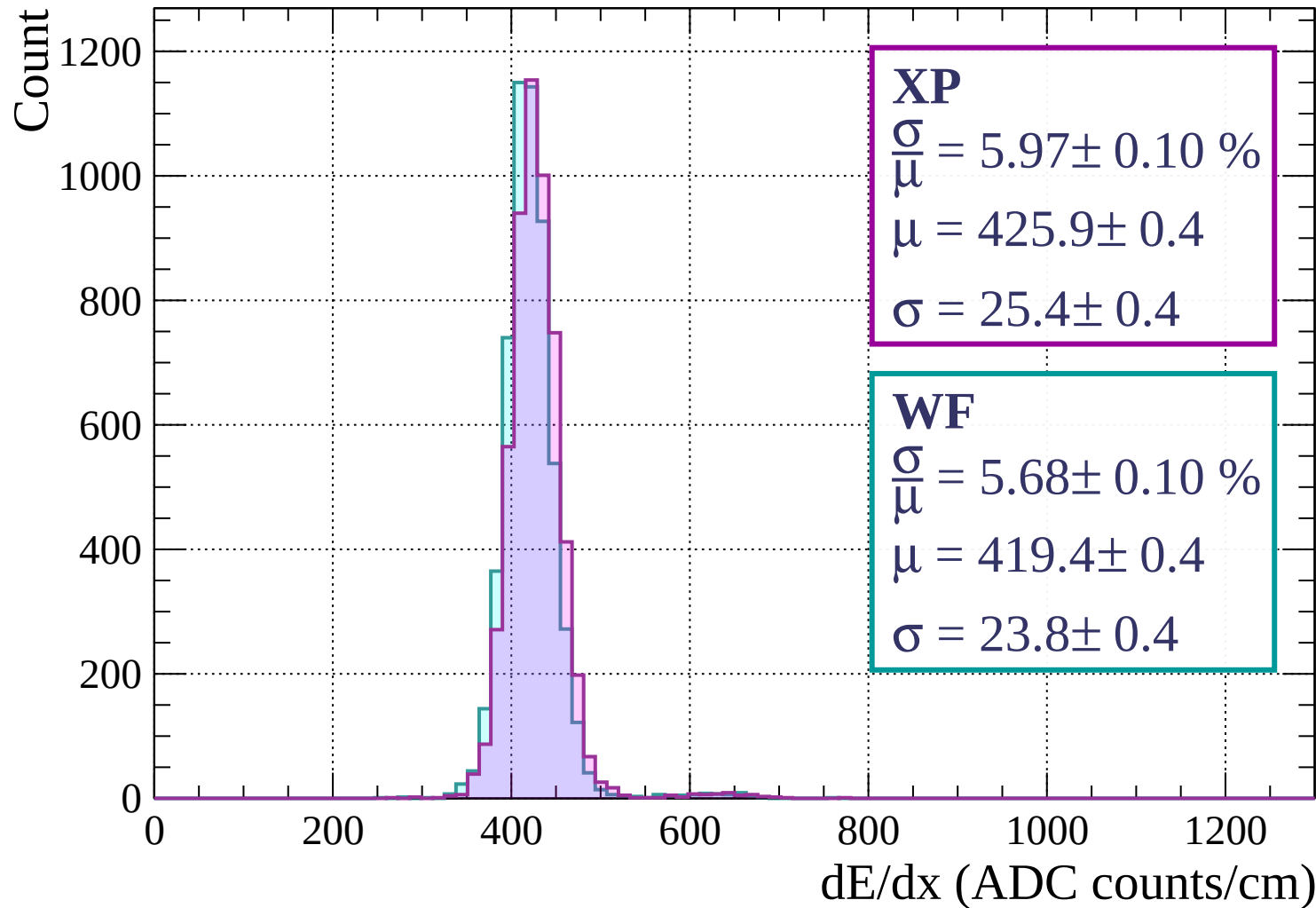
$$\frac{\sigma}{\mu} = 7.32 \pm 0.14 \%$$

$$\mu = 451.8 \pm 0.6$$

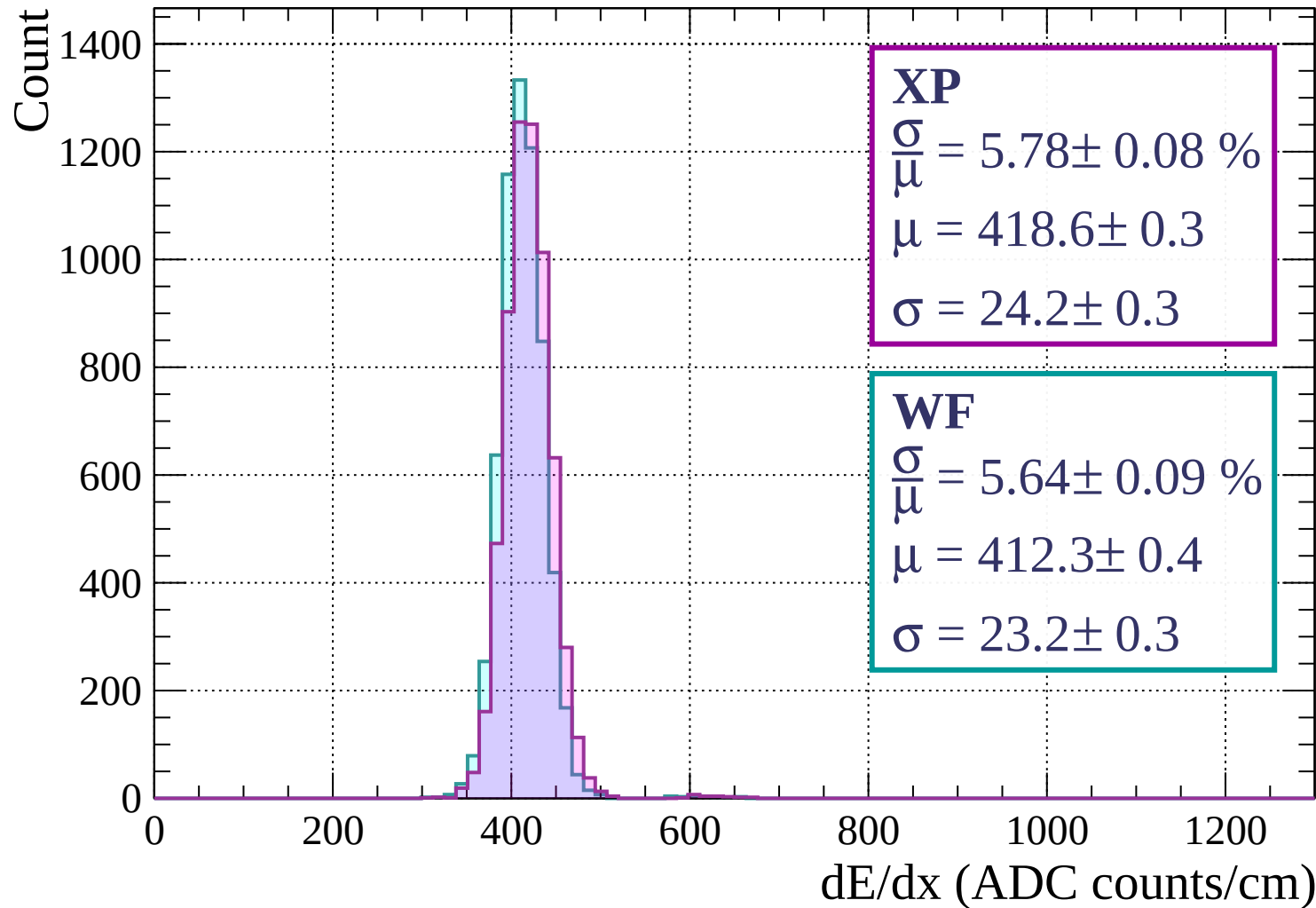
$$\sigma = 33.1 \pm 0.6$$

dE/dx (ADC counts/cm)

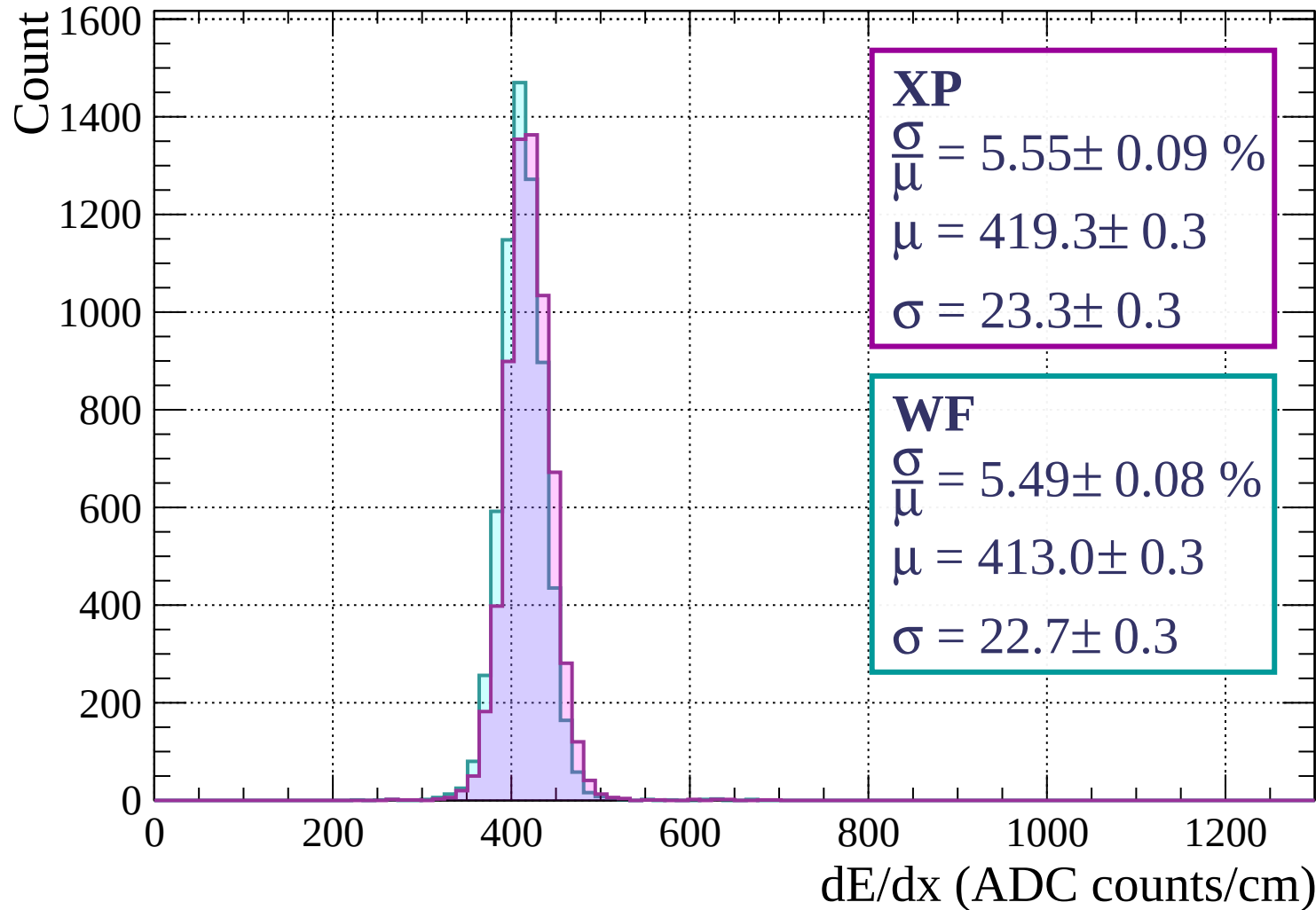
Energy loss | $240 < p < 320$



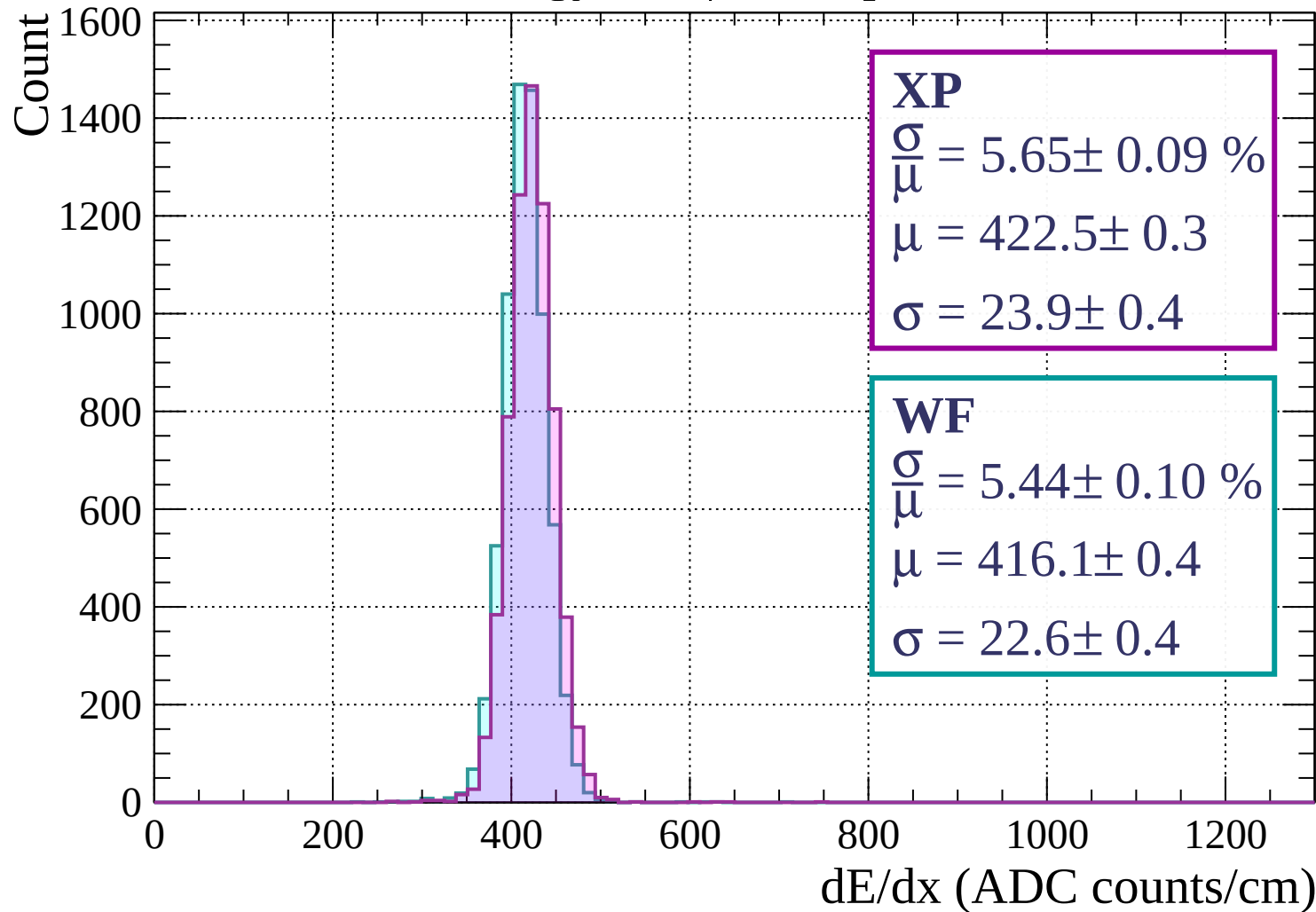
Energy loss | $320 < p < 400$



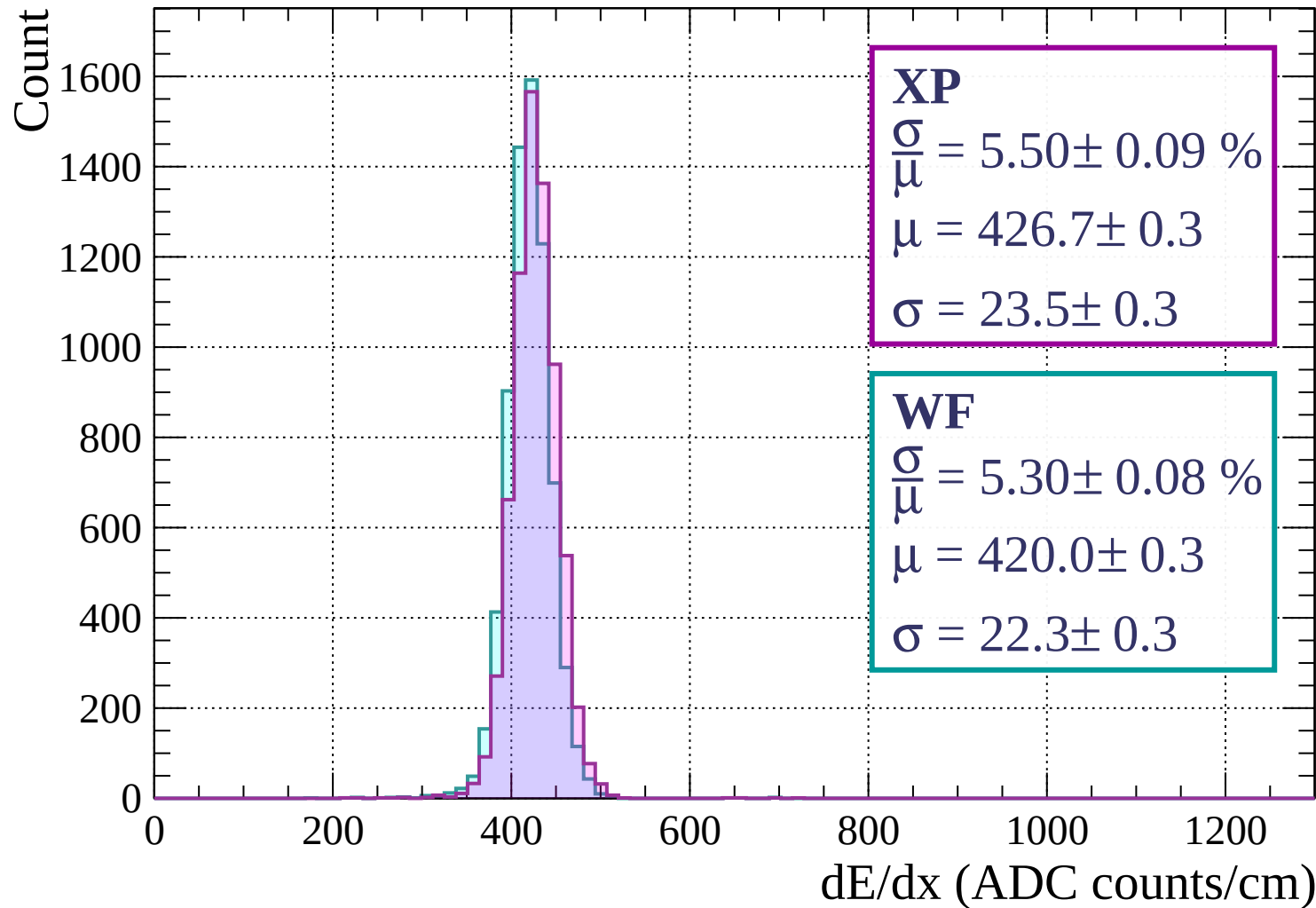
Energy loss | $400 < p < 480$



Energy loss | $480 < p < 560$



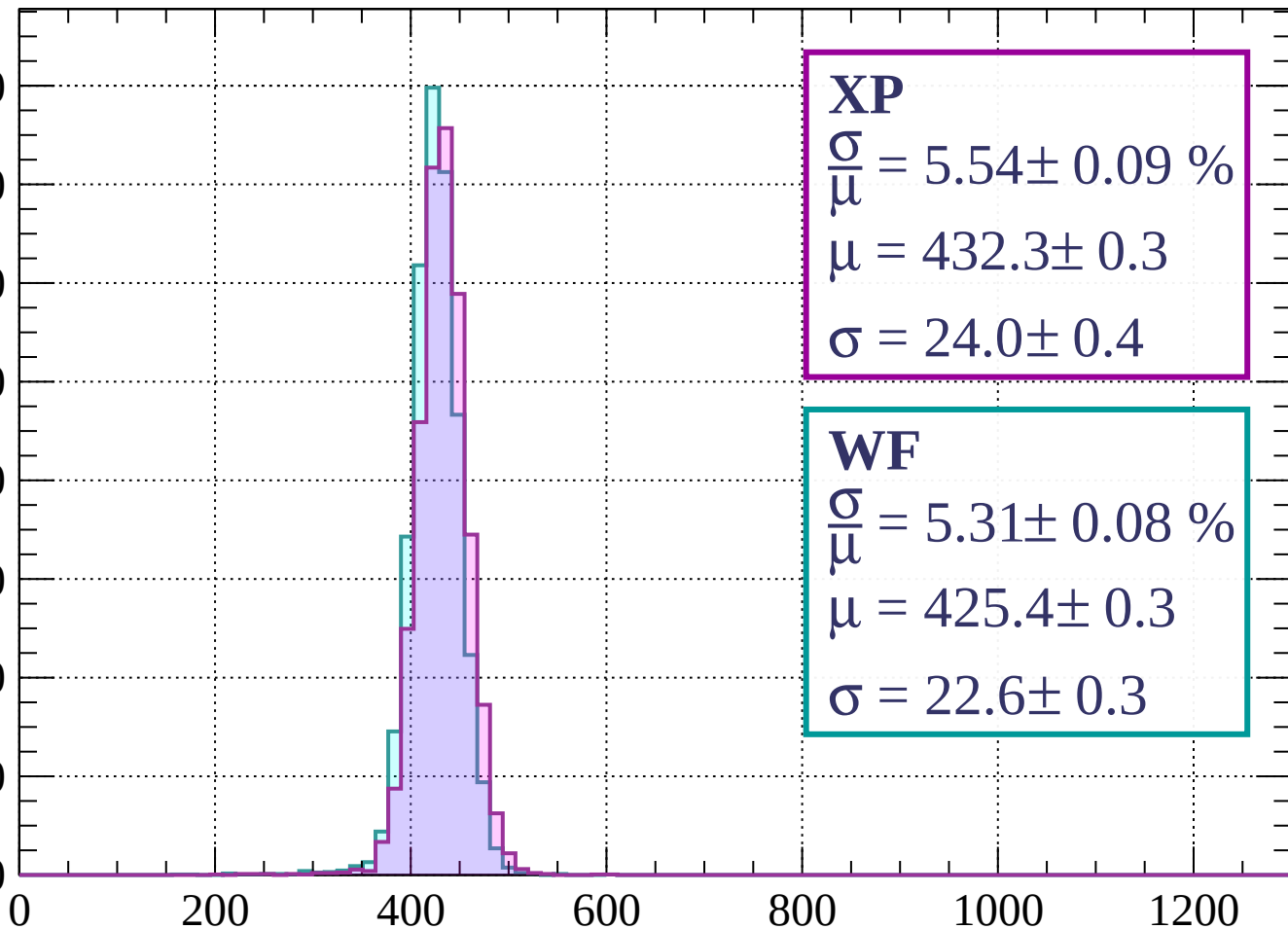
Energy loss | $560 < p < 640$



Energy loss | $640 < p < 720$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.54 \pm 0.09 \%$$

$$\mu = 432.3 \pm 0.3$$

$$\sigma = 24.0 \pm 0.4$$

WF

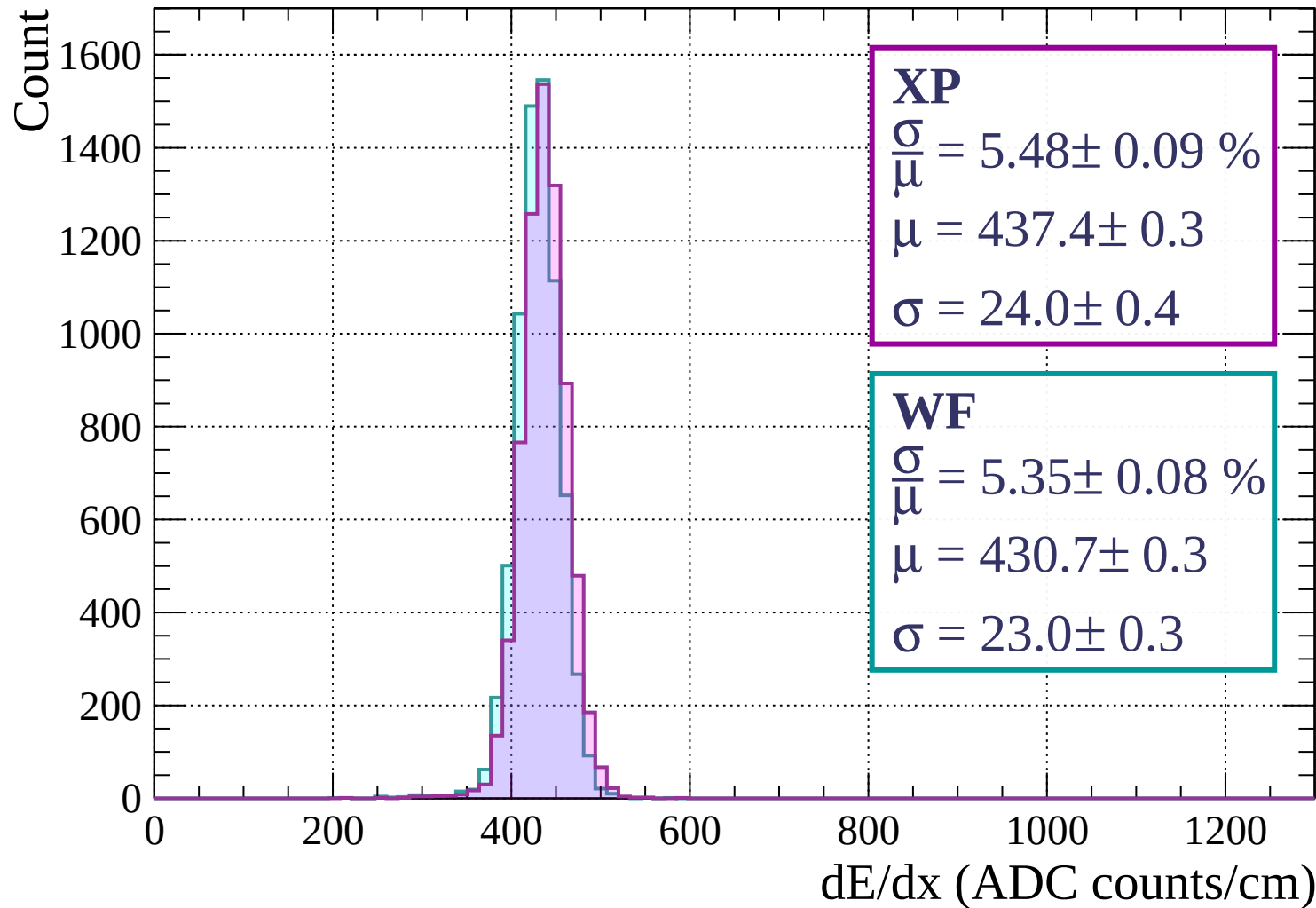
$$\frac{\sigma}{\mu} = 5.31 \pm 0.08 \%$$

$$\mu = 425.4 \pm 0.3$$

$$\sigma = 22.6 \pm 0.3$$

dE/dx (ADC counts/cm)

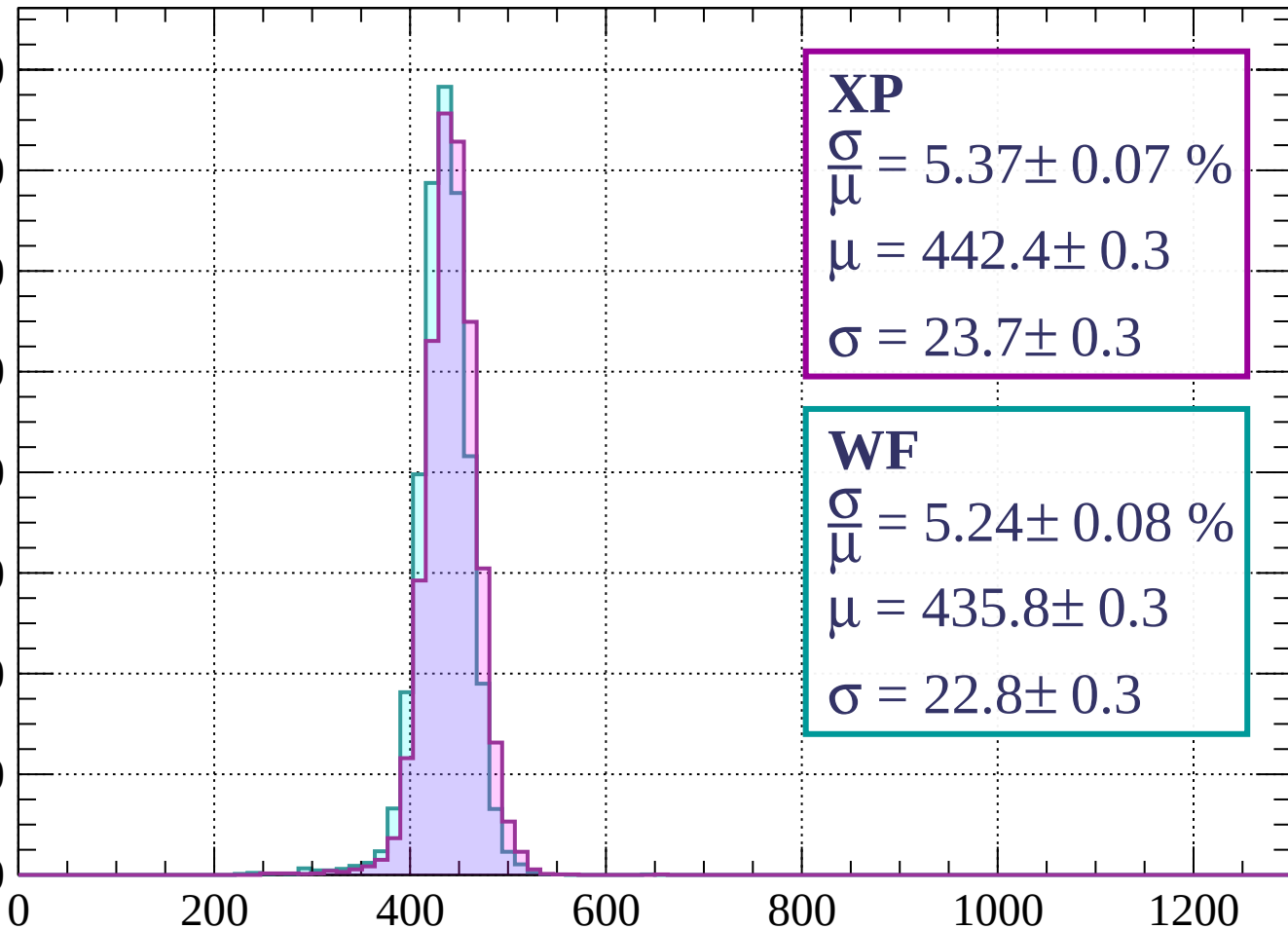
Energy loss | $720 < p < 800$



Energy loss | $800 < p < 880$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.37 \pm 0.07 \%$$

$$\mu = 442.4 \pm 0.3$$

$$\sigma = 23.7 \pm 0.3$$

WF

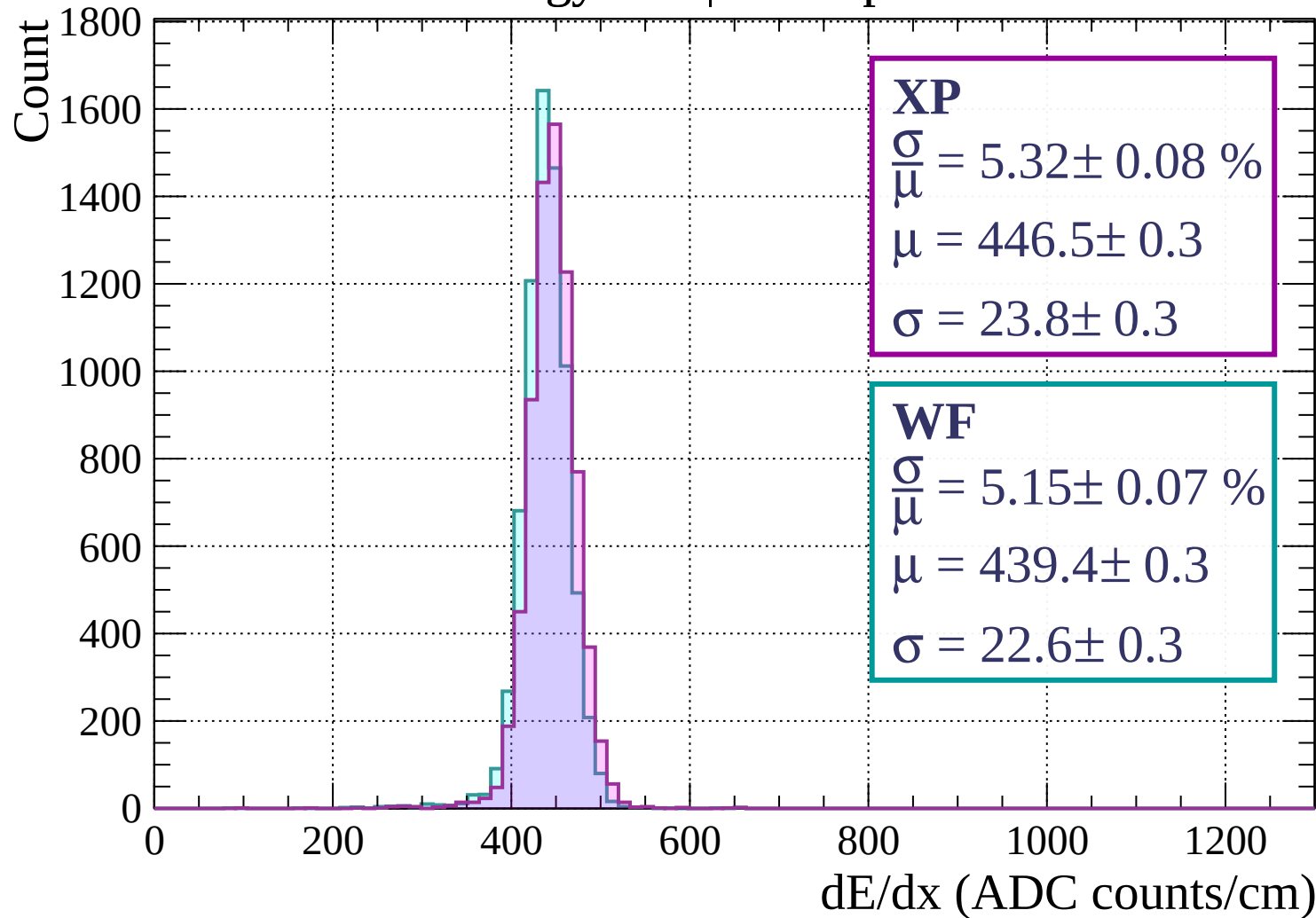
$$\frac{\sigma}{\mu} = 5.24 \pm 0.08 \%$$

$$\mu = 435.8 \pm 0.3$$

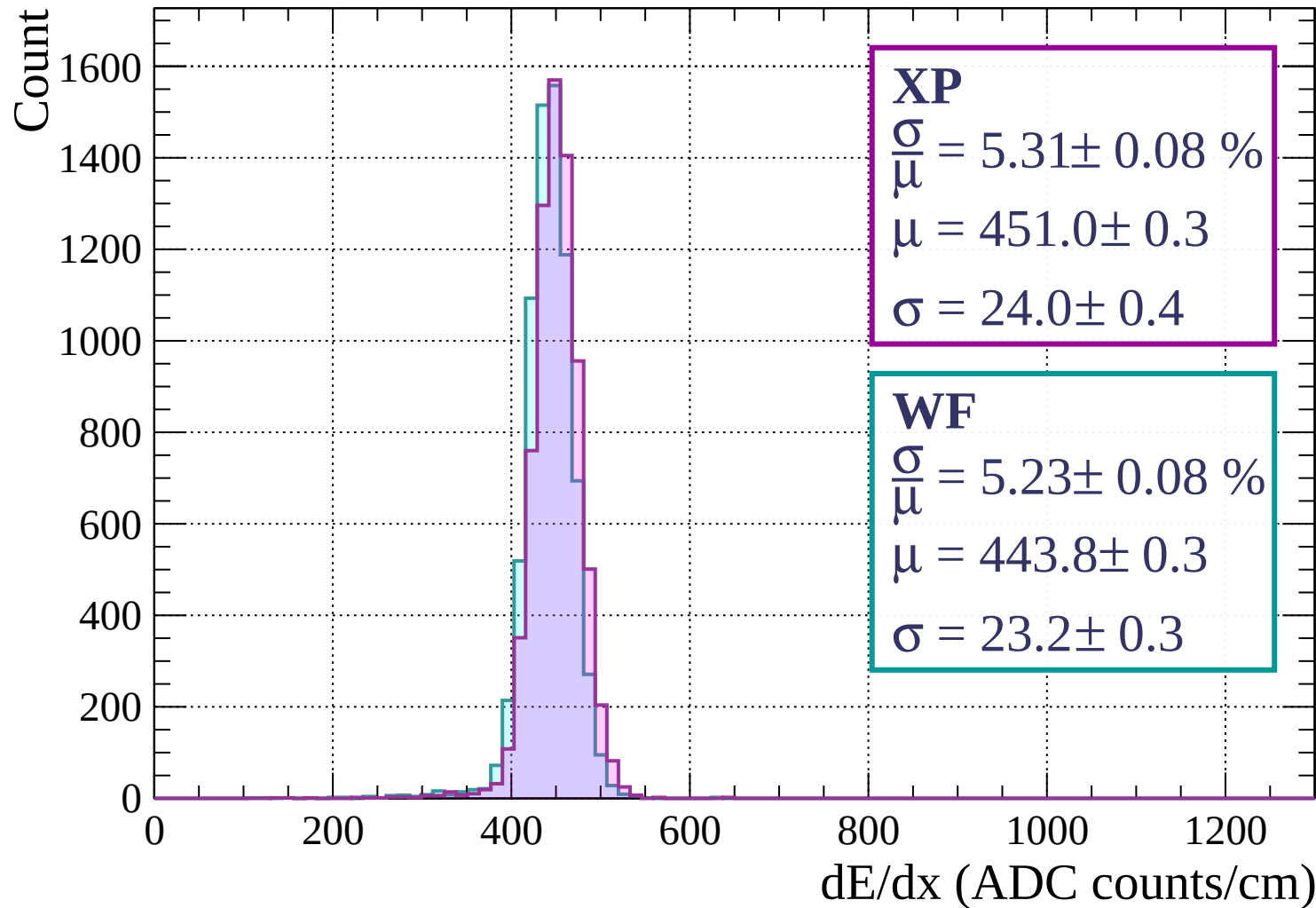
$$\sigma = 22.8 \pm 0.3$$

dE/dx (ADC counts/cm)

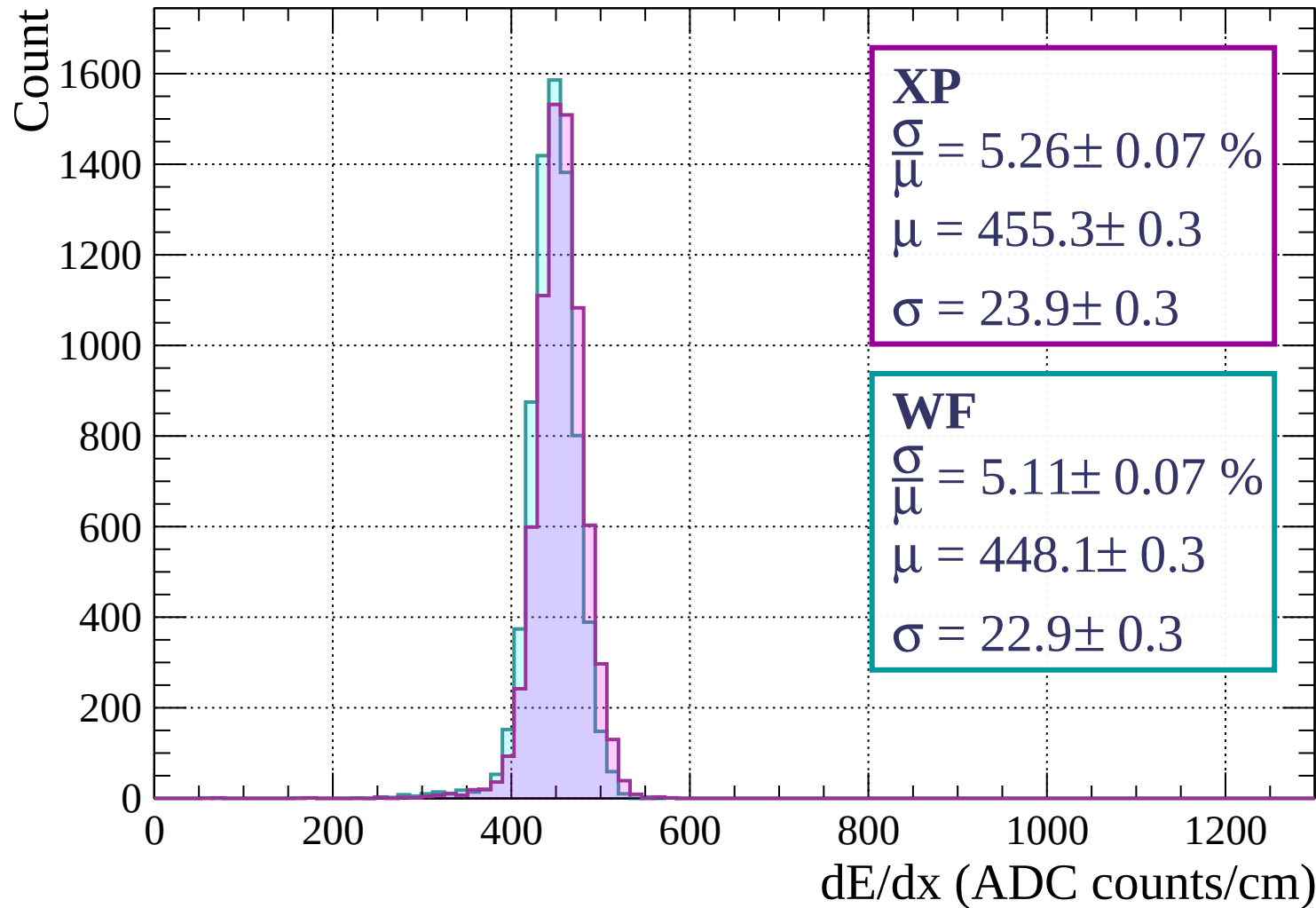
Energy loss | $880 < p < 960$



Energy loss | $960 < p < 1040$



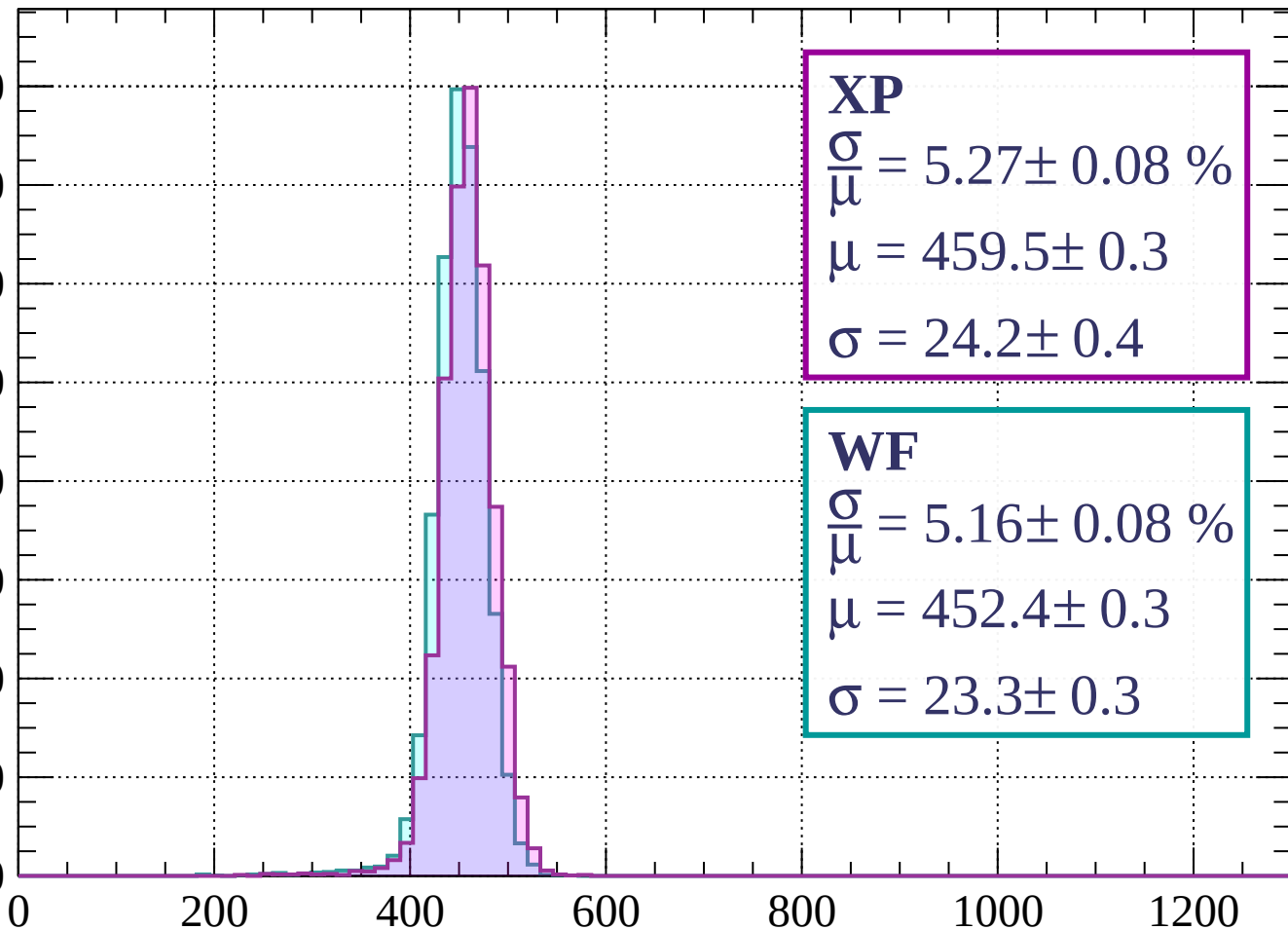
Energy loss | $1040 < p < 1120$



Energy loss | $1120 < p < 1200$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.27 \pm 0.08 \%$$

$$\mu = 459.5 \pm 0.3$$

$$\sigma = 24.2 \pm 0.4$$

WF

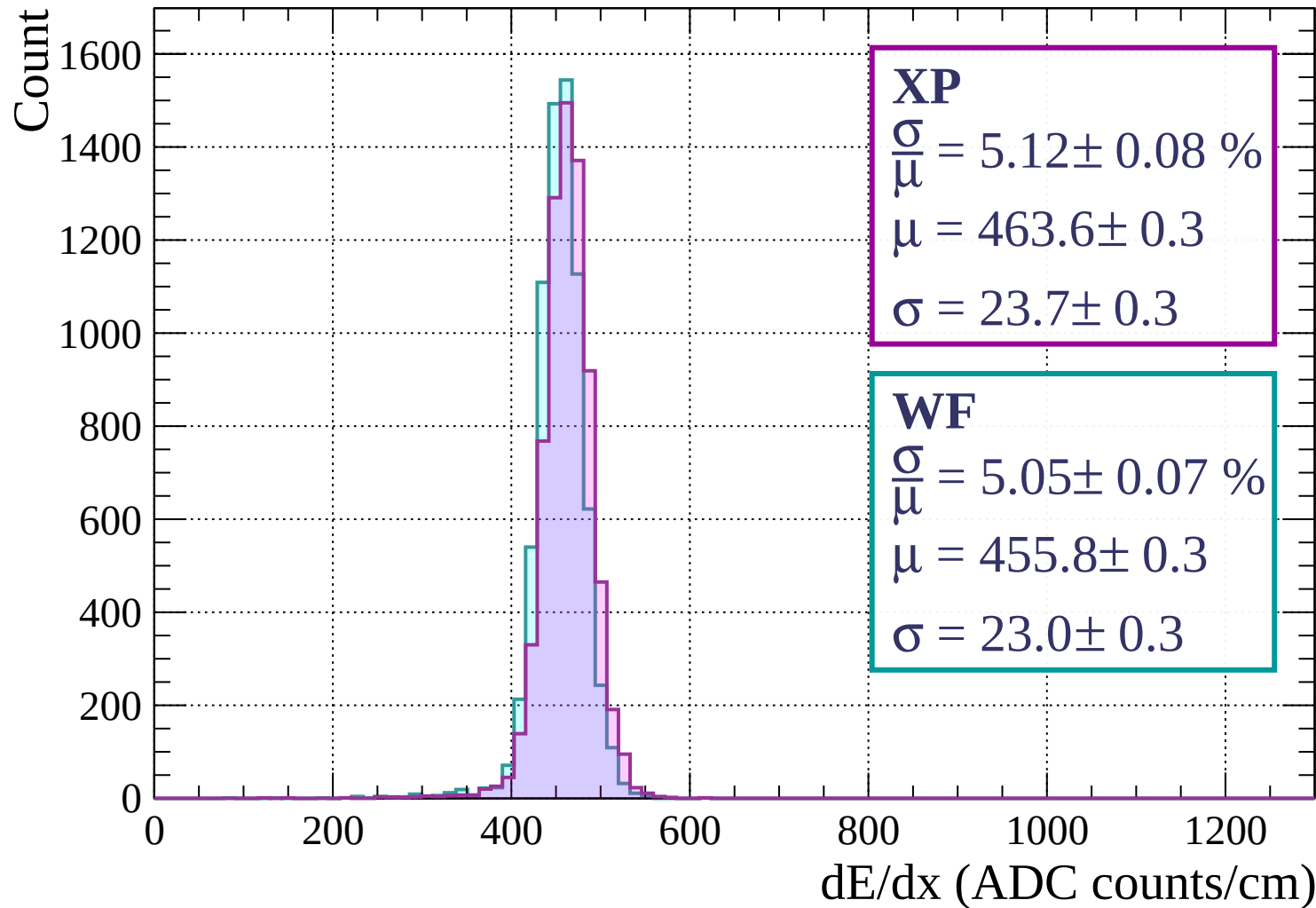
$$\frac{\sigma}{\mu} = 5.16 \pm 0.08 \%$$

$$\mu = 452.4 \pm 0.3$$

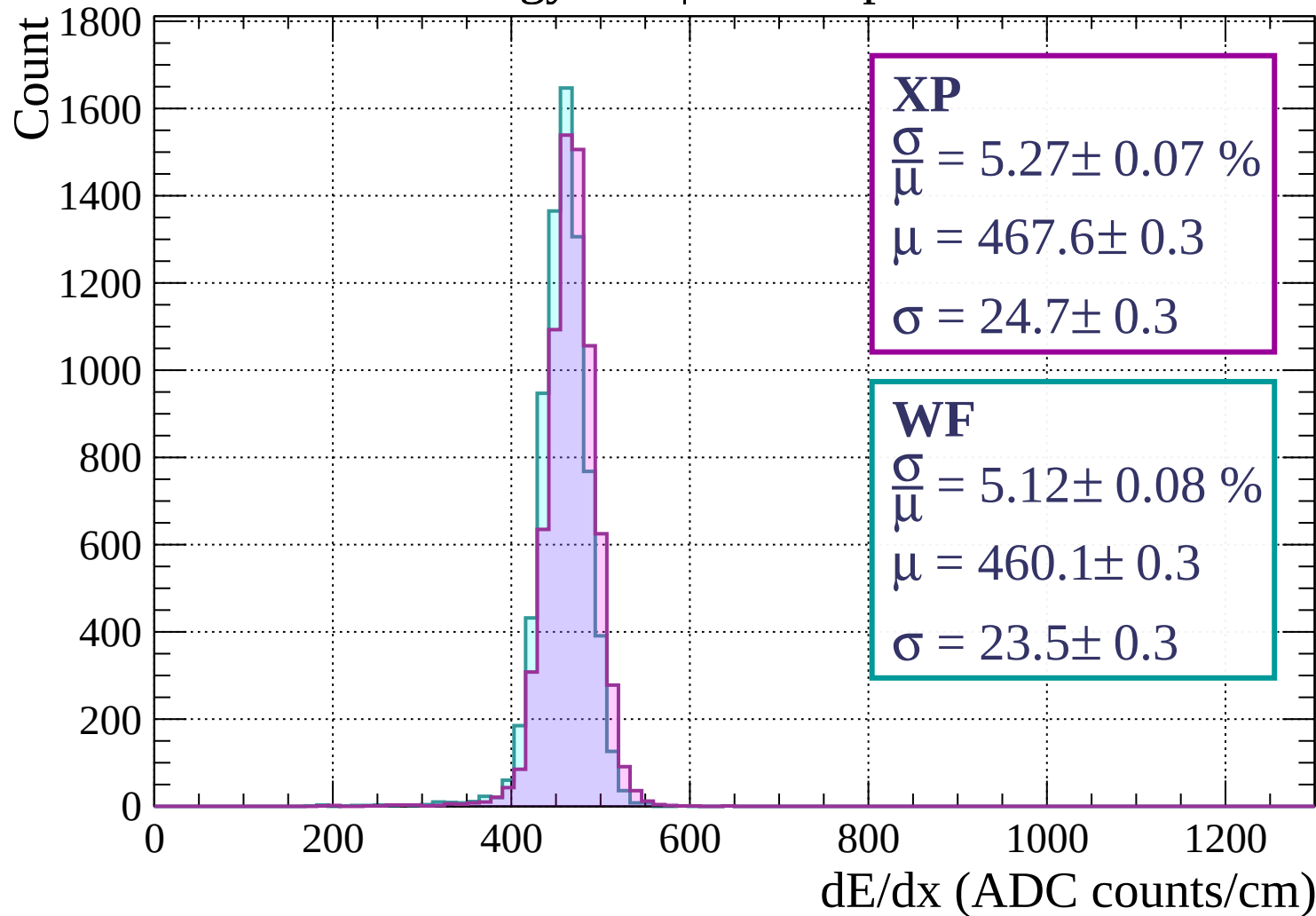
$$\sigma = 23.3 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

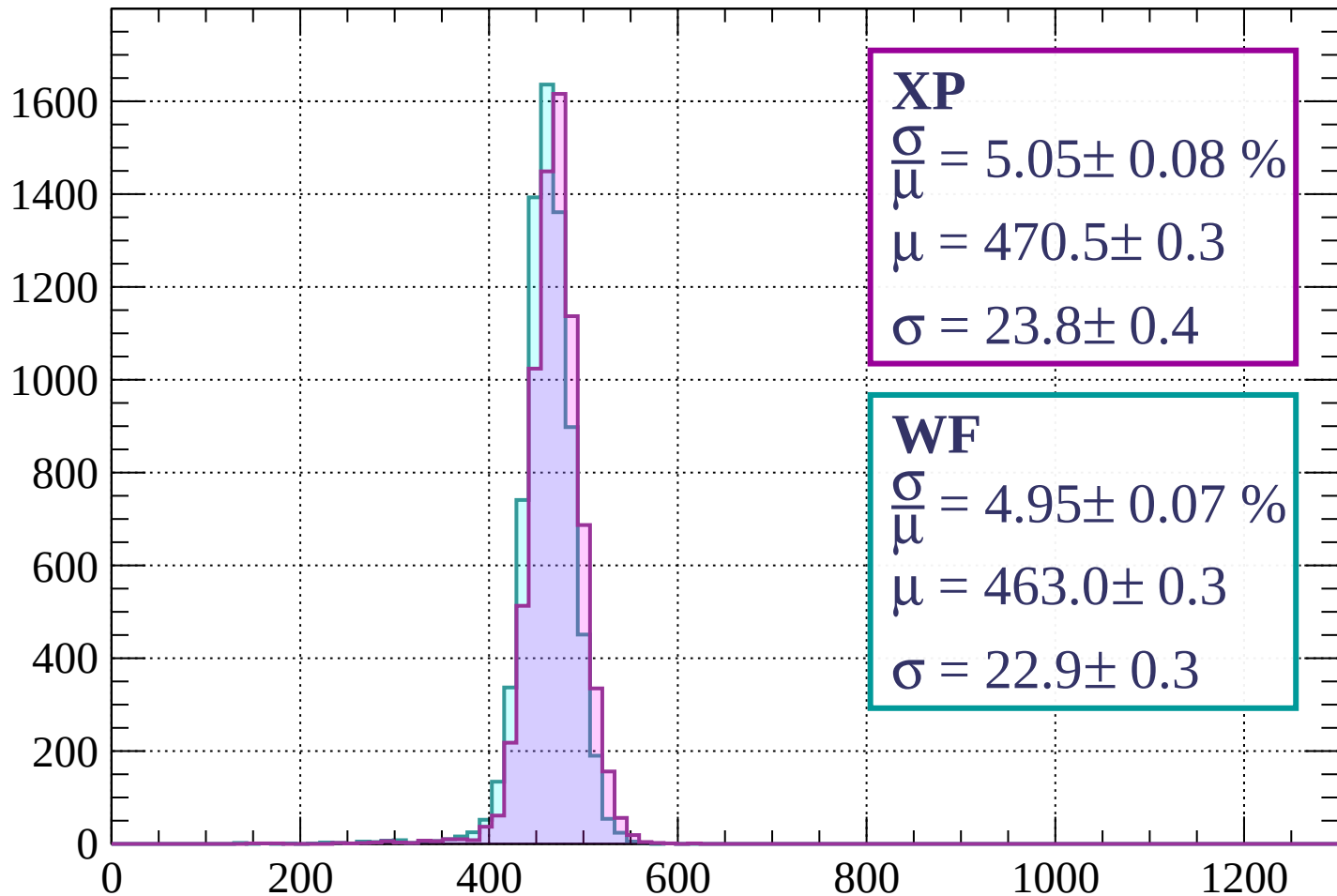


Energy loss | $1280 < p < 1360$



Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 5.05 \pm 0.08 \%$$

$$\mu = 470.5 \pm 0.3$$

$$\sigma = 23.8 \pm 0.4$$

WF

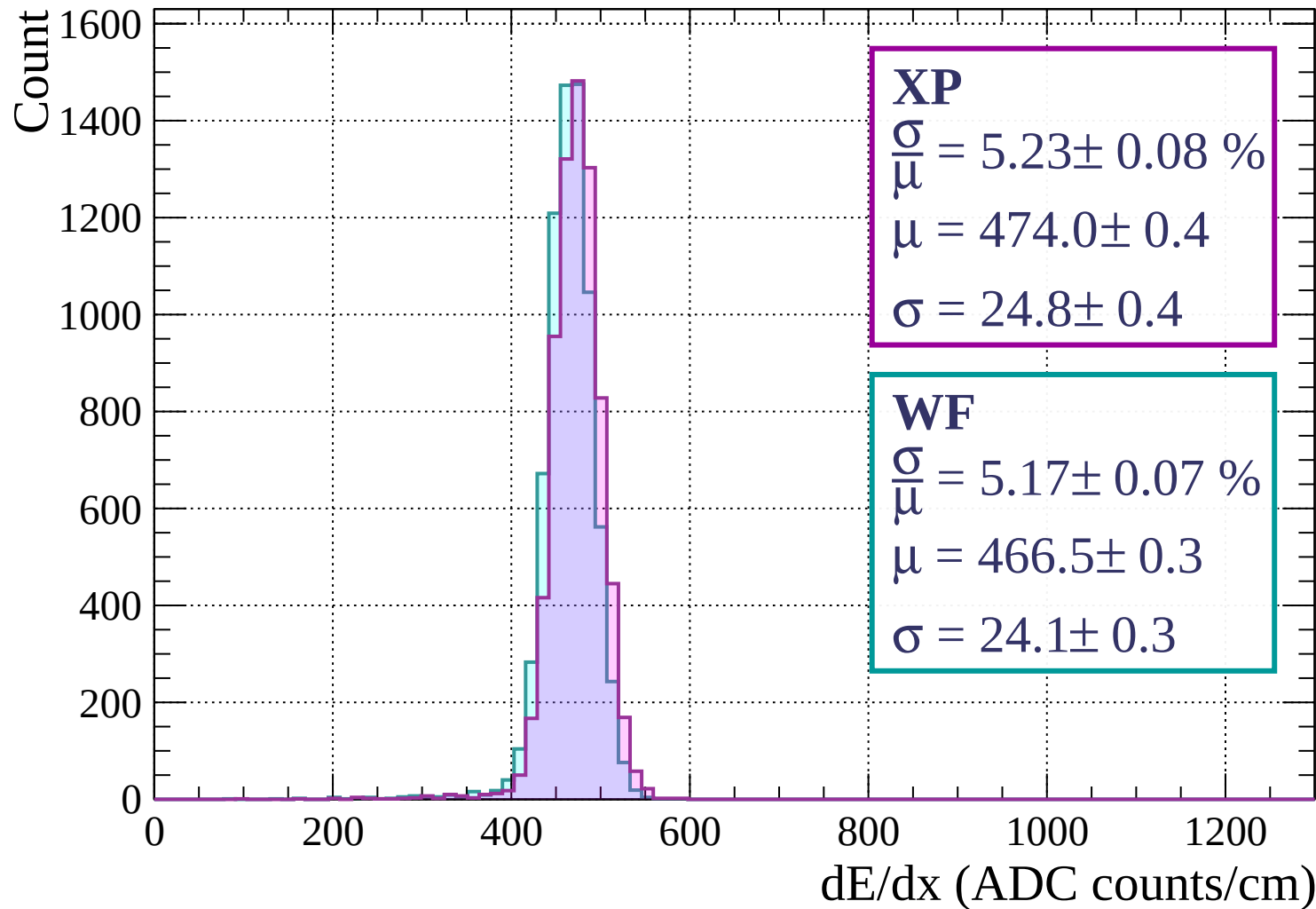
$$\frac{\sigma}{\mu} = 4.95 \pm 0.07 \%$$

$$\mu = 463.0 \pm 0.3$$

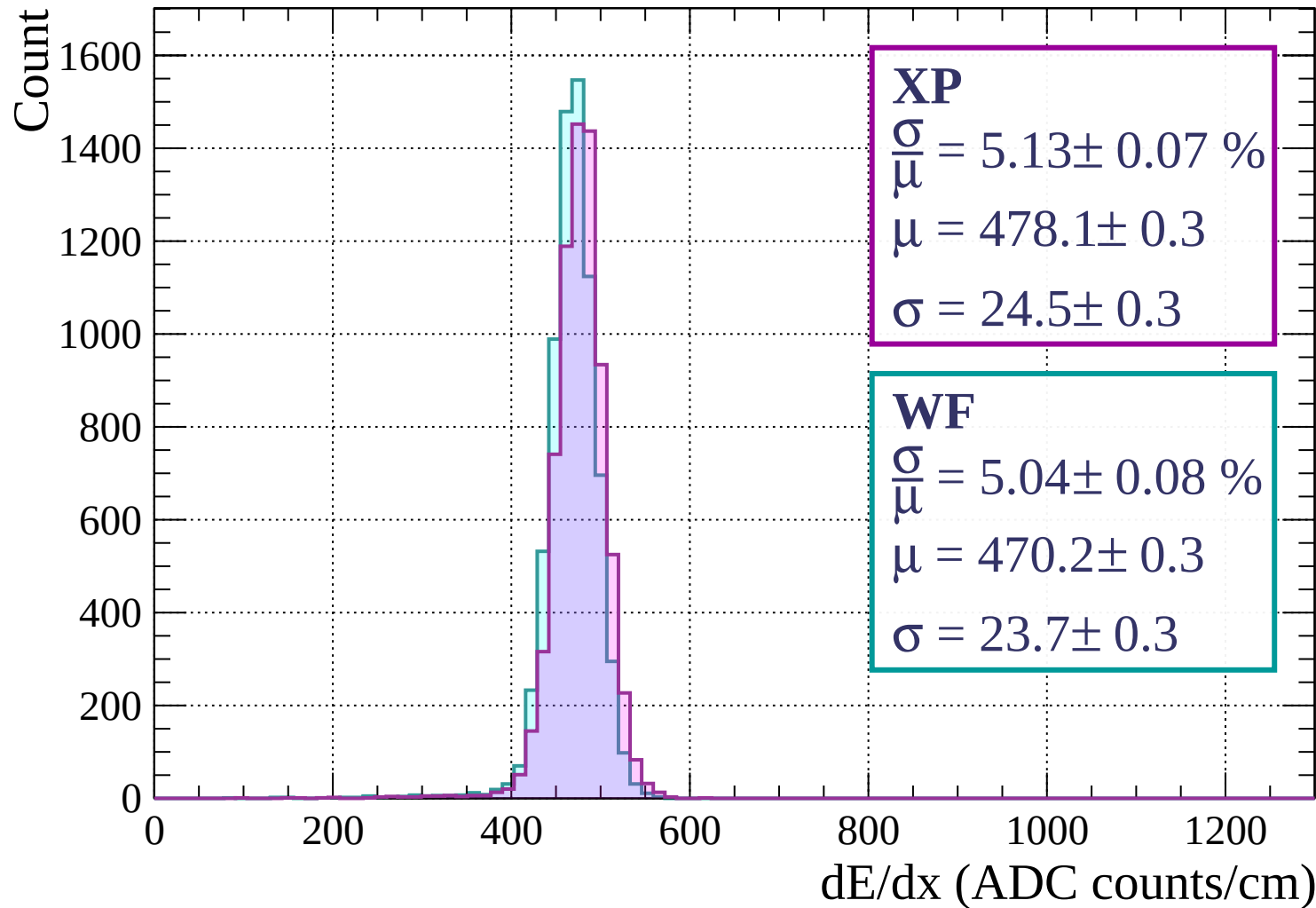
$$\sigma = 22.9 \pm 0.3$$

dE/dx (ADC counts/cm)

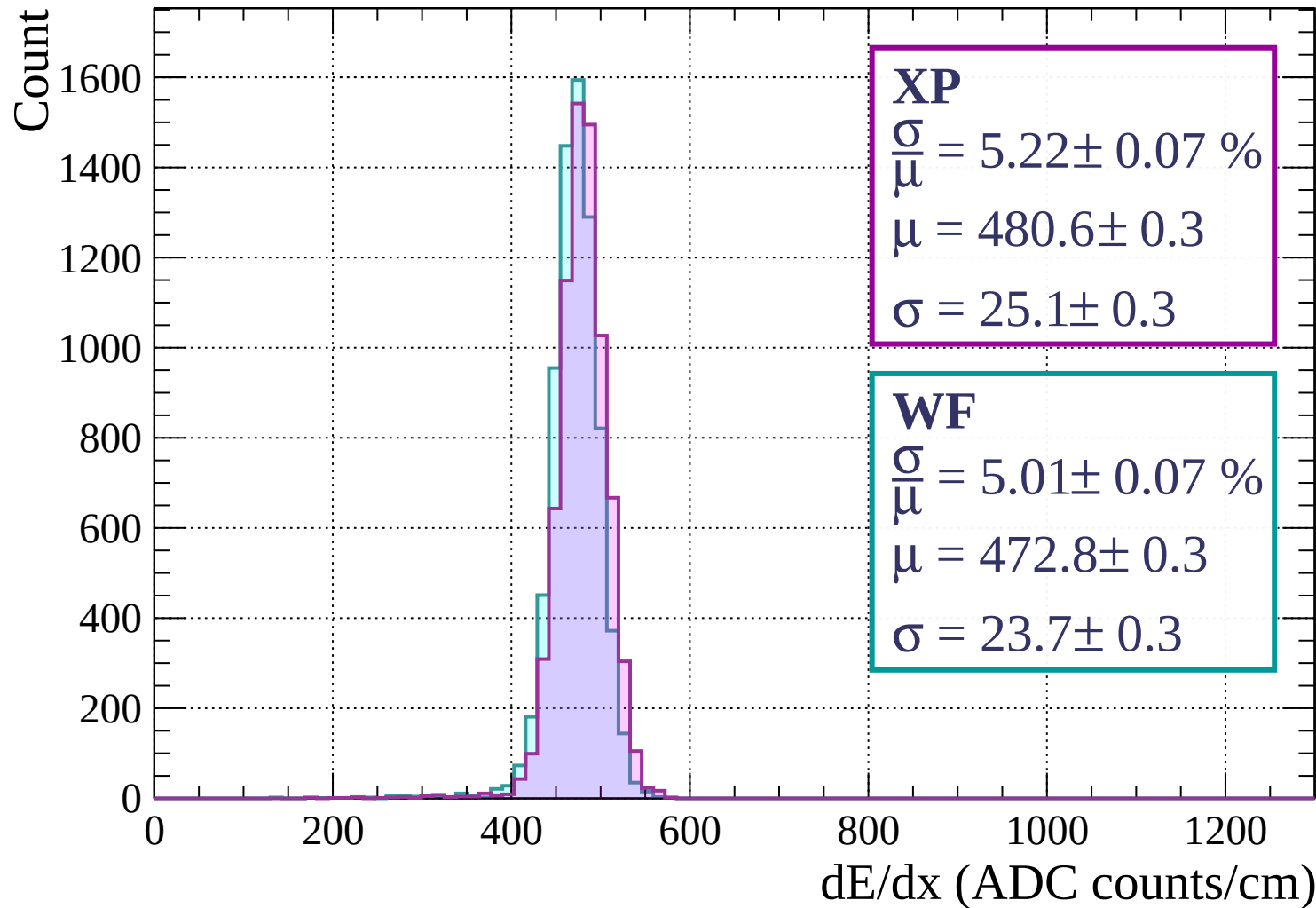
Energy loss | $1440 < p < 1520$



Energy loss | $1520 < p < 1600$

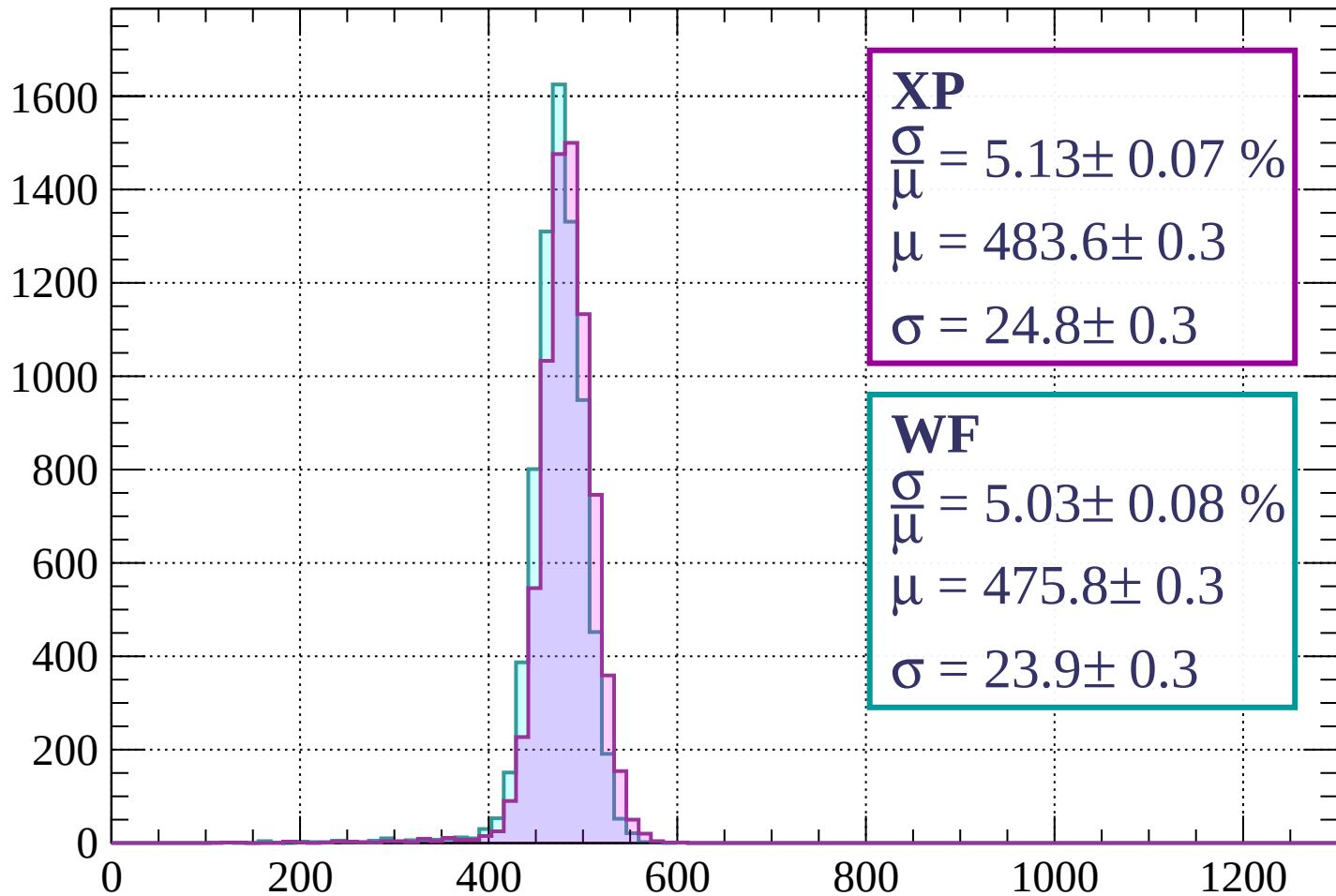


Energy loss | $1600 < p < 1680$



Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 5.13 \pm 0.07 \%$$

$$\mu = 483.6 \pm 0.3$$

$$\sigma = 24.8 \pm 0.3$$

WF

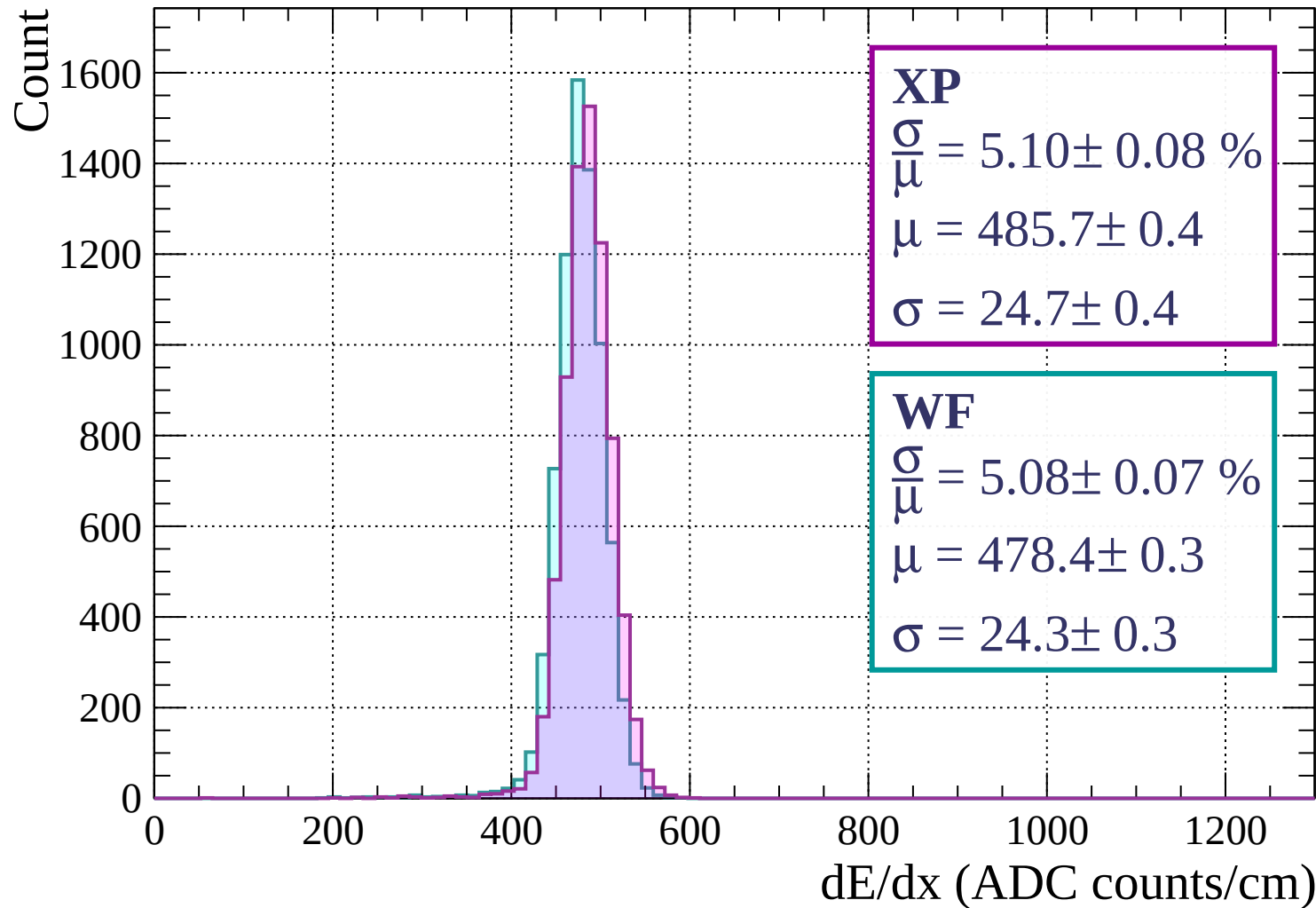
$$\frac{\sigma}{\mu} = 5.03 \pm 0.08 \%$$

$$\mu = 475.8 \pm 0.3$$

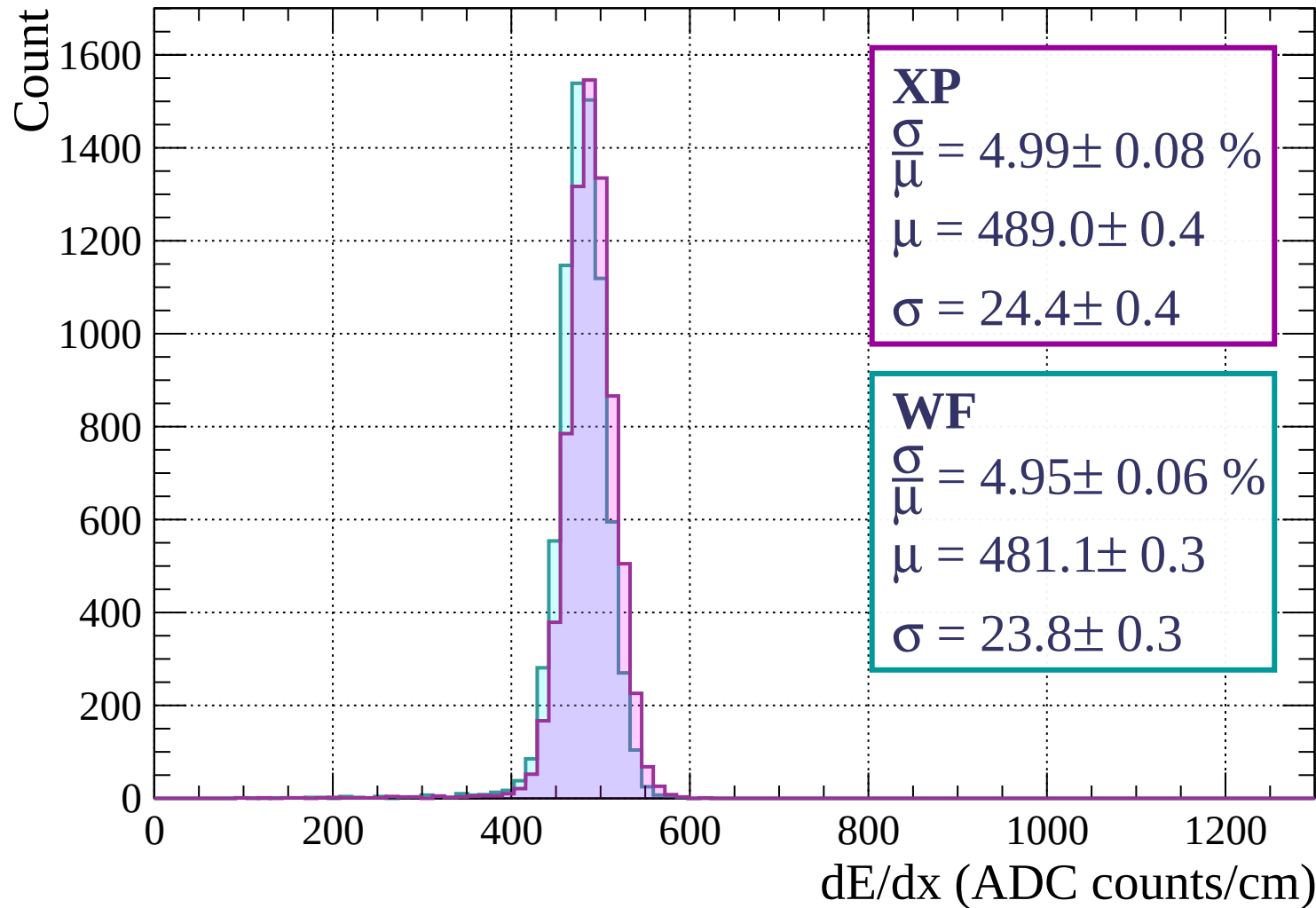
$$\sigma = 23.9 \pm 0.3$$

dE/dx (ADC counts/cm)

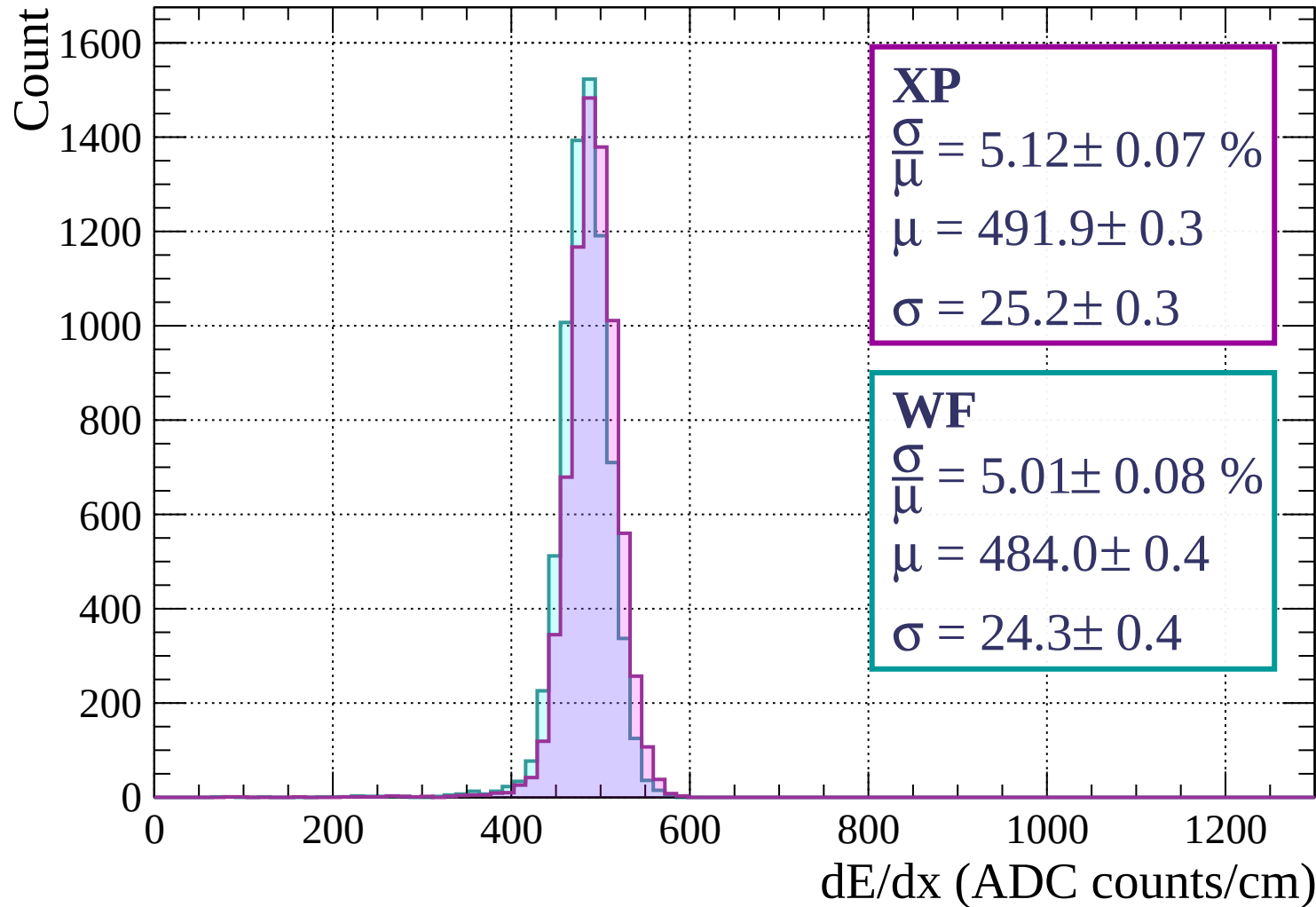
Energy loss | $1760 < p < 1840$



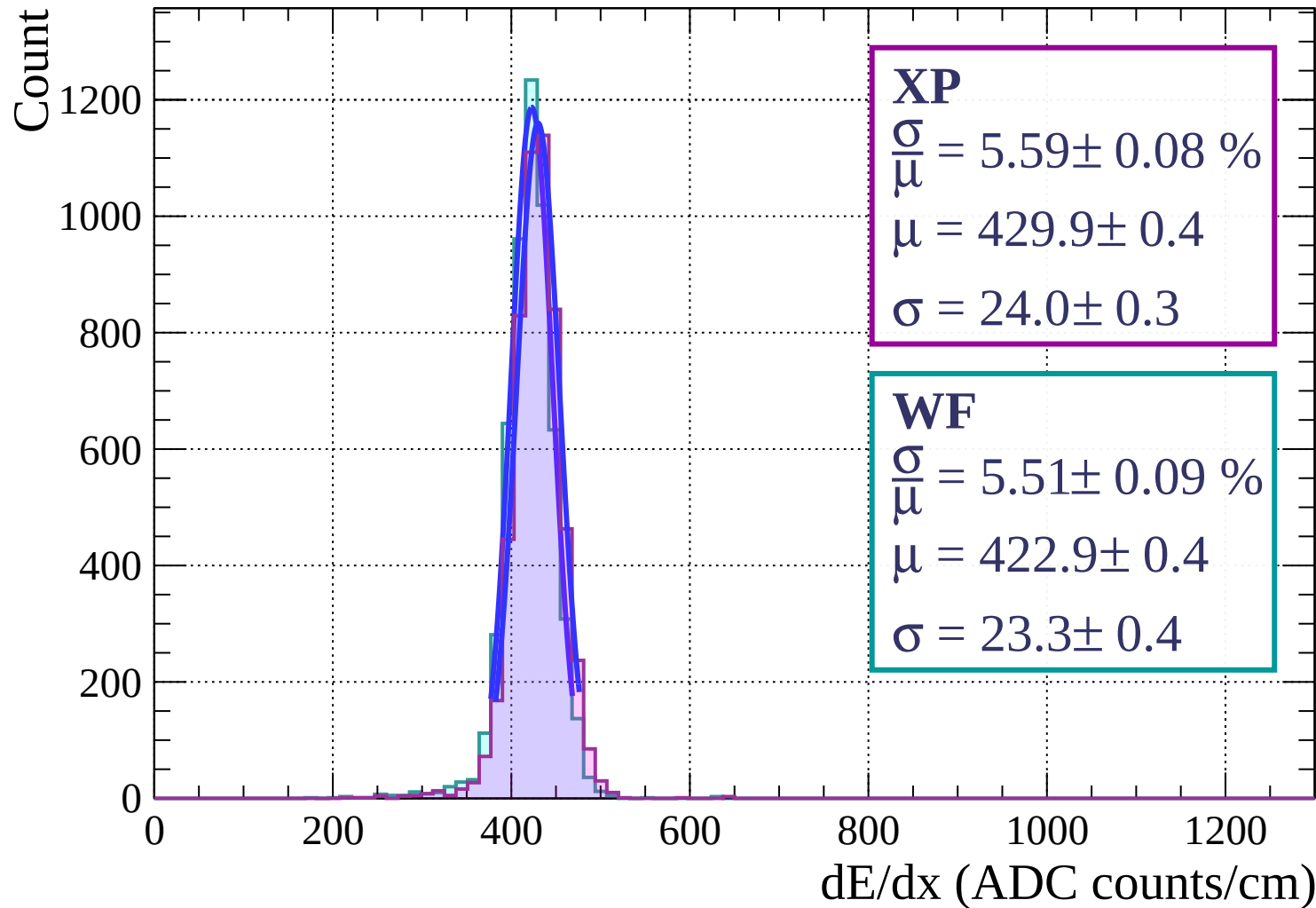
Energy loss | $1840 < p < 1920$



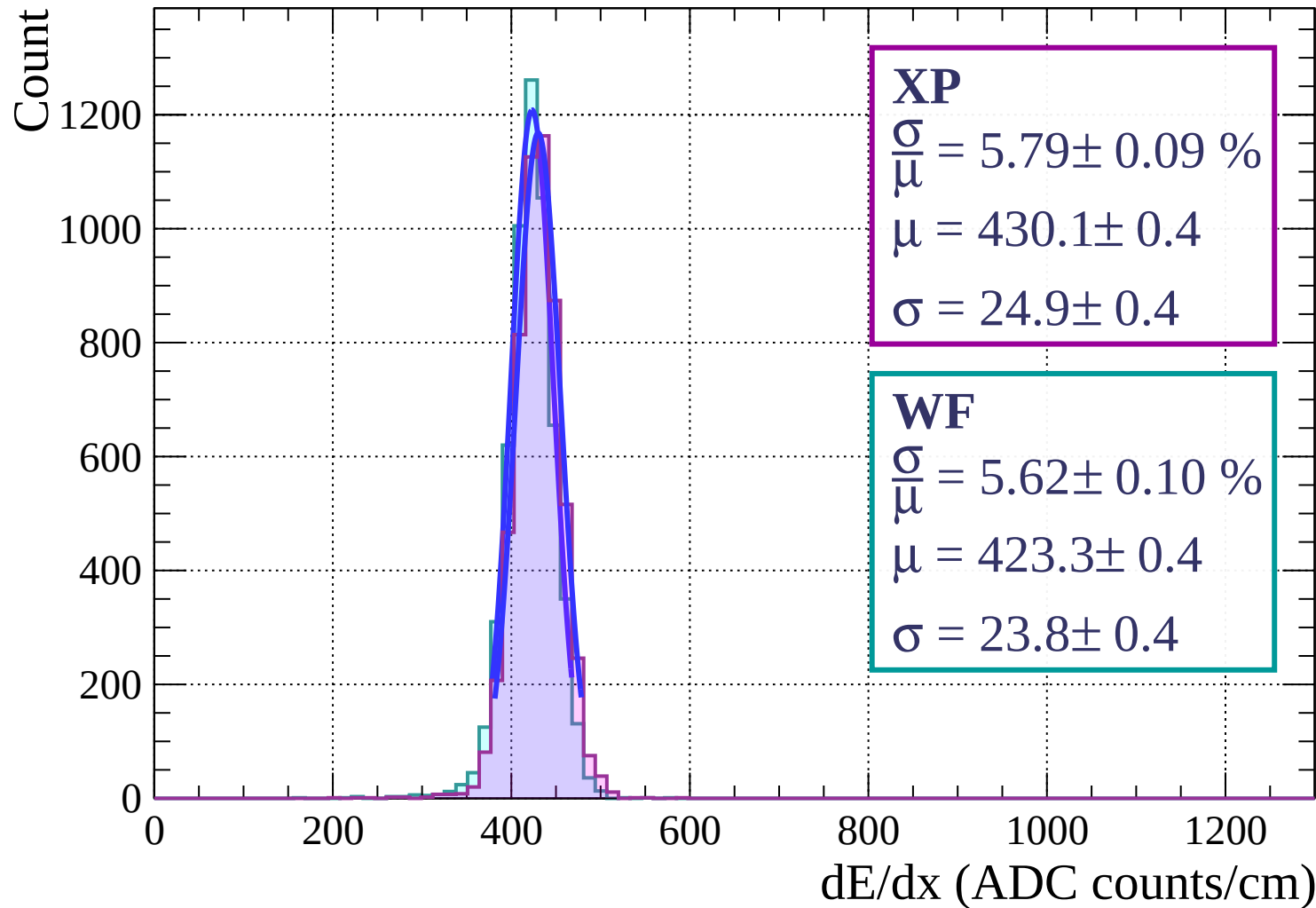
Energy loss | $1920 < p < 2000$



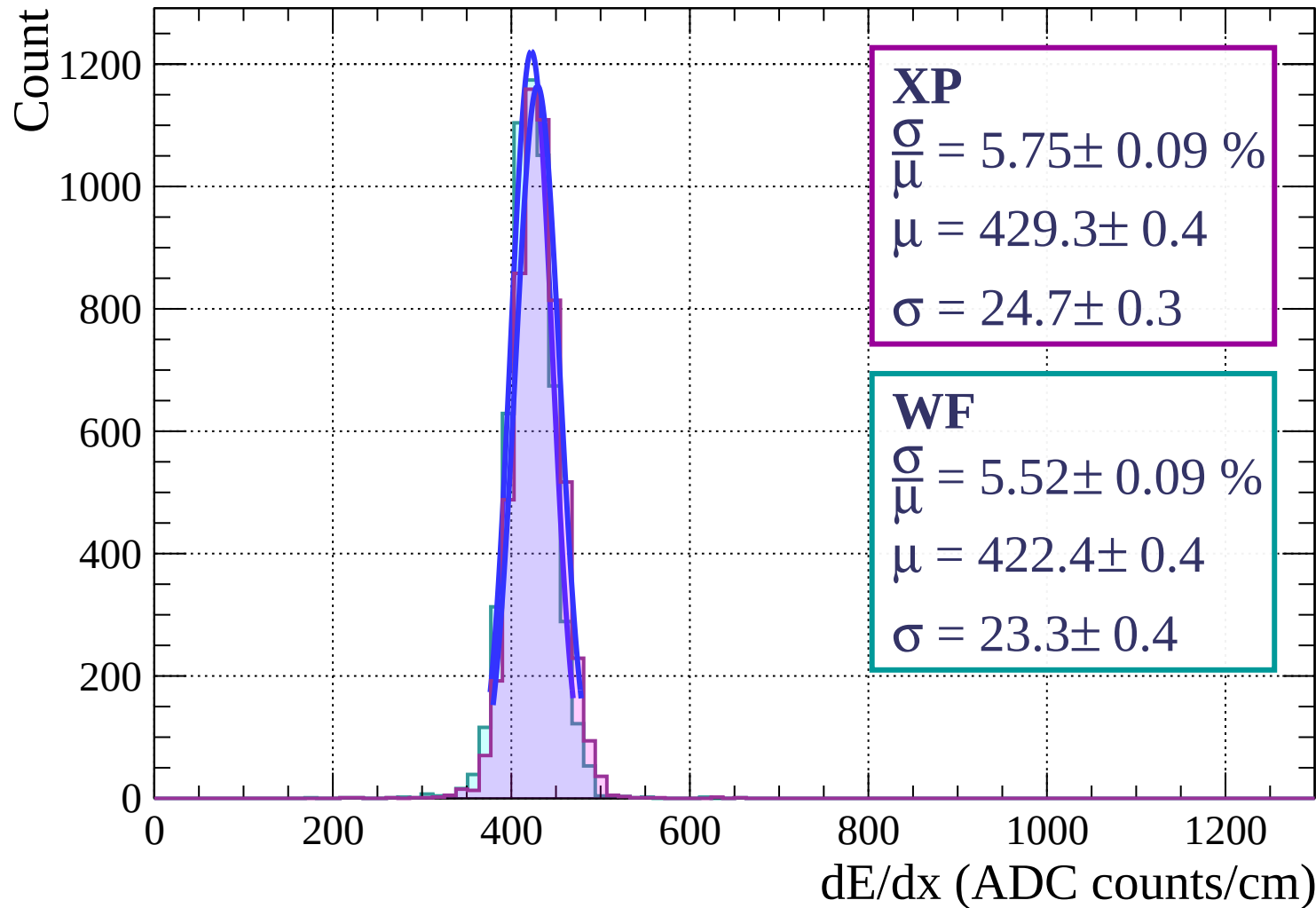
Energy loss | $0 < \phi < 3$



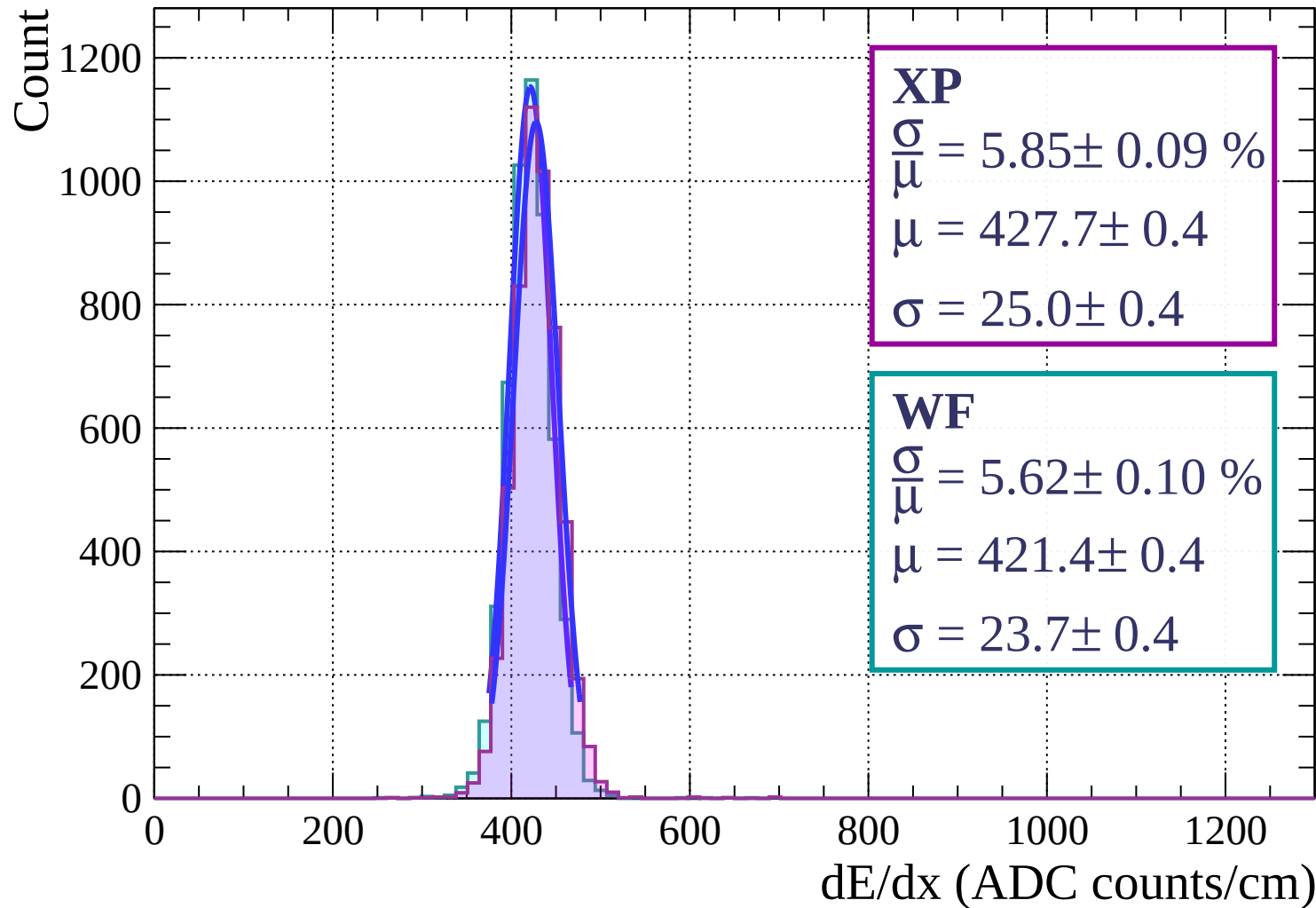
Energy loss | $3 < \phi < 6$



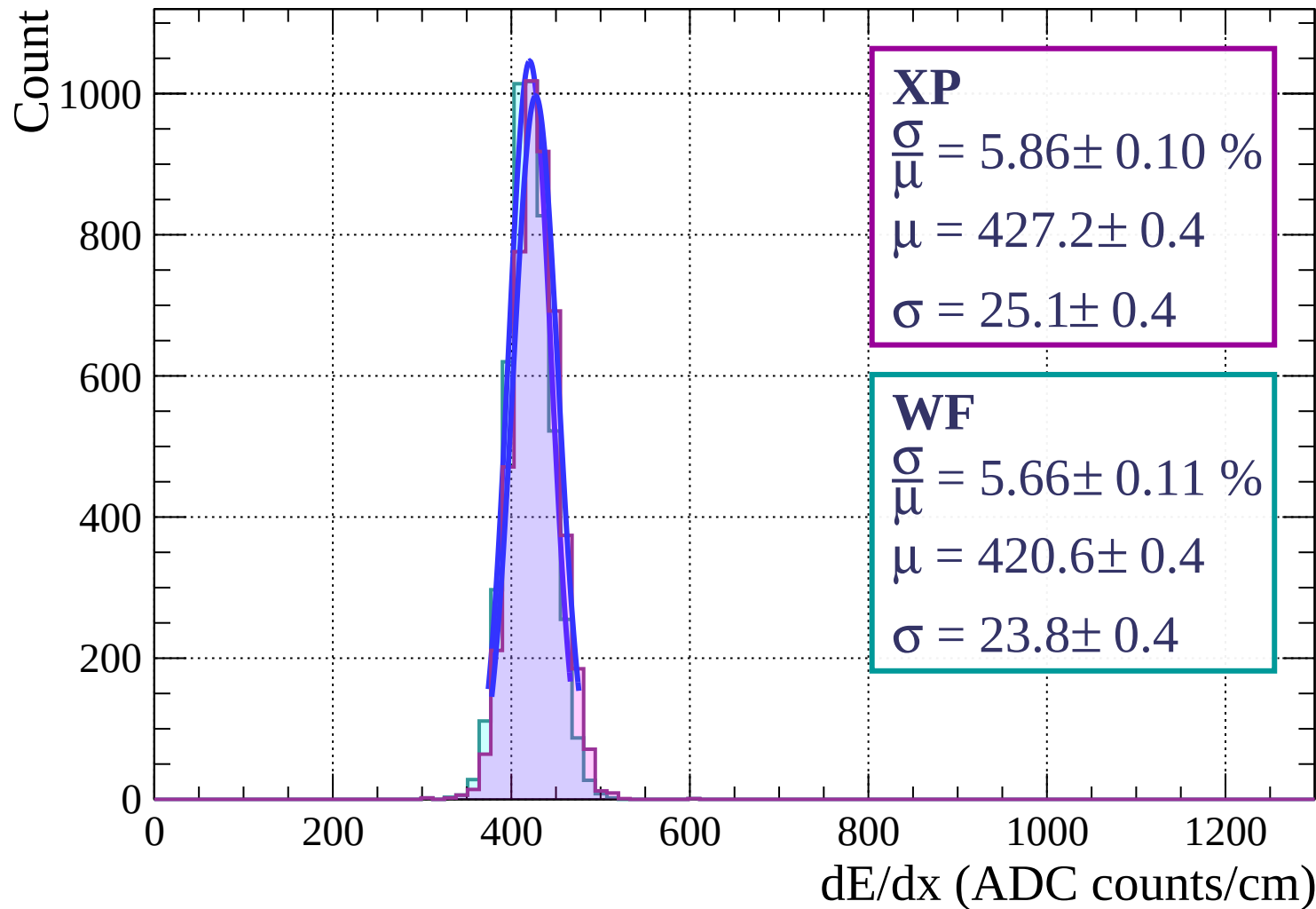
Energy loss | $6 < \phi < 9$



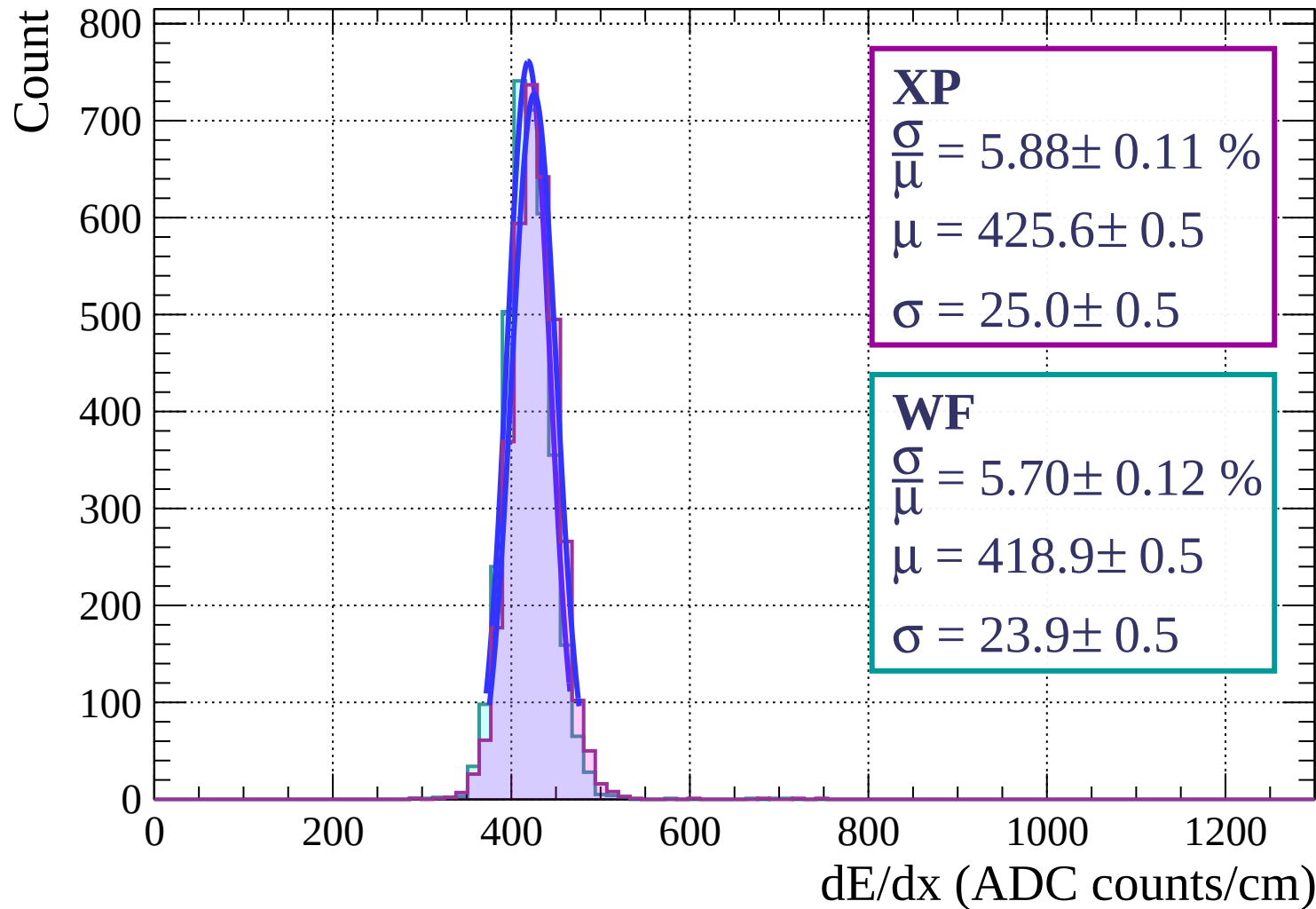
Energy loss | $9 < \phi < 12$



Energy loss | $12 < \phi < 15$



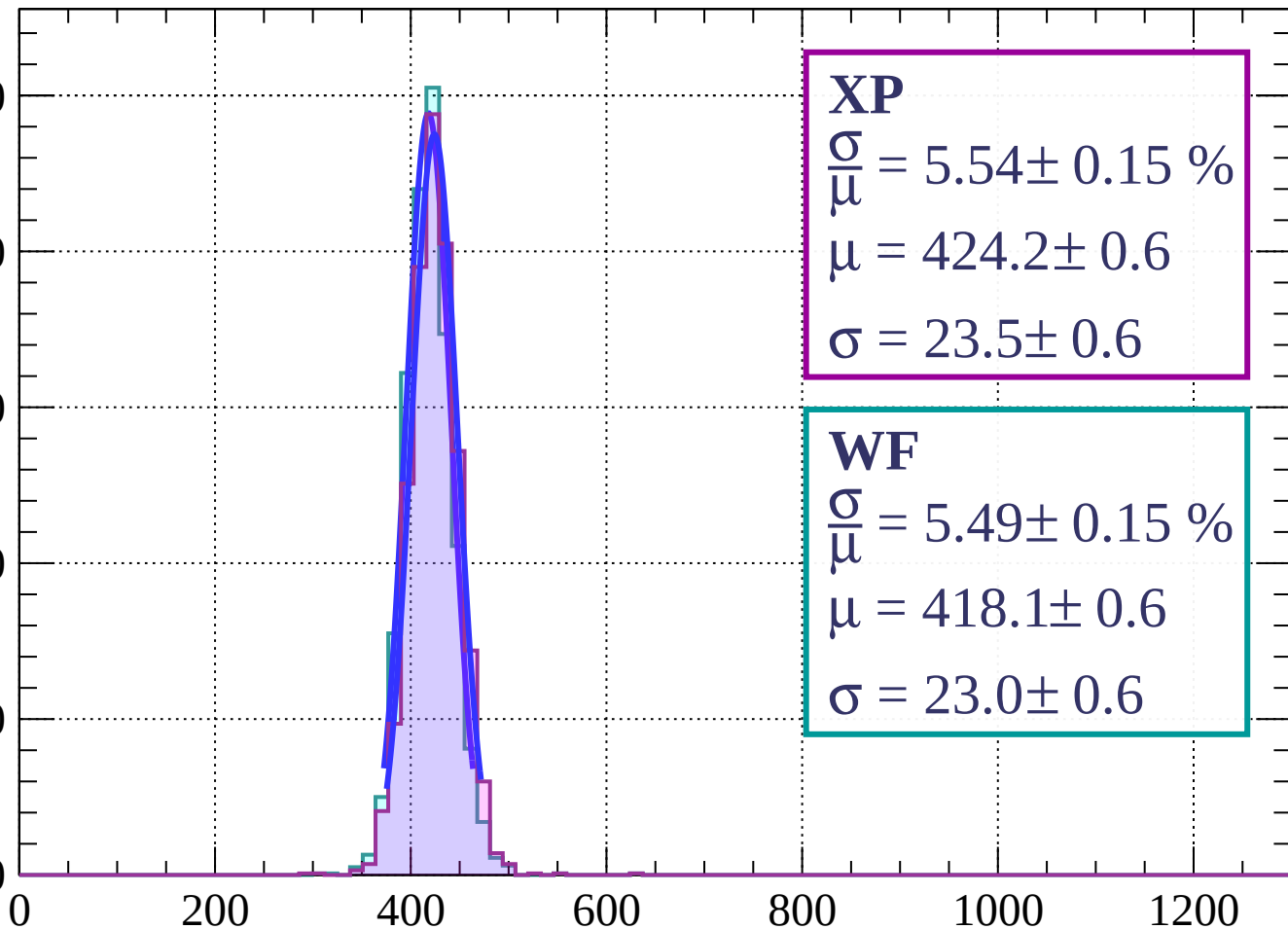
Energy loss | $15 < \phi < 18$



Energy loss | $18 < \phi < 21$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 5.54 \pm 0.15 \%$$

$$\mu = 424.2 \pm 0.6$$

$$\sigma = 23.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.49 \pm 0.15 \%$$

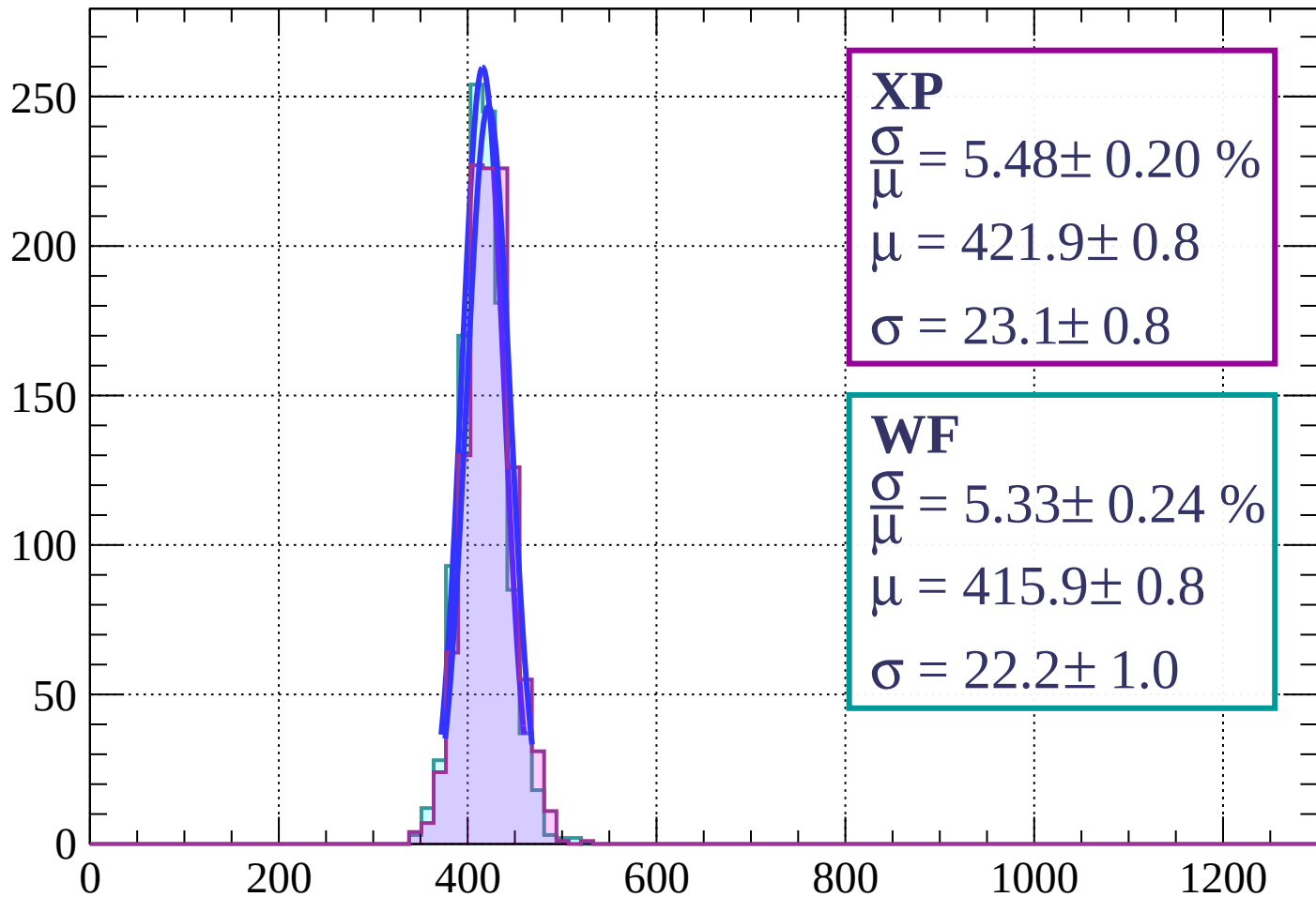
$$\mu = 418.1 \pm 0.6$$

$$\sigma = 23.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 5.48 \pm 0.20 \%$$

$$\mu = 421.9 \pm 0.8$$

$$\sigma = 23.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.33 \pm 0.24 \%$$

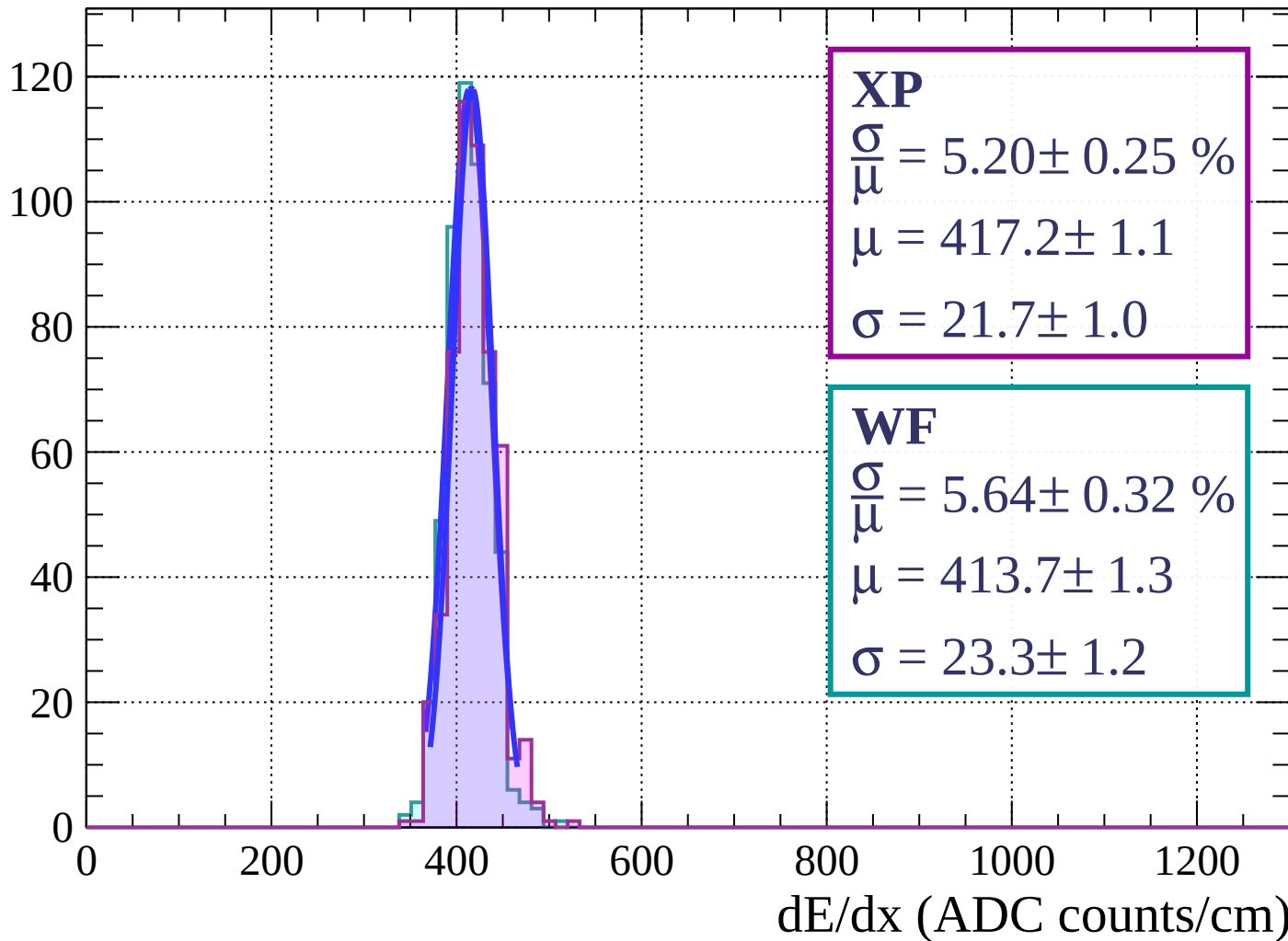
$$\mu = 415.9 \pm 0.8$$

$$\sigma = 22.2 \pm 1.0$$

dE/dx (ADC counts/cm)

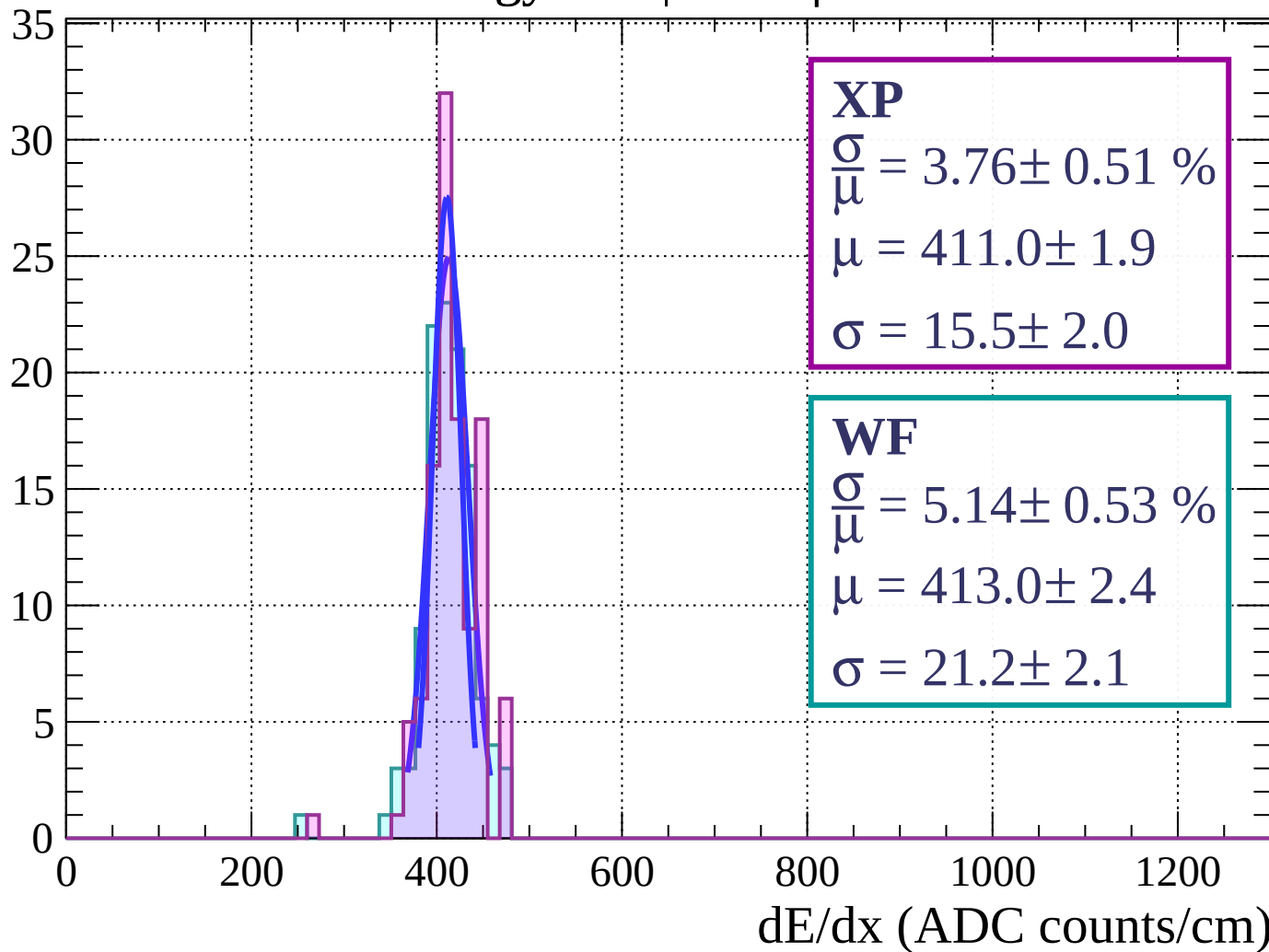
Energy loss | $24 < \phi < 27$

Count



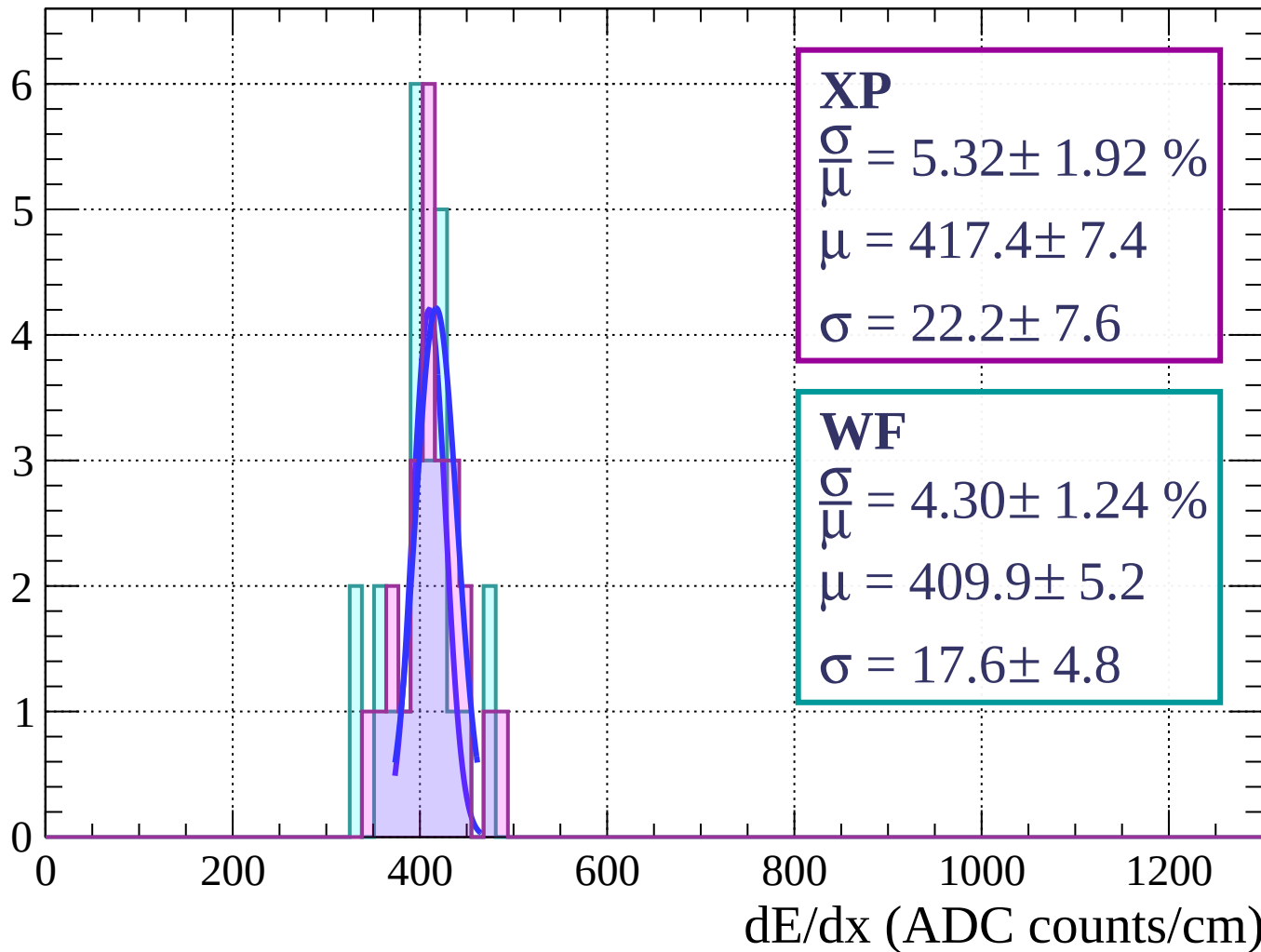
Energy loss | $27 < \phi < 30$

Count



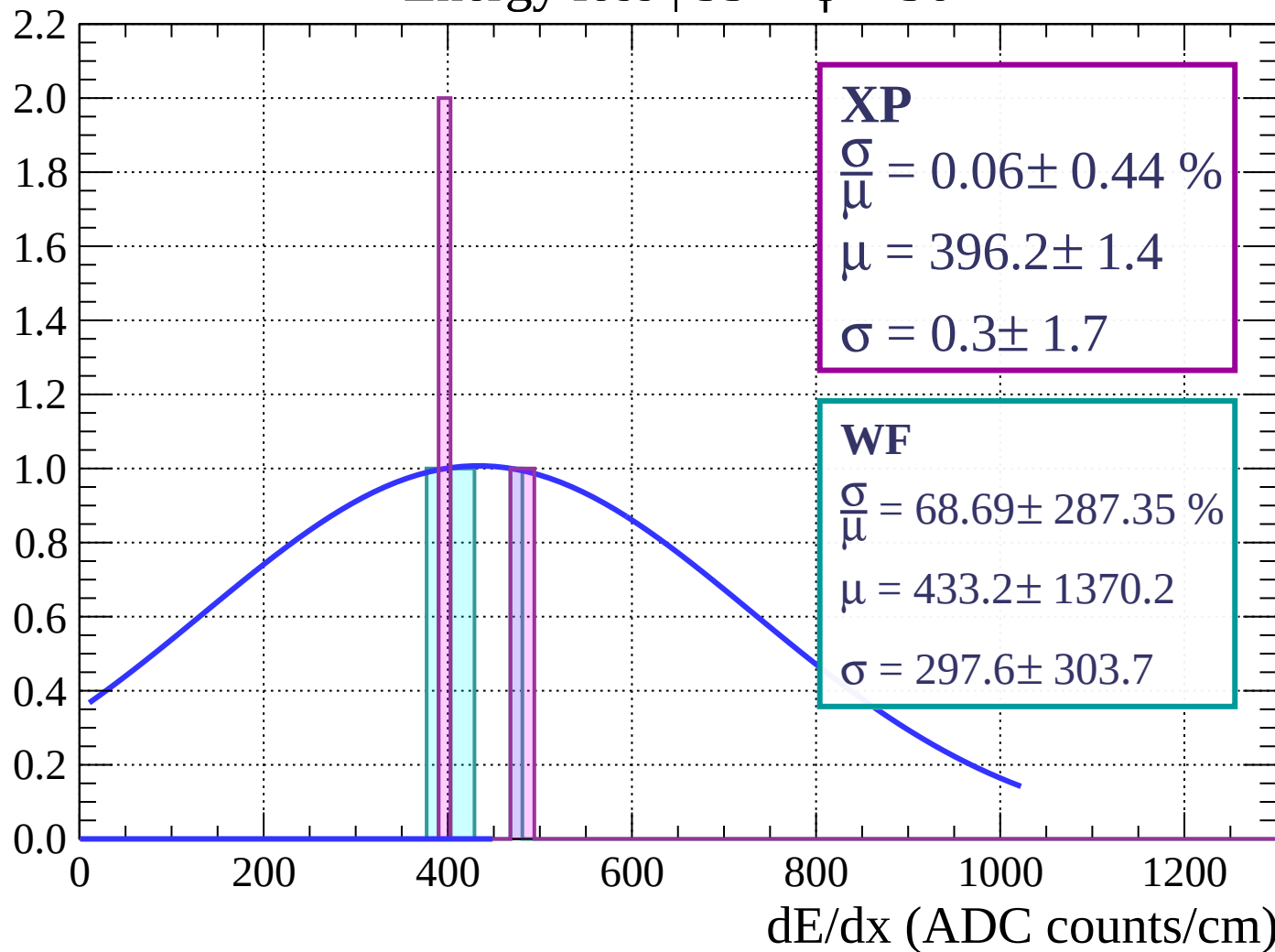
Energy loss | $30 < \phi < 33$

Count



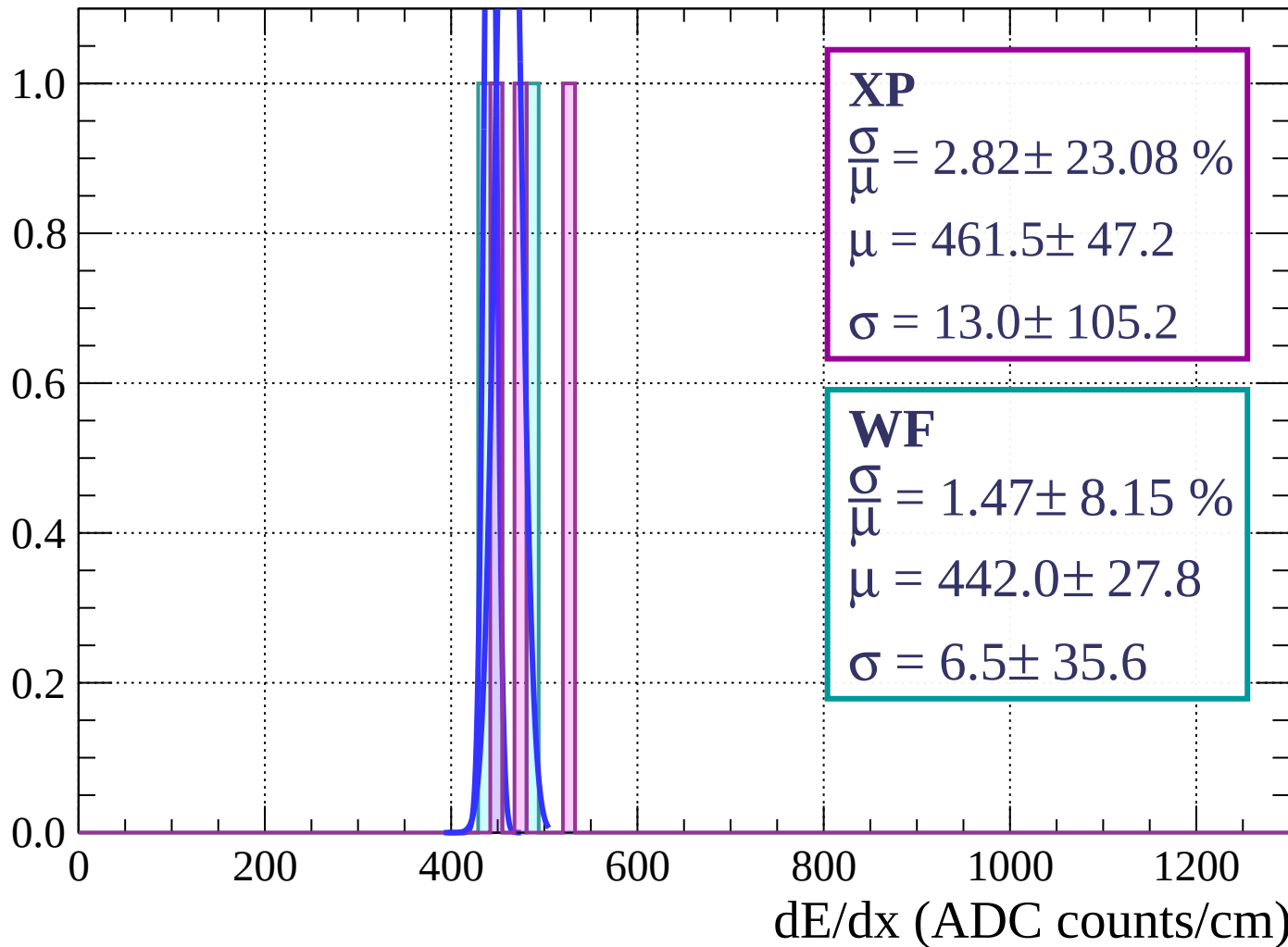
Energy loss | $33 < \phi < 36$

Count



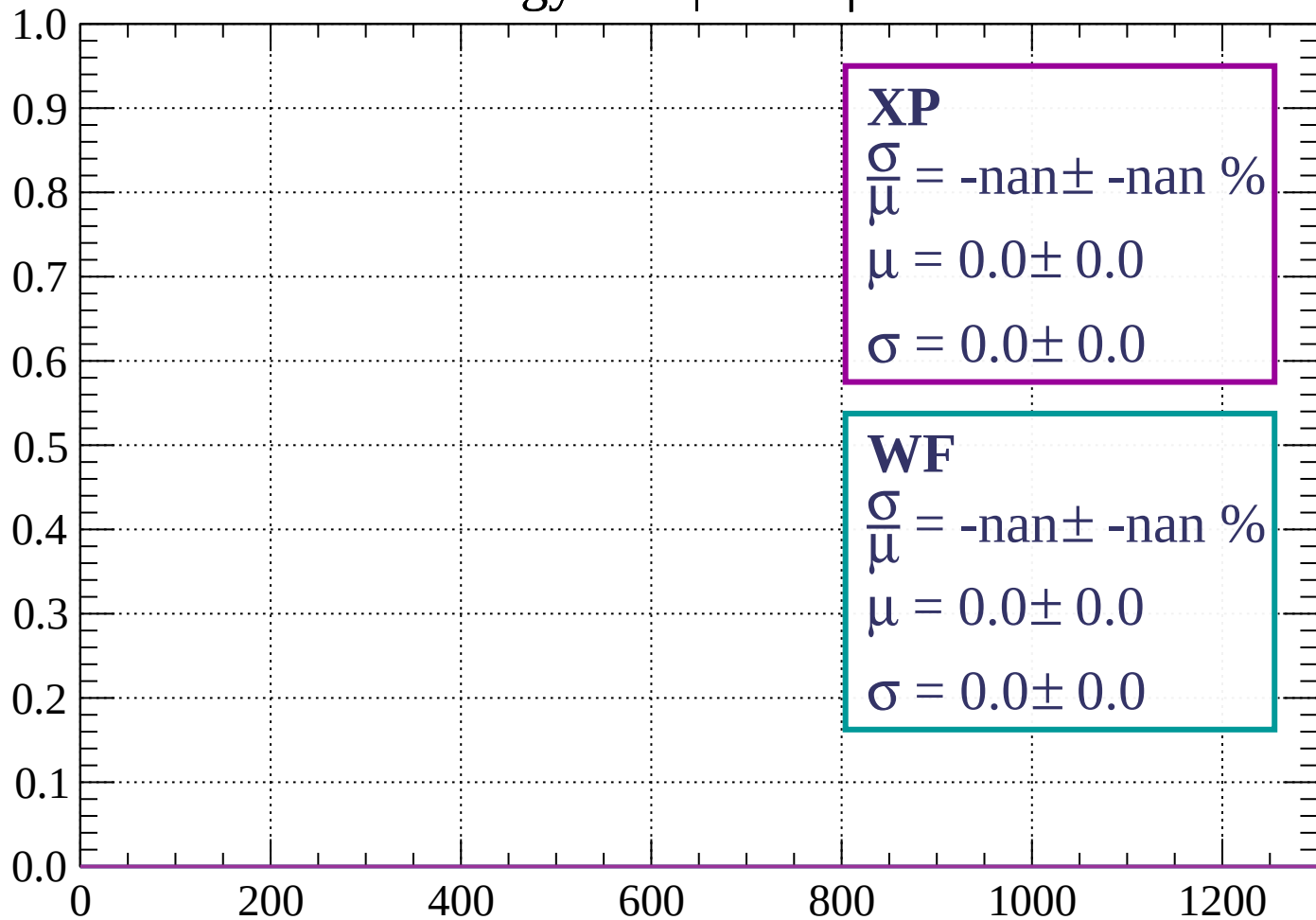
Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

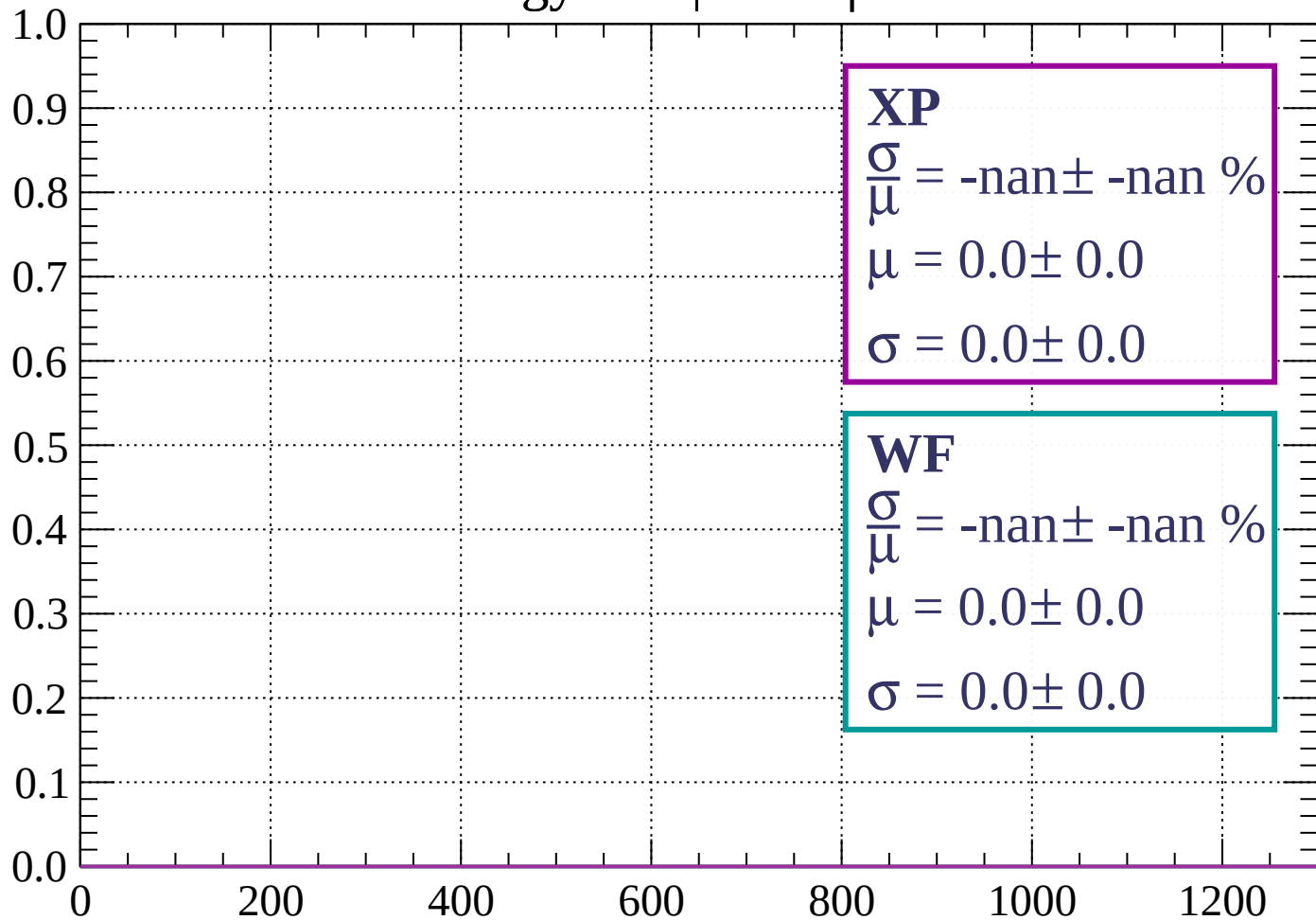
Count



dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$

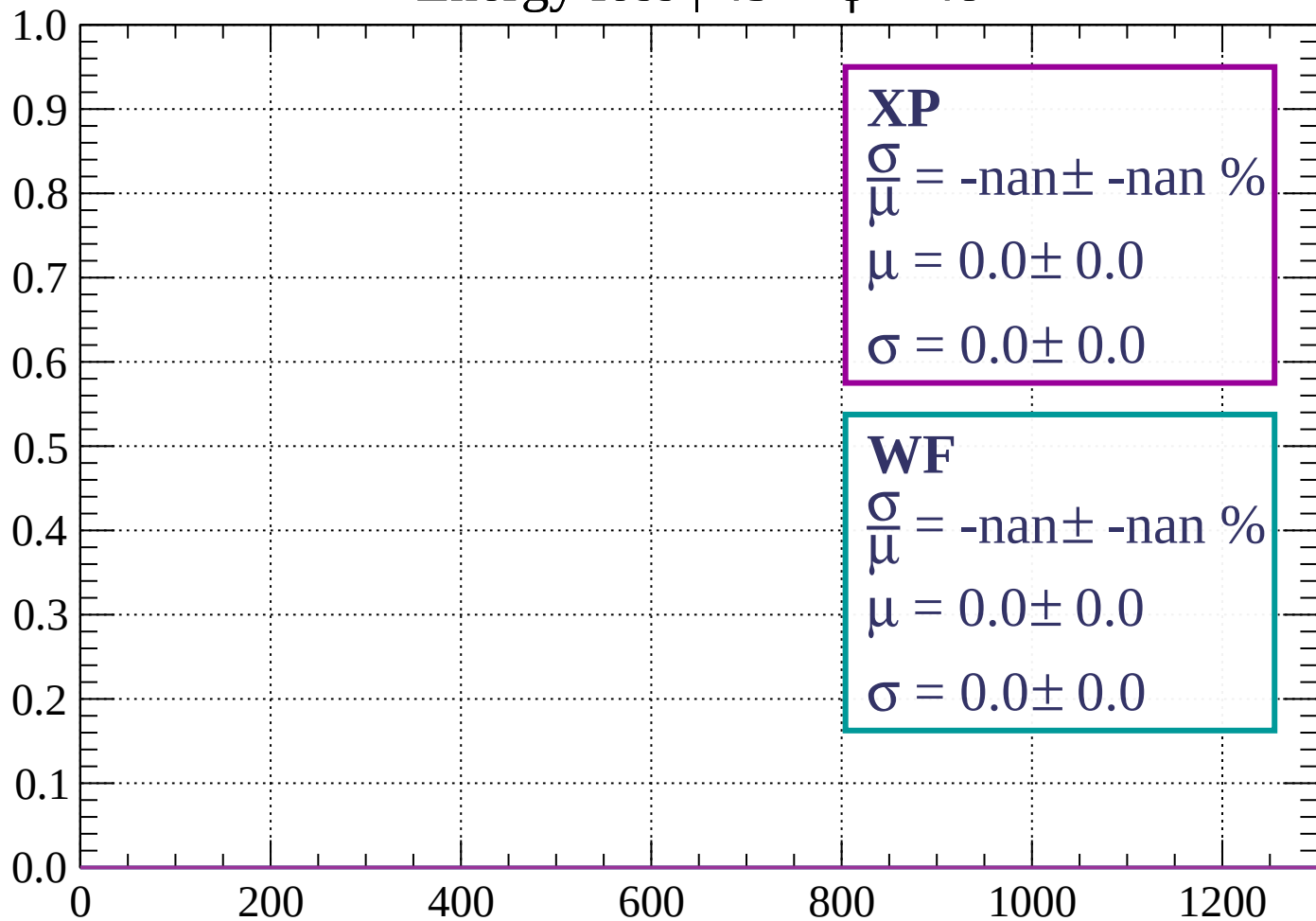
Count



dE/dx (ADC counts/cm)

Energy loss | $45 < \phi < 48$

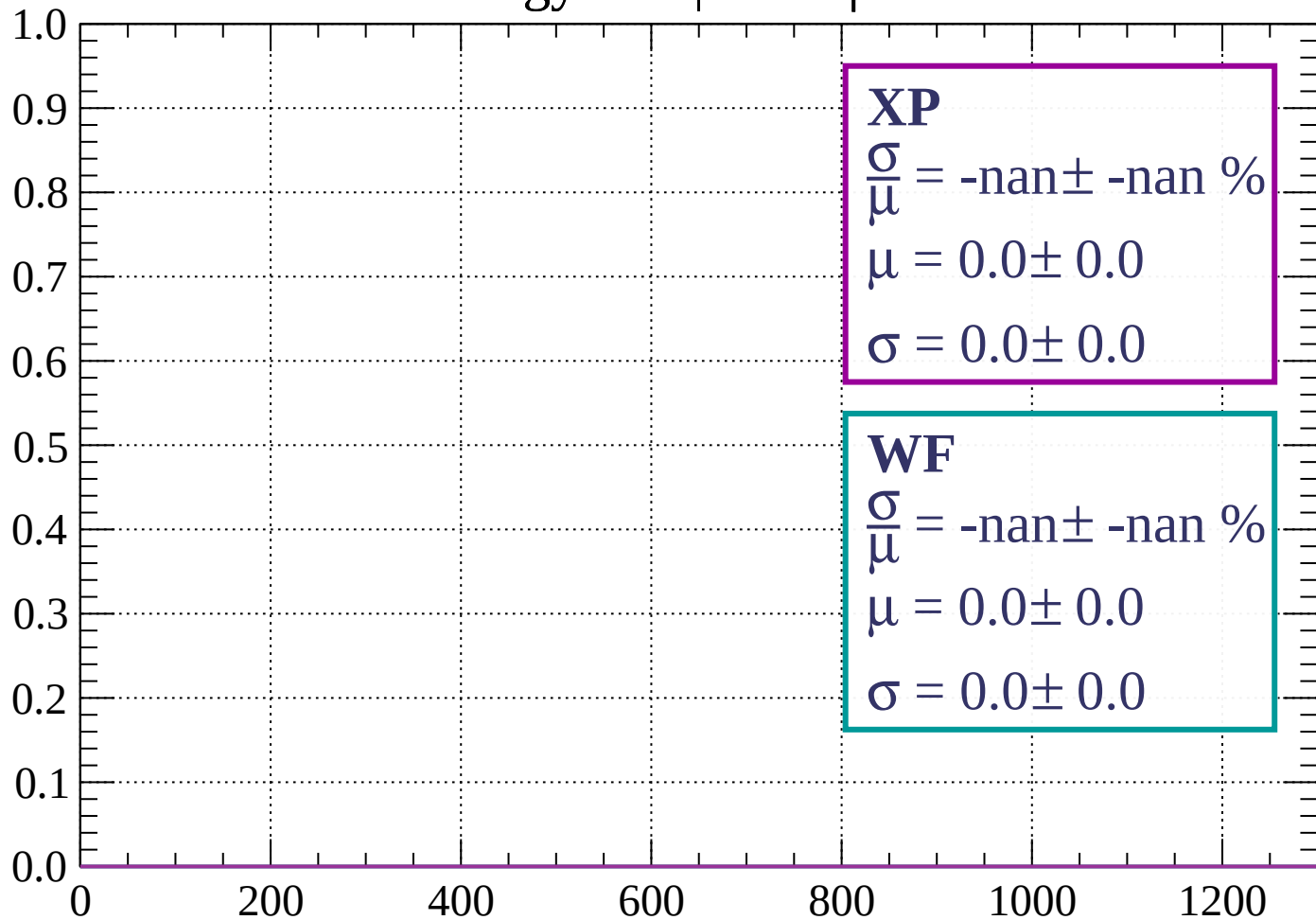
Count



dE/dx (ADC counts/cm)

Energy loss | $48 < \phi < 51$

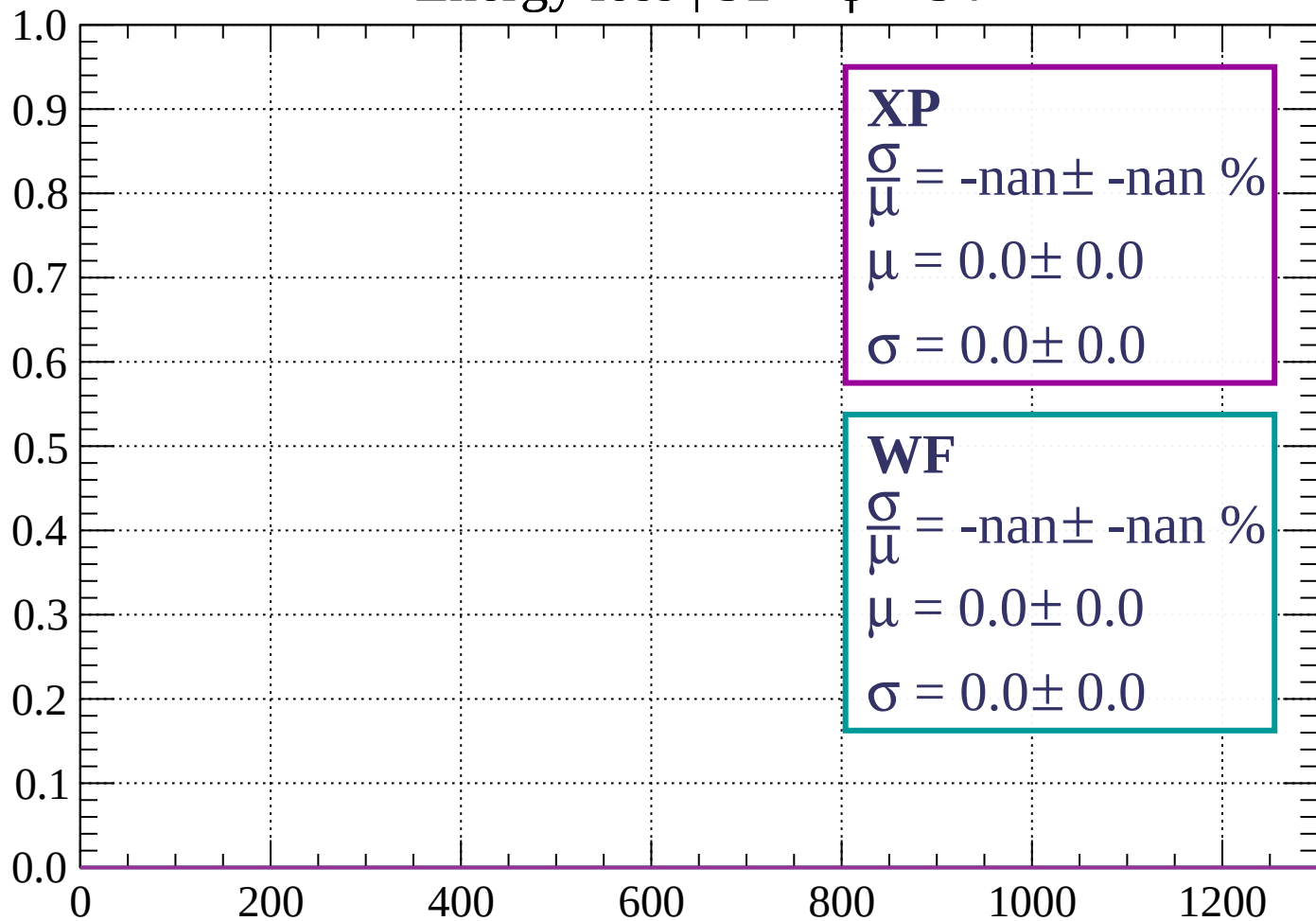
Count



dE/dx (ADC counts/cm)

Energy loss | $51 < \phi < 54$

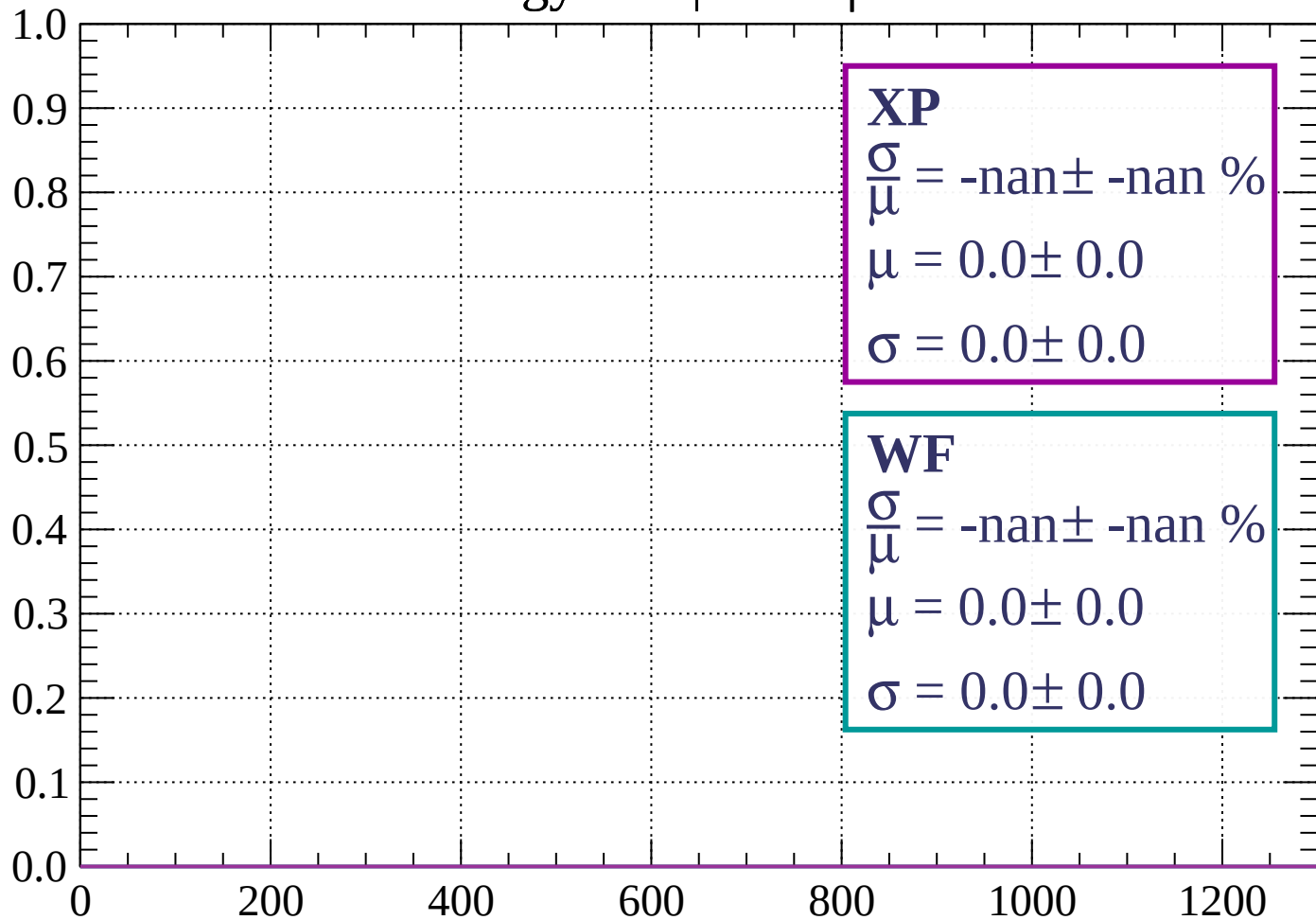
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \phi < 57$

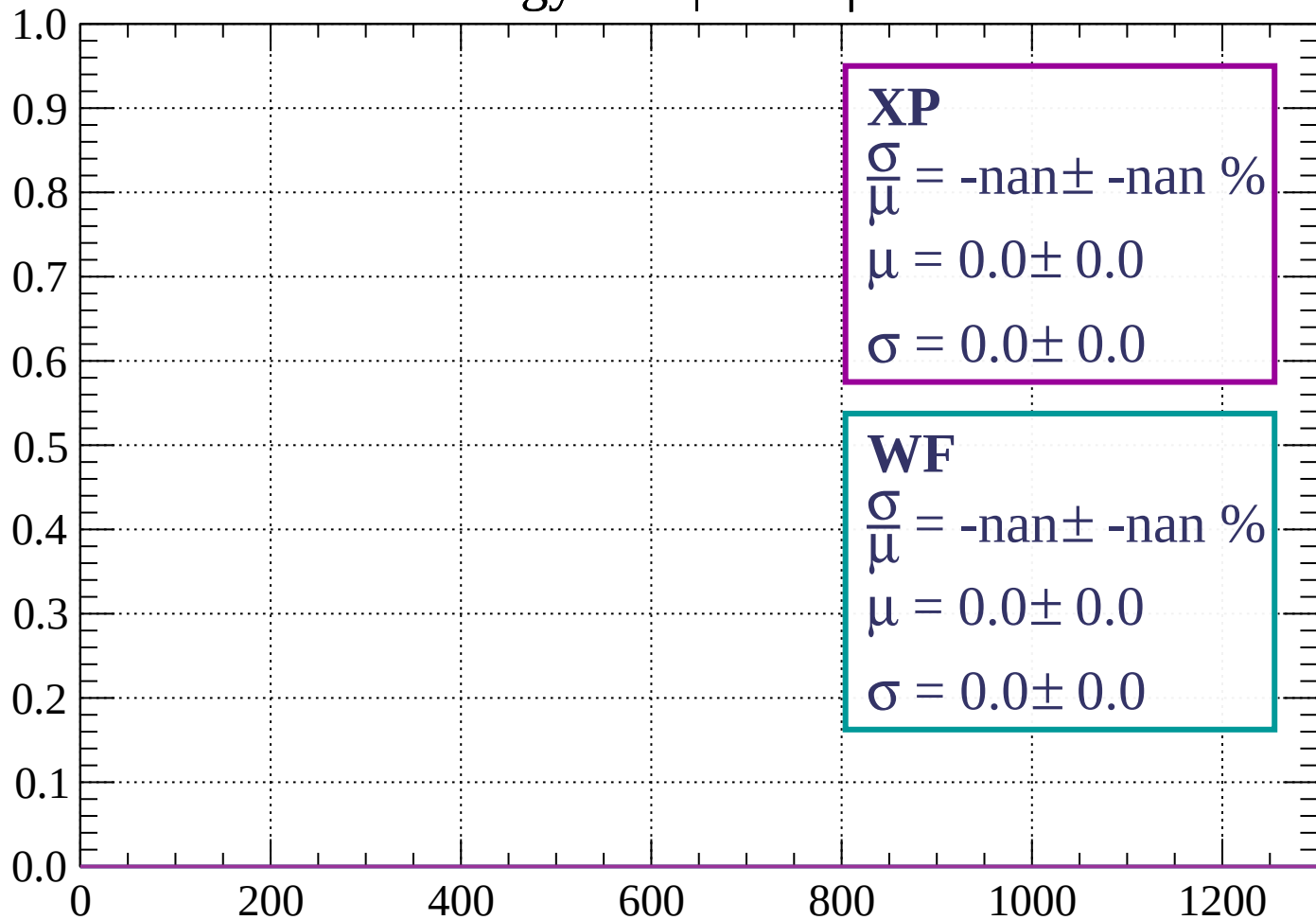
Count



dE/dx (ADC counts/cm)

Energy loss | $57 < \phi < 60$

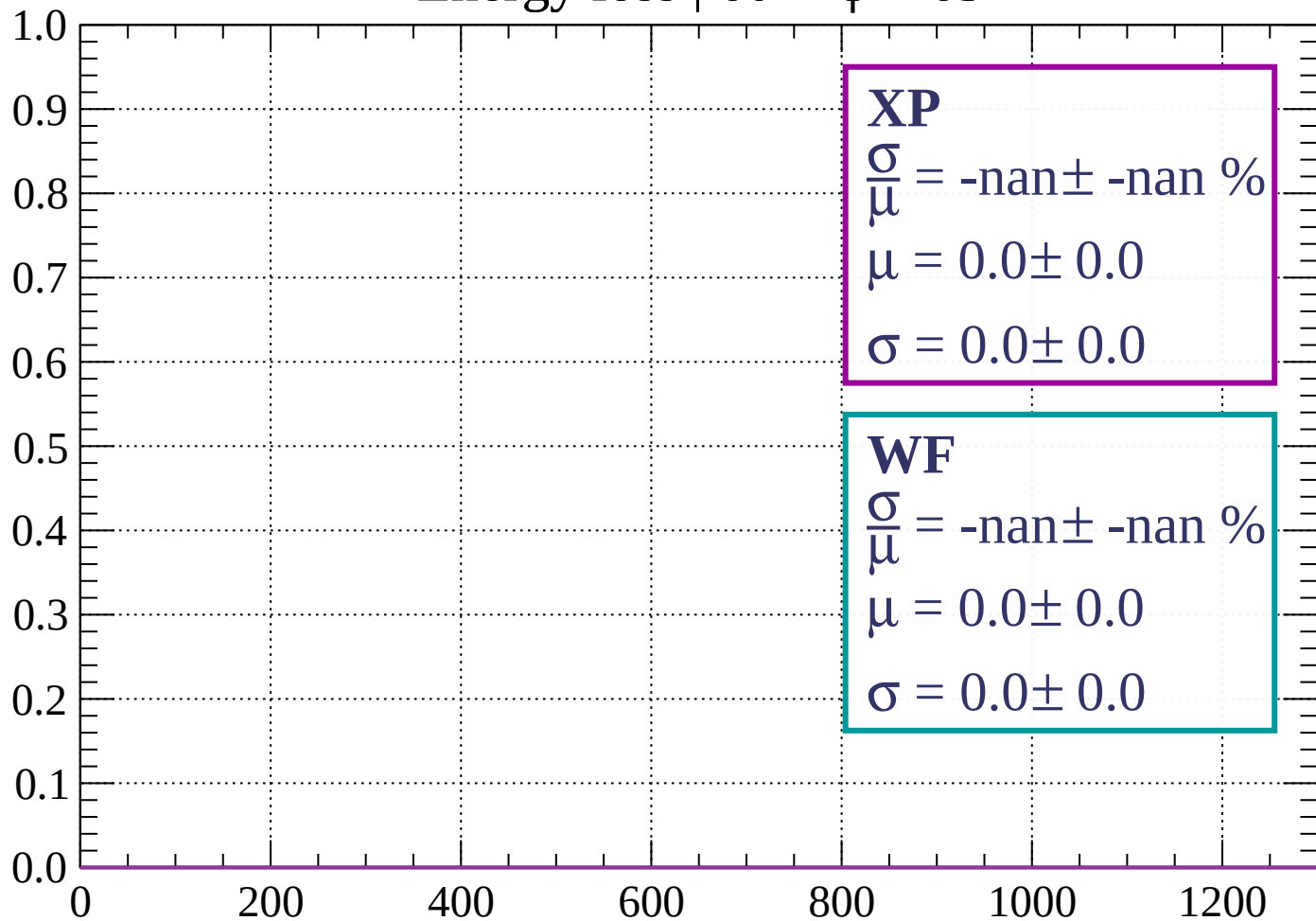
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \phi < 63$

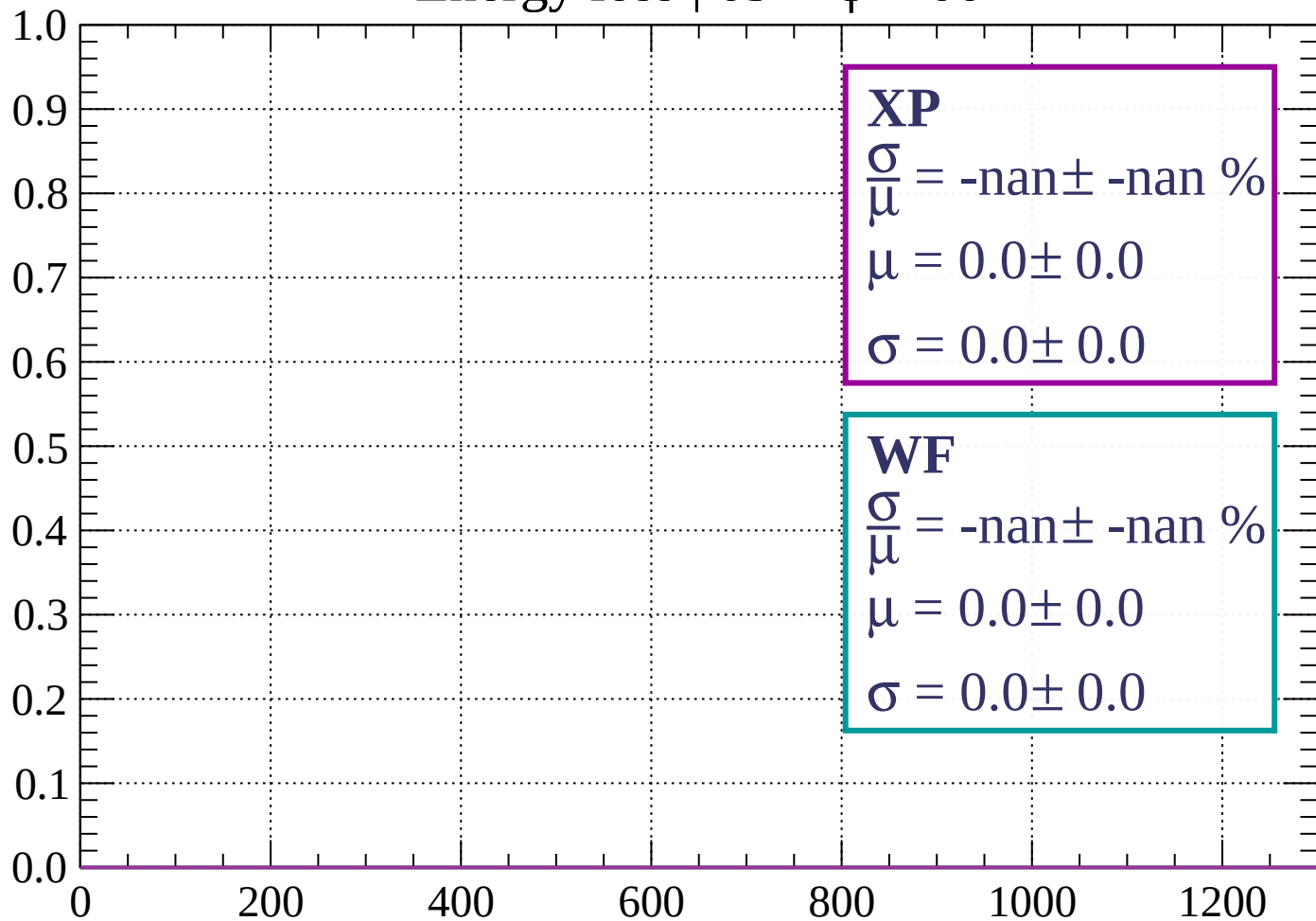
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \phi < 66$

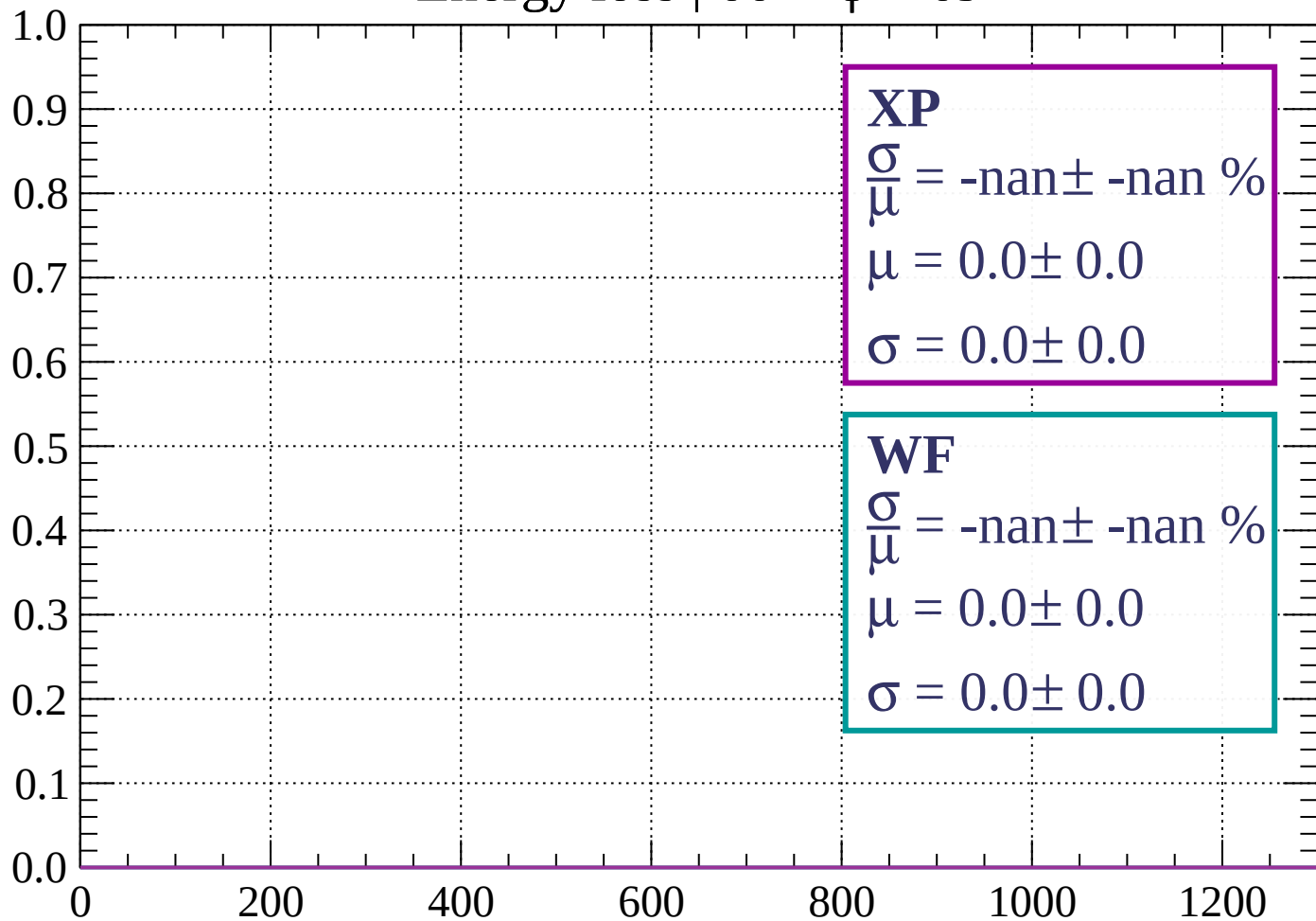
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \phi < 69$

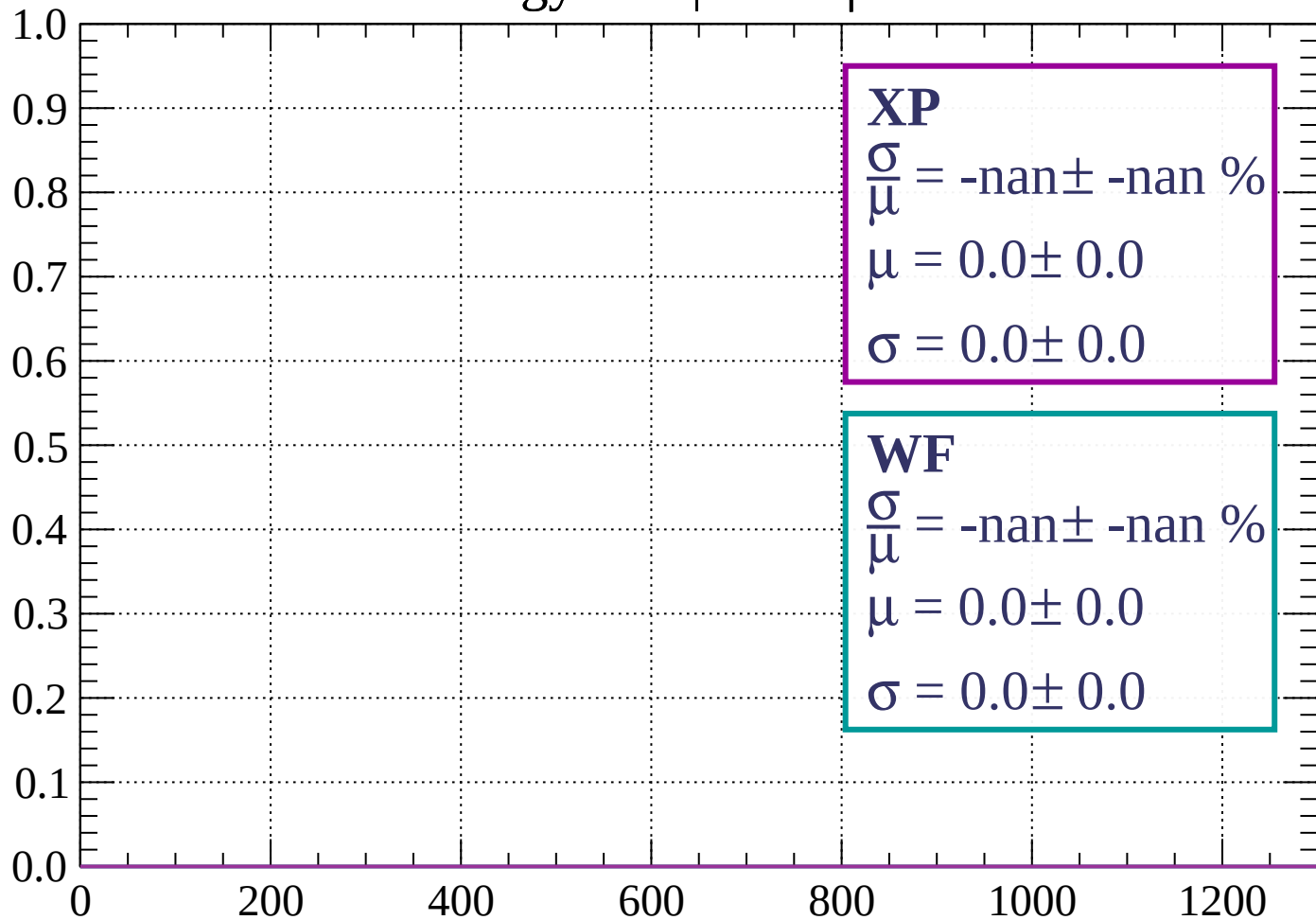
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \phi < 72$

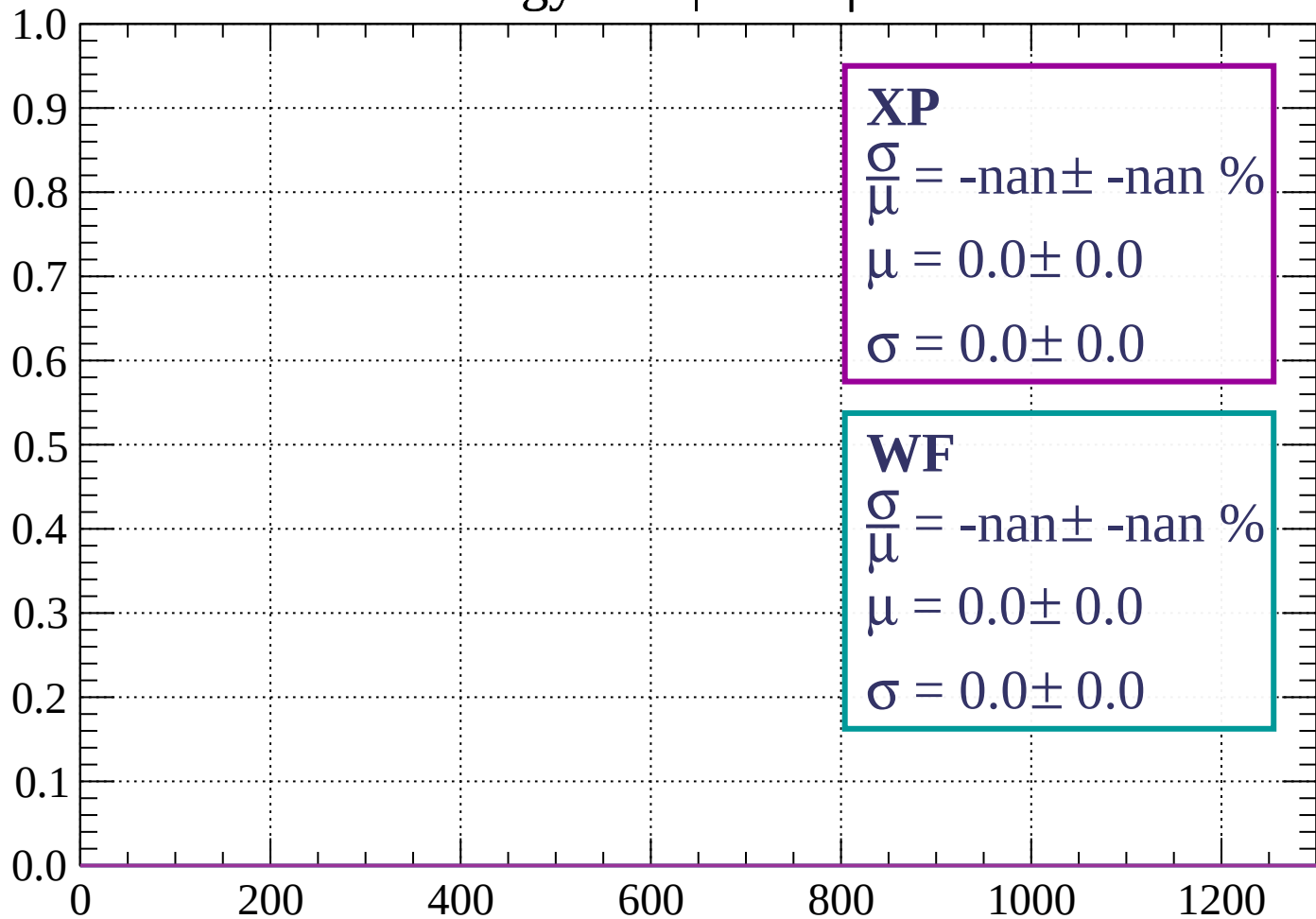
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

WF

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

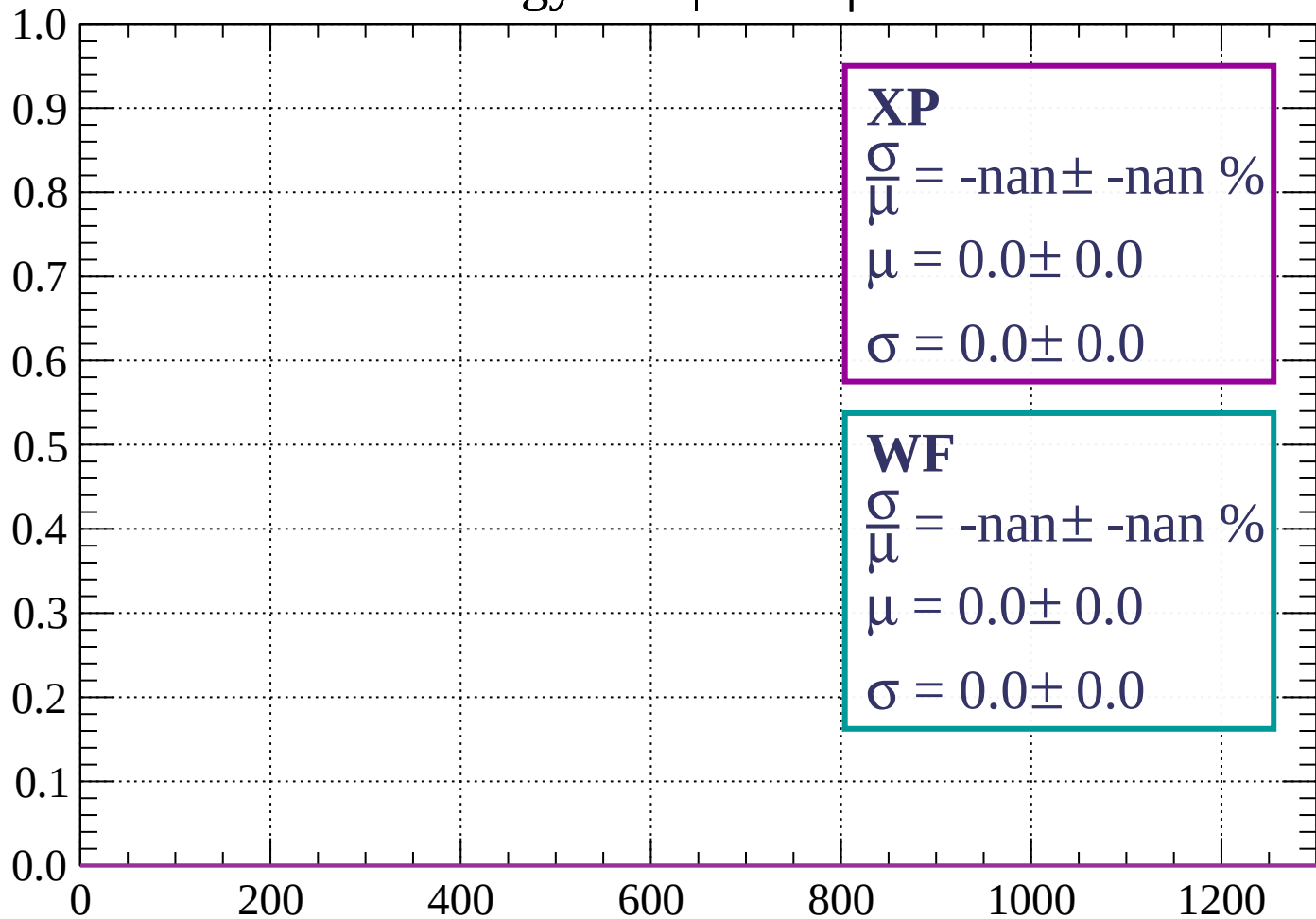
$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

dE/dx (ADC counts/cm)

Energy loss | $75 < \phi < 78$

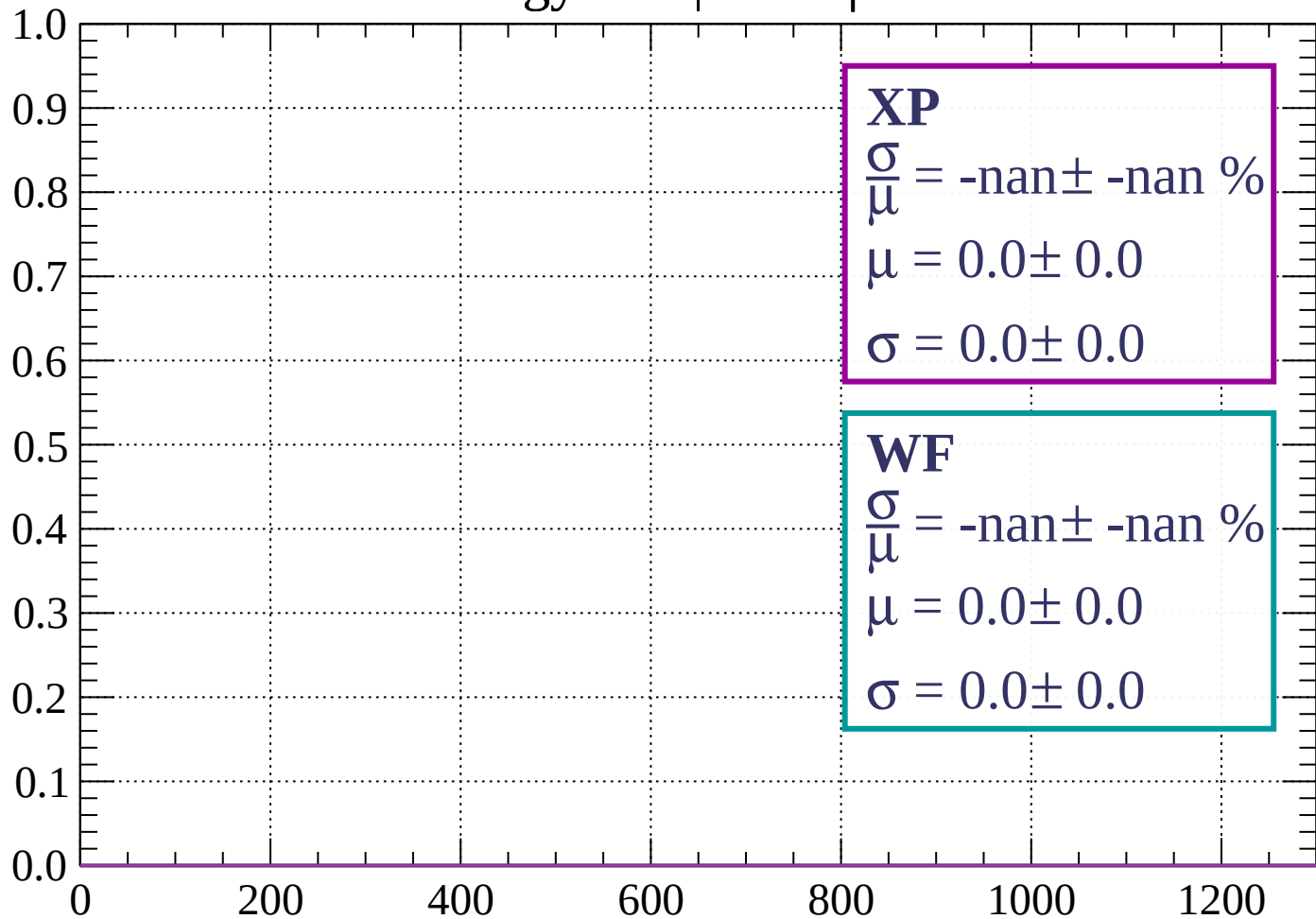
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \phi < 81$

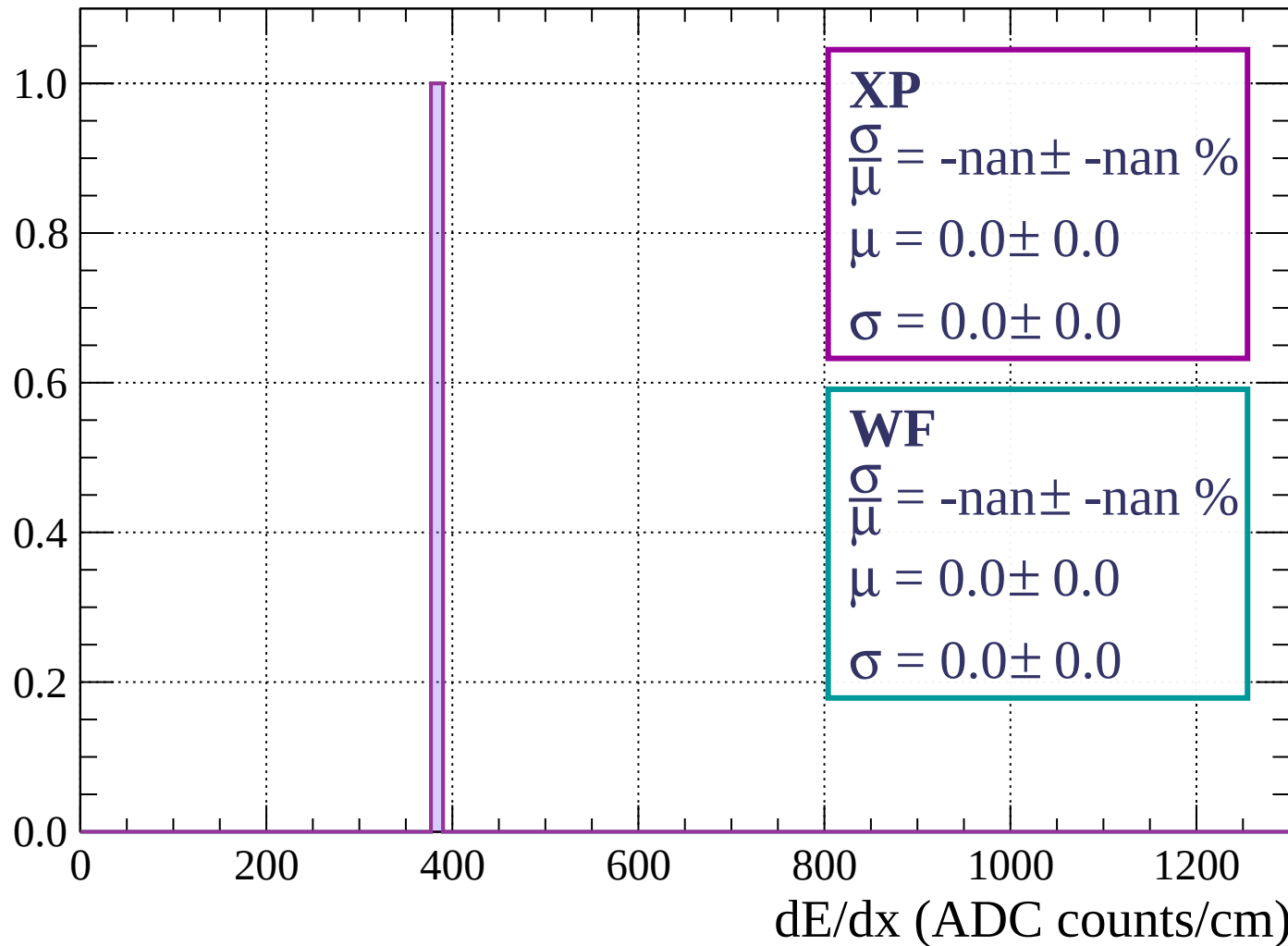
Count



dE/dx (ADC counts/cm)

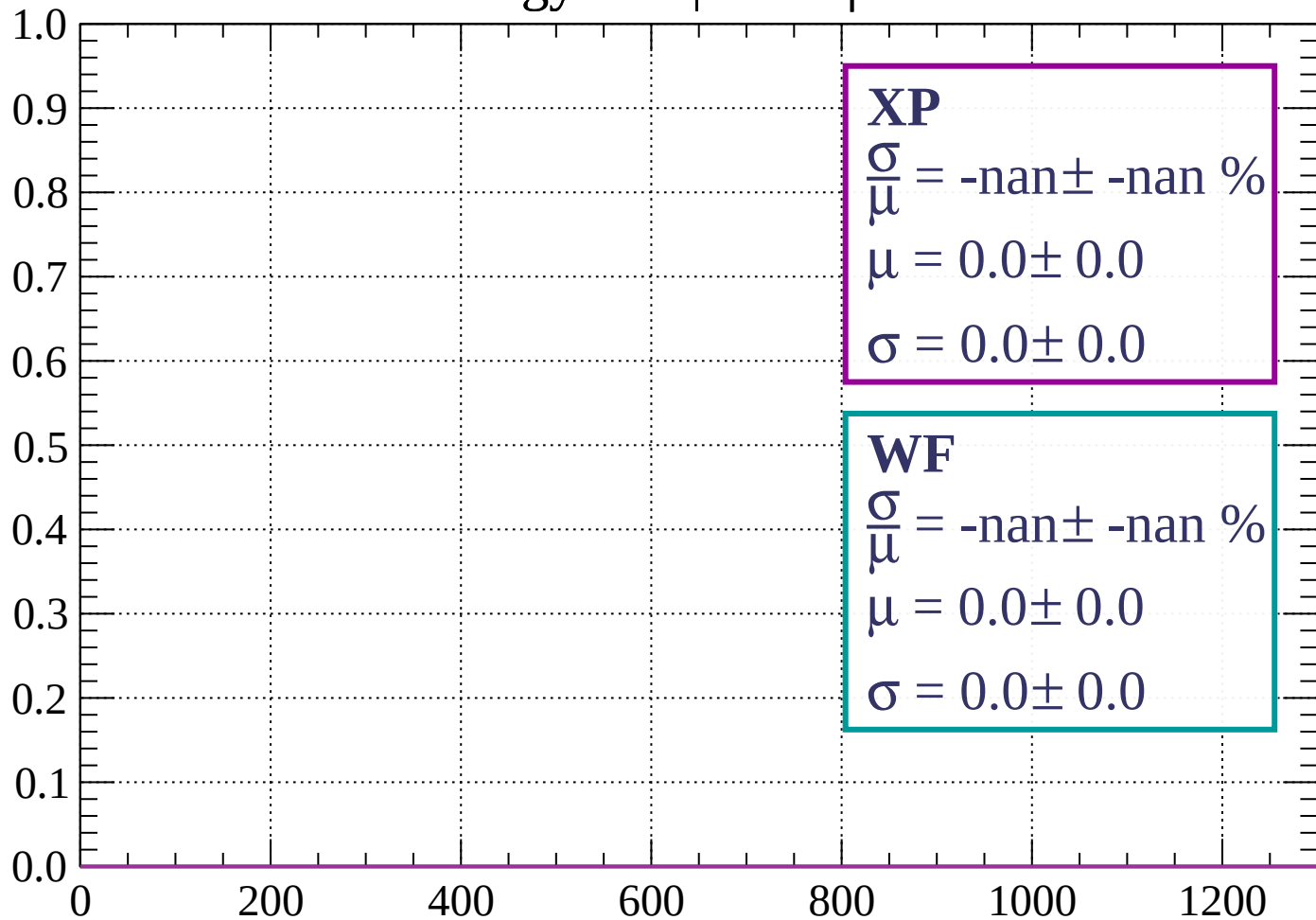
Energy loss | $81 < \phi < 84$

Count



Energy loss | $84 < \phi < 87$

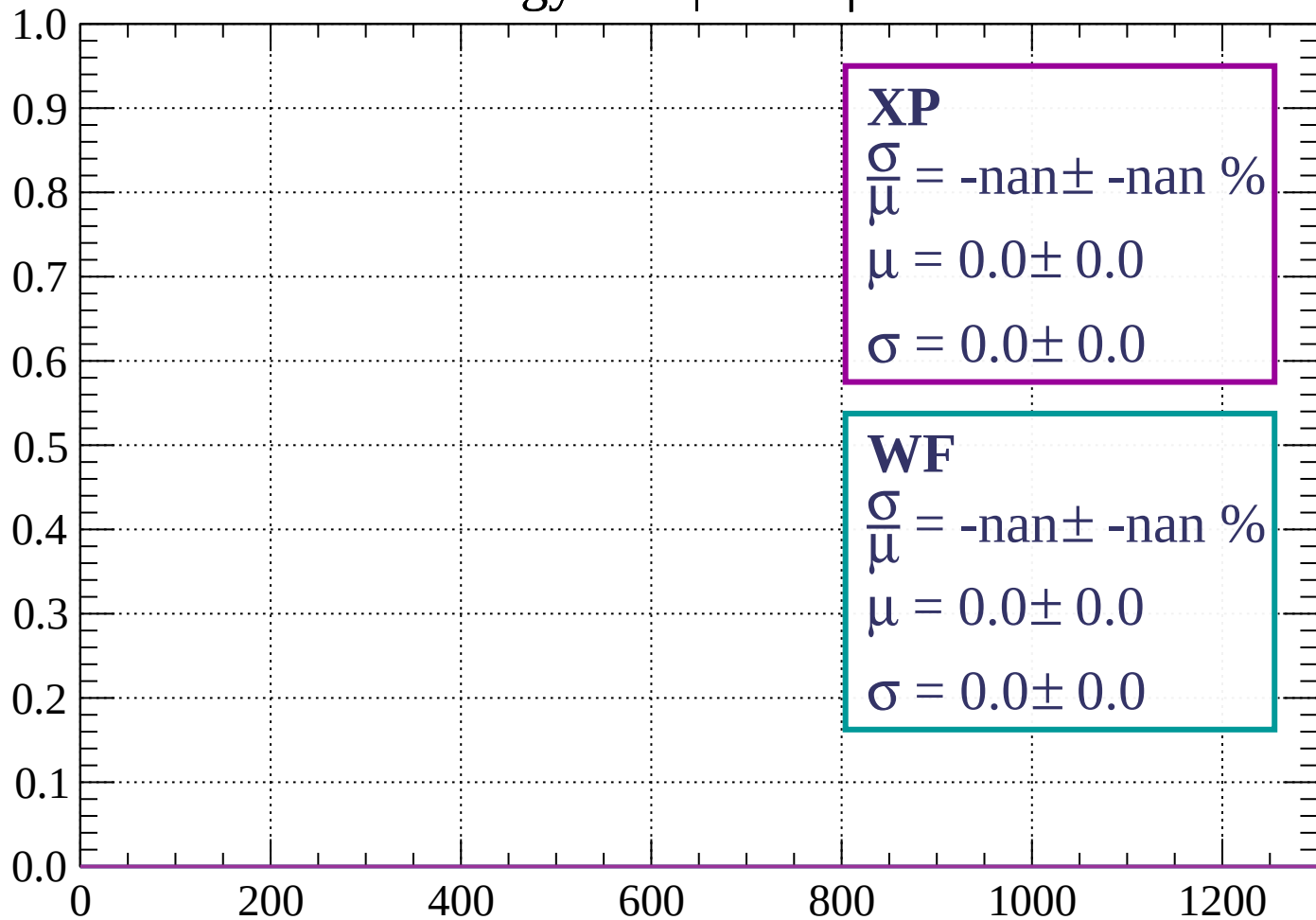
Count



dE/dx (ADC counts/cm)

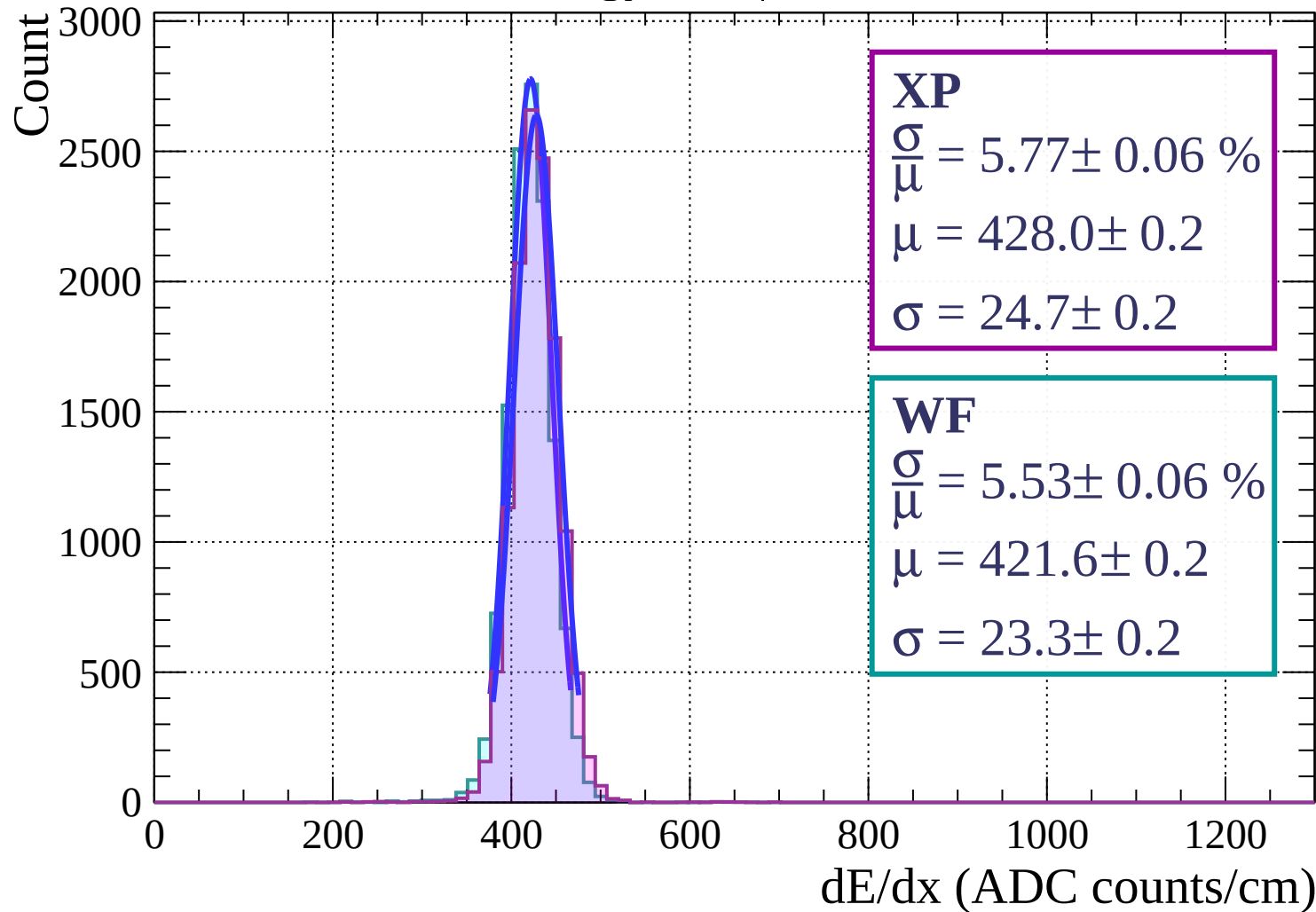
Energy loss | $87 < \phi < 90$

Count

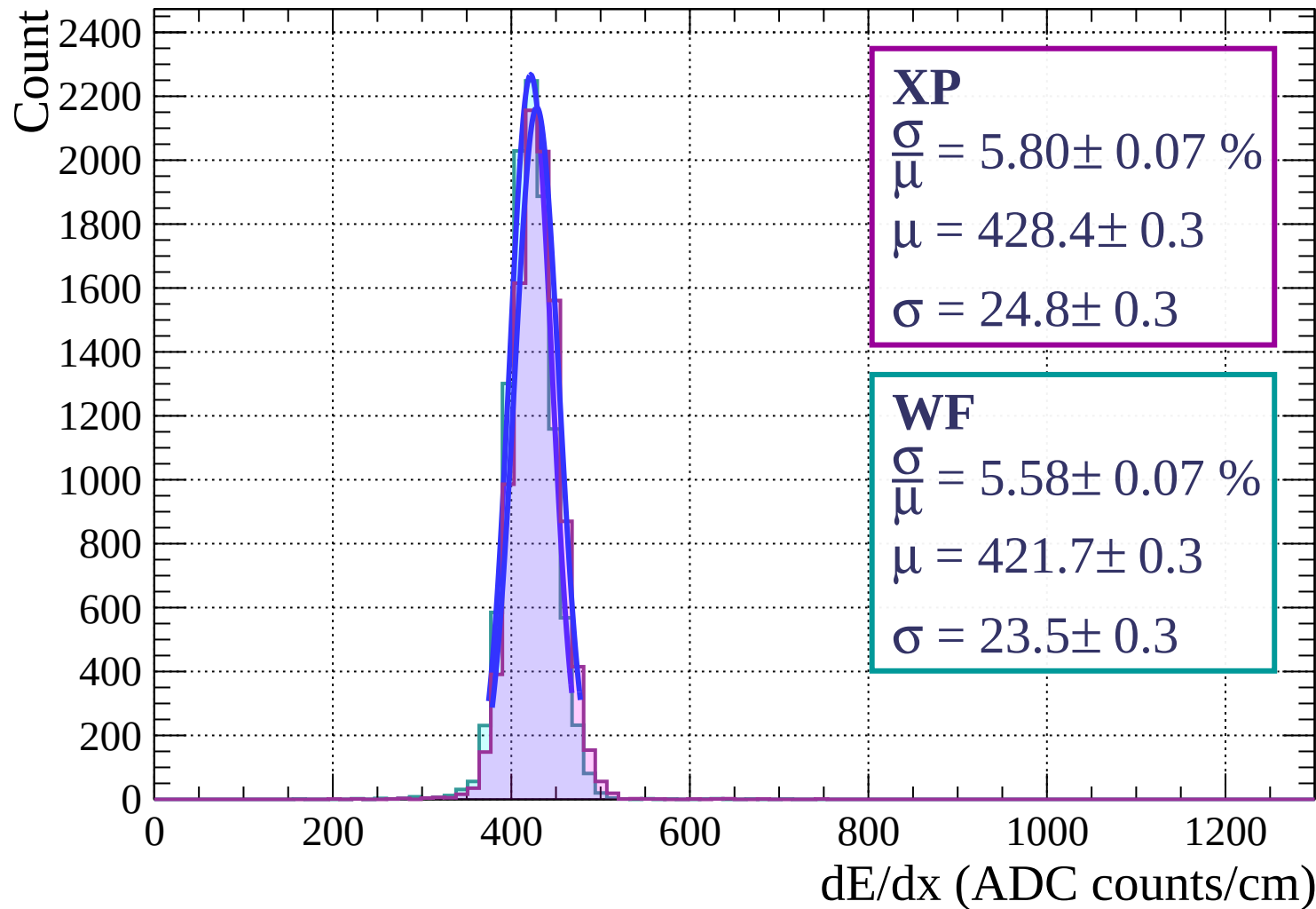


dE/dx (ADC counts/cm)

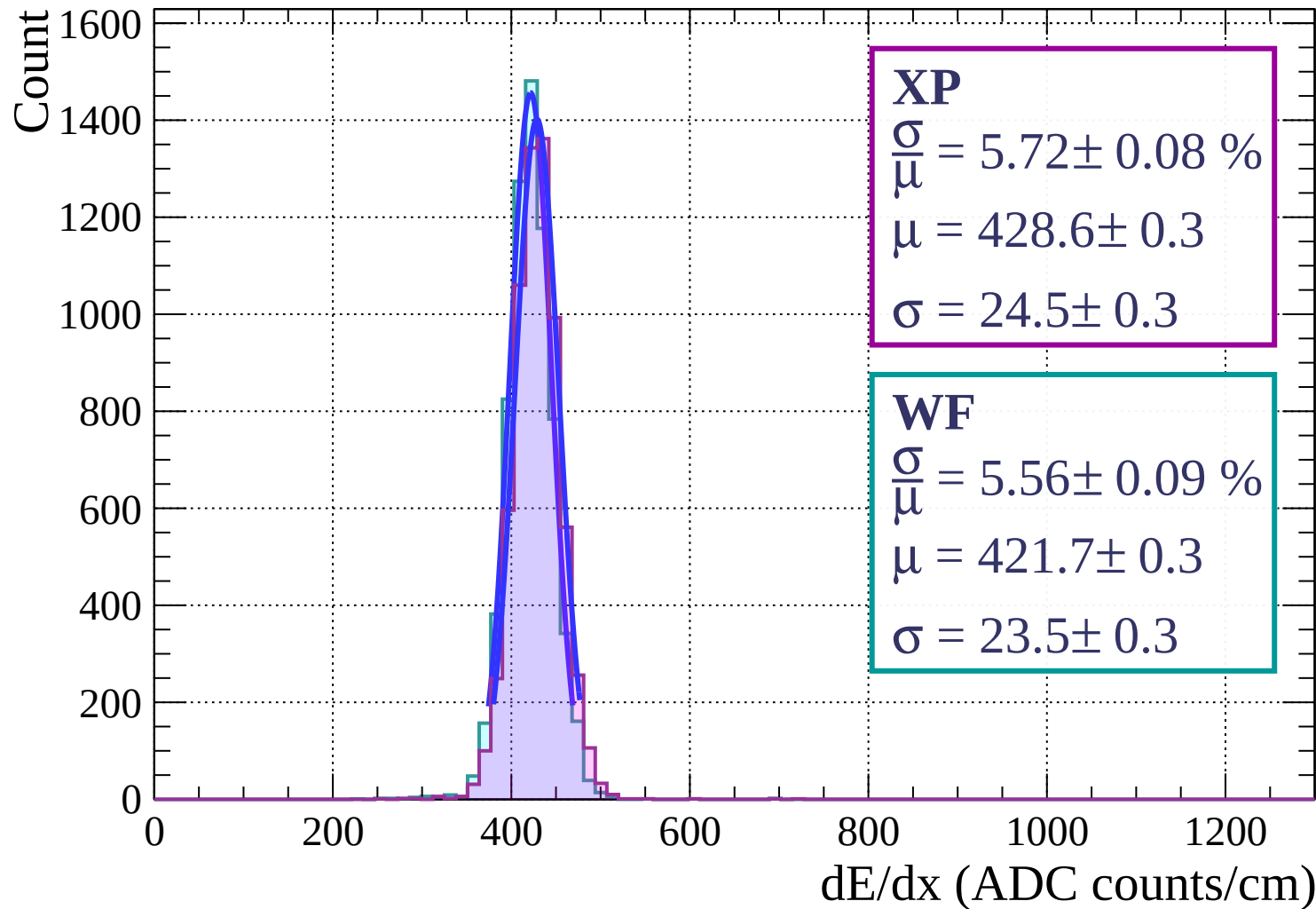
Energy loss | $0 < \theta < 3$



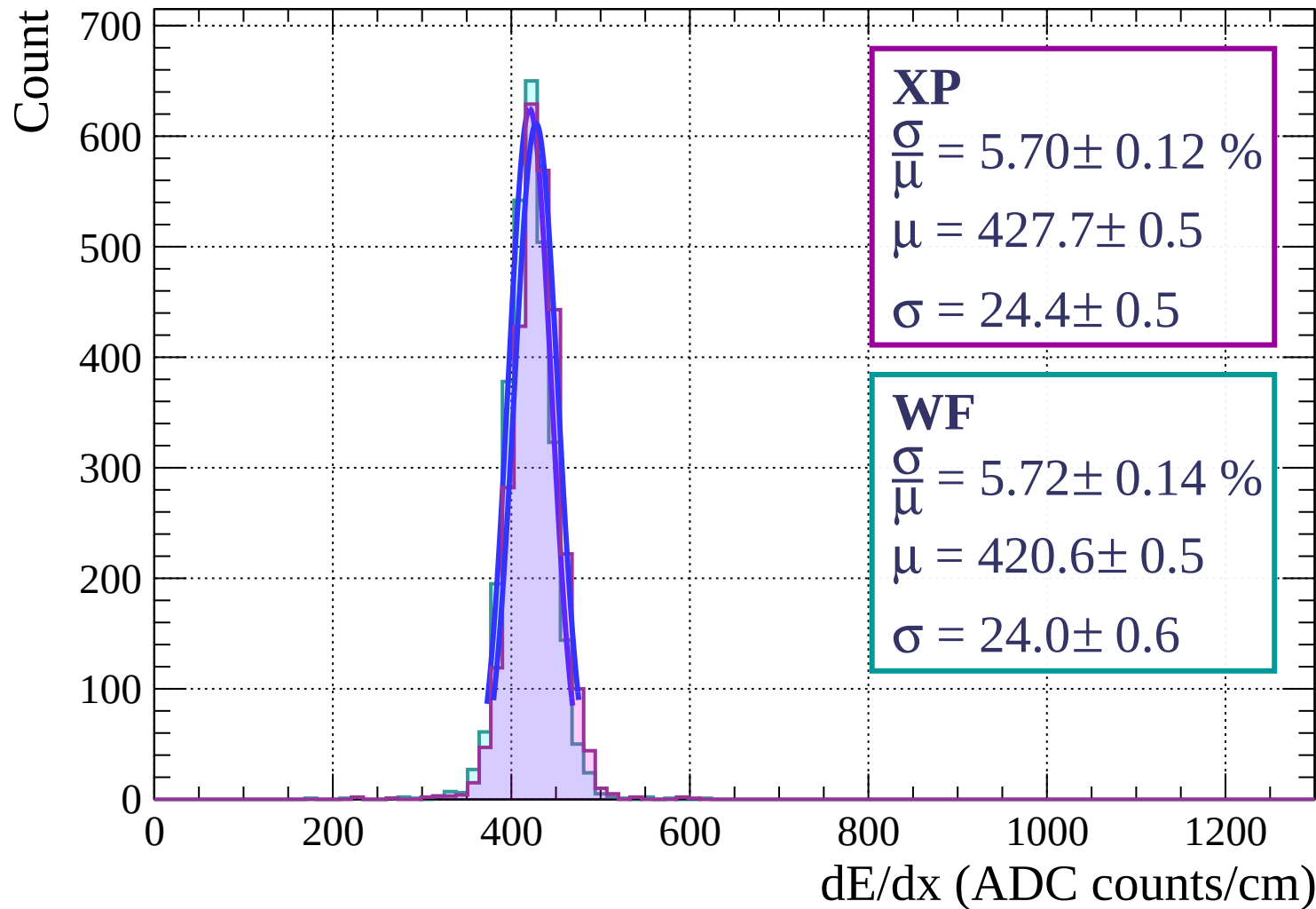
Energy loss | $3 < \theta < 6$



Energy loss | $6 < \theta < 9$

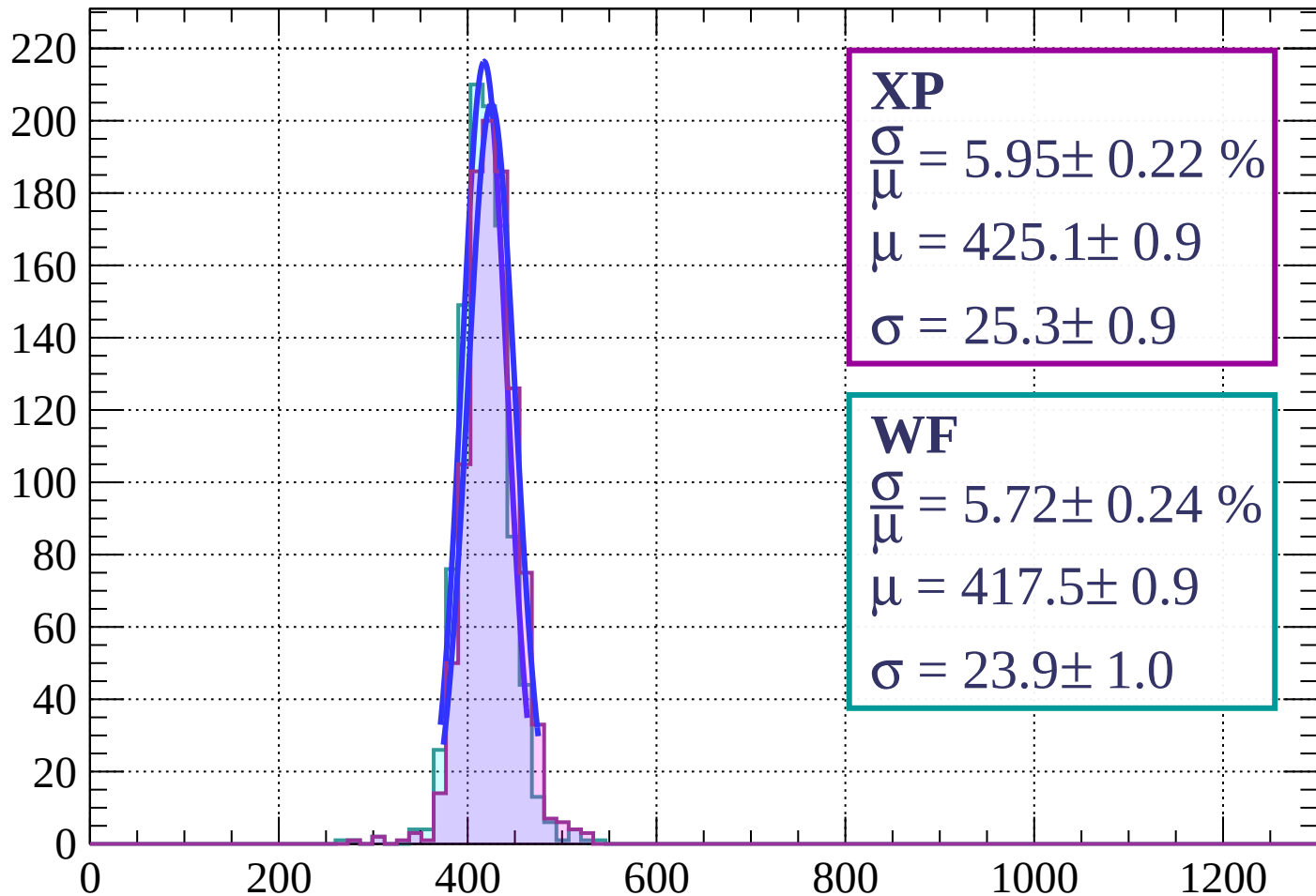


Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 5.95 \pm 0.22 \%$$

$$\mu = 425.1 \pm 0.9$$

$$\sigma = 25.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.72 \pm 0.24 \%$$

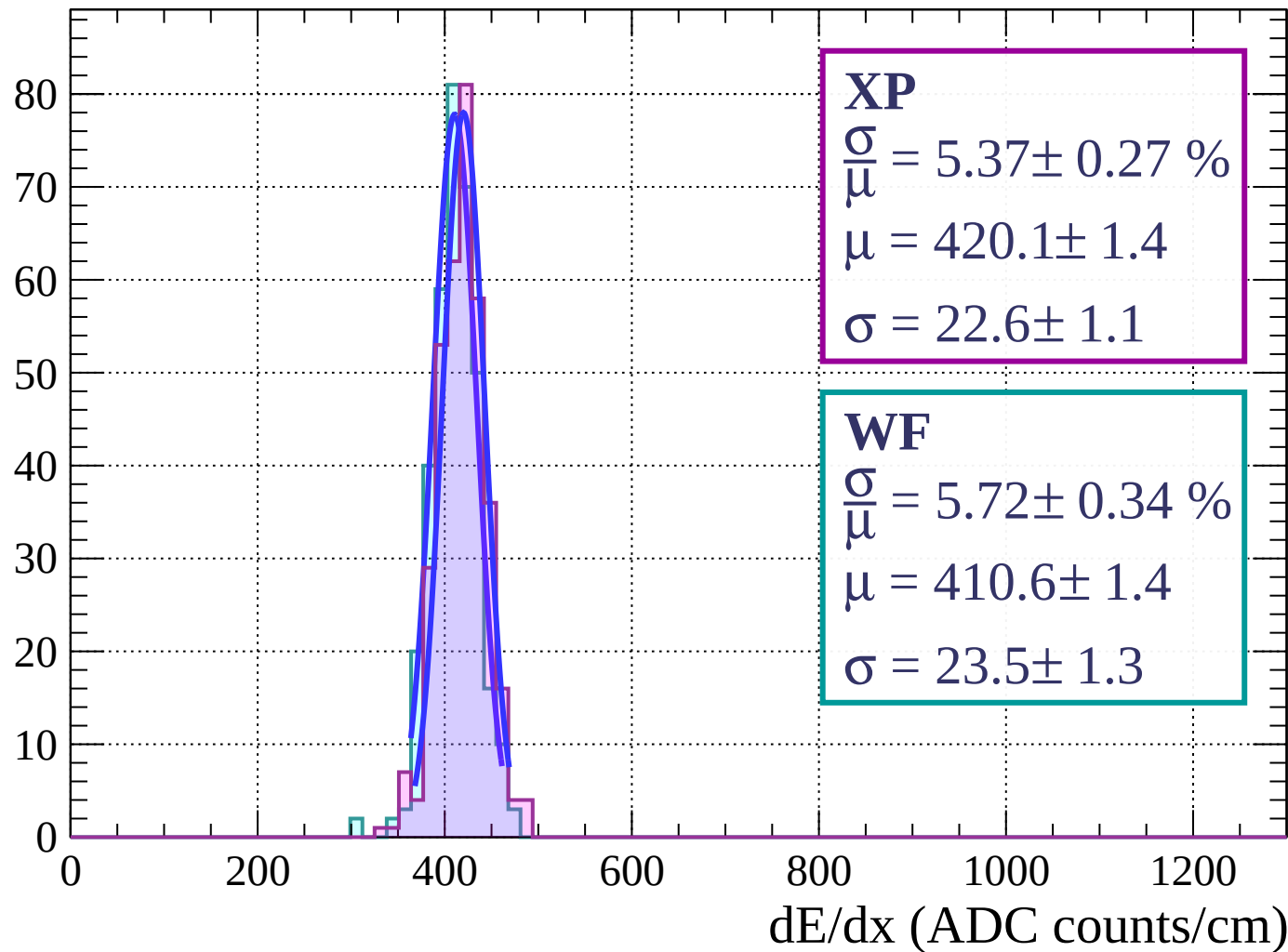
$$\mu = 417.5 \pm 0.9$$

$$\sigma = 23.9 \pm 1.0$$

dE/dx (ADC counts/cm)

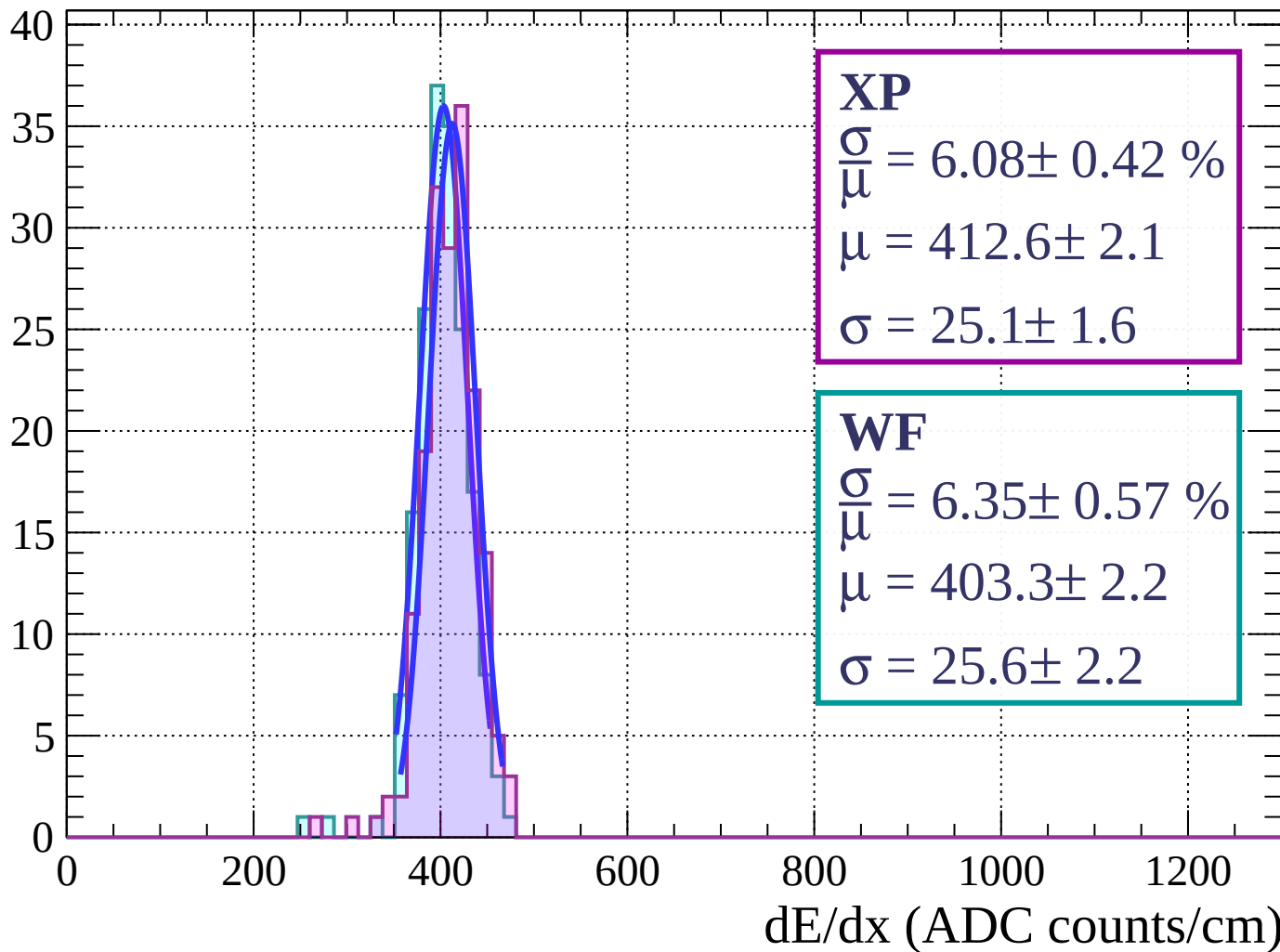
Energy loss | $15 < \theta < 18$

Count



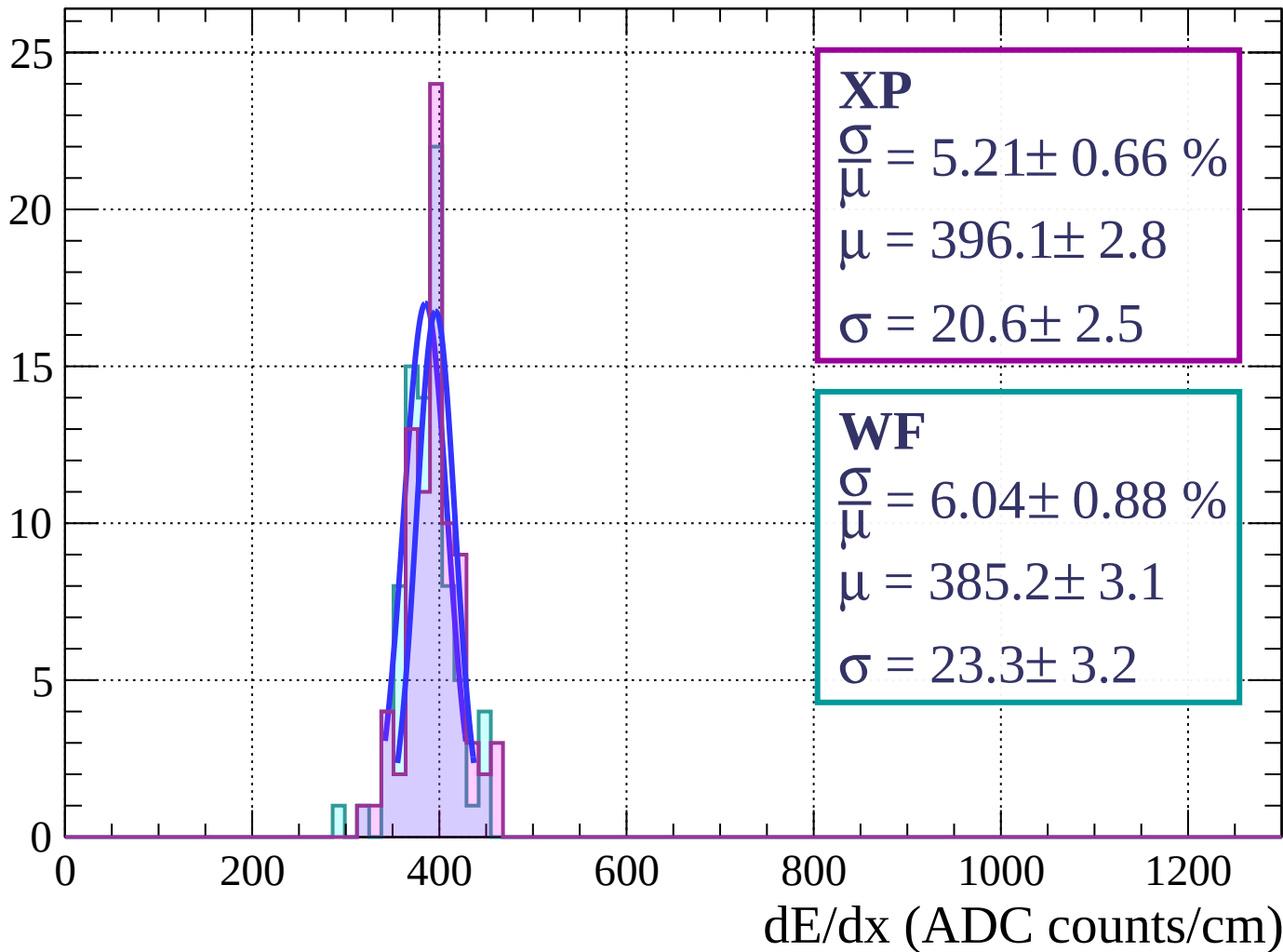
Energy loss | $18 < \theta < 21$

Count



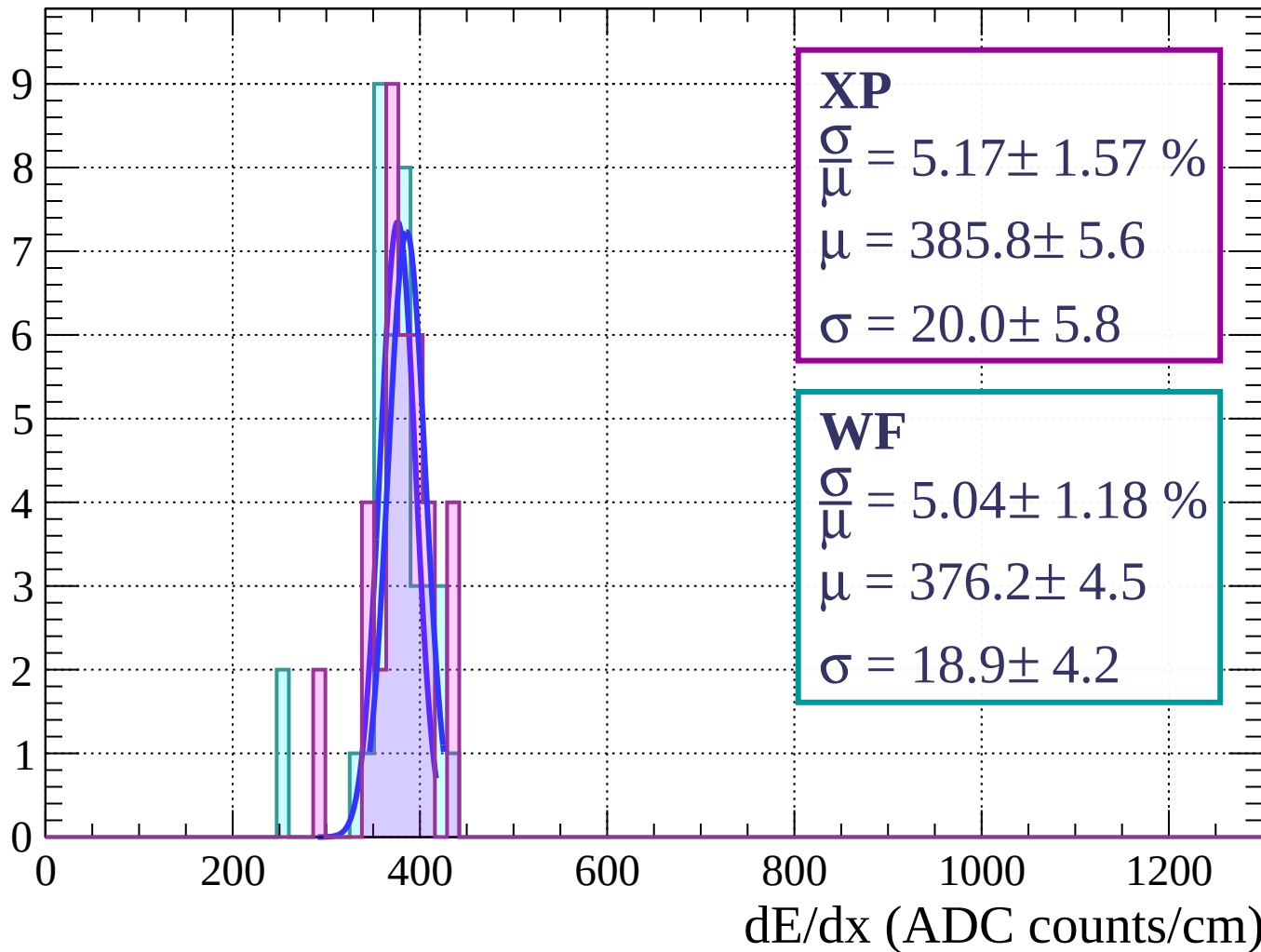
Energy loss | $21 < \theta < 24$

Count



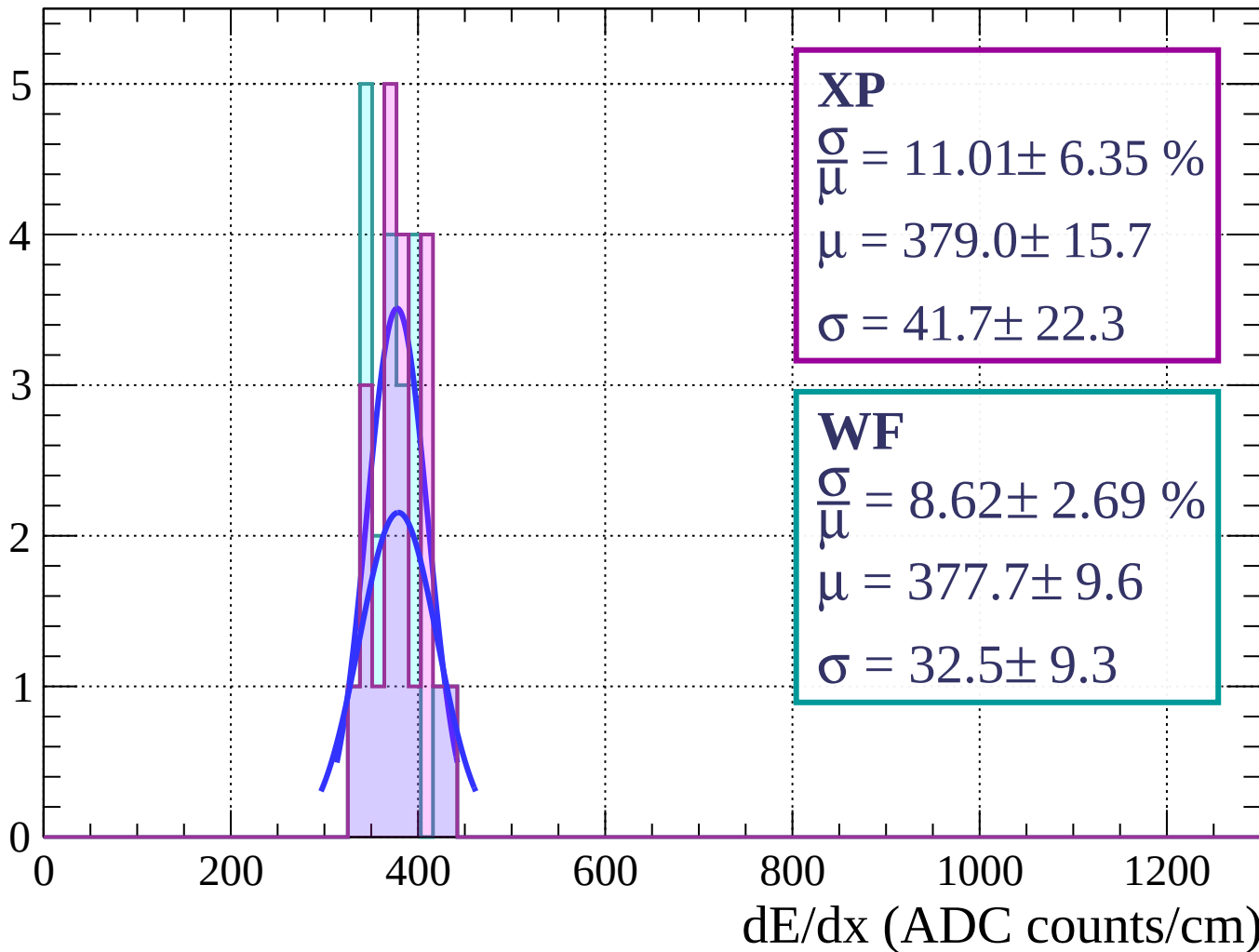
Energy loss | $24 < \theta < 27$

Count



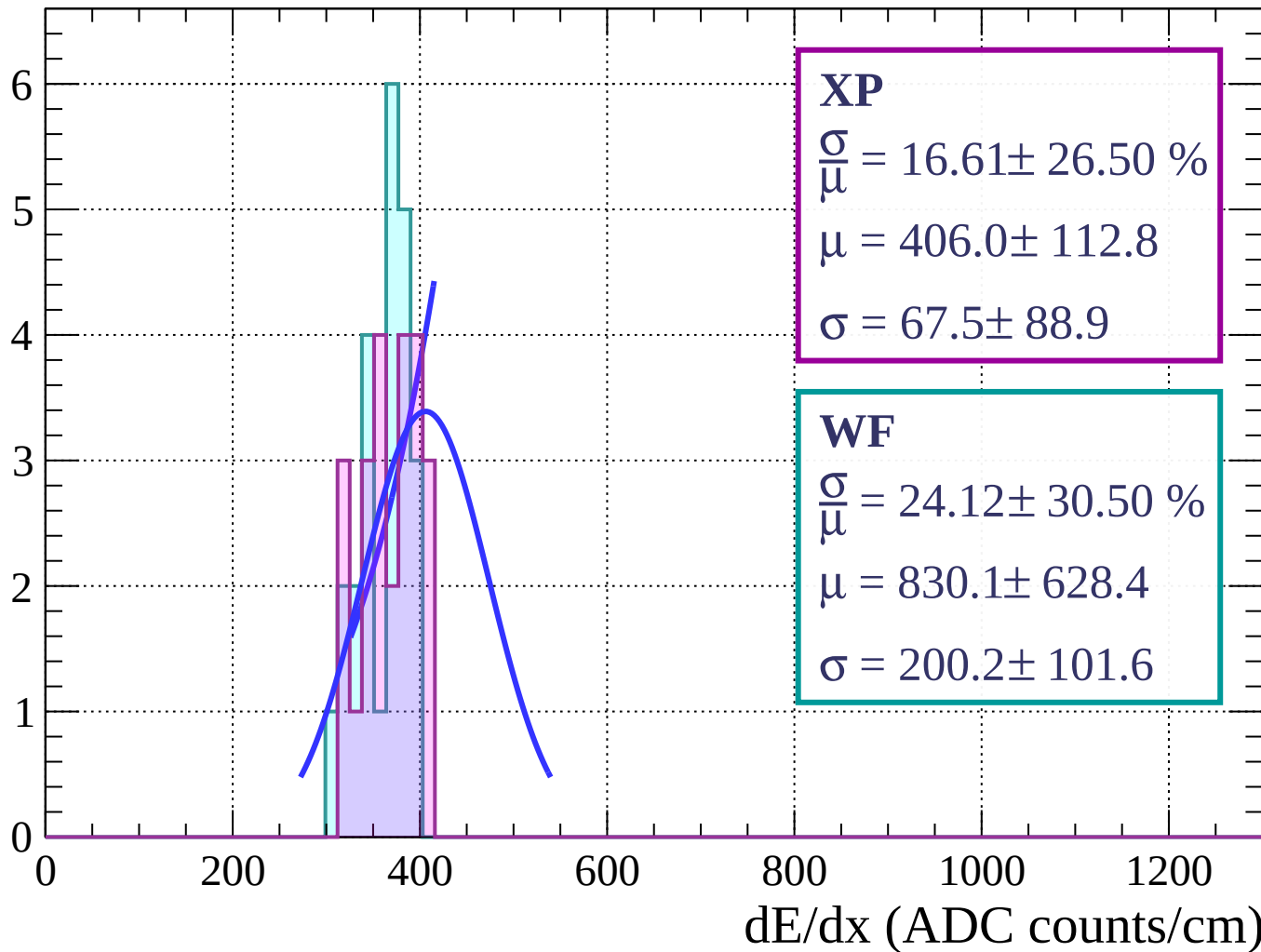
Energy loss | $27 < \theta < 30$

Count



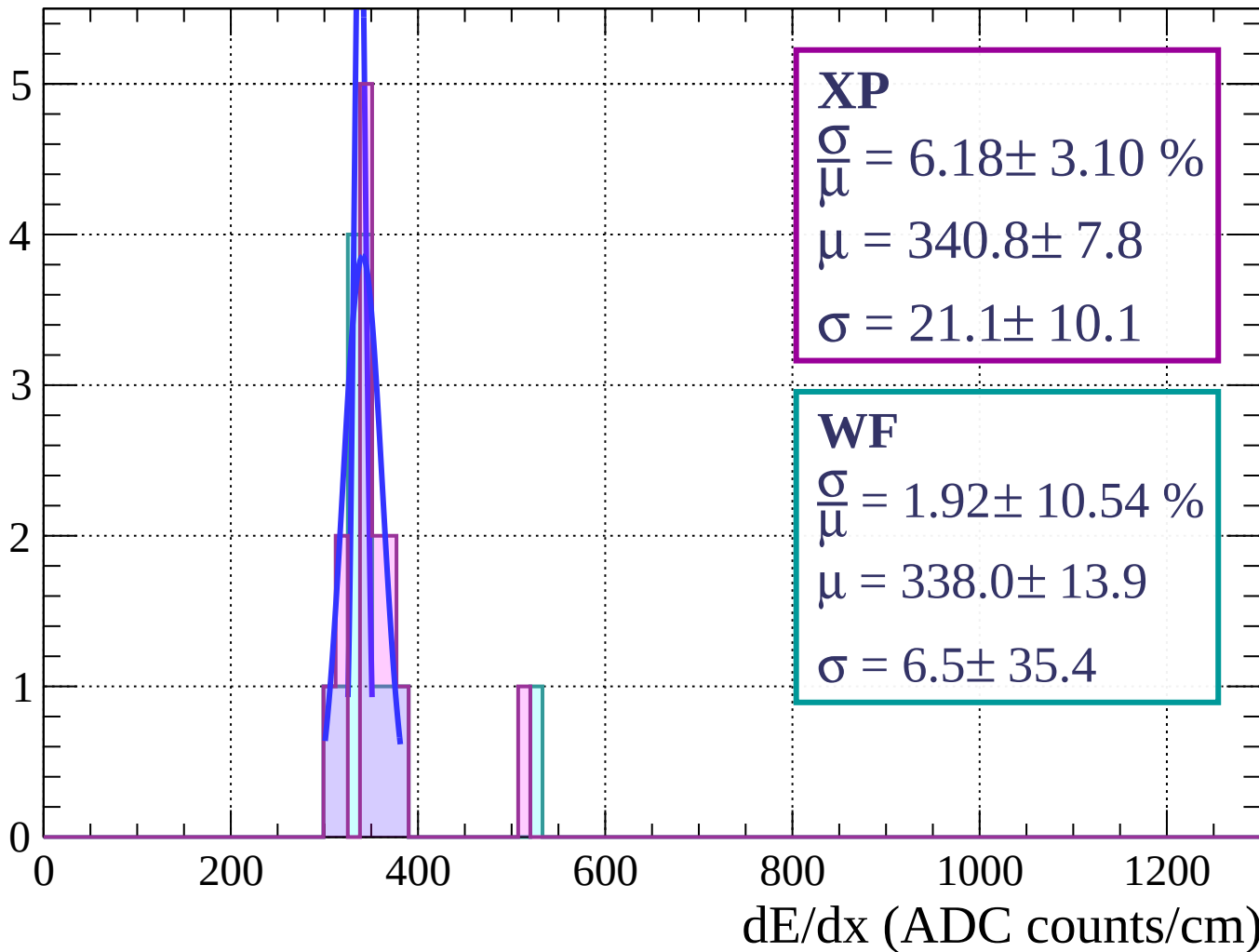
Energy loss | $30 < \theta < 33$

Count



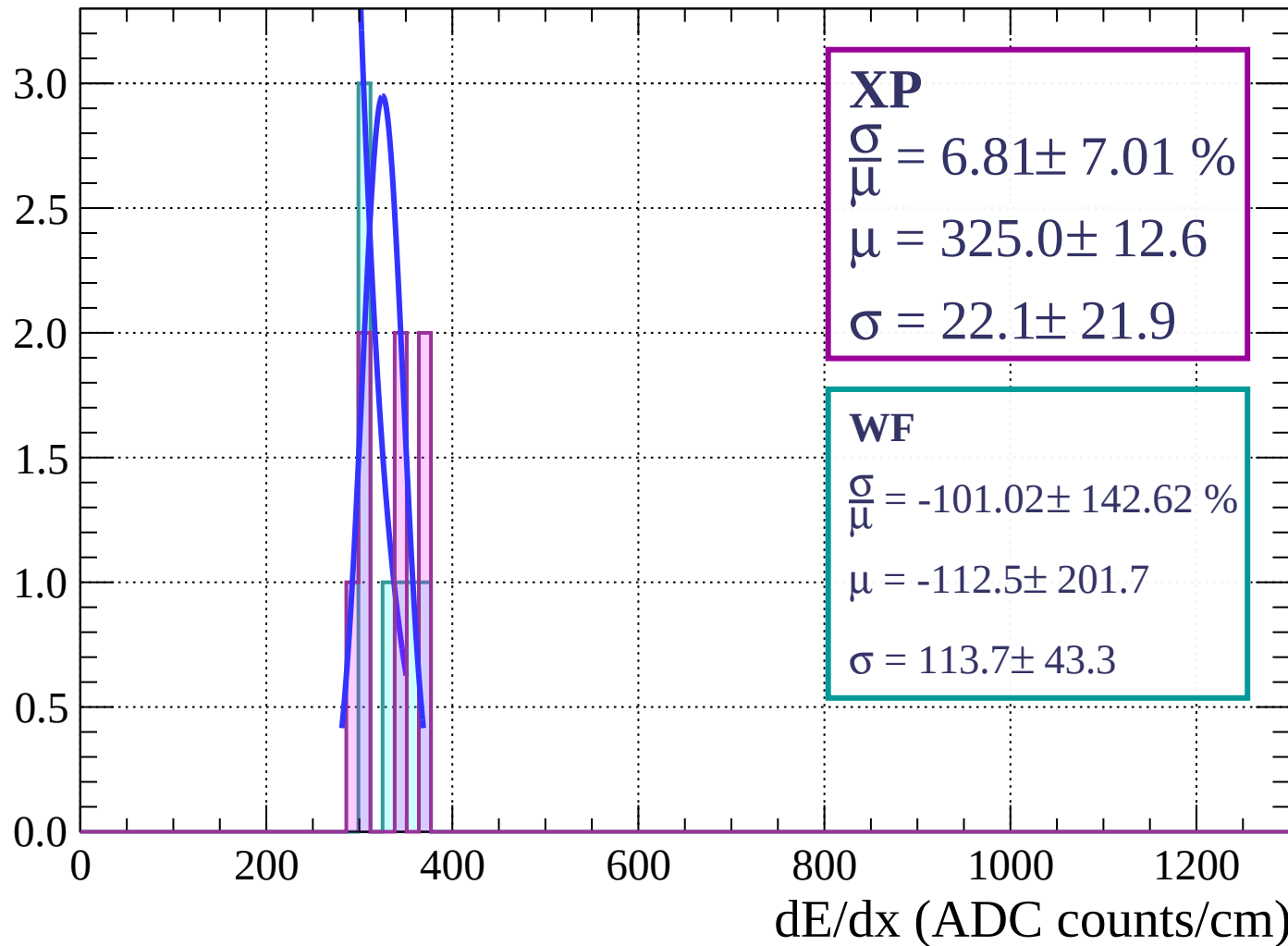
Energy loss | $33 < \theta < 36$

Count



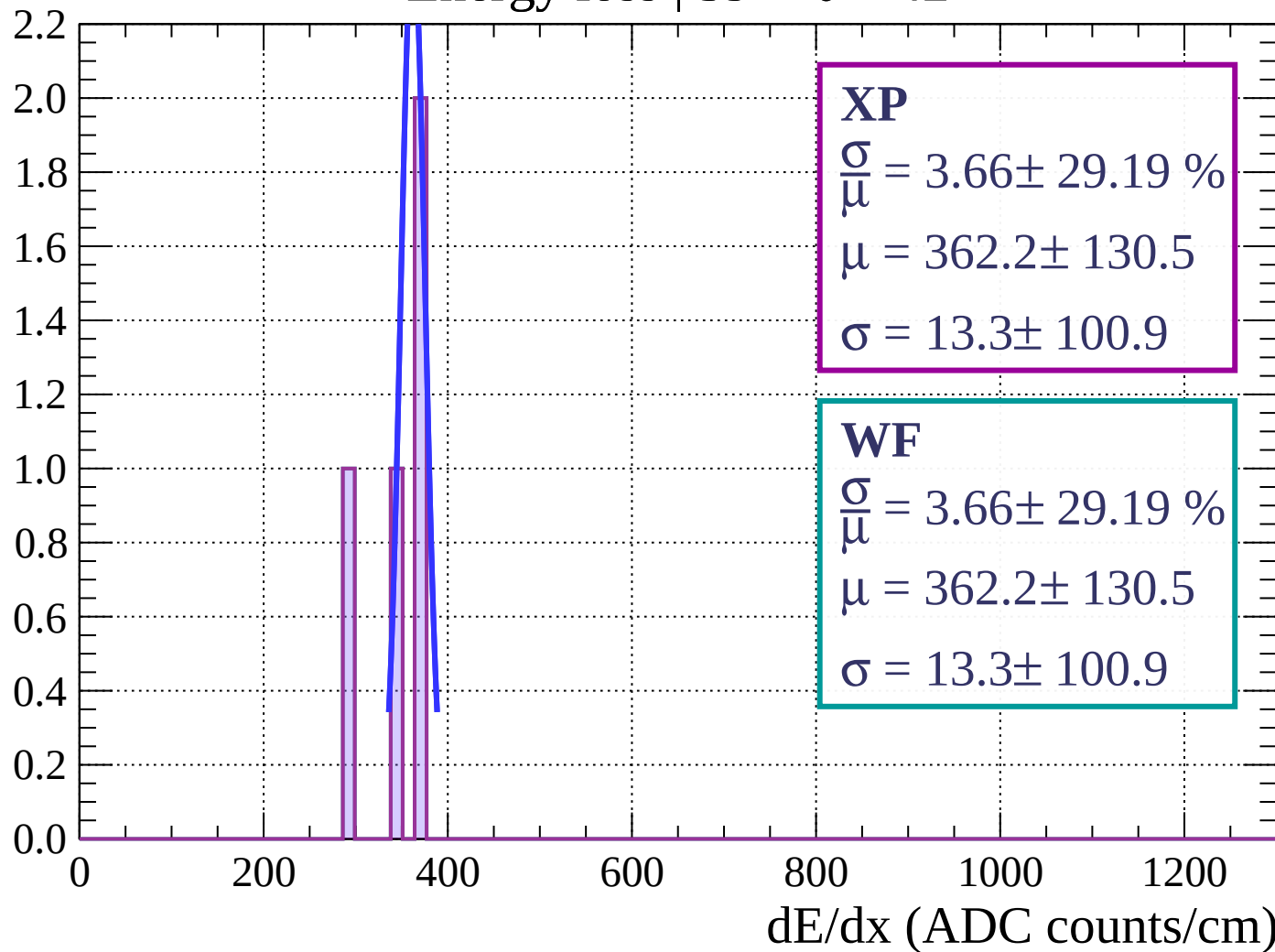
Energy loss | $36 < \theta < 39$

Count



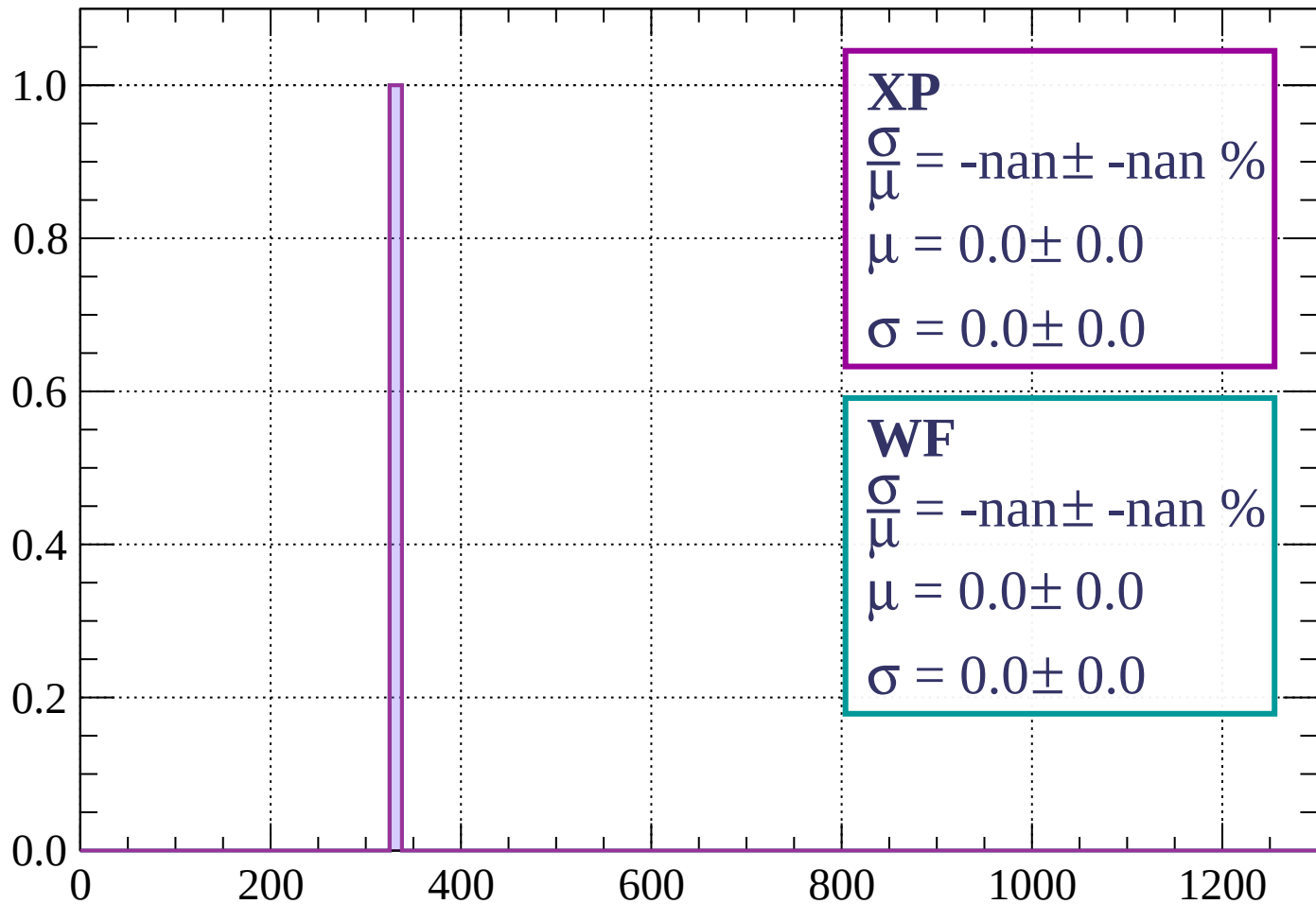
Energy loss | $39 < \theta < 42$

Count



Energy loss | $42 < \theta < 45$

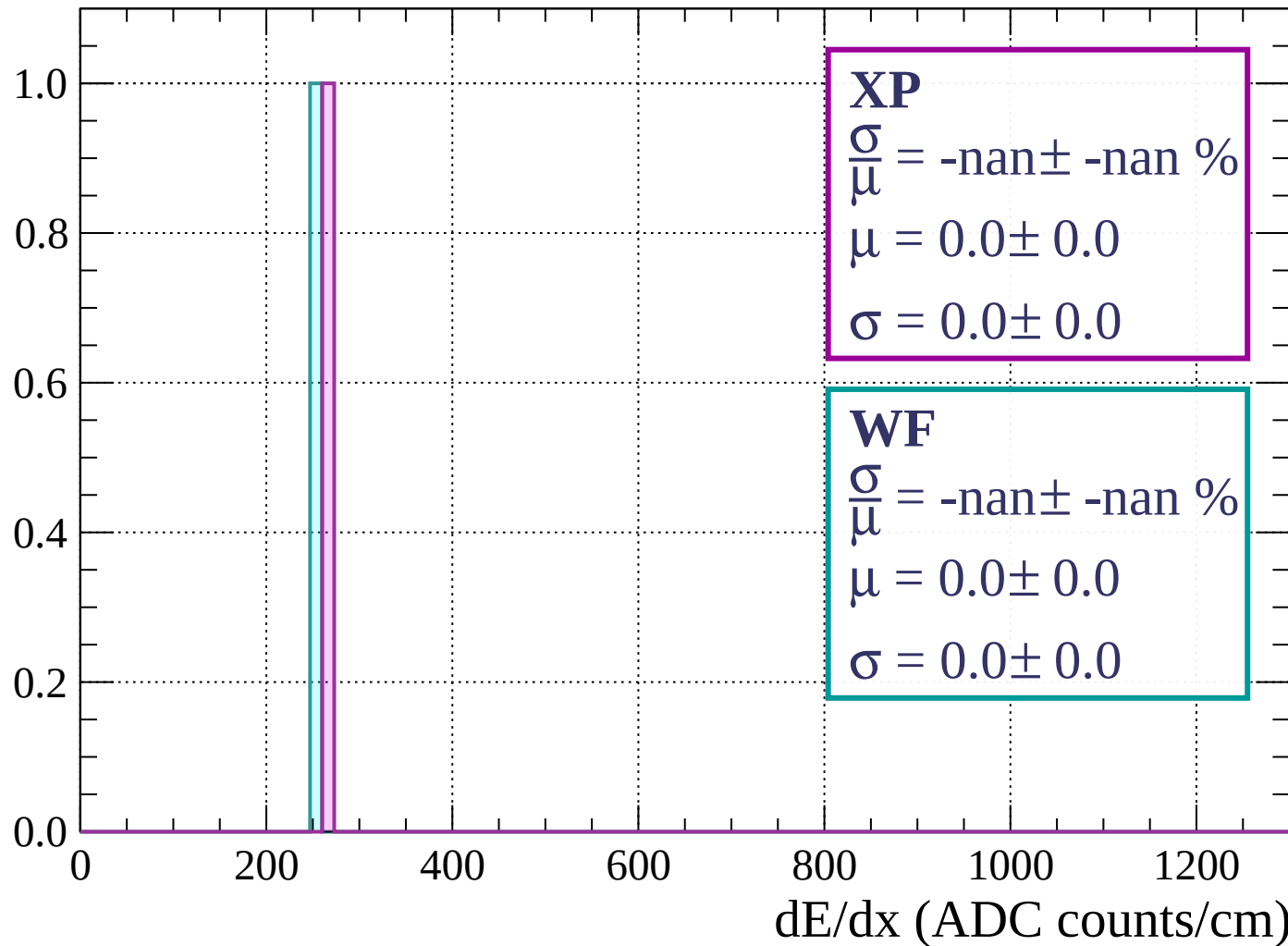
Count



dE/dx (ADC counts/cm)

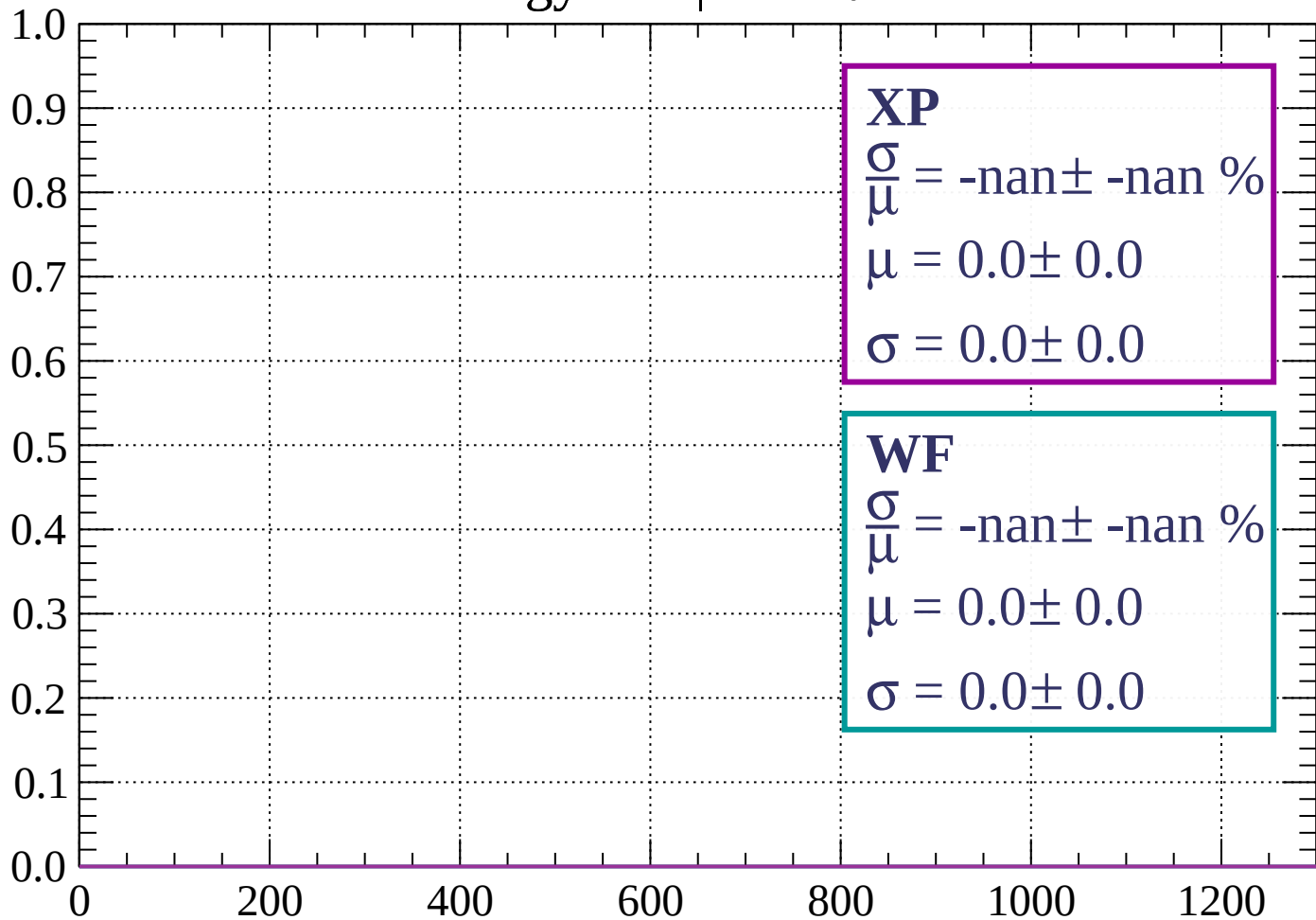
Energy loss | $45 < \theta < 48$

Count



Energy loss | $48 < \theta < 51$

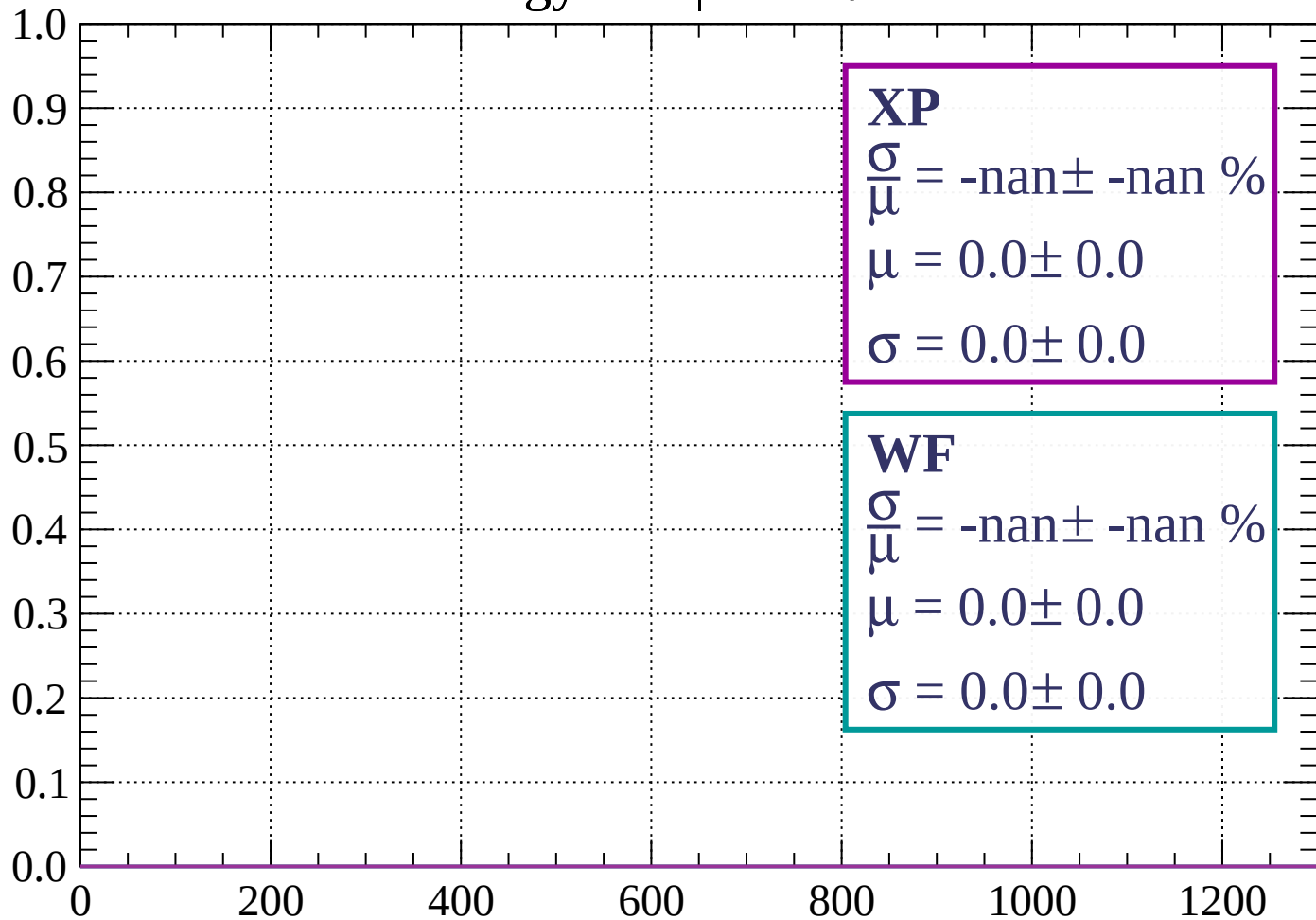
Count



dE/dx (ADC counts/cm)

Energy loss | $51 < \theta < 54$

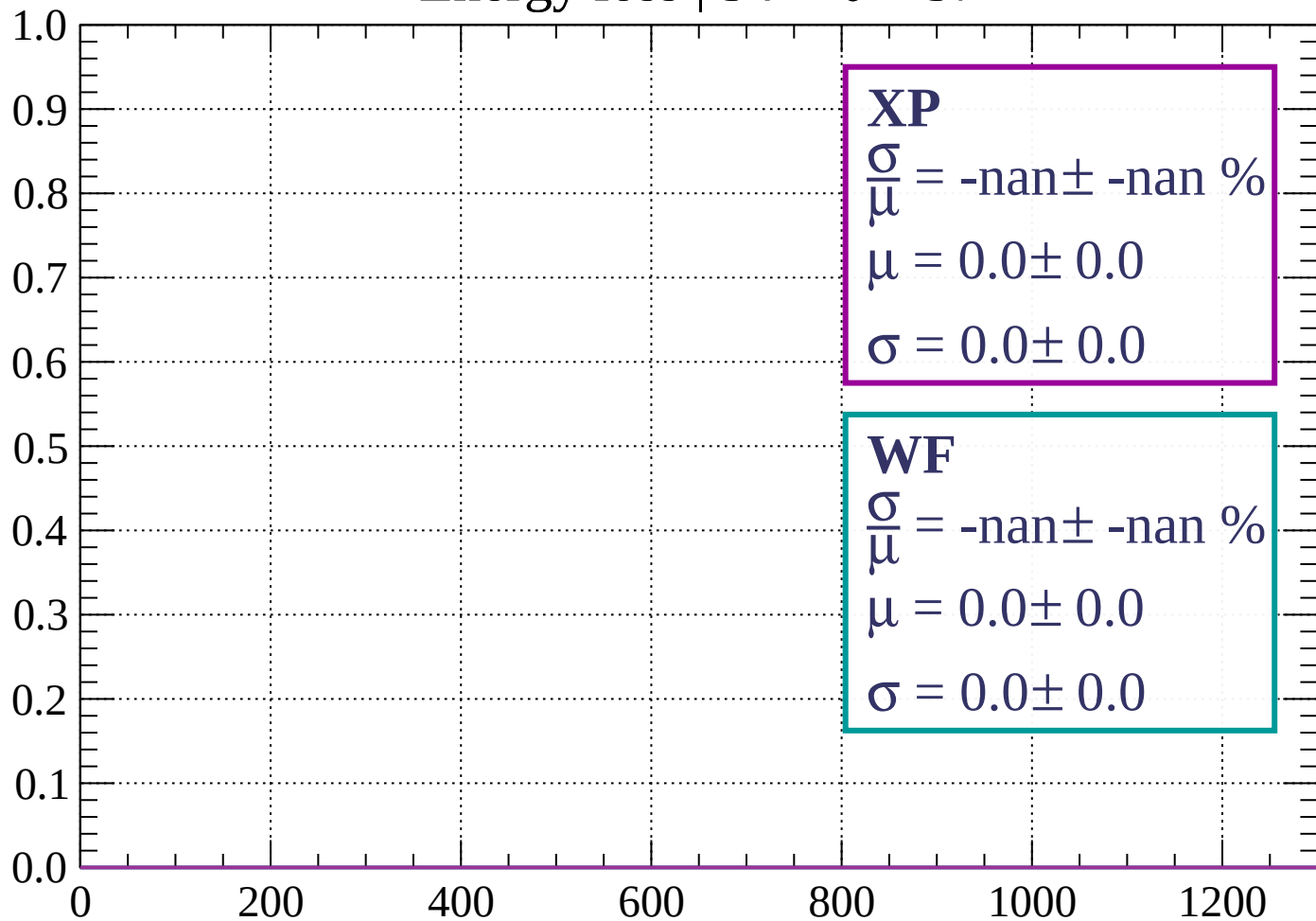
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 57$

Count



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

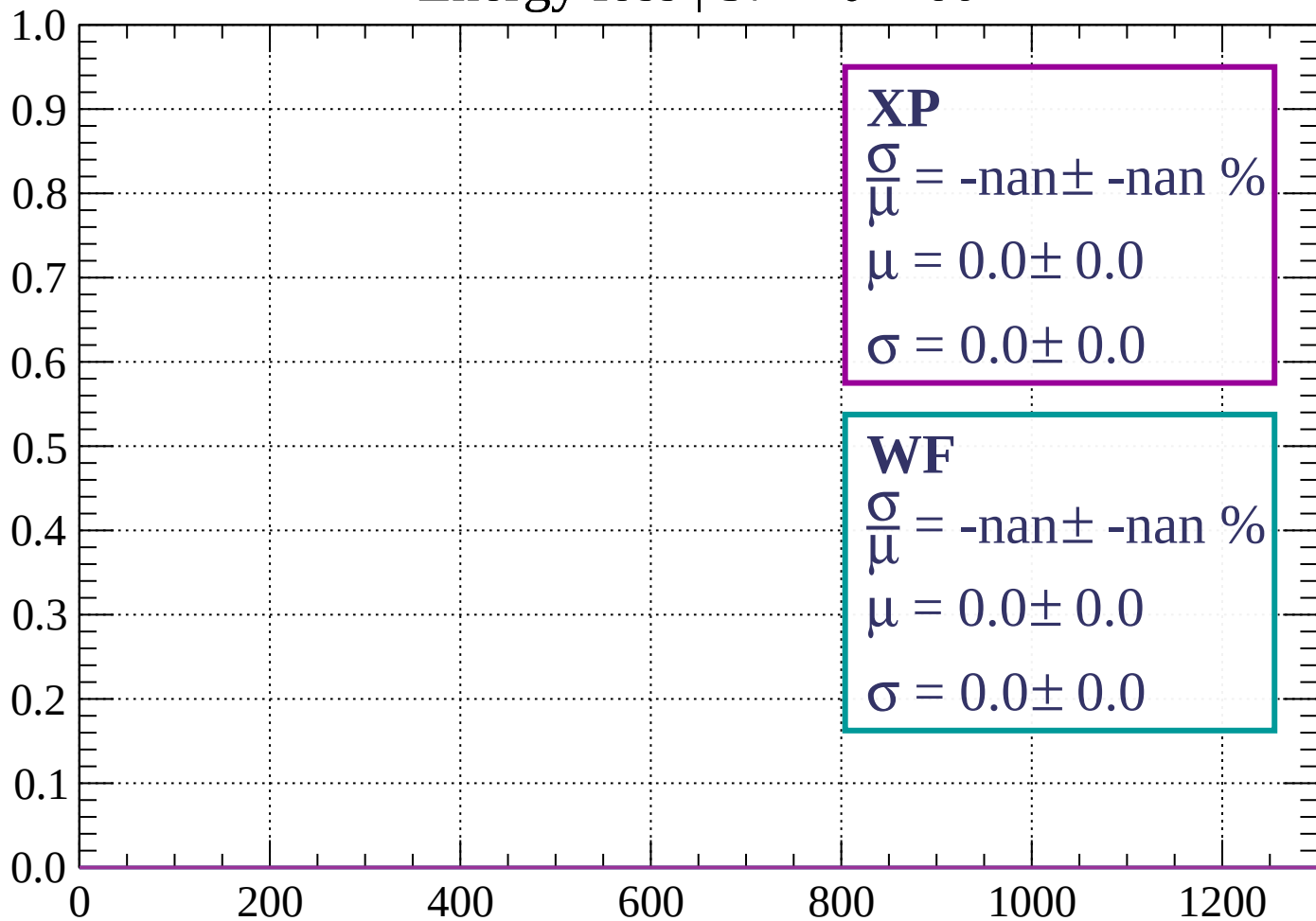
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $57 < \theta < 60$

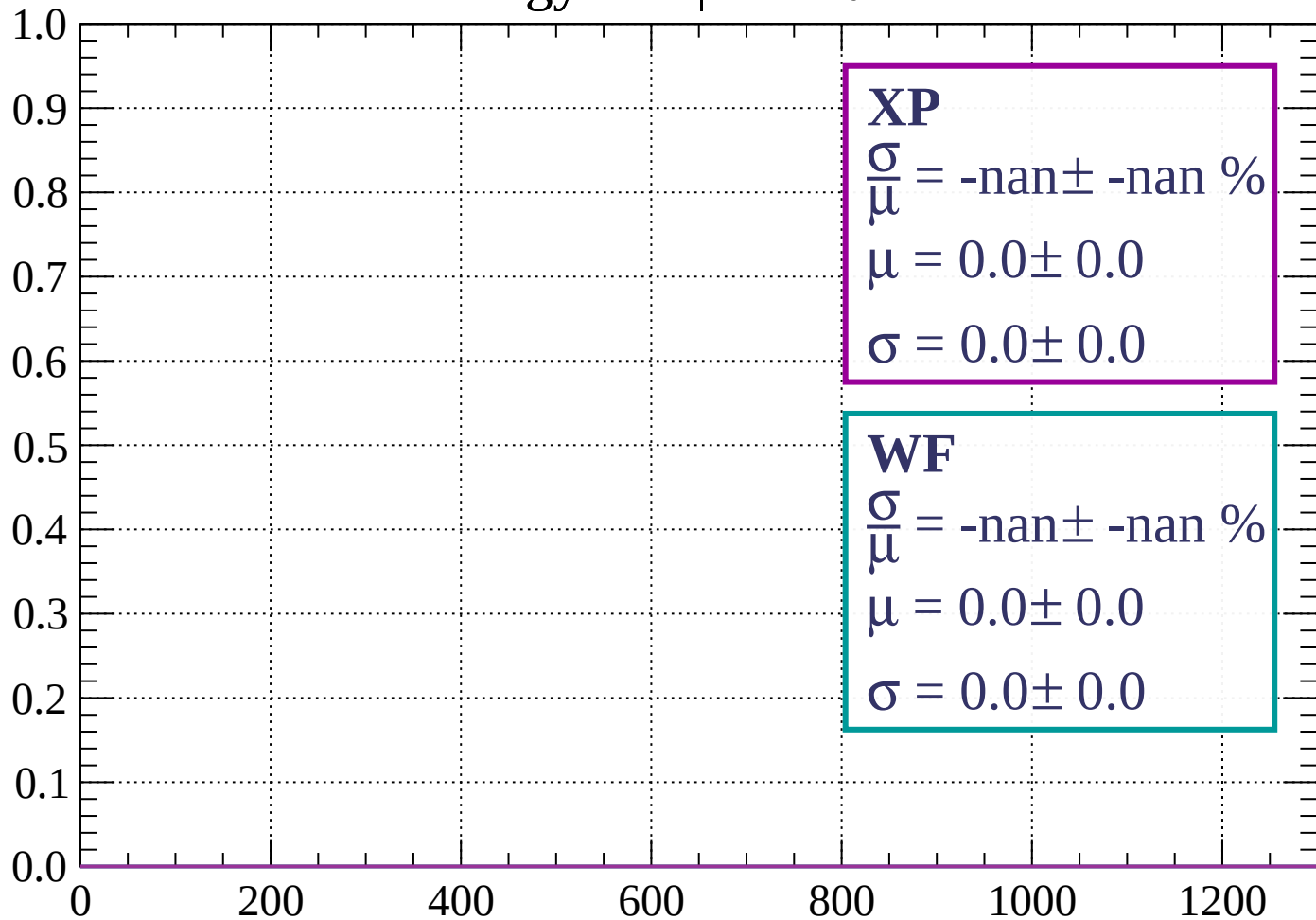
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 63$

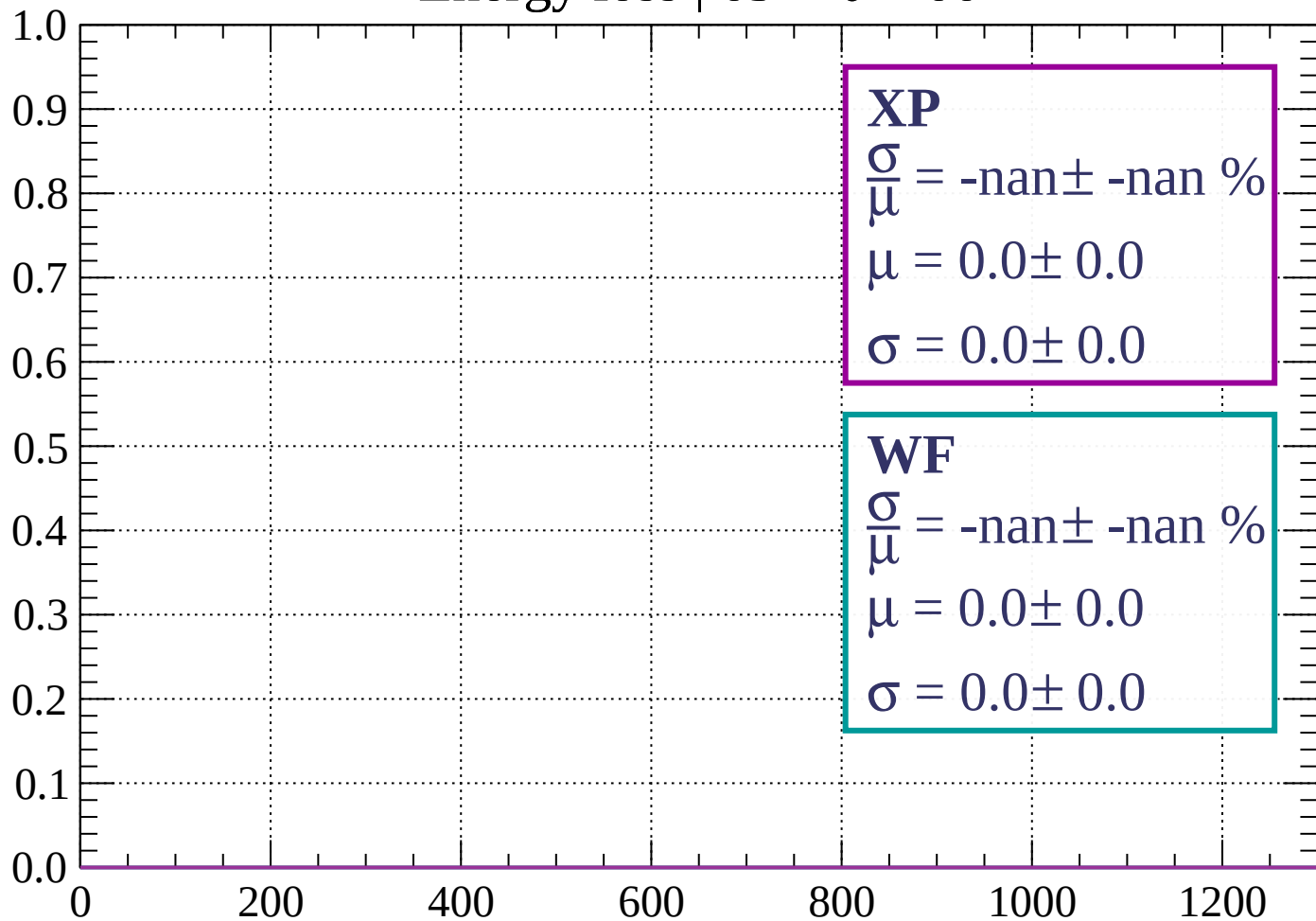
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

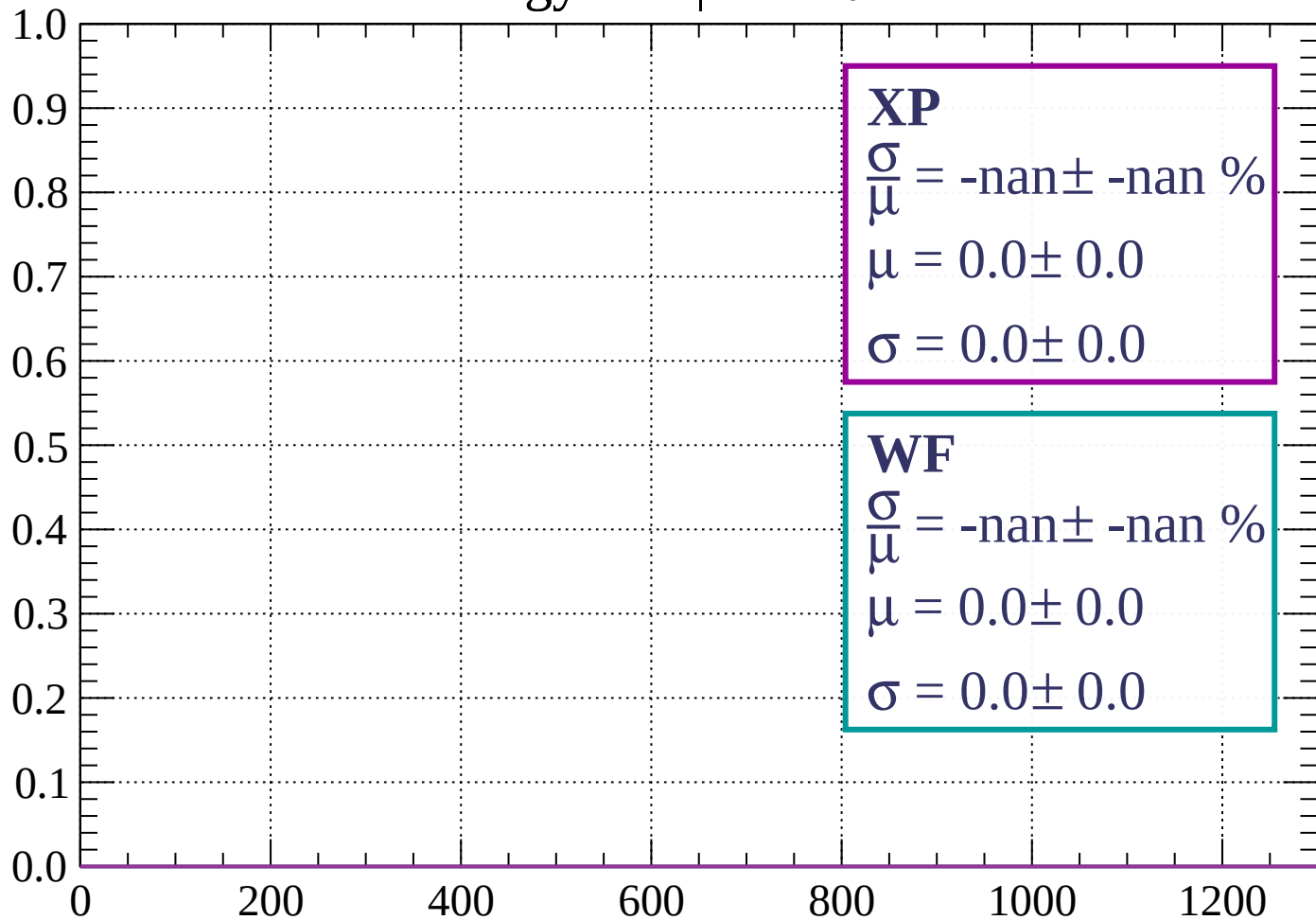
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

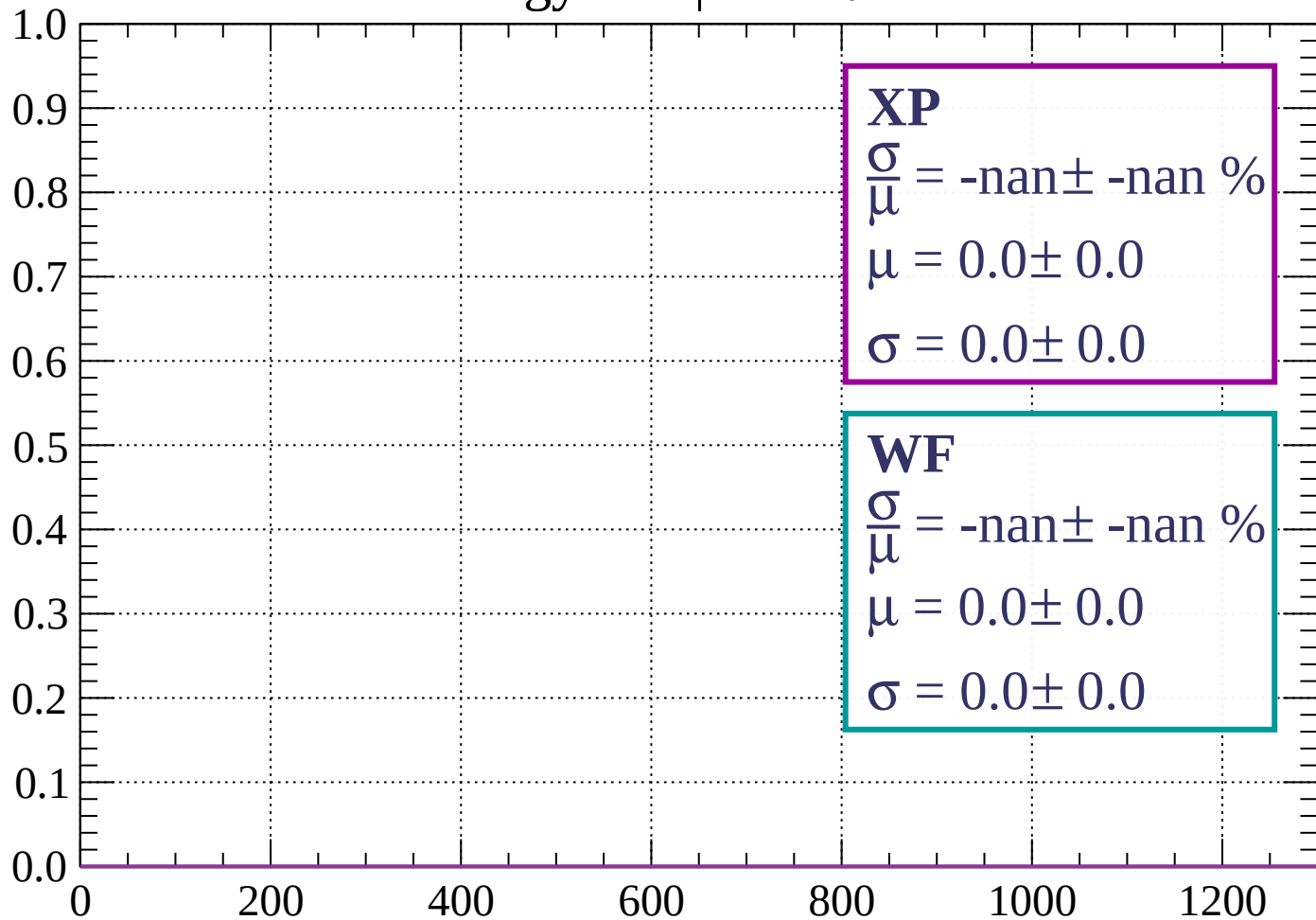
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

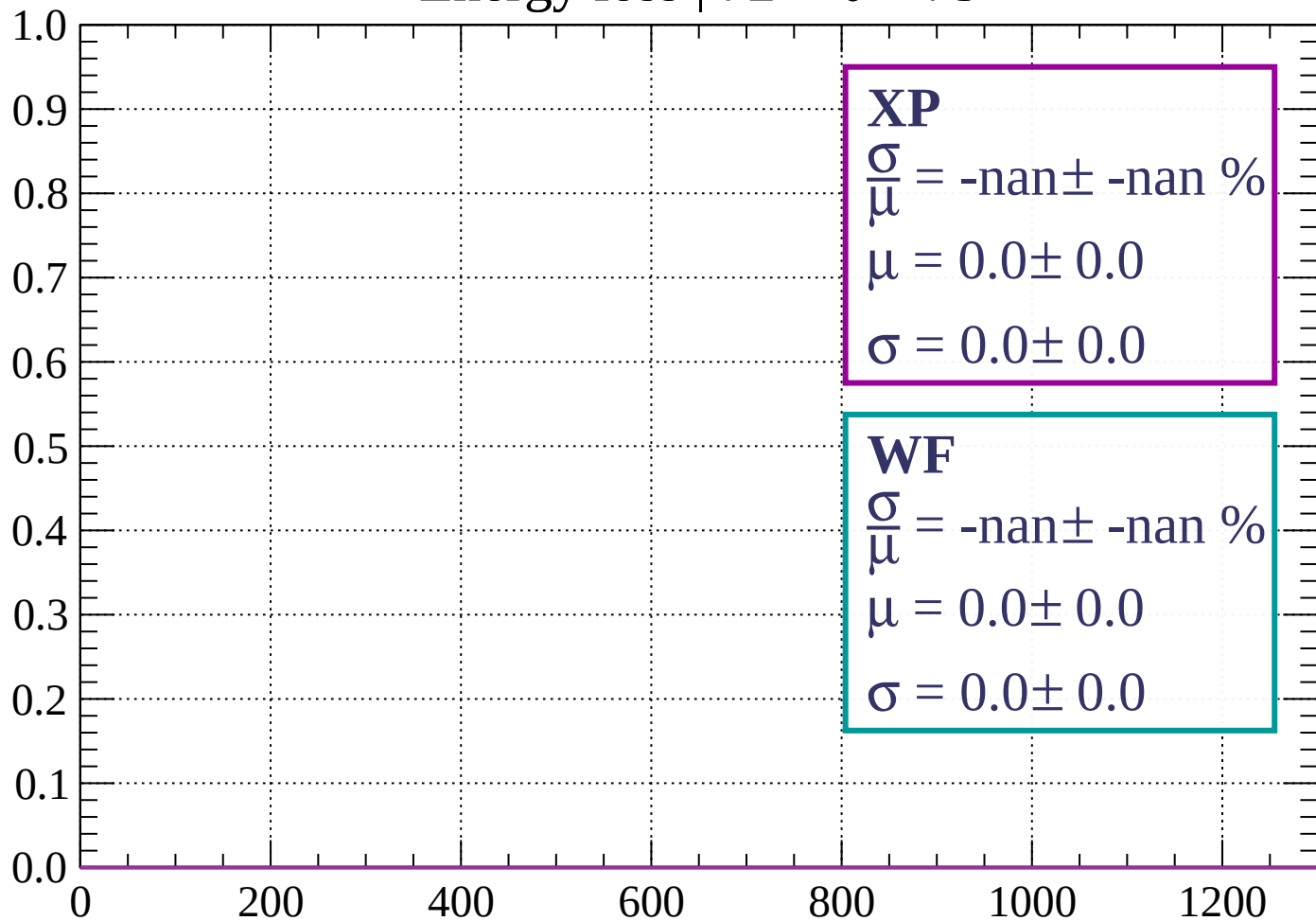
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

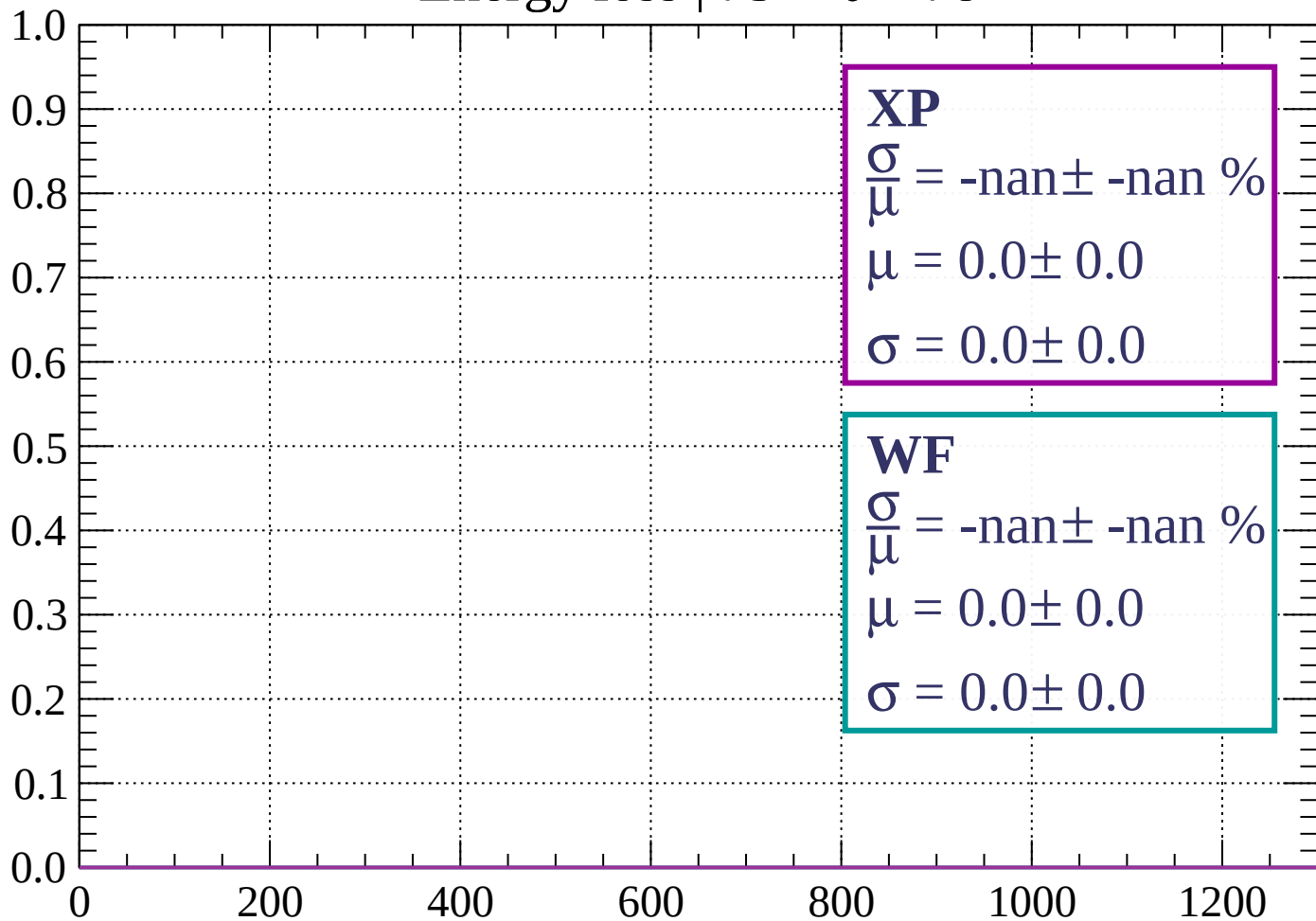
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

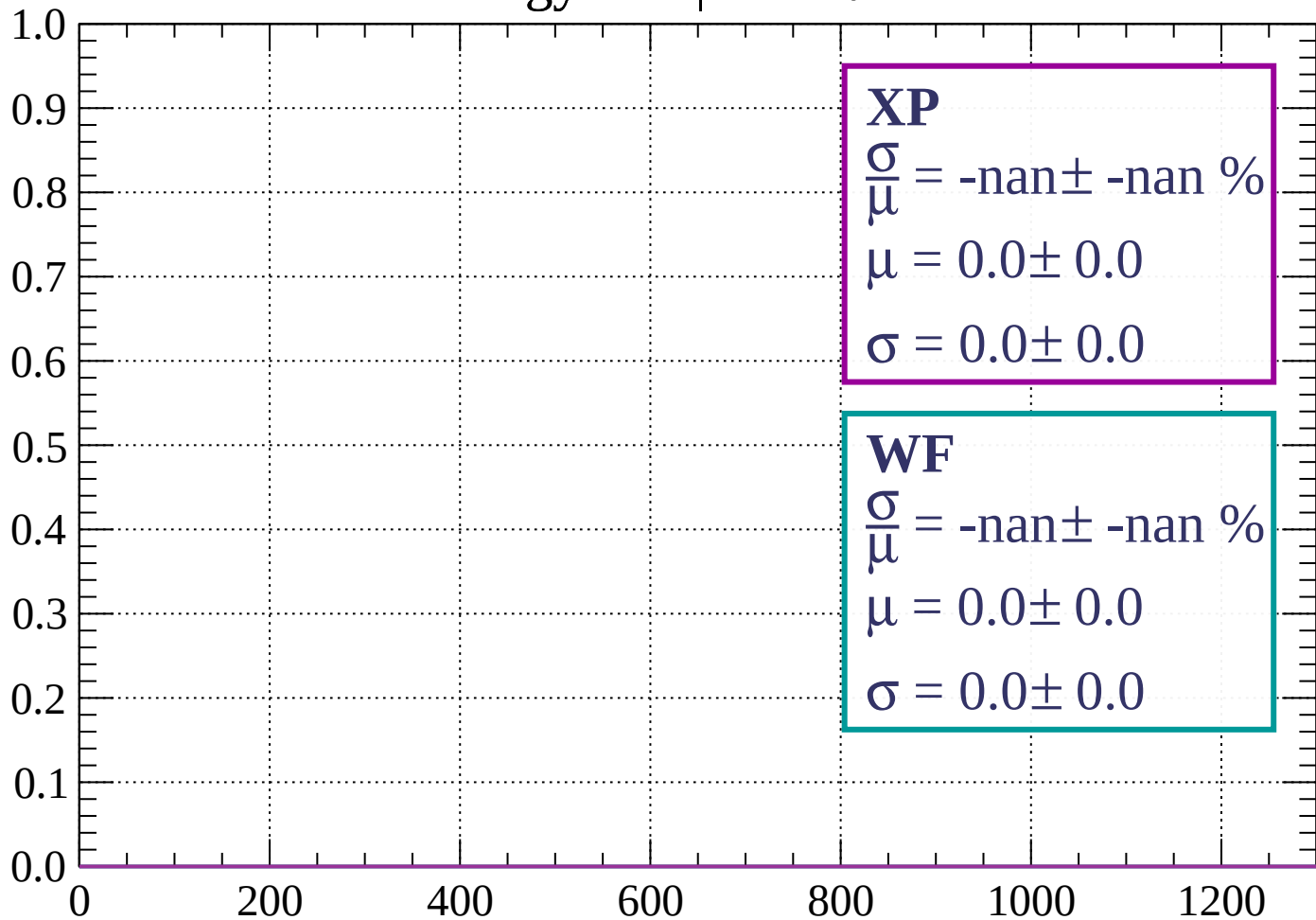
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

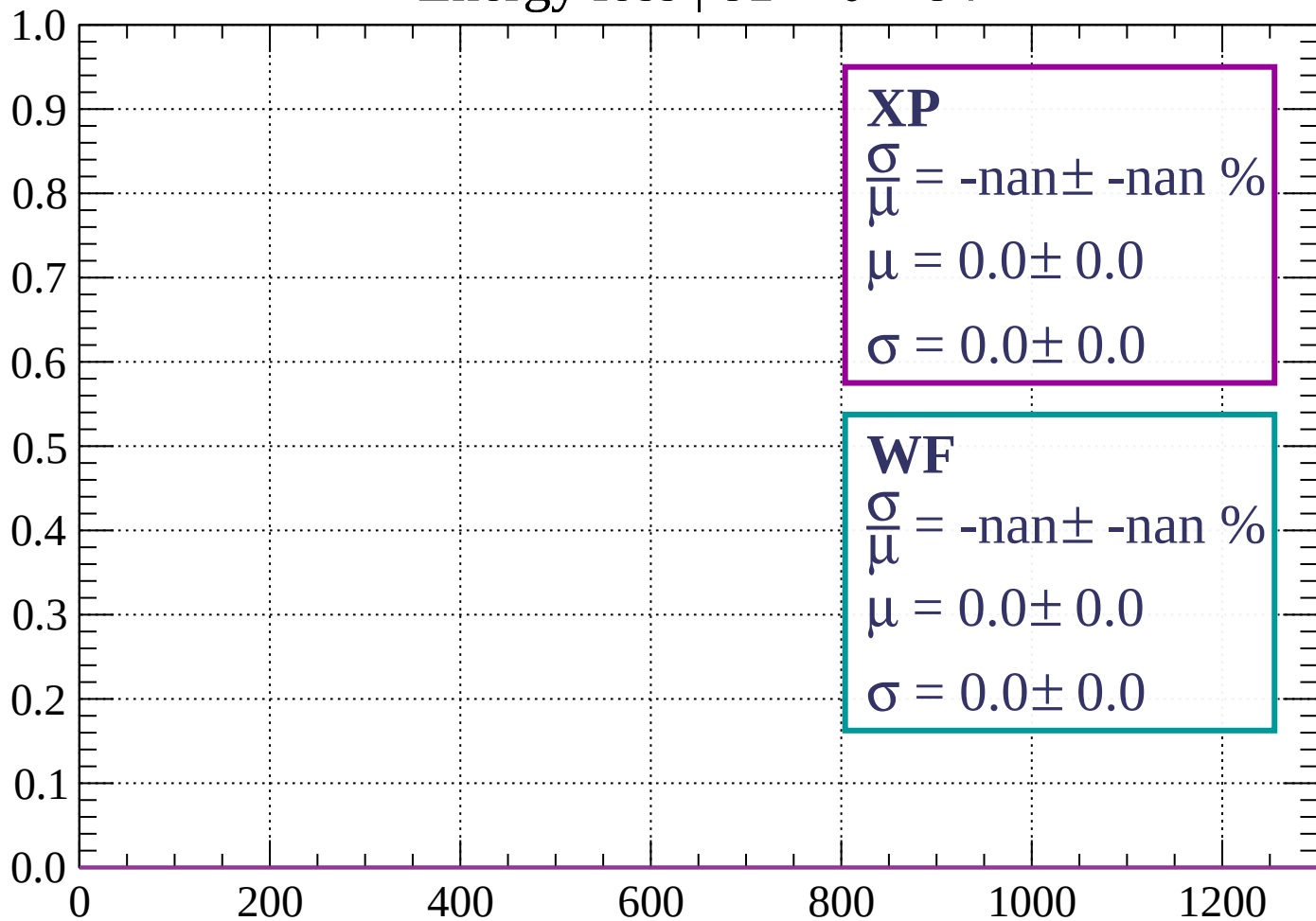
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

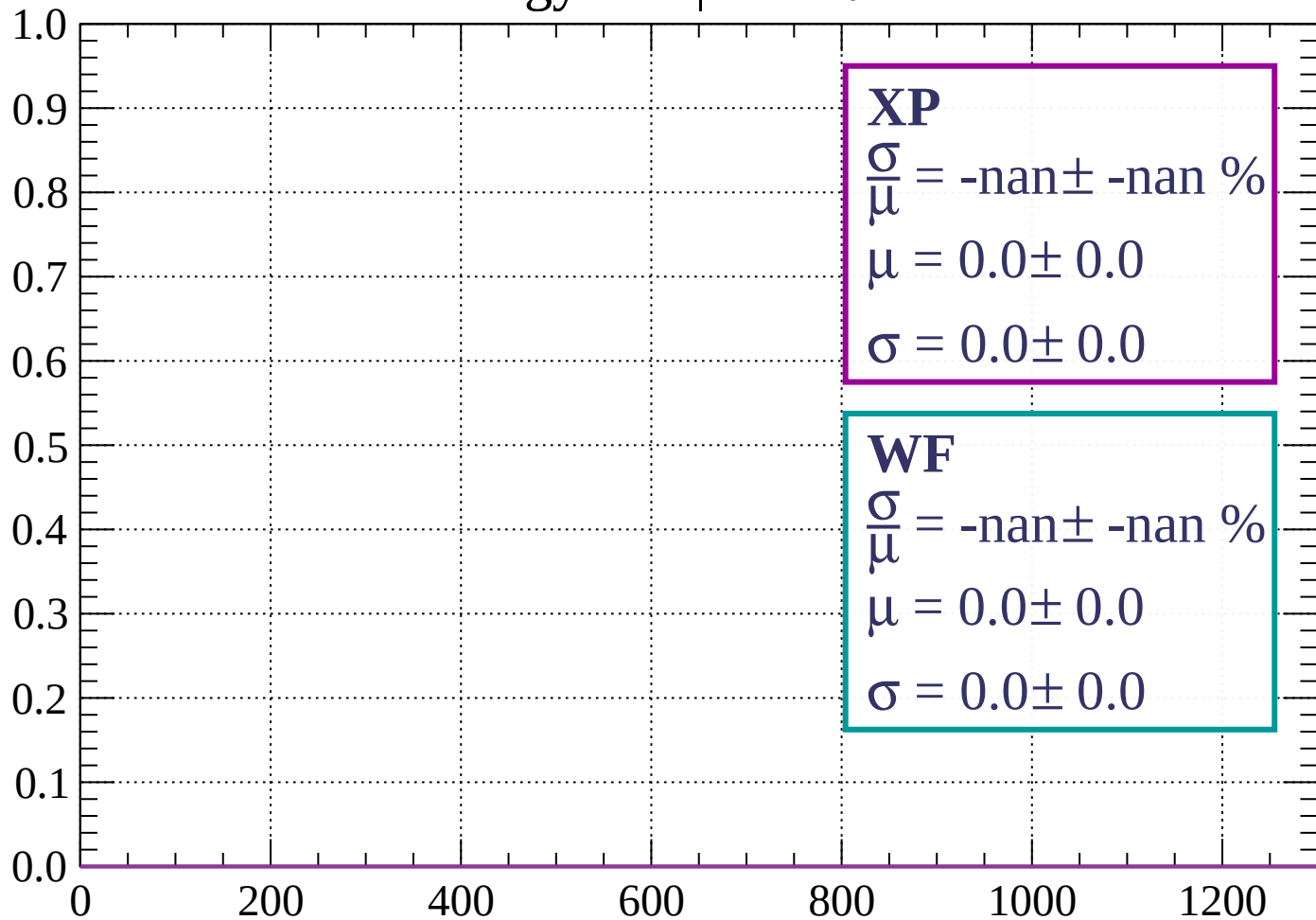
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

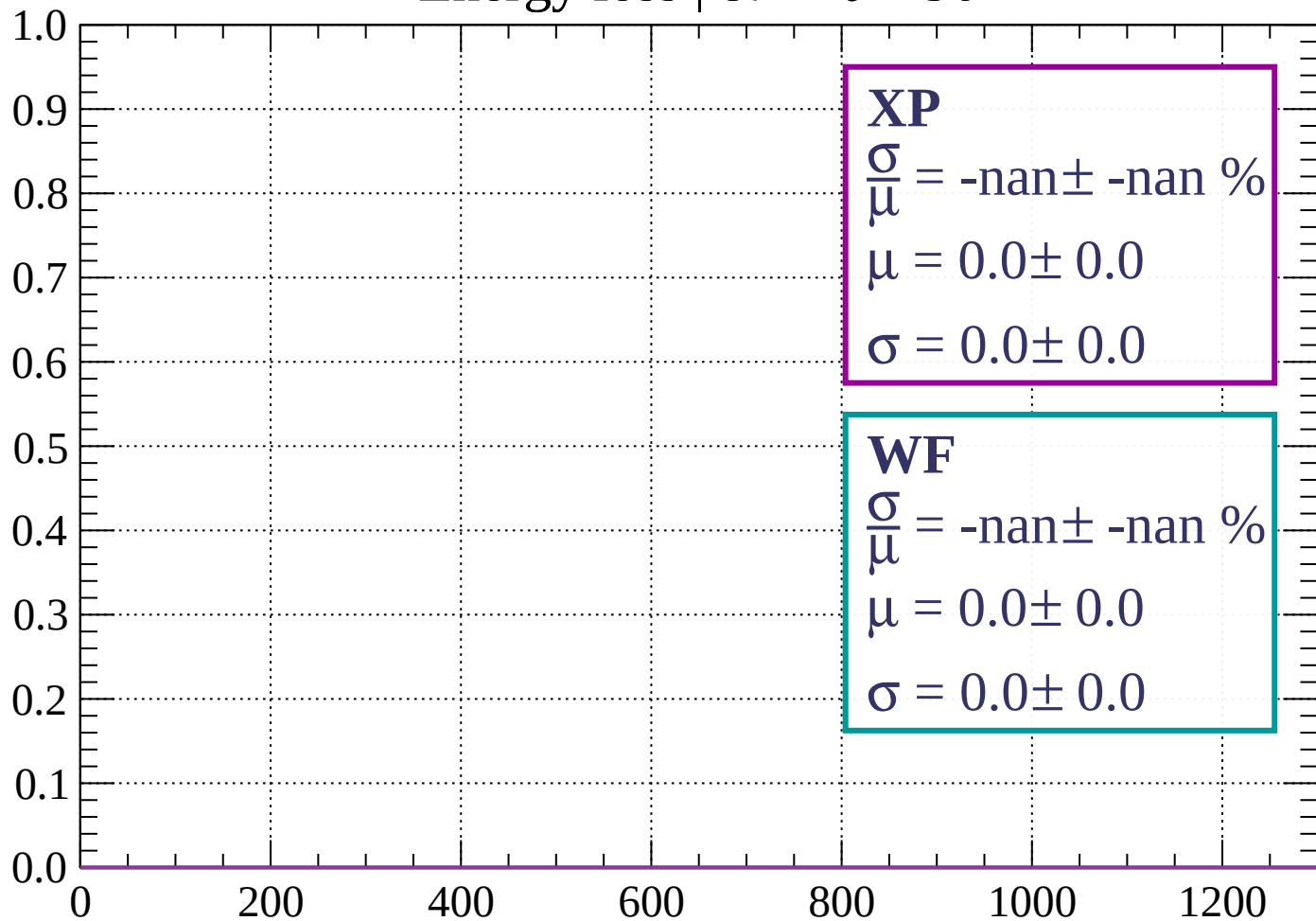
Count



dE/dx (ADC counts/cm)

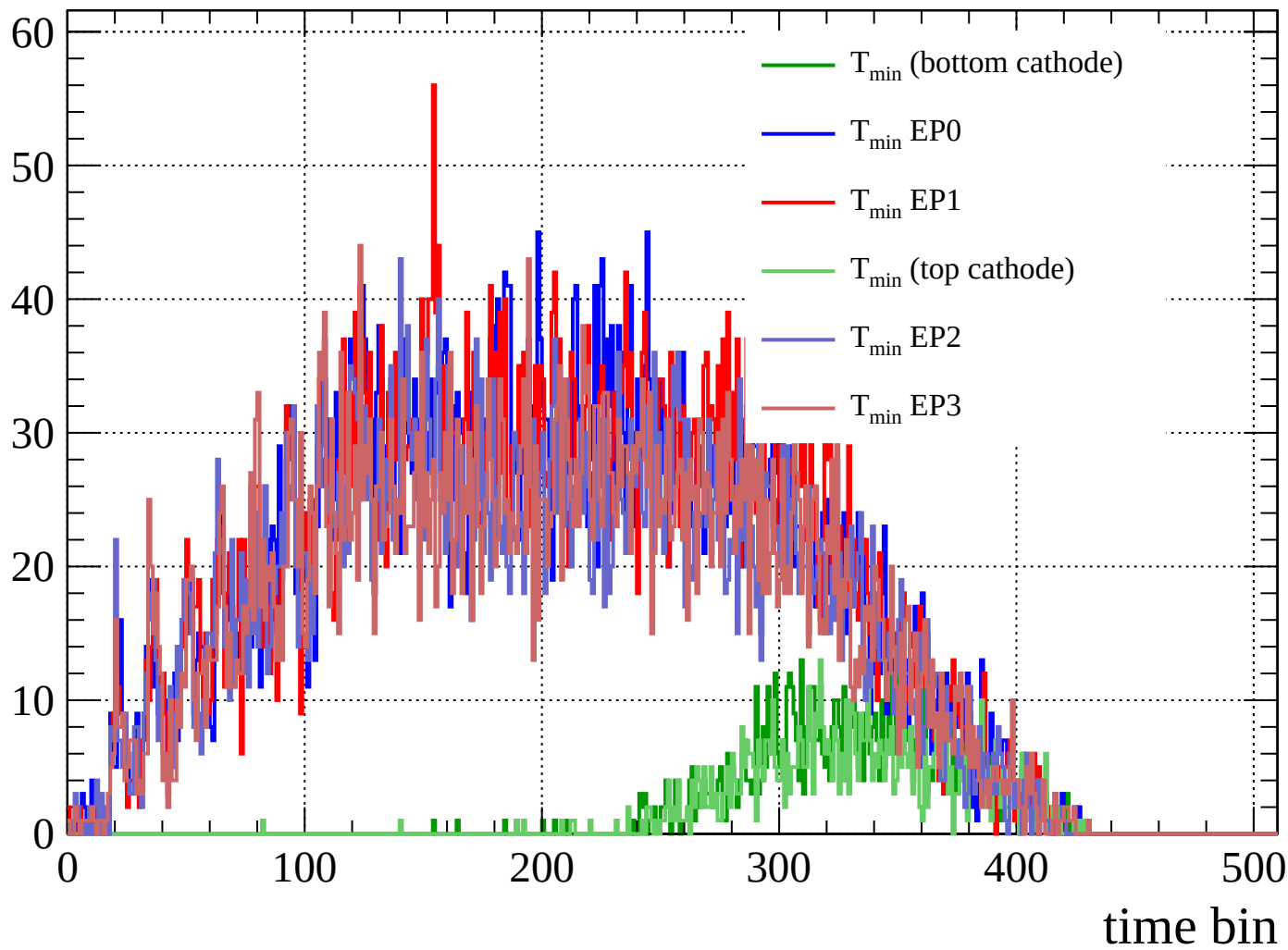
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

